Artificial Intelligence

Vrije Universiteit Amsterdam - Faculteit der Bètawetenschappen - Artificial Intelligence - 2024-2025

Artificial intelligence (AI) is the science of building and analysing intelligence demonstrated by machines. How could AlphaGo beat the best human Go player? Is it possible to build a social robots that seamlessly talk to an elderly with dementia, and help her to feel less lonely? Do you want to understand the algorithms Google applies for its search, or Netflix to recommend us the latest movies?

In the first year, students learn basics in Computer Science and Mathematics, such as programming and logic. They also get acquainted with some of the main concepts in Artificial Intelligence, such as modelling, knowledge representation and machine learning, as well as interaction between humans and machines. In the second year, students are able to choose between the Intelligent Systems track, which focuses on methods to build intelligent agents, and the Socially Aware Computing track, with an emphasis on AI in a social context, with applications in health and law. In the third year, students follow one of the many minors offered by the university or the faculty, including an educational minor (which gives them a second-degree teaching qualification for secondary school), and our Situated AI minor, but many students take the opportunity to study abroad for half a year. The final semester is mostly dedicated to the final bachelor project.

Info

Niveau	Bachelor
Taal	Engels
Duur	3 years
Vorm	Voltijd
Studiepunten	180 EC
Faculteit	Faculteit der Bètawetenschappen

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Artificial Intelligence year 1

Omschrijving

The first year of the AI Bachelor's degree programme focuses on basics of Computer Science, Psychology and Mathematics, such as programming and
logic. During this year students are introduced to some of the main concepts in Artificial Intelligence, including modelling, knowledge representation and algorithms, as well as the interaction between humans and machines.

Vakken

Naam vak	Periode	Credits	Code
English Language Test	P1	0.00EC	VU_ELT
Introduction to Artificial Intelligence	P1	3.00EC	X_401089
Introduction to Psychology and its Methods	P1	6.00EC	XB_0069
Introduction to Python Programming	P1+2	6.00EC	XB_0082
Computational Thinking	P2	3.00EC	X_400475
Intelligent Systems	P2	6.00EC	XB_0031
Project Intelligent Systems	P3	6.00EC	X_401076
Dynamic Modelling for Human-Centered Systems	P4	6.00EC	XB_0036
Logic and Sets for AI	P4	6.00EC	XB_0083
Academic Writing (BETA)	P5	3.00EC	L_ETBAALG006
History of AI	P5	3.00EC	XB_0113
Human-Computer Interaction	P5	6.00EC	XB_0013

Opleidingsdelen

- Artificial Intelligence year 1 constained choice

Artificial Intelligence year 1 constained choice

Vakken

Naam vak	Periode	Credits	Code
Applied Programming for AI	P6	6.00EC	XB_0102
Information Management	P6	6.00EC	X_401085

Artificial Intelligence year 2

Omschrijving

During the second year of the AI Bachelor's, students make a selection between the more technical track Intelligent Systems, which focuses on methods of building intelligent agents, and the track Socially Aware Computing, which focuses on human-computer interaction, and applications areas of AI in health and law. More information about the tracks can be found by clicking on the tracks.

Vakken

Naam vak	Periode	Credits	Code
Knowledge and Data	P1	6.00EC	X_400083
Linear Algebra and Calculus	P2	6.00EC	XB_0114
Multi-Agent Systems	P2	6.00EC	XB_0024
Machine Learning	P4	6.00EC	X_400154
Probability and Statistics	P4	6.00EC	XB_0115
Text Mining for AI	P5	6.00EC	XB_0085

Opleidingsdelen

- Artificial Intelligence track Intelligent Systems
- Artificial Intelligence track Socially Aware Computing

Artificial Intelligence track Intelligent Systems

Omschrijving

The Intelligent Systems track of the AI Bachelor's degree programme focuses on building intelligent systems, i.e. on developing, applying and evaluating new intelligent algorithms. You will study how computers are able to learn automatically and how they accumulate, represent and apply knowledge on a large scale. This programme will teach you three main aspects: essential informatics skills and knowledge, such as programming, databases and algorithms; a sound foundation in mathematics in order to use and evaluate new methods, such as logic, algebra or statistics; and general AI methodology and technology, such as automatic learning and building intelligent systems.

Vakken

Naam vak	Periode	Credits	Code
Data Structures and Algorithms for AI	P1	6.00EC	XB_0043
Project Conversational Agents	P3	6.00EC	XB_0101

Opleidingsdelen

- AI track IS year 2 constrained choice
- AI track IS year 2 constrained choice

AI track IS year 2 constrained choice

Vakken

Naam vak	Periode	Credits	Code
Computational Intelligence	P5	6.00EC	XB_0025
Databases	P5	6.00EC	X_401008

AI track IS year 2 constrained choice

Vakken

Naam vak	Periode	Credits	Code
Project Collective Intelligence	P6	6.00EC	XB_0026
The Law of Artificial Intelligence	P6	6.00EC	XB_0084

Artificial Intelligence track Socially Aware Computing

Omschrijving

The track Socially Aware Computing focuses on the interaction between

humans and Artificial Intelligence systems. You will learn how to

develop the tools

and programs that make people's everyday lives more pleasant, safer and

healthier. For example, you could develop an intelligent living

environment for the elderly, smart training programs for athletes, or

intelligent sensors and devices to help deal with stress. You also learn

how to develop AI systems that can effectively interact with humans.

Socially Aware Computing is very much a multidisciplinary programme. As well as basic knowledge about informatics and artificial intelligence, you will learn about other specialist areas, including psychology, health sciences and law. In addition, you will learn to investigate how people live and how to use technical skills to increase their quality of life,
and how to design a natural interaction with AI systems.
To this end, you will also learn how new technologies work, so that you can combine your medical and psychological knowledge and your AI and informatics expertise for application in theoretical and practical skills.
To do so the programme will also teach you essential informatics skills and knowledge, such as databases; mathematics
such as statistics, and general AI methodology and technology, such as machine learning.

Vakken

Naam vak	Periode	Credits	Code
Robot Interaction	P1	6.00EC	XB_0040
Project Socially Aware Computing	P3	6.00EC	XB_0042
The Law of Artificial Intelligence	P6	6.00EC	XB_0084

Opleidingsdelen

- AI track SAC year 2 constrained choice

AI track SAC year 2 constrained choice

Vakken

Naam vak	Periode	Credits	Code
AI and Law	P5	6.00EC	XB_0070
AI in Health	P5	6.00EC	XB_0020

Artificial Intelligence year 3

Omschrijving

In the third year of the AI Bachelor's, students can follow a minor offered by the university or the faculty. During this year, it is also possible
for students to take the opportunity to study abroad for half a year. In the
final semester students focus on their Bachelor projects which is meant as a conclusion of their AI Bachelor's programme.

Vakken

Naam vak	Periode	Credits	Code
Bachelor Project Artificial Intelligence	Ac. Year (sept)	15.00EC	XB_40002
Ethical AI	P4	3.00EC	XB_0117
Text Mining for AI	P5	6.00EC	XB_0085

Opleidingsdelen

- Artificial Intelligence track Intelligent Systems year 3
- Artificial Intelligence track Socially Aware Computing year 3
- Artificial Intelligence minor programmes

Artificial Intelligence track Intelligent Systems year 3

Vakken

Naam vak	Periode	Credits	Code
Automata and Complexity	P4	6.00EC	X_401049

Artificial Intelligence track Socially Aware Computing year 3

Vakken

Naam vak	Periode	Credits	Code
Cognitive Psychology for AI	P4	6.00EC	XB_0072

Artificial Intelligence minor programmes

Opleidingsdelen

- Educational minor Computer Science
- University minors
- Minor Business Analytics and Data Science
- Minor Bioinformatics and Systems Biology
- Minor Information Sciences
- Minor Situated Artifical Intelligence

Educational minor Computer Science

Opleidingsdelen

- Educatieve Minor verplicht

Educatieve Minor verplicht

Vakken

Naam vak	Periode	Credits	Code
Education Minor Didactics 1	P1	6.00EC	O_EMDID1
Education Minor Practical Training 1	P1	6.00EC	O_EMPRAK1
Educatieve Minor, Peergroep	P1+2+3	0.00EC	O_EMPEERGR
Education Minor Didactics 2	P2+3	9.00EC	O_EMDID2
Education Minor Practical Training 2	P2+3	9.00EC	O_EMPRAK2

University minors

Omschrijving

The university minors
- Are in principle accessible to all bachelor students of all faculties.
- Have an entry requirement for some minors.
- Have a fixed study load of 30 EC.
- No prior permission from your own Examination Board is required to have the 30 EC of this minor count in your graduation course package.
- In case a particular subject of the university minor is (also) part of your regular curriculum, you cannot (fully) follow this minor because courses cannot be counted twice. In that case, ask permission from the Examination Board for the filling out of the profiling space.

Opleidingsdelen

- Minor Brain and Mind
- Minor Global Health
- Minor Sustainability: Global Challenges, Interdisciplinary Solutions
- Minor Sport, Movement and Health
- Minor Business Administration - New Ways of Doing Business
- Minor Managing Digital Innovation
- Minor Economics
- Minor Islam
- Minor Digital Humanities and Social Analytics
- Minor in English
- Minor Gender and Diversity
- Minor History
- Minor Literature
- Minor The Writing Academy
- Minor Psychology and the Brain
- Minor Law and Global Justice: Climate, Internet and Migration
- Minor Technology, Law and Ethics
- Minor Visual Evidence: crafting truth and knowledge through film
- Minor Development and Global Challenges
- Minor Peace and Conflict Studies
- Minor Political Science
- Minor Philosophy

Minor Brain and Mind

Omschrijving

The purpose of this minor is to acquaint the student with different disciplines within the field of Neuroscience. The student will become familiar with the workings and functions of different types of brain cells and brain areas. The student will learn how this knowledge can be used to understand characteristics of the healthy brain (e.g., perception, attention, learning and memory, consciousness, personality), of the developing brain (pre- and postnatal), and of the diseased brain (e.g., depression, addiction, eating disorders). In addition, the students will be familiarized with recent findings from the fields of human genetics and and the challenges that characterize this field. Finally, this minor provides an introduction into recent technological advances in brain-machine interfaces, deep brain stimulation, and robotics in the context of Neuroscience. The integration of disciplines such as biology, psychology, sociology, and genetics, is central to this minor. Students learn to think critically about how knowledge of the brain and the human genome can be applied to tackle societal issues, and will reflect on the ethical sides of this research.

Intekenen

TARGET GROUP: Third year BSc students alpha and gamma topics (e.g. Sociology, Psychology, Economics, Law, Artificial Intelligence) and students from Lifesciences (e.g., Biology, Fysics, Chemistry, Medicine, Movement Science, Nutrition) with a broad interest.

ENTRY REQUIREMENTS: Bachelor: 90 ECTS, HBO: 120 ECTS

Due to overlap in curriculum, this minor is not open for students from Biomedical Sciences-VU and Health & Life Sciences-VU. These students, as well as students who plan to pursue a career in Neuroscience, are advised to sign up for the more specialized minor Biomolecular/Neurosciences. Please contact the minor coordinator dr Sophie van der Sluis (s.vander.sluis@vu.nl) for more information about participation.

For international students, we explicitly note that you will only obtain credits for the courses if you successfully pass the course exams as described in the individual course guides (i.e., participation alone is not sufficient).

Vakken

Naam vak	Periode	Credits	Code
Cognitive Neuroscience	P1	6.00EC	AB_1056

Naam vak	Periode	Credits	Code
Nature versus Nurture	P1	6.00EC	AB_1057
Brain in Trouble	P2	6.00EC	AB_1038
The Developing Brain	P2	6.00EC	AB_1059
Mind and Machine	P3	6.00EC	AB_1060

Minor Global Health

Omschrijving

New and emerging infectious diseases, changing disease patterns, ageing, rising costs of healthcare – these are problems that add complexity to the already considerable health challenges the world is facing today. Many diseases do not stop at national borders, and most health problems have social, political and economic impacts.

The world is more than ever in need of a vision of health that spans the globe. At the same time, innovative answers to these challenges emerge, like novel pharmaceuticals, neurotechnologies, genomics, mhealth (using mobile phones), field test kits that replace whole laboratories, as well as innovative funding schemes and care arrangement (e.g. community-based health care). But how can we make these answers fit the challenges that are emerging? History shows numerous, at best, not-so-effective health interventions and unintended consequences. To effectively address these complex health challenges, global health researchers need to cross-disciplinary boundaries and interact with health professionals, patients and others to gain in-depth understanding of global health problems, and to set out cohesive and strategic action to solve these problems.

In this University-wide Minor (open to students from all Bachelor programs from VU and other Universities inside and outside of the Netherlands), students from different study programs will get the opportunity to work together with various local and global communities on a real-life health challenge. The minor challenges students to cross disciplinary boundaries and make a lasting impact by designing a sustainable innovation for a complex health problem.

Important Note:

For students who are interested in the minor Global Health, we highly encourage you to either follow all 5 courses (periods1+2+3) within this minor, or atleast 4 courses (periods1+2)!

This is important because in the practical part of all courses in the minor Global Health you will use your newly acquired knowledge and apply it in a community engaged learning project that runs from period 1 to 3. Students will work in project teams on a real-life transdisciplinary global health challenge together with various local and global community partners. Students will cross disciplinary boundaries and interact with health professionals, patients and others to gain an in-depth understanding of global health problems, and to take cohesive and strategic actions to address these problems at a local level. Each course in the minor program will incorporate a different sub-project, starting in the first two courses with a broad exploration and needs analysis on the topic, followed by a more thorough investigation on a specific challenge in the next two courses, and concluding in the final course with a proposal for a sustainable innovation. Training workshops are offered throughout the minor program to support the development of specific skills such as transdisciplinary collaboration, reflection, social justice, global citizenship, dealing with diversity, creativity, leadership and adaptability.

Intekenen

This University-wide Interdisciplinary Minor is open to students from all Bachelor programs from VU Amsterdam, and other Universities inside and outside of the Netherlands. Please note the requirements below:

- This university minor is open to all students of the VU, you do not need permission of the examination board.
- To enter this minor, you must have earned 90 credits within your bachelor programme.
- Wo-bachelor students of another university can enter this minor, depending on whether their own university grants permission.
- The minor is taught in English and available for exchange students and Semester-in-Amsterdam students (from other countries).

Vakken

Naam vak	Periode	Credits	Code
Future Challenges in Global Health	P1	6.00EC	AB_1042

Naam vak	Periode	Credits	Code
Key Strategies in Disability and Neuropathy	P1	6.00EC	AB_1045
Double Burden of Disease	P2	6.00EC	AB_1109
Drivers of Change in Global Health	P2	6.00EC	AB_1108
Community-based Health Interventions	P3	6.00EC	AB_1110

Minor Sustainability: Global Challenges, Interdisciplinary Solutions

Intekenen

Students from other universities that want to register for this minor may need to indicate to which BSc programme the minor "Sustainability: Global Challenges, Interdisciplinary Solutions" belongs. Kindly fill in the BSc program "Aarde, Economie & Duurzaamheid" when registering.

Vakken

Naam vak	Periode	Credits	Code
Grand Challenges for Sustainability	P1	6.00EC	E_MFS_GCS
Sustainability and Environmental Change	P1	6.00EC	AB_1230
Global Development and Prosperity	P2	6.00EC	AB_1275
Governance of Global Sustainability	P2	6.00EC	AB_1229
Designing Solutions for Global Sustainability	P3	6.00EC	AB_1231

Minor Sport, Movement and Health

Omschrijving

De minor bestaat uit 5 cursussen uit de bacheloropleiding Bewegingswetenschappen. Studenten volgen in periode 1 de cursus Inleiding Inspanningsfysiologie. Daarnaast kiezen studenten in deze periode uit de cursussen Sportpsychologie en Revalidatie. In periode 2 worden de cursussen Sensomotorische Coordinatie en Toegepaste Inspanningsfysiologie gevolgd. In periode 3 kiezen studenten tussen de cursussen Talent and Talent Identification of Neuro- en Revalidatiepsychologie.

Intekenen

De minor staat open voor WO-studenten van buiten Bewegingswetenschappen en voor studenten die een relevante HBO-opleiding volgen zoals fysiotherapie, manuele therapie, bewegingstechnologie, ergotherapie of lichamelijke opvoeding. We verwachten daarbij dat studenten beschikken over Biologie- en Scheikundekennis op minimaal 5 HAVO-niveau. Voor HBO-studenten geldt bovendien dat zij deze minor alleen kunnen volgen als zij deelnemen aan een honourstraject en/of gemiddeld minimaal een 8 hebben gehaald voor hun bachelorvakken.

VU-studenten kunnen zich via VUnet inschrijven. Selecteer in VUnet de minor Sport, Bewegen en Gezondheid en meld je vervolgens voor de vakken uit de minor aan. Studenten van buiten de VU (HBO of WO) dienen zich als bijvakstudent in te schrijven om (vakken uit) de minor te kunnen volgen.

De aanmelddeadline voor de minor is 31 augustus. Er geldt geen maximum voor de minor: eenieder die voldoet aan de toelatingseisen kan starten.

Vakken

Naam vak	Periode	Credits	Code
Introduction to Exercise Physiology	P1	6.00EC	B_IF
Rehabilitation	P1	6.00EC	B_REVAL

Naam vak	Periode	Credits	Code
Sport Psychology	P1	6.00EC	B_SPORTPSY
Applied Exercise Physiology	P2	6.00EC	B_TIF
Sensorimotor Coordination	P2	6.00EC	B_SENSOCOR
Neuropsychology and Rehabilitation Psychology	P3	6.00EC	B_NEURREVPSY
Talent and Talent Identification	P3	6.00EC	B_TALIDENT

Minor Business Administration - New Ways of Doing Business

Omschrijving

Across all courses, a variety of traditional and innovative teaching techniques are used to create a comprehensive learning experience. After completing the minor, you will have a better understanding of how modern organizations compete in today's business world. You will also have the opportunity to develop core skills to contribute as an individual to the value and business model of your future organization!

Once you have finished this minor, you:

Have an advanced understanding of the traditional and emerging theoretical frameworks and concepts developed for studying organizations and their environment (foundational knowledge),

Are able to adopt theoretical frameworks and apply tools to real-world situations and organizations (application),

Are able to report, expose and defend your analyses and business recommendations (critical, creative, and practical thinking), in both writing (reports) and speaking (discussion and presentations), and

Are able to work in teams and properly allocate tasks among team members under time pressure (team project and project management).

Therefore, in addition to the content-centred goals and objectives, the minor fosters the development of the following skills: self-reliance, meaningful communication, creativity, and strategic thinking. Case studies, guest lectures, discussions, individual works, teamwork, and a consulting project are aimed at encouraging reflection and challenging conventional wisdom!

Finally, you have the opportunity to take a 12 EC internship as part of the minor. If you choose this option you have to arrange your internship in either in period 1 and 2, or in period 2 and 3.

If you choose to do the internship in period 1 and 2, you will follow the courses:

- Foundations in Business Administration (p1)
- Business Model Innovation (p2)
- Business Project (p3)

If you choose to do the internship in period 2 and 3, you will follow the courses:

- Foundations in Business Administration (p1)
- Business Professionals (p1)
- you may choose which one of the courses in period 2.

To prepare for it you are warmly invited to contact Career Services at the Faculty of Business and Economics. They can help you find an internship and get the best out of it.

Intekenen

"The minor is accessible for students already studying at VU Amsterdam and students from other universities. Check the admission requirements below to see if you can apply for this minor.

General requirements

At least 90 EC obtained within one university bachelor programme.

Specific admission requirements

You have basic knowledge of math and statistics, as provided in the academic core of any academic programme at

VU Amsterdam or equivalent.

Students in the BSc programmes Economics and Business Economics, and (International) Business Administration are excluded from participating in this university minor."

Vakken

Naam vak	Periode	Credits	Code
Business Professionals	P1	6.00EC	E_MUB_BPROF
Foundations of Business Administration	P1	6.00EC	E_MUB_FBA
Internship Minor Business Administration - New Ways of doing Business	P1+2, P2+3	12.00EC	E_MUB_IMBA
Business Model Assessment	P2	6.00EC	E_MUB_BMA
Business Models: Traditional and Emerging Perspectives	P2	6.00EC	E_MUB_BMT
Business Project	P3	6.00EC	E_MUB_BPROJ

Minor Managing Digital Innovation

Omschrijving

The opportunities of the digital era are essentially unlimited.

Innovative technologies may completely change how business and design

processes are set up, while new directions for fruitful start-ups are

countless. This calls for new and strategic ways of organising these

opportunities to innovate in the digital world. If you are interested in

new, exciting ways to organise for digital innovation, if you want to

learn how new digital technologies such as big data, 3D printing and

robotization change the way of working in your own field of expertise;

if you are interested in how to design and organise pervasive digital

technologies, if you would like to start your own Spotify, Uber or

Airbnb in your own specific discipline and would like to learn how to do

so; if you are interested in new professional, organisational and

managerial insights related to digital innovation, this minor is for

you.

This minor is a 30 EC programme taught in English. The programme

consists of five courses taught during the first semester of the third

year of your Bachelor program. It is also possible to opt for an internship in period 1 and 2 or in period 2 and 3.

If you choose to do the internship in period 1 and 2, this internship isinstead of the courses :

- Introduction to Digital Innovation (p1) and
- New Ways of Working (p2)

If you choose to do the internship in period 2 and 3, this internship isinstead of the courses:

- For period 2 you may choose which of the two courses you skip,
- Emerging Technologies for E-Business and Online Commerce (p3)

Students in the Bachelor programmes (International) Business

Administration are excluded from participating in this university minor.

Intekenen

Obtained at least 90 EC in a Bachelor's programme.

Vakken

Naam vak	Periode	Credits	Code
Introduction to Digital Innovation	P1	6.00EC	E_MUM_IDI
Strategic Management of Technology and Innovation	P1	6.00EC	E_MFEN_SMTI
Internship Minor Managing Digital Innovation	P1+2, P2+3	12.00EC	E_MUM_IMDI
AI for Business	P2	6.00EC	E_MUM_AIBI
New Ways of Working	P2	6.00EC	E_MUM_NWW
Emerging Technologies for E-Business and Online Commerce	P3	6.00EC	E_MFEO_ETEOC

Minor Economics

Omschrijving

How to deal with the economic impacts of pandemics such as COVID-19 or of the war in Ukraine? How do gender and ethnic discrimination affect the Dutch labor market? Are crypto currencies only for criminals or also affecting the real economy? What are the impacts of climate change on the livelihoods of populations in low- and middle-income countries?

Economics is about more than finance, banks and money alone: it addresses some of the most pressing problems of today. The choices we make in our (economic and social) policies can affect our world in very real ways. In the University Minor in Economics you will learn to understand and tackle societal challenges from an economic perspective. You will be able to grasp why economies grow or stay behind, why economic crises occur, what determines welfare and wellbeing, and why inequalities in many domains of life persist. The Minor offers a strong foundation to critically assess the news coverage through an economic lens, and we will jointly search for solutions for local, national and local issues we face every day.

The minor in Economics is a 30 EC programme of 5 compulsory courses,

taught in English.

More information is also available on our general Canvas page

https://canvas.vu.nl/courses/27772.

Notes:

- Students in the BSc programmes Economics and Business Economics (EBE) as

well as Econometrics are excluded from participating in this University

Minor.

- Students in the BSc programme International Business Administration

(IBA) can follow this minor only on top of their regular programme.

- Note for students Business Administration (Bedrijfskunde, BK) and

International Business Administration (IBA, on top of their regular

programme): The course ‘An Introduction to Economics’ is quite similar

to the B1-course ‘Economics’, resp. ‘Economics for the Global Era’ of

your BK/IBA programme. Therefore, instead of the course ‘Introduction to

Economics’ you will follow the course ‘Health Economics’ (from the minor

Health Care Management).

Opleidingsdelen

- Minor Economics – regular track
- Minor Economics – track for BK and IBA

Minor Economics – regular track

Omschrijving

This regular track is meant for all students except students Business Administration (Bedrijfskunde, BK) and International Business

Administration (IBA); they can follow the adjusted track

for BK and IBA.

Note: Students in the BSc programmes Economics and

Business Economics (EBE) as well as Econometrics are excluded from

participating in this university minor.

Vakken

Naam vak	Periode	Credits	Code
An Introduction to Economics	P1	6.00EC	E_MUE_AIE
Economic Challenges for Minor Students	P1	6.00EC	E_MUE_ECMS
Choices, Inequality and Welfare	P2	6.00EC	E_MUE_CIW
The Economics of Crises	P2	6.00EC	E_MUE_TEC
Applying Economics	P3	6.00EC	E_MUE_APEC

Minor Economics – track for BK and IBA

Omschrijving

This track is exclusively meant for students Business Administration (Bedrijfskune, BK) and International Business Administration (IBA).

Note: Students in the BSc programme International

Business Administration (IBA) can follow this minor only on

top of their regular programme.

Vakken

Naam vak	Periode	Credits	Code
Economic Challenges for Minor Students	P1	6.00EC	E_MUE_ECMS
Health Economics	P1	6.00EC	E_MFHC_HEC
Choices, Inequality and Welfare	P2	6.00EC	E_MUE_CIW
The Economics of Crises	P2	6.00EC	E_MUE_TEC
Applying Economics	P3	6.00EC	E_MUE_APEC

Minor Islam

Vakken

Naam vak	Periode	Credits	Code
Introduction Qur'an and Sunna	P1	6.00EC	G_BATRSPC123
Islam and European Culture	P1	6.00EC	G_BATRSAL005
Hadith Studies	P2	6.00EC	G_BATRSAL026
Islamic Theology/Kalam	P2	6.00EC	G_BATRSAL027
Islamic Ethics	P3	6.00EC	G_BATRSAL049

Minor Digital Humanities and Social Analytics

Omschrijving

All courses are compulsory.

Introduction to Python for Humanities and Social Sciences (L_AABAALG075): If you already have programming skills (students computer science and informatics) you can choose an alternative course from a selection of Humanities and Social Science courses, after consultation with the coordinator of the minor.

Vakken

Naam vak	Periode	Credits	Code
Introduction to Digital Humanities and Social Analytics	P1	6.00EC	L_AABAALG076
Introduction to Python for Humanities and Social Sciences	P1	6.00EC	L_AABAALG075
Data Science: Visualization and Analytics in R	P2	6.00EC	S_DSVAR
Interpreting Information in Text by Humans and Machines	P2	6.00EC	L_PABAALG005
Digital Humanities and Social Analytics in Practice	P3	6.00EC	XB_0015

Minor in English

Vakken

Naam vak	Periode	Credits	Code
Global English	P1	6.00EC	L_ETBAETK209
Minor English: Grammar and Writing 1	P1	6.00EC	L_ETBAALG007
Minor English: Pronunciation and Presentation	P2	6.00EC	L_EABAALG006
Minor English: Writing 2	P2	6.00EC	L_ETBAALG005
Minor English: English in my own Discipline	P3	6.00EC	L_ETBAALG008

Minor Gender and Diversity

Omschrijving

The courses Theorizing Gender and Intersectionality (W_TGI), Sexual Health: Threats and Opportunities (AB_1294) and Religions and Gender (G_BATRSAL054) are mandatory. To complete the minor, you choose two additional courses from The Personal is Political: life-world and biographies of the Othered (L_AABAALG068), American Film (L_ELBAELK208) and Identity, Diversity and Inclusion (S_IDI).

Vakken

Naam vak	Periode	Credits	Code
Sexual Health: Threats and Opportunities for minor GAD	P1	6.00EC	AB_1294
Theorizing Gender and Intersectionality	P1+2+3	6.00EC	W_TGI
American Film	P2	6.00EC	L_ELBAELK208
Identity, Diversity and Inclusion	P2	6.00EC	S_IDI
The Personal is Political: life-world and biographies of the Othered	P2	6.00EC	L_AABAALG068
Religions and Gender	P3	6.00EC	G_BATRSAL054

Minor History

Vakken

Naam vak	Periode	Credits	Code
European Cultural History	P1	6.00EC	L_GABAMKD103
Imagining the Dutch: themes in Dutch History	P1	6.00EC	L_GCBAALG003
Decolonizing Europe: History and Memory	P2	6.00EC	L_GCBAGES200
Sub-Saharan Africa and the World, 1800-present	P2	6.00EC	L_GABAGES226
Oral History & Biography	P3	6.00EC	L_AABAGES208

Minor Literature

Vakken

Naam vak	Periode	Credits	Code
Creative Writing I	P1	6.00EC	L_NABALES103
Literature from the World 1	P1	3.00EC	L_AABAALG091
Writer's Palette: Rhetoric	P1	3.00EC	L_AABAALG092
Writer at Work	P2	6.00EC	L_NNBAALG002
Writers at Work I: 1100 - 1720 (minor)	P2	6.00EC	L_AABAALG093
Literature from the World 2	P2+3	6.00EC	L_AABAALG094

Minor The Writing Academy

Vakken

Naam vak	Periode	Credits	Code
Creative Writing I	P1	6.00EC	L_NABALES103
Literature from the World 1	P1	3.00EC	L_AABAALG091
Writer's Palette: Rhetoric	P1	3.00EC	L_AABAALG092
Multatuli from a Multicultural Perspective	P2	6.00EC	L_AABAALG083
Stories from the Rijksmuseum	P2	6.00EC	L_AABAALG085
Literature from the World 2	P2+3	6.00EC	L_AABAALG094

Minor Psychology and the Brain

Omschrijving

Knowledge about the psyche and our brain is growing rapidly. Weekly articles and books appear with groundbreaking insights into the functioning of our brain and its effect on our behavior. This knowledge changes the world, with ever-increasing effects on marketing, justice, technology, computers, our food and the economy. It gives us insights into where and why we differ from each other, and helps us to explain certain group processes in society. Knowledge about the link between our psychology and our brain are a must for anyone who wants to understand why we do what we do.

Goal :

The minor Psychology and the brain introduces students to the disciplines that study behavior and the brain. The minor gives students an overview of biological approaches in psychology and of cognitive and clinical neuroscience. They are introduced in the way of conducting research in these fields. The learning objectives are:

a. to provide the knowledge to make an informed judgment about claims made -both inside and outside of science- about the psyche and brain,

b. to impart the skills to needed to conduct research into psyche and brain.

Target audience:

The minor is attractive to students with a general interest in human behavior and how it links to our biology (genes, brain structure and function) with prior knowledge of statistics (e.g. as offered in bachelors in social sciences, economics, exact and biomedical sciences).

NB: This minor is NOT accessible to students who have completed or follow the Bachelor Psychology.

Entry requirements:

-At least 90 EC obtained within a bachelor programme.

-At least 6 EC obtained in statistics courses.

-Participation in the Biological Psychology course is required before participation in the Neuropsychology course.

Vakken

Naam vak	Periode	Credits	Code
Evolutionary Psychology	P1	6.00EC	P_BEVOLPS
Introduction Psychology (UM)	P1	6.00EC	P_UINLPSY
Behaviour Genetics in Psychology (UM)	P2	6.00EC	P_UBEGEPS
Biological Psychology (UM)	P2	6.00EC	P_UBIOPSY
Neuropsychology (UM)	P3	6.00EC	P_UNEUPSY

Minor Law and Global Justice: Climate, Internet and Migration

Omschrijving

The regulation of the internet, the international mobility of people and the management and causation of climate change are amongst the most hotly debated issue of our times. These issues transcend national and regional borders, which not only complicates questions of accountability, effectivity but also the access to justice for the legal subjects involved. They raise questions of the control of national and international exercises of power, of the relationship between international law, the nation state and capitalism but also of the effects of colonialism in contemporary North/South relations. This minor contains courses on the law relating to migration, climate change, the internet, as well as a selection of other current topics in transnational law and justice which vary from year to year (past topics including fertility tourism, selling citizenship, the Covid-19 pandemic, and geoengineering).

If you are pursuing a career in European and international law or in the field of domestic or international migration, this is an excellent minor for you. The minor prepares you for the master programs European and International Law, International Migration and Refugee Law, International Technology Law or Internet, Intellectueel Eigendom en ICT.

After completing this minor, the student has knowledge of the core of the legislation concerning the three topics, has gained insight in the most important critique and analysis of this legislation (from a legal, policy-orientated, sociological, anthropological and/or philosophical perspective), and is capable of critically judging proposed changes. For each of the topics the student knows which actors play a role in making rules and policy, how states work together (or not), the consequences of this (lack of) cooperation and the future perspective for transnational regulations in migrations, climate change and internet. Knowledge of

these ‘case studies’ and the theory involved also enables student to

independently reflect on other areas of transnational problems, such as

security.

Vakken

Naam vak	Periode	Credits	Code
Human Rights and the Border	P1	6.00EC	R_HumRB
Internet Governance	P1	6.00EC	R_InternGov
Climate Change Law	P2	6.00EC	R_TL-TP
Human Rights and Citizenship	P2	6.00EC	R_HumRC
Current Issues	P3	6.00EC	R_Currissues

Minor Technology, Law and Ethics

Vakken

Naam vak	Periode	Credits	Code
Governance and Regulation of Emerging Technologies	P1	6.00EC	R_GRET
Robot Law and Artificial Intelligence	P1	6.00EC	R_RLAI
Data Analytics and Privacy	P2	6.00EC	R_DAP
Philosophy and Neuroethics	P2	6.00EC	W_BA_PNEU
Law and Ethics of Reproductive Technologies	P3	6.00EC	R_LERT

Minor Visual Evidence: crafting truth and knowledge through film

Omschrijving

In the minor you will learn how knowledge is produced through documentary film, covered by two main aspects. Firstly, you will study visual literacy that is required to critically watch and analyze documentaries. And secondly, you will develop the practical skills that are required to conduct research through filmmaking.

At the end of the minor, you…:

· can critically reflect on the multiple truth claims presented in film as well as the cinematographic and narrative methods and evidence used to underpin such truth claims;

· recognize the ways in which audiovisual evidence is strategically deployed in different disciplines;

· distinguish and compare different truth claims in these genres and disciplines and their different purposes;

· critically reflect on the positionality of film makers as well asyour own positionality as a viewer and a maker;

· understand how academic writing and film deal with the limits of representation;

· collaboratively produce a research-based film on a current issue and demonstrate relevant skills related to the pre-production, production, and post-production stages of filmmaking.

Vakken

Naam vak	Periode	Credits	Code
Introduction to the minor Visual evidence: learning how to look	P1	6.00EC	R_IMVE
Showed trials: Truth telling in documentary film and court	P1	6.00EC	R_STTT
AudioVisual Anthropology	P2	6.00EC	S_AVA
Seeing is Believing: Film and Religion and the Construction of Sacredness Beyond the Limits of Representation and Articulation.	P2	6.00EC	G_BATRSAL090
Filmmaking in practice: pre-production, production and post-production	P3	6.00EC	R_FIP

Minor Development and Global Challenges

Vakken

Naam vak	Periode	Credits	Code
Development and Globalization: Anthropological Perspectives	P1	6.00EC	S_DG
Environment and Development	P1	6.00EC	S_ED
Global Political Economy	P2	6.00EC	S_GPE
Nation and Migration	P2	6.00EC	S_NM
Urban Studies	P2+3	6.00EC	S_UBS

Minor Peace and Conflict Studies

Omschrijving

The Minor is made up of the following courses:

- Philosophy and the Ethics of Violence: Peace, War, and Terrorism
- Political Violence and the Human condition
- Conflict and Peacebuilding: Local and global perspectives
- The Law and Politics of Fencing the Use of Force
- Peace and Conflict: From Theory to Practice

Vakken

Naam vak	Periode	Credits	Code
Philosophy and the Ethics of Political Violence: Peace, War and Terrorism	P1	6.00EC	S_PEV
Political Violence and the Human Condition	P1	6.00EC	S_PVHC
Peace and Conflict: From Theory to Practice	P1+2+3	6.00EC	S_PC
Conflict and Peacebuilding: Local and Global Perspectives	P2	6.00EC	S_CPB
The Law and Politics of Fencing the Use of Force	P2	6.00EC	S_LPFUF

Minor Political Science

Vakken

Naam vak	Periode	Credits	Code
International Relations and Global Governance	P1	6.00EC	S_IRGG
State, Power and Conflict	P1	6.00EC	S_SPC
Minor Seminar in Political Science	P1+2+3	6.00EC	S_MSPS
EU Governance in an International Context	P2	6.00EC	S_EUGIC
Global Political Economy	P2	6.00EC	S_GPE

Minor Philosophy

Vakken

Naam vak	Periode	Credits	Code
Great Minds I	P1	6.00EC	W_BA_MND1
Political Philosophy	P1	6.00EC	W_BA_PP
Ethics	P2	6.00EC	W_BA_ETEN
Philosophy of Science	P2	6.00EC	W_BA_SCIE
Great Minds II	P2+3	6.00EC	W_BA_MND2

Minor Business Analytics and Data Science

Omschrijving

In the minor Business Analytics & Data Science you deepen your knowledge on various aspects in the field of data science. The minor consists of a number of advanced courses in that field and complements the corresponding bachelor programme in Business Analytics in the direction of data collection and data-driven solution methods. The

minor is concluded with the hands-on project Data Wrangling, in which you learn how to scrape data.

The minor consists of 30 EC in total and is a choice of 5 courses from the list below subject to the following constraints:

- out of the courses XB_0043 Data Structures and Algorithms for AI (period 1) and XB_0091 Data Structures and Algorithms for CS (period 2), you can choose at most one.
- Students enrolled in the BSc Artificial Intelligence and BSc Computer Science programs are not permitted to count the courses Data Science and Algorithms for CS/AI towards the Business Analytics and Data Science minor, as these courses are already part of the core curricula of those two Bachelor programs. Students enrolled in the BSc Artificial Intelligence and BSc Computer Science programs have to choose one additional course from the other courses in period 1, besides Strategic Management of Technology and Innovation. Please see the courses mentioned under: Minor Business Analytics and Data Science for Artificial Intelligence/Computer Science. Please note that the information regarding studies AI and CS only applies to the academic year 2024-2025.

The recommended programme includes

- two of the period 1 courses,
- two of the period 2 courses except you can not choose for Data Structures and Algorithms for CS,
- the period 3 course.

If you would like to choose more than 2 courses in period 2, please discuss with your academic advisor.

Opleidingsdelen

- Minor Business Analytics and Data Science
- Minor Business Analytics and Data Science for Artificial Intelligence / Computer Science

Minor Business Analytics and Data Science

Vakken

Naam vak	Periode	Credits	Code
Data Structures and Algorithms for AI	P1	6.00EC	XB_0043
Strategic Management of Technology and Innovation	P1	6.00EC	E_MFEN_SMTI
Data Analytics and Privacy	P2	6.00EC	R_DAP
Data Structures and Algorithms for CS	P2	6.00EC	XB_0091
Information Retrieval	P2	6.00EC	X_400435
Logistics Analysis	P2	6.00EC	X_401084
Data Wrangling	P3	6.00EC	XB_0014

Minor Business Analytics and Data Science for Artificial Intelligence / Computer Science

Vakken

Naam vak	Periode	Credits	Code
Advanced Network Programming	P1	6.00EC	XB_0048
Calculus 1	P1	6.00EC	X_400635
Equational Programming	P1	6.00EC	X_401011
Introduction to Reinforcement Learning	P1	6.00EC	XB_0118
Robotics	P1	6.00EC	XB_0120
Strategic Management of Technology and Innovation	P1	6.00EC	E_MFEN_SMTI
Data Analytics and Privacy	P2	6.00EC	R_DAP
Information Retrieval	P2	6.00EC	X_400435
Logistics Analysis	P2	6.00EC	X_401084
Data Wrangling	P3	6.00EC	XB_0014

Minor Bioinformatics and Systems Biology

Omschrijving

The minor Bioinformatics and Systems Biology allows students to gain essential data science skills such as programming and calculus. Moreover, it introduces current research in Bioinformatics and Systems Biology. There are three courses (12 EC) that are compulsory for every student in the minor programme. In addition, 12-18 EC will of courses will be assigned to you depending on your primary BSc programme. The courses that are assigned include: Calculus, Programming in Python and Biochemistry. The minor is 30 EC total, see the following page for detailed information on the courses: https://canvas.vu.nl/courses/37187

It is essential to email the minor coordinator (bscminorbsb.beta@vu.nl) to check your list of courses before starting the minor. To gain admission to the MSc Bioinformatics & Systems Biology you need to complete all the courses as agreed by the minor coordinator.

Information on registration for the BSc minor can be found here:
https://canvas.vu.nl/courses/37187/pages/registration-for-the-bsc-minor-and-premaster

Opleidingsdelen

- Minor Bioinformatics and Systems Biology HBO Bioinformatics / Biotechnology
- Minor Bioinformatics and Systems Biology Biomedical Sciences / Biology / HLO
- Minor Bioinformatics and Systems Biology electives
- Minor BSB constrained choice Artificial Intelligence
- Minor Bioinformatics and Systems Biology compulsory courses

Minor Bioinformatics and Systems Biology HBO Bioinformatics / Biotechnology

Vakken

Naam vak	Periode	Credits	Code
Calculus 1	P1	6.00EC	X_400635
Calculus	P1+2	6.00EC	X_400617

Minor Bioinformatics and Systems Biology Biomedical Sciences / Biology / HLO

Vakken

Naam vak	Periode	Credits	Code
Calculus 1	P1	6.00EC	X_400635
Calculus	P1+2	6.00EC	X_400617
Introduction to Programming (PYTHON)	P2	6.00EC	X_401096

Minor Bioinformatics and Systems Biology electives

Omschrijving

List of elective courses (choose courses up to 12 ECTS in total)

Vakken

Naam vak	Periode	Credits	Code
Knowledge and Data	P1	6.00EC	X_400083
Extended Research Questions in Bioinformatics	P1+2	6.00EC	XB_0074
From Protein to Cell	P2	6.00EC	AB_1052
Information Retrieval	P2	6.00EC	X_400435
Mechanics and Thermodynamics in the Cell	P2	6.00EC	X_422589
Metabolomics	P2	6.00EC	AB_1291

Minor BSB constrained choice Artificial Intelligence

Vakken

Naam vak	Periode	Credits	Code
Calculus 1	P1	6.00EC	X_400635
Calculus	P1+2	6.00EC	X_400617
Biochemistry	P2	6.00EC	AB_1137

Minor Bioinformatics and Systems Biology compulsory courses

Vakken

Naam vak	Periode	Credits	Code
Basics of Systems Biology	P1	3.00EC	XB_0078
Research Questions in Bioinformatics	P1	3.00EC	XB_0073
Principles of Bioinformatics	P3	6.00EC	X_401094

Minor Information Sciences

Omschrijving

The courses in the minor are meant to introduce students to sustainability, from the perspective of sustainable development and PPP (People-Planet-Profit), and help students integrate their analyses into the design of innovative solutions for global sustainability challenges in a systematic and creative manner. During the programme, students will take one mandatory course from the School of Business and Economics that is meant to help them develop an understanding of the core Business aspects relevant to Information Science. Further, students will choose four out of the six available constrained-choice courses from the Faculty of Sciences, Faculty of Humanities, and School of Business and Economics. These courses are meant to further strengthen their knowledge in the business aspects of Information Science, consolidate their technical background, introduce the role of data digitalization in the humanities and social sciences, and provide them with an introduction to data representation and processing.

Intekenen

The minor Information Sciences is open to all students in the Bachelor programmes of Computer Science and Artificial Intelligence. For more information concerning the educational programme, contact the study advisor.

The minor is coordinated by Emitzá Guzmán. She can be contacted through. email: e.guzmanortega@vu.nl or telephone: +31 20 59 88881.

Vakken

Naam vak	Periode	Credits	Code
Strategic Management of Technology and Innovation	P1	6.00EC	E_MFEN_SMTI
Information Retrieval	P2	6.00EC	X_400435
Sustainable Business Processes	P2	6.00EC	E_MFS_SBP
Designing Solutions for Global Sustainability	P3	6.00EC	AB_1231

Opleidingsdelen

- Minor Information Sciences constrained choice

Minor Information Sciences constrained choice

Omschrijving

Students need to choose one 6 EC course among those available in the constrained choice.

Vakken

Naam vak	Periode	Credits	Code
Environment and Development	P1	6.00EC	S_ED
Grand Challenges for Sustainability	P1	6.00EC	E_MFS_GCS

Naam vak	Periode	Credits	Code
Introduction to Digital Humanities and Social Analytics	P1	6.00EC	L_AABAALG076

Minor Situated Artifical Intelligence

Omschrijving

5 courses for a total of 30 ECTS, jointly taught by AI members as coordinators and co-coordinators.

1. Introduction to Reinforcement Learning

- Topics: History and related concepts from psychology, Markov Decision Processes, Bellman equations, Model-free vs. model-based reinforcement learning, exploration/exploitation tradeoff, Q learning, overview of deep RL, applications of RL
- 6 ECTS
- Period 1
- Coordinator: Shujian Yu and Vincent Francois-Lavet

1. Conversational AI

- Topics: Grounding language, understanding conversations, management of dialogues (NLU), mental models, QA, storytelling, large language models, Applications (e.g., chatbot technologies), Embodiment in conversational agents, embodied QA
- 6 ECTS
- Period 2
- Coordinators: Filip Ilievski and Piek Vossen

1. Robotics

- Topics: Mechanics (forward and inverse kinematics, control and dynamics), Perception (Computer Vision, Odometry, Localization and Mapping), Cognition (Path planning, ML for robotics), Practicals with ROS operating system.
- 6 ECTS
- Period 1
- Coordinators: Kevin Sebastian Luck and Kim Baraka

1. Formal Logic for AI

- Topics: Modal and description logics, model theory, automated reasoning, model checking, ontology-based database access
- 6 ECTS
- Period 2
- Coordinators: Vera Stebletsova and Patrick Koopman

1. Project Situated AI

- Topics: weekly assignments and demos covering different areas, including reinforcement learning, natural language processing, logic, and robotics.
- 6 ECTS
- Period 3
- Coordinator: Michel Klein

Intekenen

This minor is only available to BSc AI students of VU Amsterdam. In this first iteration, the minor will be tailored to extend the present AI curriculum and thus deepen the technical expertise of AI students. We expect students to have sufficient conceptual understanding and technical skills in programming, building intelligent systems, and human-computer interaction. BSc AI students at the VU can register for the courses of this minor via the VU dashboard. There are no other prerequisites for admission and there is no upper limit on the number of students

that will be admitted.

Vakken

Naam vak	Periode	Credits	Code
Introduction to Reinforcement Learning	P1	6.00EC	XB_0118
Robotics	P1	6.00EC	XB_0120
Conversational AI	P2	6.00EC	XB_0119
Formal Logic in AI	P2	6.00EC	XB_0116
Project Situated AI	P3	6.00EC	XB_0121

Artificial Intelligence Honours programme

Omschrijving

The Honours Programme (HP) is an annotation for the bachelor programmes in the field of Computer Science. The HP focuses on deepening the

understanding of scientific and technology processes, on gaining

expertise in state-of-the-art knowledge in specific fields, and on

practicing advanced skills in relevant fields of science and technology.

Students in the HP can take deeper and more diverse courses, and can

also

conduct their own research project under the guidance of an expert in

the field. Topics cover the entire spectrum of computer science and

artificial intelligence, and also inter-departmental topics. Students

can apply for the HP at the end of the first year of the Bachelors.

Opleidingsdelen

- Artificial Intelligence/ Computer Science/ Information Science Honours Components
- Departmental Honours Courses Science Faculty
- Interdepartmental Honours Courses

Artificial Intelligence/ Computer Science/ Information Science Honours Components

Omschrijving

The Honours Programme (HP) is an annotation in the Computer Science Department of the Faculty of Sciences. The HP focuses on deepening the understanding of scientific and technology processes, on gaining knowledge on state-of-the-art methods in specific fields, and on practicing advanced skills in relevant fields of science and technology. Students in the HP can take more advanced and diverse courses, and can also conduct their own research project under the guidance of an expert in the field. Topics cover the entire spectrum of computer science and artificial intelligence, and also inter-departmental topics. Students can apply for the HP at the end of the first year of their Bachelor.

Vakken

Naam vak	Periode	Credits	Code
Honours Research Project 1	Ac. Year (sept)	6.00EC	X_HP003
Honours Research Project 2	Ac. Year (sept)	6.00EC	X_HP004
Honours Research Project 3	Ac. Year (sept)	6.00EC	X_HP006
HP Research Community and Practice	Ac. Year (sept)	6.00EC	X_HP010

Departmental Honours Courses Science Faculty

Vakken

Naam vak	Periode	Credits	Code
Impact of Algorithms on Human Lives		6.00EC	X_HP015
Project Science and Tech-based Entrepreneurship		6.00EC	A_HP007
Climate Repair	P4	6.00EC	A_HP006
Five O'Clock Neurosciences (Honours)	P5	6.00EC	A_HP003

Interdepartmental Honours Courses

Omschrijving

The interdepartmental courses of the Honours Programme are taught mainly in the evening by lecturers from Vrije Universiteit, the University of Amsterdam and Amsterdam University College, as well as guest lecturers from the Netherlands and abroad. The classes are small and you will be expected to give presentations, write papers and make an active contribution to discussions.

As an honours student, you have to choose at least 12 credits of interdepartmental honours courses.
Go to https://vu.nl/en/education/more-about/interdepartmental-courses for more information and links to all courses offered by VU, UvA and AUC.
For more information about registering for interdepartmental honours courses, visit
https://vu.nl/en/education/more-about/course-registration

Courses

Academic Writing (BETA)

Vakcode	L_ETBAALG006
Studiepunten	3.00
Periode	P5
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. G.A. Dreschler
Examinator	dr. G.A. Dreschler
Betrokken Docenten	dr. G.A. Dreschler, J.K. Branum, dr. A. Martonfi, dr. U.P. Caci MA
Onderwijsvormen	Lecture, Seminar

Doel vak

The aim of this course are to improve your academic writing skills. More specifically, you will:

gain insight into the specific characteristics of formal written English;

learn the conventions of using sources through quoting, referring and paraphrasing;

develop your academic skills with regard to text structure, linguistic accuracy, and clarity of expression.

In addition, you will be able to apply this knowledge by writing a text about an academic topic, which uses academic sources appropriately; has no major grammatical errors; uses a clear text structure; and has a style which is suitable for academic texts.

Inhoud vak

In the course, we will focus on generic skills that are relevant for academic writing: creating an appropriate formal

style and using standard academic phrases and creating a clear structure for your text. You will write a text about a topic from your discipline, based on academic sources, using a referencing style that is standard in your discipline. We will also pay attention to general principles of good writing, such as guiding the reader through the text and creating a good flow in the text. We will address these topics in such a way that you will be able to apply these principles in any text that you write after following the course.

In addition, you will get individual feedback on your own texts, which will allow you to gain insight into your own strengths and weaknesses in terms of linguistic accuracy (grammar, punctuation, spelling and vocabulary).

The subjects covered are the following:

- Academic style: formality, use of academic phrases

- Quoting, referring and avoiding plagiarism

- Structure: introduction, thesis statement, paragraph structure, conclusion

- Linguistic accuracy: grammar, vocabulary and punctuation

- Clarity of formulation: linking language, clause combining

Aanvullende informatie onderwijsvormen

Lectures (1 hour a week) and seminars (2 hours a week).

Toetsvorm

A series of short writing assignments (25%) and a final, longer text (75%).

Literatuur

Reading materials will be made available via Canvas.

Aanvullende informatie doelgroep

First-year students of Artificial Intelligence

Overige informatie

This course has obligatory attendance (80% of all classes)

Advanced Network Programming

Vakcode	XB_0048
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. D. Bonetta
Examinator	dr. D. Bonetta
Betrokken Docenten	B. Chandrasekaran, dr. D. Bonetta, ir. J.J.R. Donkervliet MSc
Onderwijsvormen	Lecture, Seminar, Practical

Doel vak

The objective of this course is to teach students about advanced and practical concepts in networking technologies. Upon completion of this course, students will be able to:

- Understand end-host networking concepts [Knowledge and understanding]
- Learn about Linux networking concepts and tools (tcpdump, ethtool, arping, ping, numactl) [Knowledge and understanding] [Applying knowledge and understanding]

- Explain network stack design and implementation challenges associated with multi-core systems and high-speed networks (100+ Gbps) [Knowledge and understanding] [Applying knowledge and understanding]
- Analyze various network stack designs (kernel, userspace, application-specific) and APIs (e.g., sockets, RDMA and friends) on their trade-offs, performance gains, and implementation efforts [Applying knowledge and understanding] [Making judgements]
- Understand how to manage modern networks and how popular applications (e.g., video streaming) are built and run on the Internet [Knowledge and understanding] [Making judgements]
- Learn about ongoing and emerging research challenges in networking technologies [Knowledge and understanding] [Lifelong learning skills]
- Apply the knowledge gained into building a practical implementation of TCP/IP protocol stack, and explain networking concepts [Applying knowledge and understanding] [Lifelong learning skills] [Communication]

Inhoud vak

The course covers recent advancements in networking technologies in general-purpose computing (e.g., data center, cloud computing). The course is split into two themes: End-host networking and network infrastructure.

In the end-host networking theme, students will learn about networking concepts (MTU, Segmentation, stateful and stateless Offloading, RSS, SGE, multiqueue), internals of the Linux networking stack, tools (tcpdump, ethtool, arping, ping, numactl), challenges associated with delivering high-performance network I/O operations (100+ Gbps bandwidth, millions of network operations/sec, many/multi-core scalability), alternate ways of building networking stacks in the userspace, design of packet processing stacks, and introduction to the high-performance Remote Direct Memory Access (RDMA) technology.

The second theme of the course will be focused on the recent development in network infrastructure and its management. Students will learn the details about the protocols running on our Internet and how the Internet is managed with new approaches like software-defined networking, and how to utilize these new technologies to build popular Internet applications/services.

Aanvullende informatie onderwijsvormen

2x Lectures per week, Practical coding group project (with 2 or 3 students) with 1 lab session per week for QA (deliverables: code and report)

Toetsvorm

Exam (50%, 50 points), Practical Assignments (50%, 50 points over 5 assignments) -- resits only available for the exam. Both exam and practical assignment needs to be passed separately. Their cut off is 25 points for both. A total of (25 + 25 =) 50 / 100 points will give you a grade of 6.0/10.

Students are graded on both the completion of the assignment as well as their understanding of the material (based on the oral discussion/interview and report/code submitted).

Literatuur

Slides (provided in the class), project handbook, and further online reading material/links.

Aanvullende informatie doelgroep

Minor Deep Programming

Aanbevolen voorkennis

IMPORTANT: There is no support to do the project work in any other language than C/C++ (not python supported). We get asked this every year.

Operating systems (userspace, kernel, processes, memory management), System Architecture (I/O devices, interrupts, CPU caches), Computer Networks (TCP/IP, Ethernet, Packet switching networks, ARP, ICMP protocols), good knowledge of C/C++ programming (important) with sockets, Linux build and coding environment (bash, CMake, gdb, gcc, make and friends). Courses that cover this background systems knowledge include X_400487 Computer Networks, X_405067 Operating Systems, XB_40009 Computer Organization. **It is strongly**

recommended that you must finish these courses before you take this course.

In case you lack these skills and/or knowledge, we can provide additional reading material and lab sessions, however, you need to study them by yourself outside the class hours. Keep in mind this is not a course to learn how to program in C/C++ or learn about networking concepts like how to program sockets or what is TCP/IP or ARP. This is an advanced course that focuses on low-level details of designing and programming a working networking stack.

AI and Law

Vakcode	XB_0070
Studiepunten	6.00
Periode	P5
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	C. Gerritsen
Examinator	C. Gerritsen
Betrokken Docenten	C. Gerritsen, dr. M. Galič LLM
Onderwijsvormen	Study Group, Lecture

Doel vak

(a) Subject-specific learning outcomes (knowledge and understanding)
Upon completion of the course the student should have basic knowledge of:
• The relation between AI and Law
• Key applications of AI within Law
• Ethical questions related to the use of AI within the legal domain

(b) Academic learning outcomes (applying knowledge and understanding; making judgements)
Upon completion of the course the student should be able to:
• Read, understand, analyse and reflect upon interdisciplinary literature relating to AI applications in the field of law
• Analyse, interpret and employ scientific knowledge in a written assignment on a basic level
• Make a reasoned choice for an AI application and reflect on the potential legal consequences of this application

(c) Social and communication learning outcomes (applying knowledge and understanding; communication; lifelong learning skills)
Upon completion of the course the student should be able to:
• Work in a focused and systematic way
• In a scientifically substantiated manner clearly and convincingly present viewpoints and findings to different target groups in written form

Inhoud vak

Innovation in computational technologies have given rise to a whole range of new developments, such as autonomous cars and drones, predictive policing and automated warfare. Each of these developments has significant legal, ethical and policy implications. With autonomous driving, for example, this became particularly obvious after a pedestrian was killed for the first time by an autonomous vehicle on a test drive. How will autonomous vehicles affect the law, particularly once they are authorized for regular usage, and take the streets in large numbers? Can it even be considered ethical to have machines think by themselves? Another, closely related, AI application – automated weapons – raises potentially even more pressing issues. What are the legal and ethical implications of autonomous weapons that determine themselves when to fire, without a human being involved in the decision process? Examples like these show the tremendous impact that AI will have on the legal field.

Aanvullende informatie onderwijsvormen

Lectures and tutorials. The tutorials are mandatory.

Toetsvorm

Written exam and practical assignment (individually or in groups of 2). They both count 50% towards the

final grade and both components need to passed with a minimum grade of

5.5

The practical assignment can be redone if not sufficient (< 5.5). There

will be a resit for the written exam if not sufficient (<5.5).

Literatuur

Literature will be made available through Canvas

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 2)

AI for Business

Vakcode	E_MUM_AIBI
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. W.A. Günther
Examinator	dr. W.A. Günther
Betrokken Docenten	dr. W.A. Günther, F.A.M. Balocco
Onderwijsvormen	Hoorcollege, Werkgroep

Doel vak

Academic & Research Skills

- Approaching AI processes from a socio-technical perspective

- Developing the ability to critically reflect on and examine theories and concepts

Knowledge

- Defining, describing and recalling the basic concepts, constituent components, principles and theories underlying the development and implementation of AI solutions for business

- Understanding the impact of AI solutions on organizations

Bridging Theory & Practice

- Choosing, applying, and evaluating AI concepts, principles and solutions to solve business problems and recognize opportunities for creating innovative products and services

- Recognizing and mitigating challenges to effective and responsible AI development and implementation

Social skills

- Being able to work in teams

- Being able to communicate project results through presentation and writing

- Being able to connect to both technical and managerial audiences

Inhoud vak

Organizations are increasingly trying to leverage Artificial Intelligence (AI) to solve business problems and create innovative products and services. However, AI is also prone to a number of limitations and challenges. In this course, students will learn about:

- basic AI concepts, techniques, and solutions

- how organizations utilize AI to improve and innovate their business

- the impact of AI on organizational processes and work

- major challenges to the effective and responsible development and implementation of AI in organizations

- relevant concepts, theories and frameworks that help to understand how organizations may overcome challenges related to the development and implementation of AI

The course primarily has a managerial focus.

Aanvullende informatie onderwijsvormen

Lectures, guest lectures and workshops

Toetsvorm

Group project

Individual take-home essay exam

Both components are mandatory for passing the course.

Vereiste voorkennis

Period 1 courses of the minor Managing Digital Innovation

Literatuur

Book: Waardenburg, L., Huysman, M., & Agterberg, M. (2021) “Managing AI Wisely”, New Horizons in Business Analytics, Cheltenham, UK: Edward Elgar Publishing.

Various papers that will be made available through Canvas.

Aanvullende informatie doelgroep

This course is part of the minor Managing Digital Innovation.

Toelichting Canvas

Use of Canvas is mandatory. All the updates regarding the course content, schedules, assignments and grading will becommunicated on Canvas.

AI in Health

Vakcode	XB_0020
Studiepunten	6.00
Periode	P5
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. A.C.M. ten Teije
Examinator	prof. dr. A.C.M. ten Teije
Betrokken Docenten	prof. dr. A.C.M. ten Teije, prof. dr. J.E. Bosmans, prof. dr. F.A.H. van Harmelen
Onderwijsvormen	Seminar, Lecture

Doel vak

The aim of the course is to make the students familiar with a number of Artificial Intelligence techniques (Knowledge and Understanding) and how they can be applied to the medical field (Applying Knowledge and Understanding, Making Judgments). Furthermore, students should have capability to work in teams (Communication).

Inhoud vak

The course will be structured into thematic modules. In each of these modules an AI technique will be introduced, and then applied in the medical field. In the assignments, both the AI-technique and a medical topic have to be applied in a practical, hands on, way.

In a nutshell, we plan to cover the topics:

- medical ontologies (eg. SNOMED, MESH, UMLS, ICD10)
- indicators for medical care (cost and quality of care)
- modelling of clinical guidelines
- machine learning

Beside these four modules there are a couple of guest lectures, on natural language processing for medical texts, bioinformatics, and ethical impact.

Aanvullende informatie onderwijsvormen

There will be lectures and practical sessions, plus significant time for self-study and practical work.

Toetsvorm

Practical group assignments for each module, and an individual exam. Exam (50%) and practical assignments (50%) form the final grade. For both parts the grade needs to be sufficient (5.5 or higher) to obtain a final grade. No resit is possible for the practical assignments.

Literatuur

Selected scientific papers.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 2)

American Film

Vakcode	L_ELBAELK208
Studiepunten	6.00
Periode	P2
Vakniveau	300

Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. R.V.J. van den Oever
Examinator	dr. R.V.J. van den Oever
Betrokken Docenten	dr. R.V.J. van den Oever
Onderwijsvormen	Seminar

Doel vak

Students become acquainted with different theorizations of spectatorship in relation to film characters.

By the end of the course, students can:

- Explain selected concepts from film theory.

- Apply these concepts to the interpretation of selected films.

- Do all this in clear and concise written English.

Inhoud vak

Central to this course is the question: How do spectators relate to film characters? Film theory provides various answers, ranging from forced identification with a protagonist to reading against the grain of a film.

Theories and films discussed in previous years include:

- The male gaze and *Vertigo* (Alfred Hitchcock, 1958): spectators identify with a male protagonist who lustfully gazes at objectified women.

- Sentimental pedagogy and *Philadelphia* (Jonathan Demme, 1993): through emotional manipulation, films can enforce ideological beliefs onto spectators.

- Character engagement and *Psycho* (Alfred Hitchcock, 1960): spectators develop a cognitive moral evaluation of film characters.

- Queer connotation and *Gentlemen Prefer Blondes* (Howard Hawks, 1953): in response to Hollywood censorship, spectators read film characters as denotatively straight yet connotatively queer.

Both theories and films are subject to change.

Aanvullende informatie onderwijsvormen

- We meet once a week for four hours.

- Attendance is mandatory.

- Meetings are held on campus; no hybrid alternative is offered.

- Each meeting consists of group discussions, exam instruction/practice, and a lecture portion.

- Prior to each meeting, students complete a Canvas assignment.

- Devices with access to social media (this includes laptops) are not allowed in class. If an exception needs to be made for medical reasons, students must contact the academic advisor prior to the start of the course.

Toetsvorm

- Successful completion of the course is contingent on timely submission of all Canvas assignments and attendance at all meetings.

- The full final grade for the course is based on a take-home exam consisting of short essay questions. The take-home exam will be held over three days in week 8 of period 2 (the week of 16 December 2024). **English writing proficiency is a key aspect of the assessment.**

Vereiste voorkennis

Students must have participated in the period 1 courses of either the Schoolvakminor English Literature or the minor package Gender and Diversity. No exceptions will be made.

Literatuur

To be announced in the course syllabus, which will be posted on Canvas well in advance of the start of the course.

Aanvullende informatie doelgroep

This course is part of the Schoolvakminor English Literature and the minor package Gender and Diversity.

An Introduction to Economics

Vakcode	E_MUE_AIE
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	prof. dr. P.A. Gautier
Examinator	prof. dr. P.A. Gautier
Betrokken Docenten	prof. dr. P.A. Gautier, L. Klein
Onderwijsvormen	Study Group, Lecture

Doel vak

This course will introduce you to the fundamentals and basic tools of economic analysis. It will teach you the perspective of the economist: viewing reality as the result of choices and the trade-offs that these choices reflect (Academic Skills; Knowledge).
In terms of Knowledge you will learn the basic theories and models of micro- and macroeconomics. You will also learn the basic analytical tools used in the analysis of economic data (Quantitative Skills).
Finally, the course will introduce you to the most important institutions
of the global economy. The course will also show you how the tools and theories can be applied to real life examples taken from the world economy. (Bridging Theory and Practice).

Inhoud vak

Topics to be discussed are:
- Consumers, sellers and Incentives;
- Perfect competition, Trade;
- Externalities;
- Labor market/ human capital/unemployment;
- Economics of Information;
- Game theory/ Auctions;
- Economic Growth;
- Business Cycles;
- Credit Markets

Aanvullende informatie onderwijsvormen

(Online) Lectures and Tutorials.

Toetsvorm

Written exam (MC) - Individual assessment
Assignment(s)

Literatuur

Acemoglu, Daron, David Laibson and John A. List, Newest edition (older editions are also allowed), Economics, Harlow, Essex, Pearson Education Ltd. ISBN 13: 978-1-292-07920-2, incl. access code MYECONLAB.

Applied Programming for AI

Vakcode	XB_0102
Studiepunten	6.00
Periode	P6
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M. Verano Merino MSc
Examinator	dr. M. Verano Merino MSc
Betrokken Docenten	dr. M. Verano Merino MSc, B. Postma
Onderwijsvormen	Lecture, Practical

Doel vak

The course prepares students for future projects that involve programming within the bachelor programme. Students learn to apply programming concepts in a medium sized project.

The course introduces students to web technologies such as HTML, CSS and JavaScript, and commonly used tooling in software development, e.g. Git. (Knowledge and understanding), as well as using and understanding RESTful services.

After completing the course, students will be able to plan, develop, and implement a web application that involves data manipulation and visualization requirements. (Applying knowledge and understanding) (Making Judgements)

Inhoud vak

In this course, students will implement a complete pipeline of information visualization including data acquisition, parsing, cleaning, representation, and interaction. The course builds upon the course Introduction to Python Programming, as well as on the course Human-Computer Interaction in Period 5.

Students will learn how to choose and use different data structures (e.g., lists, dictionaries, tuples, matrices). They also will learn how to choose the right way of presenting information depending on the target audience and type of data.

Aanvullende informatie onderwijsvormen

A combination of a few lectures and a lot of practical work. Students are expected to be full time available for this course.

Toetsvorm

The student’s knowledge will be assessed based on their group project’s implementation, their individual contribution to the development based on version control statistics, and weekly meetings, and a final coding interview.

The final grade for this course consists of the following parts:

- Project Proposal (team) → Pass/fail
- Implementation (team) → 60%
- Degree of participation (individual) (20%)
- Attendance,
- Contribution to the project.
- Git Activity,
- Project report/reflection (individual and team components) → 10%
- Final demo presentation (team) → 10%
- Coding interview (individual) → Pass/fail

To obtain a passing grade for the course, you must pass the coding interview.

Vereiste voorkennis

Students should have completed Introduction to Python Programming (XB_0082) with a sufficient grade before they can start this course.

Literatuur

The following literature will be used:

- Judith Donath, Alex Dragulescu, Aaron Zinman, Fernanda Viégas, and Rebecca Xiong. 2010. Data portraits. In ACM SIGGRAPH 2010 Art Gallery (SIGGRAPH '10). Association for Computing Machinery, New York, NY, USA, 375–383. https://doi.org/10.1145/1836786.1836793
- Giorgia Lupi. Data Humanism: The Revolutionary Future of Data Visualization. In Printmag, January 30, 2017. https://www.printmag.com/article/data-humanism-future-of-data-visualization/
- Edward R. Tufte. 2001. The visual display of quantitative information. Graphics Press, USA.
- Jacoba Urist. From Paint to Pixels. In The Atlantic, May 14, 2015. https://www.theatlantic.com/entertainment/archive/2015/05/the-rise-of-the-data-artist/392399/
- Andy Kirk. 2016. Data Visualisation: A Handbook for Data Driven Design. Sage Publications Ltd.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 1)

Aanbevolen voorkennis

Human Computer Interaction (XB_0013) is highly recommended as background knowledge.

Applying Economics

Vakcode	E_MUE_APEC
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	prof. dr. M.P. Pradhan
Examinator	prof. dr. M.P. Pradhan
Betrokken Docenten	dr. P.R. Koster, M.J. van der Meijden BSc, prof. dr. M.P. Pradhan
Onderwijsvormen	Study Group, Lecture

Doel vak

The objective of this course is to develop your capability to

report your findings in a clear, logical and informative way. Specific learning outcomes upon completion of this course are:

• you are able to analyse a relevant (economic) policy issue, to motivate the relevance of the issue, and to formulate an appropriate research question;

• you are able to locate relevant economic theory in the literature and to apply it correctly in order to analyse the policy issue and to identify the economic rationale of potential or actual policy responses;

• you are able to identify, interpret and compare empirical findings from the economic literature to describe the policy issue, and/or the behavioural response of the market and government actors, and/or the impact of these responses;

• you have developed a critical attitude to the relevance and shortcomings of empirical data compared to theoretical requirements, and have become aware of limitations in insights that can be gained from theoretical reasoning alone when addressing real-life issues;

• you are able to present your findings clearly to academic expert and non-expert audiences;

• you are able to work independently, while incorporating relevant feedback into their work;

• you are able to give constructive feedback to peers.

Inhoud vak

In this intensive four-week course, you work in one of a selected number of policy areas (for example international financial systems and banking regulation, macro policy, development and growth, environment, urban/transport, health, human capital, competition policy, industrial policy). You will write an economic policy-oriented research paper focusing on policy analysis, design and/or evaluation.

Aanvullende informatie onderwijsvormen

One introductory lecture followed by weekly working groups organized by policy area. The course closes with a plenary presentation where all groups present. All have mandatory attendance.

Toetsvorm

Grade is based on individual participation and contributions in tutorial sessions, and a group grade based on the group policy research paper and presentation of research. A sufficient individual contribution in the first 2 weeks is required to participate in rest of the course.

Literatuur

Various theoretical and empirical academic papers (dependent on the

student’s selected topic), based on suggested and own literature search.

Aanbevolen voorkennis

All courses from period 1 and period 2:

"An Introduction to Economics", "Economic Challenges", "Choices,

Inequality and Welfare" and "The Economics of Crises"

AudioVisual Anthropology

Vakcode	S_AVA
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. E. van Roekel
Examinator	dr. E. van Roekel
Betrokken Docenten	dr. P.G.A. Versteeg, dr. E. van Roekel
Onderwijsvormen	Lecture

Doel vak

Obtained skills after completion of the course:

1. Historical understanding of the role of sound and image in anthropology and their analytical value and limitations

2. Comprehension of the impact of positionality in the production and viewing of audiovisual projects

3. Practical skills to gather and manipulate audiovisual data

4. Basic knowledge of the practical and ethical dilemmas of ‘doing’ audiovisual ethnography

During the course students will gain understanding of the anthropological theories on and ethnographic approaches to image and sound. Students will furthermore familiarize with different genres and seminal works in audiovisual anthropology and learn to analyse and work with these various media from anthropological perspectives. Students

will also acquire basic skills how to design, develop, and present ethnographic data by means of a small collective film project. In tandem, they will learn about the use and impact of these media vis-á-vis academic productions and disseminations of anthropological knowledge and learn to critically reflect on the political dimension of choices involved in anthropological audiovisual production.

Inhoud vak

Image and sound are increasingly being implemented in both academic and applied research, also in anthropology. It is now difficult to imagine doing qualitative research without incorporating the virtual lives of our interlocutors, or, more practically, without a smartphone that captures sound and image instantly. Besides text, anthropologists now increasingly produce audiovisual media about their ethnographic findings from the field. By turning image and sound into documentaries and essay films, photo essays, or (virtual) art installations, they actively challenge the limits of academic representation. These new developments are shaping our methodologies and our ways of crafting anthropological knowledge. But audiovisual anthropology is not new. The gathering of audiovisual material and the audiovisual representation of ethnographic data have a prevailing history in anthropology.

In this course, we will explore the history and variety of audiovisual approaches in anthropology and analyse how different forms of image and sound work as tools for cultural exchange and situated understanding. In the course, we will also investigate in detail the role of positionality of film makers in the way film is being made as well as our own positionality as a viewer and a maker. Specifically, we will watch and analyse new and classical work in anthropological filmmaking and collectively produce a short film. The course therefore not only deals with anthropological theories on audiovisual representations, but also on the practical dilemmas and the ethical concerns of ‘doing’ audiovisual ethnography.

Aanvullende informatie onderwijsvormen

Interactive lectures, film viewings, practical labs, group discussions

Toetsvorm

One written assignment (individual) and one short audiovisual project (collective)

Literatuur

The Looking Machine (2019) by David MacDougall (available as e-book at VU-library), and additional anthropological articles and ethnographic films that will be announced on Canvas before the start of the course.

Aanvullende informatie doelgroep

This course is part of the minor 'Visual Evidence. Crafting truth and knowledge through film'

Apart from regular bachelor students, the course is also available for exchange students and students from other universities/faculties.

Overige informatie

You will work in groups on a short film for which you need per group a smartphone and laptop to install the required editing software

Aanbevolen voorkennis

Basic knowledge of anthropology is recommended.

Automata and Complexity

Vakcode	X_401049
Studiepunten	6.00
Periode	P4
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. J. Endrullis
Examinator	dr. J. Endrullis

Betrokken Docenten	dr. J. Endrullis
Onderwijsvormen	Lecture, Seminar

Doel vak

The first part of the course, on automata and languages, deals with the. concepts of formal language, grammar, and automaton. Two types of

languages are covered: regular and context-free languages. Regular. languages in the form of regular expressions are ubiquitous in computer science. They are used in search queries and text manipulation.

Context-free languages and grammars (e.g. the Backus–Naur form) are the prevailing standards for describing programming languages. We also discuss parsing algorithms to determine whether a given string is in a context-free language. The automata-theoretic counterparts for regular and context-free languages are finite automata and the more powerful pushdown automata.

The students will learn to:

(a) design finite automata and create regular expressions for a given regular language, [Knowledge and understanding] [Applying knowledge and understanding]

(b) apply algorithms to translate between finite automata, right-linear grammars, and regular expressions, [Knowledge and understanding] [Applying knowledge and understanding]

(c) apply algorithms to make automata deterministic and minimal, [Knowledge and understanding] [Applying knowledge and understanding]

(d) design pushdown automata and create context-free grammars for a given context-free language, [Knowledge and understanding] [Applying

knowledge and understanding]

(e) apply algorithms to translate between pushdown automata and context-free grammars, [Knowledge and understanding] [Applying knowledge and understanding]

(f) use pumping lemmas to reason about whether a language is regular or context-free, [Applying knowledge and understanding] [Making judgements]

(g) apply algorithms for parsing context-free languages. [Knowledge and understanding] [Applying knowledge and understanding]

In the second part of the course, on computability theory, the central question is: what computations can be performed on a computer? To reason about computability, we introduce the mathematical model of Turing machines and discuss the Church-Turing thesis. We discuss examples of undecidable problems: the halting problem and the Post correspondence problem. It is shown how the undecidability of new problems can be shown by reduction from a known undecidable problem. Finally, we discuss a classification of decidable/computable problems into important complexity classes, notably P, NP, and NP-complete problems, together with the corresponding reduction arguments.

The students will learn to:

(h) reason whether a given problem is decidable (computable) or undecidable (not computable), [Making judgements]

(i) understand the classification of decidable problems in the complexity hierarchy (e.g. P, NP, EXP) of, and [Knowledge and understanding]

(j) reason about the complexity of a problem via the reduction to/from problems with known complexity. [Making judgements] [Communication] [Lifelong learning skills]

Inhoud vak

This course treats automata & formal languages and computability theory.

The student gets acquainted with important notions and algorithms regarding formal languages, automata, grammars, compilers, computability, and complexity.

This course addresses foundational questions in computer science:

- What can be computed? What are the limitations to what computers can do?

- How much time and memory does solving a problem require?

- What is a (programming) language?

- How can languages be recognized by computers (automata)?

- Which problems can be solved by what kinds of automata?

This course conveys the important idea that certain problems cannot be solved by computers. A computer scientist must be able to reason whether a given problem is computable (decidable) or not. Moreover, a computer scientist should be able to reason what language/complexity class a given problem belongs to, and hence what kind of automata/algorithms are needed to tackle the problem.

Aanvullende informatie onderwijsvormen

4 hours per week lectures; 4 hours per week exercise classes.

Toetsvorm

The homework is mandatory for qualifying for the exam (70% of the homework points to qualify for the exam). In case at least 90% of the

homework points is obtained, 0.5 bonus point is awarded for the final grade. At the end of the course there is a final exam.

The overall grade is the grade of the final exam plus the possibly 0.5 bonus point obtained for the homework. (The bonus is only added for students that pass the exam with a grade of at least 5.5.)

There is no resit opportunity for the homework.

Literatuur

Peter Linz, An Introduction to Formal Languages and Automata, Jones & Bartlett, 4th or 5th edition

Aanvullende informatie doelgroep

BSc Computer Science (year 3)

Bachelor Project Artificial Intelligence

Vakcode	XB_40002
Studiepunten	15.00
Periode	Ac. Year (sept)
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M. Cochez
Examinator	dr. M. Cochez
Betrokken Docenten	S.M.A. Zaghow
Onderwijsvormen	Lecture

Doel vak

After completing this course, the student:

• has demonstrated its individual ability to apply knowledge from at

least one of the key areas of Artificial Intelligence (Apply knowledge and understanding) to a relevant problem, resulting in a tangible outcome;

• has demonstrated an analytical and critical way of thought and its ability to apply logical reasoning, with clear identification of assumptions and conclusions (Making judgements) (Apply knowledge and understanding);

• has shown to be able to manage one's own learning and development, including time management and organizational skills (Apply knowledge and understanding);

• has shown its competence in searching and critically processing professional literature in Artificial Intelligence (Making judgements) (Lifelong learning skills)

• has learned to present and communicate the results obtained during the project;

(Communication)

• has reflected on its own style of thought and working methods (Lifelong learning skills);

Inhoud vak

The Bachelor Project is meant as a conclusion of AI bachelor studies.

Each student will select a project based on available topics and supervisors. A student can select from an assortment of topics offered by

the department, suggest his/her own topic, or do a project within a company.

Part of the project consists of the study and review of literature.

The Bachelor Project AI is concluded with an individual report/thesis.

Aanvullende informatie onderwijsvormen

After an orientation meeting, the student selects a supervisor and a topic. The work, carried out on individual basis, results in a written report (thesis) and an oral presentation at the annual AI Bachelor Seminar. The student performs a number of reviews of final concept-reports by fellow students.

Toetsvorm

Grading is done via a rubric available on Canvas.

- evaluation of student work (50%)

- final report (40%)

- final oral presentation (10%)

Vereiste voorkennis

Students can only enroll for the Bachelor Project Artificial Intelligence XB_40002 after

obtaining 6 ECs for the course Introduction to Python Program XB_0082.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 3)

Toelichting Canvas

On Canvas you can find the Bachelor Project manual with detailed information, as well as the list of possible topics and supervisors.

Basics of Systems Biology

Vakcode	XB_0078
Studiepunten	3.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M. Gralka
Examinator	dr. M. Gralka
Betrokken Docenten	prof. dr. B. Teusink, dr. M. Gralka
Onderwijsvormen	Lecture, Practical

Doel vak

The properties of cells and higher levels of organisation emerges out of molecular interactions within huge networks. Examples of such molecular-reaction networks are: metabolism, signal transduction, and gene networks. Systems biology studies the principles of such networks using quantitative experiments, computer modelling and theory.

The aim of this course is to introduce the students to the approach taken in systems biology. We will study a number of theoretical findings of systems biology about the dynamics and regulation of molecular networks inside living cells using small "toy" mathematical models and basic principles of biochemistry and physics.

Inhoud vak

The following topics will be discussed:
- system definition
- balances of system variables
- mass-action kinetics and basic enzyme kinetics
- thermodynamic equilibrium and steady state
- dynamics of simple systems
- protein complex formation
- ultrasensitivity of small signalling circuits
- supply and demand as coarse-grained method to dissect regulation of metabolism
- robust integral feedback control
- role of positive and negative feedback

The practicals involve answering exercises and analysing results of mathematical models.

Aanvullende informatie onderwijsvormen

Lecture/Q&A sessions (2h/week), computer practicals and work groups (2h/week), 7 weeks in total.

Toetsvorm

Written exam (open book). Learned techniques will have to be applied to new systems.

Literatuur

Syllabus and selected papers provided via Canvas

Aanvullende informatie doelgroep

Students interested in the intricate workings of biological systems and not afraid of basic equations and computer models describing the properties of dynamic systems.

Overige informatie

This course is part of the regular MNW curriculum but also open for students who want to follow it as part of a minor (e.g. bioinformatics and systems biology)

Aanbevolen voorkennis

Basic knowledge of mathematics, (bio-)chemistry, and cell biology.

Computer scientists or mathematicians can follow the course but will

need to plan time early on to quickly catch up with the biology (links

to material will be provided).

Behaviour Genetics in Psychology (UM)

Vakcode	P_UBEGEPS
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. B. Sauce Silva
Examinator	dr. B. Sauce Silva
Betrokken Docenten	dr. E. van Bergen, dr. B. Sauce Silva, prof. dr. M. Bartels, dr. R. Pool, dr. J. van Dongen
Onderwijsvormen	Lecture

Doel vak

To gain knowledge on the latest research and state of affairs within the field of behavioural genetics applied to psychology.

Inhoud vak

Behaviour genetics is the field that uses genetic methods to investigate why certain traits run within families. Family members may be alike in personality, behaviour, cognition, psychological health, and lifestyle because they share their home environment but also because they are genetically related to each other. This course will focus on the influence of genes and the environment, and their interplay, on individual differences between people. We will discuss various research methods in behaviour genetics, including twin and family studies, and genome-wide association studies.

Note that before 2022, this course was called "Twin Research in Psychology".

Aanvullende informatie onderwijsvormen

14 lectures (fully on campus), weekly online assignments, and self-study.

Toetsvorm

The multiple-choice exam will consist of questions covering the literature and lectures. To complete the course you must pass the exam, hand in the essay writing activity, review 2 blogs written by other students, and have completed and submitted via CANVAS at least 80% of the other assignments.

Literatuur

The textbook is recommended but not mandatory. Use the book as supporting material: to see more examples, to find background information, to get references, etc.

Behavioral Genetics (7th Edition). Editors Knopik, Neiderhiser, DeFries,

and Plomin. New York, NY: Worth publishers (ISBN: 978-1464176050).

Aanvullende informatie doelgroep

Students in the University Minor "Psychology and the Brain".

This course is NOT meant for BSc VU Psychology students. If you're a student in the Psychology department, you CANNOT take this course.

Biochemistry

Vakcode	AB_1137
Studiepunten	6.00
Periode	P2
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. ir. Y.J.M. Bollen
Examinator	dr. ir. Y.J.M. Bollen
Betrokken Docenten	dr. ir. Y.J.M. Bollen, prof. dr. B. Teusink, S. Palacin i Bonson
Onderwijsvormen	Study Group, Lecture, Practical, Written partial exam, Computer lab

Doel vak

The general aim of this course is to generate a basic, molecular,

understanding of essential components, mechanisms, forces and

processes that support cellular function, as a basis for later courses in

molecular life sciences. After taking this course, the student can:

• Identify important biochemical building blocks and how they are

assembled into macromolecules.

• Describe the forces and interactions that determine the structure and

composition of biological macromolecules.

• Explain what determines the direction of a biochemical reaction, how

enzymes work, how enzymes are regulated and which role membranes

have in cells.

• Explain how macromolecules function in networks within living cells

and how they cooperate in order to execute important life processes,

with particular attention for cellular energetics and the concepts,

coherence and regulation of metabolic processes in a human cell.

• Apply mathematical techniques as learned in secondary school to solve

biochemical problems.

• Use simple mathematical models of biochemical processes and explain the relevance of the outcome.

• Apply various biochemical and biophysical techniques with attention for biological and chemical safety rules.

• Setup a biochemical experiment, keep record of the results, analyze them and present them in a written report.

Inhoud vak

During the course, the following topics will be covered:

• The concept of Gibbs energy and changes in Gibbs energy during reactions and interactions of molecules.

• Biological macromolecules and the building blocks they are composed of, with particular attention for proteins.

• The most important roles of proteins, in particular catalysis and its regulation.

• The structure of biological membranes and relevant transport processes across them.

• The concepts of catabolism and coupled energy transformations.

• The quantitative analysis of cellular processes and the required mathematical tools.

The following learning pathways are incorporated into this course:

Academic skills

Mathematic skills

Bioinformatic skills

Aanvullende informatie onderwijsvormen

Lectures: 32 hours. Tutorials: 14 hours. Practical: 22 hours. Self study: 100 hours. This course is taught using the "flipped classroom" concept. This means that students need to study the available material (on Canvas) first, before coming to the lecture on the respective topic. The lectures will have an interactive format focussed on answering questions, applyingknowledge and solving problems.

Toetsvorm

The grade for this course is composed out of three parts: one grade for knowledge and understanding of theory tested in a digital exam (50%), one grade for the application of mathematics to solve biochemical problems tested in a written (paper) exam (20%),
and one grade for the practicals, based on a written report (30%).
Explanatory notes:
(i) The grade for the practicals is based on the written report, provided that the student has actively participated during the practical

training. In the case of an insufficient grade, the report can be corrected one time based on the feedback provided. The grade for a corrected report cannot exceed 6.0.
(ii) In order to pass the course, sufficient grades (5.5 or higher) must be obtained for all three parts of the course.

Literatuur

Marks' BASIC Medical Biochemistry, A clinical approach
5th or 6th edition
Michael Lieberman & Alisa Peet
Wolters Kluwer USA
ISBN-13: 978-1-4963-8772-1
ISBN-10: 1-4963-8772-4

Aanvullende informatie doelgroep

Compulsory course for first-year BSc Biomedical Sciences students.

Overige informatie

Presence and preparation are compulsory for tutorials and practicals.
During laboratory practicals, wearing a lab coat is obligatory.

Afwijkende intekenprocedure

You need to register yourself for the course via VUnet, including lectures and (partial) exam(s). The faculty will register you for all remaining teaching methods.

Please note that if one of the grades is insufficient, you need to register for the resit exam in order to obtain your new course grade, even if it is the report of the practicals that was insufficient.

Biologische Psychologie (UM)

Vakcode	P_UBIOPSY
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. D. van t Ent
Examinator	dr. D. van t Ent
Betrokken Docenten	dr. D. van t Ent
Onderwijsvormen	Hoorcollege

Doel vak

Inzicht verwerven in de structuur en functie van het zenuwstelsel en de rol van het zenuwstelsel in (ab)normaal gedrag.

Inhoud vak

Begrippen uit de biologie aansluitend bij de processen die men in de

psychologie bestudeert. Aan de orde komen structuur en organisatie van

het centrale en perifere zenuwstelsel, neurotransmissie, psychofarmaca

en de biologische mechanismen achter waarnemen, motoriek, emoties, slaap en de

hogere cognitieve functies (geheugen, spraak). Tijdens de

colleges wordt tevens ingegaan op neurologische stoornissen (Parkinson,

Broca's afasie, Alzheimer etc.) en de biologie van gedragsstoornissen

(slaapstoornissen, psychosen, angst, depressie, verslaving).

Aanvullende informatie onderwijsvormen

Hoorcollege

Toetsvorm

Tentamen

Literatuur

Carlson and Birkett - Physiology of Behavior

(13th Ed). Pearson. Verkrijgbaar in de VU boekhandel.

Brain in Trouble

Vakcode	AB_1038
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. H.K.E. Vervaeke
Examinator	dr. A. Baldi
Betrokken Docenten	dr. H.K.E. Vervaeke, prof. dr. T.J. de Vries
Onderwijsvormen	Study Group, Computer lab, Lecture

Doel vak

The goal of this course is to deepen understanding of the etiology, expression and treatment of (psychiatric) brain disorders, as well as models used in preclinical science. Students will be encouraged to critically analyze the impact of brain disorders on society.

Learning outcomes:

After completion of this course, the student is able to:
1) Explain the contribution of genetic and environmental factors to complex multifactorial diseases such as mental traits and mental disorders
2) Elaborate on the etiology of addiction, ADHD, obsessive-compulsive disorder, eating disorders, mood disorders and anti-social personality disorder
3) Elaborate on the various treatment options for psychiatric disorders
4) Explain gene-environment interactions and discuss some examples
5) Elaborate on the differential role of the environment in the etiology of traits / disorders according to the ‘differential-susceptibility hypothesis’
6) Summarise, understand and apply the results of current neurobiological research into behavioural interventions to increase brain health and mental well-being
7) Critically reflect on the impact of mental disorders on society
8) Critically reflect on the boundaries between normal (healthy) and abnormal (ill) behaviour and the implications for society
9) Verbally defend a position on the various topics of this course (class discussions, ACADEMIC SKILL)

Inhoud vak

The focus of this course is on the etiology of mental disorders, such as addiction, ADHD, obsessive-compulsive disorder, eating disorders, mood

disorders and anti-social personality disorder, with special attention to the nature-nurture discussion. Various treatments options for these conditions will be discussed, including the use of pharmacological agents, behavioral therapy and deep brain stimulation. Students will be challenged to critically reflect on the boundaries between normality and abnormality and the implications for society.

First Theme: addiction and impulsivity

What is addiction? Is addiction truly a brain disorder? Do genes play a role in addiction? How does society view illicit drug use and addiction? Are all drugs equally harmful? How to treat addiction? Is ADHD a real mental disorder, or a cultural construct used to bring deviant or socially undesirable behavior under medical surveillance and control? Is it a good idea to treat children who have been diagnosed ADHD, with psychostimulant medications? What is the role of pharmaceutical companies? Do sugar and food additives elicit hyperactive behavior? Are there any advantages in having ADHD?

Second Theme: obsessive compulsive disorders, eating disorders and cognitive enhancement

Can you treat OCD with Deep Brain Stimulation? Is our Western beauty ideal at the root of eating disorders? Is the individual to blame for being obese? Is it ethical to improve your mental performance by cognitive enhancers?

Third Theme: mood & social behaviours

Is depression a real brain disorder or an inability of our culture to accept sadness as an integral part of life? Do genes play a role in the etiology of major depressive disorder and bipolar disorder? What is the efficacy of pharmacotherapy and behavioral therapy? What is the role of pharmaceutical companies?
Is there a neural basis to antisocial behavior? If biology and circumstance conspire to prime certain individuals toward violence, how much responsibility do people really bear for their actions? Are violent delinquents worth treating? Should brain imaging / genetic profiling be used in legal cases? Can neuroscience assist in determining responsibility? If neural circuitry underlying morality is compromised, is it morally wrong to punish prisoners? Are there positive aspects to psychiatric disorders?
Next, the healthy mind will be the focus. What is the current state of neurobiological research into behavioral interventions to increase brain health and mental well-being?

Aanvullende informatie onderwijsvormen

Lectures (30 hours), computer practical (2 hours), class discussions (2 hours)

Course coordinators are Hylke Vervaeke and Taco de Vries

Toetsvorm

Written exam (combination of MC-questions and open-end questions) (75%) and class discussions (25%), each at least grade 5.5

Vereiste voorkennis

This course is part of the minor Brain and Mind. University students need at least 90 ECTS to be eligible for (courses in) this minor. HBO students can follow (courses in) this minor if they have at least 120 ETCS.

Literatuur

"Foundations Of Behavioral Neuroscience" by N.R. Carlson (Pearson

Education (US)), 8th edition.

Extra literature on Canvas

Aanvullende informatie doelgroep

Part of minor Brain and Mind
Open to students from all educational backgrounds (e.g., exact, social, life and economic sciences) with an interest in the brain and mind.
Due to curriculum overlap, students from Biomedical Sciences and Health and Life Sciences from the VU cannot follow this course.

Overige informatie

Central Academic Skill: Debating and discussing

Afwijkende intekenprocedure

Groups for Class Discussions via Canvas

Aanbevolen voorkennis

The courses 'Cognitive Neuroscience' and 'Nature vs. Nurture' from the minor 'Brain & Mind'

Business Model Assessment

Vakcode	E_MUB_BMA
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. G. Tumer-Alkan
Examinator	dr. G. Tumer-Alkan
Betrokken Docenten	dr. G. Tumer-Alkan, dr. V. Duplat
Onderwijsvormen	Study Group, Lecture

Doel vak

This course has three main learning objectives:

· Gain knowledge of basic concepts and theories in corporate finance in order to assess the business models.

· Provide standard answers to hypothetical cases, e.g. through solving textbook exercises

· Apply obtained knowledge in corporate finance to real life cases, e.g. interpret financial information and provide input for corporate decision making, and improve strategic thinking.

Inhoud vak

The course will start with an introduction of business assessment approaches and basic concepts. We will start with an introduction to corporations, and proceed with financial statement analysis, financial decision making, investment decision rules, capital budgeting, and raising equity capital, etc. The focus is on applying concepts and theories to real-life situations during lectures, and providing students with feedback on their exercises and cases in the tutorials. We will explain the basic concepts and theories in the lectures, and apply to relevant exercises and cases in the tutorials.

Aanvullende informatie onderwijsvormen

Lectures and tutorials

Attendance to both lectures and tutorials is highly recommended to pass the course

Toetsvorm

Your final course grade will be based on your written exam and assignments. More details are available in the course manual.

Literatuur

“Corporate Finance” by Jonathan Berk and Peter DeMarzo, 4th edition.

Overige informatie

Taking the other P1 and P2 courses in the minor (New Ways of Doing Business) is strongly recommended.

Aanbevolen voorkennis

P1 courses in the minor

Business Models: Traditional and Emerging Perspectives

Vakcode	E_MUB_BMT
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	prof. dr. T.P. Moliterno
Examinator	prof. dr. T.P. Moliterno
Betrokken Docenten	prof. dr. T.P. Moliterno
Onderwijsvormen	Study Group, Lecture

Doel vak

Though the combination of lectures, case discussions, guest lectures, teamwork, and an individual essay, students will encounter and engage with core concepts surrounding the dimensions of competitive advantage, industry-level competitive dynamics, business model design, and business model innovation. This combination will help students develop a competence in understanding and analyzing business models and identifying opportunities for business model innovation in today’s dynamic business environment.

After successfully completing Business Models: Traditional and Emergent Perspectives, students will:

- understand the core theories and ideas surrounding business-level strategy
- understand how traditional and emergent business models implement a business-level strategy
- be able to effectively apply their conceptual understanding of the course concepts in real-world settings
- be able to draw on evidence to put forward well-founded and substantiated points of view, both in spoken and written formats
- gain comfort and competence in discussing business concepts in unprepared/real-time discussions and debates
- improve their skills in working effectively with teammates from with diverse backgrounds
- gain competence in allocating tasks among team members under time pressure
- develop their critical thinking and creativity in their problem-solving approach
- hone the critical thinking need to analyze and evaluate complex organizational problems and make well-reasoned decisions

Inhoud vak

The central concern of strategic management is understanding what drives firm performance. Why do some firms outperform others? Why do some prosper and grow, while others in the same industry die? In short, how—and why—do firms create value? The firm’s strategy is its approach to gaining a competitive advantage and its business model is way it executes on that strategy to create, deliver, and capture value .

This course begins with an in-depth exploration of ideas around business-level competitive advantage, value creation, and industry-level competitive dynamics. We build on these foundations to explore how business models

address these competitive realities. In this way, the course provides a "deep-dive" into the relationship between an organization's strategy and its business model. Students will learn about both traditional and emergent business models and explore their connection with theories of business-level strategy.

Aanvullende informatie onderwijsvormen

This course takes an active blended learning approach that leverages both synchronous (face-to-face & on-campus) and asynchronous (remote & on-line) modes of teaching and learning. Students are expected to fully engage in both the synchronous and asynchronous classes. Remote participation in the synchronous on-campus sessions is not an option.

The first lecture every week will comprise one or more “knowledge clips” that students will watch on-line. These knowledge clips are short videos that will introduce the topics and lay the foundation for the material students will explore that week. Note that the material in this course is cumulative, so the ideas in any one week’s knowledge clips will build on, and require an understanding of, the material from prior weeks. To get the full value from these knowledge clips, the readings assigned for these sessions must be read before watching the videos. The material presented in the knowledge clips will assume that students have read the assigned materials. After completing the readings and watching the knowledge clips, students will take a comprehension quiz.

On-campus lecture sessions will provide students with the opportunity to explore more fully the material presented in the week’s knowledge clips. These classes will be interactive: there will be no direct lecturing, and the content for each session will be determined by the students' questions, interests, and results from the comprehension quiz. Students should come to the class with the questions, observations, thoughts, and points of clarification that occurred to them when they did the readings and watched the knowledge clips for the week. Guest speakers may also scheduled during these sessions. Attendance is not required for these sessions, but is strongly advised in order for the student to get the full benefit of the course. These sessions will not be recorded or streamed. Students actively participating during these sessions may earn extra participation points for the week.

The tutorials combine case studies and team presentations/work. Students should come to the tutorials having fully prepared the case and/or the other readings assigned for the session. These will be discussion-based sessions, where the students are expected to do most of the talking: active in-class participation is required and graded. Since much of the work we will do in the tutorials is team-based, students must come to the tutorial they for which they registered and attend that same tutorial for the whole course: there will be no exceptions to this. Attendance is mandatory for all tutorials. Students are allowed one unexcused absence from the tutorials. Each additional absence will result in a 0.5-point deduction from the student’s final grade for the course.

One distinguishing aspect of this course is its interactive nature. Students are expected to be active participants in all class discussions. As such this course provides the opportunity for students to become more comfortable with, and develop their competency in, an important skill for career success: engaging in unscripted and organic group discussions and debates.

Toetsvorm

There are both individual and team-based assessments.

Individual: class participation, weekly quizzes, essay (NOTE: there is no exam &in-class participation is graded)

Team: company analysis, presentations, and projects TBA

Vereiste voorkennis

There are no mandatory entry requirements. Students are STRONGLY encouraged to take the first two courses in the "New Ways of Doing Business" minor program: this course will build on some of the material covered in those courses. For students not taking the prior courses in the minor, some familiarity with, and interest in, modern business will be helpful.

Literatuur

Required readings will include on-line resources and articles available through the VU library. There is no required textbook.

Aanvullende informatie doelgroep

This course is in the New Ways of Doing Business Minor in Business Administration.

Toelichting Canvas

Outside of the classroom, Canvas will be used as the main mode of communication between the instructor and students.

Business Professionals

Vakcode	E_MUB_BPROF
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	drs. A.C. Guldemond
Examinator	drs. A.C. Guldemond
Betrokken Docenten	drs. A.C. Guldemond
Onderwijsvormen	Study Group, Lecture, Instruction course

Doel vak

In the course Business Professionals, the focus is on the human element in the business modeling paradigm. The overall objective is gain knowledge about business models and management from the perspective of the professional.
In addition to academic skills, the course emphasises professional skills, including creativity, communication, team work and reflexivity. This is covered both in the theory part as wel as in the training part (tutorials) of this course.
In particular, when students complete this course, they will:
• Understand the profiles of key business professional roles such as chief executive officers, marketing, finance, human resources, operations and technology executives
• Be able to apply ideas about professionals for a reflection on their own background, personal role and career development as a (future) business professional
• Be able to formulate and analyze business modelling problems from the perspective of the business professional
• Be able to report verbally and in written report on assignments in the course.

Inhoud vak

During the course students will explore cases and theories about the contribution of professionals in management and organization. Guiding questions are: Who are the people behind the key strategic decisions for the business model of an established firm or a new business venture? What functions, behaviors and capabilities are required for successful collaboration on the design and implementation of new business models? The content of the course entails an even-handed appreciation for theory and practice.

Aanvullende informatie onderwijsvormen

Lectures and tutorials --- Attendance to both, lectures and tutorials, is highly recommended to pass the course
In the first part of the course, lectures start with an introduction to (management) professionals; their task, responsibilities, and activities. Throughout the tutorials, students have the opportunity to

apply the theoretical frameworks introduced in the lectures. To this end, the tutorials combine assignments, case studies and round-table discussions. Students are expected to actively contribute to the group's experience and learning.

Toetsvorm

Written exam, assignments, presentation

Literatuur

A selection of articles, cases and support materials and the following book.
Book: Hitt, M. A., Miller, C. C., & Colella, A. (2017). Organizational behavior. John Wiley & Sons

Overige informatie

Taking the other P1 and P2 courses in the minor Business Administration is strongly recommended

Business Project

Vakcode	E_MUB_BPROJ
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. L. Glasbeek
Examinator	dr. L. Glasbeek
Betrokken Docenten	dr. L. Glasbeek
Onderwijsvormen	Study Group, Lecture

Doel vak

Upon successful completion of the course, students have developed themselves in the following five areas:

(1) Academic and Research Skills

- Students can examine management challenges from different perspectives ("analysis").

- Students can recognize fundamental structures and leave out irrelevant information ("abstraction").

- Students can put forward well-founded, substantiated points of view, both in spoken and written format ("argumentation").

- Students can design and execute a management consultant research project and develop meaningful interpretations and presentations of their findings.

(2) Bridging Theory and Practice

- They can identify and apply relevant management theories to organizational challenges and meaningfully formulate practical and innovative recommendations to overcome these.

(3) Social Skills

- They can productively and independently engage in teams and, where applicable, with external organizations.

- They can put the needs of others before their own.

- They can lead others and communicate with integrity and conviction.

(4) Broadening Your Horizon

- They can consider society at large when addressing complex business issues.

(5) Self-Awareness

- They can critically reflect on their actions, responsibilities, and personal development.

Inhoud vak

The course's main objective is to familiarize students with some of the most significant challenges of conducting a management consultancy project. The course encourages a critical appraisal of various

theoretical perspectives in the business administration domain and empirically investigates an applied research question. By designing and executing a management consultancy project, students become acquainted with the challenges that researchers face when considering management at large. Lastly, and importantly, students will learn to communicate their views succinctly, both verbally and through a written research report.

Aanvullende informatie onderwijsvormen

Lectures and tutorials.

Toetsvorm

Management Consultancy Report – Team assessment.

Intermediate Progress Reports & Presentations – Team assessment.

Professional Reflection – Individual assessment.

Vereiste voorkennis

University Minor Business Administration Courses: Foundations of Business Administration, Business Professional, Business Model Innovation, and Business Model Assessment.

Literatuur

Selection of articles.

Calculus

Vakcode	X_400617
Studiepunten	6.00
Periode	P1+2
Vakniveau	100
Onderwijstaal	Nederlands
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	L.E. Toussaint
Examinator	L.E. Toussaint
Betrokken Docenten	L.E. Toussaint
Onderwijsvormen	Werkcollege, Deeltentamen schriftelijk, Hoorcollege

Doel vak

In deze cursus leert de student:

• vergelijkingen en ongelijkheden op te lossen,

• te werken met speciale functies (waaronder goniometrische functies en hun inversen, exponentiële functies en logaritmen),

• berekeningen en vergelijkingen op te lossen met complexe getallen,

• limieten te bepalen,

• technieken voor differentiëren en toepassingen (waaronder impliciet differentiëren, Taylorpolyomen, berekenen van extremen),

• verschillende integratietechnieken (waaronder substitutie, partiële integratie en breukspitsen),

• verschillende typen differentiaalvergelijkingen op te lossen,

• werken met reeksen. In het bijzonder het berekenen van Taylorreeksen, de meetkundige reeks, machtreeksen, termsgewijze differentiatie en integratie van machtreeksen en vinden van de convergentiestraal van een machtreeks met behulp van de ratio test.

Inhoud vak

Deze cursus behandelt reële functies van één variabele, waarbij we zonder rekenmachine werken. Aan de orde komt:

• Oplossen van vergelijkingen en ongelijkheden.

• Werken met speciale functies en met inversen van functies. In het bijzonder behandelen we goniometrische functies en hun inversen (arcsinus, arccosinus en arctangens), exponentiële functies en logaritmen.

• Berekenen van limieten (ook met behulp van l'Hôpital).

• Differentiëren van functies en toepassingen: definitie met behulp van een limiet, rekenregels, afgeleiden van standaardfuncties, raaklijnen en normalen, Taylorpolynomen, extreme waarden en buigpunten.

• Integreren van functies: primitieven van standaardfuncties, integratietechnieken zoals substitutie, partiële integratie en breuksplitsen.

• Oplossen van verschillende typen differentiaalvergelijkingen. In het bijzonder eerste orde differentiaalvergelijkingen (scheiden van variabelen en integrerende factor) en lineaire tweede orde

differentiaalvergelijkingen met constante coëfficiënten (ook inhomogeen). Het college bevat enkele toepassingen van differentiaalvergelijkingen op fysisch, biologisch en chemisch gebied.

Complexe getallen worden geïntroduceerd. We leren optellen, aftrekken, delen, vermenigvuldigen en machtsverheffen met complexe getallen, zowel in rechthoekige coördinaten als in poolcoördinaten. Bij dit laatste gebruiken we de modulus en argument notatie en complexe e-macht. Eenvoudige complexe vergelijkingen worden opgelost.

Ook leert de student werken met reeksen. In het bijzonder gaat het om het berekenen van Taylorreeksen, werken met de meetkundige reeks, werken met machtreeksen, waaronder termsgewijze differentiatie en integratie van machtreeksen en vinden van de convergentiestraal van een machtreeks met behulp van de ratio test.

Aanvullende informatie onderwijsvormen

4 uur hoorcollege per week, 2 uur werkcollege per week.

Toetsvorm

Twee deeltentamens (elk 40%) en wekelijkse digitale toetsen (20%). Om te slagen voor het vak moet bovendien het gemiddelde van de twee deeltentamens hoger zijn dan 5,0.

Indien de student de cursus niet via de deeltentamens haalt, moet een hertentamen over de volledige stof gedaan worden. De deeltentamens kunnen niet apart herkanst worden. In dit geval tellen het hertentamen voor 80% en de wekelijkse digitale toetsen voor 20%. Ook nu moet voor het hertentamen tenminste een 5,0 worden gehaald om te slagen voor het vak. Indien het cijfer voor het hertentamen hoger is dan het cijfer voor de digitale toetsen, dan is het eindcijfer gelijk aan het hertentamencijfer.

Literatuur

Adams, Robert A and Essex, Christopher, Calculus: a complete course,

10th Edition, Pearson, 2021.

Aanvullende informatie doelgroep

1SBI, 1MNW, 1FAR, Minor Bioinformatics and Systems Biology.

Overige informatie

Dit vak wordt gegeven in het Nederlands.

Dit vak maakt deel uit van de Minor Bioinformatics and Systems Biology.

Aanbevolen voorkennis

VWO wiskunde A of B

Calculus 1

Vakcode	X_400635
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. T.O. Rot
Examinator	dr. T.O. Rot
Betrokken Docenten	dr. T.O. Rot
Onderwijsvormen	Lecture, Seminar, Written partial exam

Doel vak

At the end of this course the student is able to

a) calculate limits, using several methods like l'Hopitals rule or the squeeze theorem

b) calculate derivatives and find local extreme values

c) calculate integrals, using several methods like the substitution method, integration by parts, partial fraction expansion

d) verify if a function is continuous, differentiable, Riemann-integrable

e) calculate and apply a Taylor polynomial

f) formulate and apply several important theorems for continuous and/or differentiable functions, like the Intermediate Value Theorem, the Mean Value Theorem and the Fundamental Theorem of Calculus

Inhoud vak

Real functions of one variable. Topics that will be treated are:

1) Preliminaries, Real functions, Trigonometric functions

2) Limits, Continuity, Intermediate Value Theorem

3) Transcendental Functions, Inverse Functions

4) Differentiation, Chain Rule, Mean Value Theorem

5) Applications of Differentiation, Extreme Values, l'Hôpital's Rule, Taylor Polynomial

6) Integration, Fundamental Theorem of Calculus, Improper Integrals

Aanvullende informatie onderwijsvormen

Lectures (3 x 2 hours per week) and exercise classes (2 x 2 hours per week).

Toetsvorm

There is a midterm exam and a final exam. The midterm exam makes up for 40% and the final exam makes up for 60% of the final grade. There is one retake which covers all topics for the midterm exam and the final exam. Midterm exam or final exam cannot be retaken separately!

Literatuur

Adams and Essex, Calculus: A Complete Course, 10th edition, Pearson 2021, ISBN-13: 9780135732588. Older editions of the book are also fine, but please make sure that the numbering of the exercises match.

Aanvullende informatie doelgroep

1 BA

Choices, Inequality and Welfare

Vakcode	E_MUE_CIW
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. S. Hochguertel
Examinator	dr. S. Hochguertel
Betrokken Docenten	M. Mangan, dr. S. Hochguertel
Onderwijsvormen	Study Group, Lecture, Written partial exam

Doel vak

This course focuses on the interaction between the market mechanism and government policy. Policy may aim to correct market failure and work towards maximization of social welfare, which embodies trading off interests of many groups in society, both with respect to efficient use of scarce resources, and (in)tolerance to inequality or poverty. Conversely, the market mechanism is a powerful tool that can set free innovative dynamics, with implications for growth and inequality. This course helps you understand individual and social choice, consequences for market outcomes, and the foundations of policy decisions at a deeper and analytical level. We cover a broad set of topics, reflecting pressing issues in present-day complex societies. We foster understanding of core concepts and applications in economic modeling and statistical data analysis, using a novel playful and intuitive approach that leverages interactive graphical visualization tools.

We pursue two types of learning outcomes:

ACADEMIC AND RESEARCH SKILLS

After successfully completing the course Choices, Inequality and Welfare, you are able:

- to understand microeconomic models, their predictions and sensitivity to policy parameters

- to formulate the economic rationale for policy intervention and to develop policy options from economic theories;

- to apply tools of theoretical and empirical economic modeling;

- to interpret economic data in light of policy evaluation methods.

BRIDGING THEORY AND PRACTICE

After successfully completing the course Choices, Inequality and Welfare, you are able:

- to discuss the role of economic policy in the context of both market failures and government objectives to adjust market outcomes;

- to evaluate existing and potential policy options, both in theory and in practice;

- to show a critical attitude to existing theoretical and empirical policy analyses of current economic problems.

Inhoud vak

This course is geared at understanding the role of government policy from a predominantly microeconomic perspective. Not only will individual actors make choices that influence their own well-being, but government may pursue maximization of welfare for society as a whole, taking into account social inequalities between households.

We start with an analysis of individual choice and ask under what conditions a model society can achieve allocative efficiency. Competition is part of the answer. We then bring social choice considerations into play, and review the need for policy to address market failures and inequality, trading off efficiency and equity. Fighting poverty, or addressing the causes and consequences of climate change are among the core examples that we study. But government also needs to provide social insurance where private markets fail, and it needs to regulate monopolistic industries where competition cannot be introduced. In addition, we study empirical frameworks to evaluate policies, and apply them using economic data.

The following fields are prime candidates for discussion:

• taking into account the trade-off between efficiency and equity when formulating policy goals in an interconnected world;

• measuring inequality of incomes and the role of distributional policy;

• development and trade: analysis of living standards and poverty, provision of legal and political frameworks, trade protection and policy;

• environment: externalities from pollution, regulation and tax solutions; public goods and free rider behavior; climate policy challenges and trade effects;

• statistical methods of economic data analysis, such as linear regression, and policy evaluation methods

• social insurance and asymmetric information: health & disability insurance, adverse selection and moral hazard, welfare payments;

• competition policy and regulation: imperfect competition, market power, cartels, price-discrimination, regulation and de-regulation.

Aanvullende informatie onderwijsvormen

2 Lectures and 1 tutorial per week.

Toetsvorm

Exams (midterm and final); weekly problem sets.

Vereiste voorkennis

This course is on level 300 (advanced). It assumes a basic knowledge of mathematics at the high school level (differential calculus) and basic statistics skills as provided in the academic core of many Bachelor of Science programs at Vrije Universiteit Amsterdam or equivalent.

Literatuur

Main text: online lecture notes; various academic papers and ancillary textbook chapters, to be announced on Canvas or in the course manual.

Aanvullende informatie doelgroep

Third-year bachelor students of any major (except Economics and Business Economics; and Econometrics).

Overige informatie

This course is an integral part of the University Minor Economics; participants gain strongly from attending the

entire minor program. This course prepares for Applying Economics. We use tailor-made applications with computer-based tools and modern statistical software.

Aanbevolen voorkennis

The course builds on previous courses in the Minor Economics program, in particular: An Introduction to Economics and Economic Challenges for Minor Students. Familiarity with contents of those courses is assumed. Please attend the brief Math Refresher of the Minor Economics prior to the start of the semester (end of August) to check your high school level maths and basic statistics skills. You can prepare for the Refresher using our computer-based online self-test. For those who want to practice a bit more after the Refresher, we offer additional practice assignments before the course starts.

Climate Change Law

Vakcode	R_TL-TP
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	C. Kaupa
Examinator	C. Kaupa
Betrokken Docenten	C. Kaupa, dr. D. Rossati LLM
Onderwijsvormen	Lecture

Doel vak

The course analyzes climate change as an evolving, transnational legal phenomenon.
Students will learn to work across different legal fields (ranging from international and human rights law to private and economic law) and different jurisdictions (including international, European, national and local regulation), and to handle legal questions in the context of complex and shifting economic, political, social and ethical debates.
Students will be encouraged to participate in the course of the lectures, with the
goal of developing the sort of critical and analytical skills conducive to the practice of transnational law, and to understanding transnational global developments. The course approach matches the faculty's "Law in Action" profile.

Eindtermen: 1-5, 7-14, 16, 18-22.
Zie voor de eindtermen het Onderwijs- en Examenreglement Bachelor Rechtsgeleerdheid.

Inhoud vak

Climate change is one of the most pressing issues the world faces in the 21st century. It is also a particularly complex and interesting problem from a legal perspective: this is because climate change affects multiple jurisdictions (from the international to the local level), numerous areas of law (ranging from international to private law) and multiple actors (ranging from governments and international organizations to multinational businesses, NGOs and private citizens). Moreover, complex scientific, economic, political, social and ethical questions feed into the legal processes.

Analyzing the interaction of different legal fields:
Greenhouse gases originate from a broad range of activities, including energy production, industry and transport to agriculture. These are regulated in, or otherwhise affected by, numerous fields of law, such as international law, European and national economic law, private law, environmental law, international trade and investment law and human rights law. Tackling climate change therefore requires understanding how these various legal fields interact.

Analyzing how different jurisdictions interact:
Climate change is a transnational phenomenon, having local causes, but creating global effects: consequently, the problem must be addressed at the same time at a global scale, by regional organizations (such as the EU), at the national and at the regional level (e.g. cities). The course will look at how these different jurisdictions interact.

Understanding the role of different legal actors:
Climate change is not only a concern for national governments and international organizations. The European Union, as a regional organization, has long been an important actor in this field; moreover, non-state actors play an important role as well: multinational businesses, NGOs and private citizens aim to influence the regulatory process, most notably by bringing lawsuits. The course will analyze the activities of these different actors.

Understanding the context of climate change law:
Climate change has complex scientific, economic, political, social and ethical dimensions: for example, given that the emission of greenhouse gases is related to many different business sectors, a transition towards a low-carbon society will likely transform the existing economy in significant ways. This will inevitably create „losers“ along the way (e.g. coal and oil companies), who may aim to slow down the transition, thereby posing difficult economic and political questions. Or, to give another example, as greenhouse gas emissions are related to consumption, they are mainly attributable to the wealthy parts of the global population; however, climate change disproportionately affects poor populations in developing countries, and therefore raises complex ethical issues. In this course, we will study how scientific, economic, political, social and ethical questions feed into the legal process.

The course will cover:
Part 1: the science, economics and politics of climate change;
Part 2: Climate change as a global issue; the international climate change regime (e.g. Paris Agreement), international law, human rights law and international trade and investment law;
Part 3: European and national legislation (e.g. Emissions Trading System)
Part 4: Lawyering for change (e.g. lawsuits against governments and businesses in the US and in Europe)

Aanvullende informatie onderwijsvormen

Lectures, in-classroom research tasks, guest lectures, home assignments, group research project

Toetsvorm

Smaller written and oral assignments throughout the course and a final

written assignment.

Literatuur

The literature will be announced on Canvas.

Aanvullende informatie doelgroep

Apart from regular students, the course is also available for:
Students from other universities/faculties
Exchange students
Contractor (students who pay for one course)

Overige informatie

The following course objectives are only available in Dutch:

Eindtermen bachelor Rechtsgeleerdheid

De afgestudeerde bachelor beschikt over een fundamenteel academisch werk- en denkniveau;
-heeft kennis van en inzicht in de kernleerstukken van de hoofdonderdelen van het geldende recht (in het bijzonder het Nederlandse privaatrecht, staatsrecht, bestuursrecht, strafrecht en internationaal en Europees recht), alsmede de systematiek daarvan, met inbegrip van recente ontwikkelingen
-heeft kennis van en inzicht in het internationale en het Europese recht in hun verhouding tot het nationale recht
-heeft elementaire kennis van Engelse juridische terminologie
-beseft dat het recht zich ontwikkelt en manifesteert in een maatschappelijke context
-heeft kennis van de grondslagen van het (Nederlandse) recht, rechtshistorische en rechtsfilosofische aspecten en heeft besef van de eigen aard van de rechtsbeoefening

De afgestudeerde bachelor beschikt over de volgende (juridische) vaardigheden:
Analytische vaardigheden
-lezen, begrijpen en analyseren van juridische, rechtswetenschappelijke en rechtstheoretische teksten en betogen, waaronder jurisprudentie en wetgeving
-kritisch reflecteren op regelgeving, rechtspraak en literatuur, onder meer vanuit rechtshistorisch, rechtsvergelijkend en rechtsfilosofisch perspectief; is in staat om te reflecteren op de grenzen van het vakgebied
-reflecteren op de eigen maatschappelijke verantwoordelijkheid in de maatschappelijke context waarin het recht functioneert
-is in staat om juridische argumentatiestructuren te analyseren en op te zetten

Probleemoplossende vaardigheden
-selecteren van juridisch relevante feiten uit een feitencomplex
-selecteren van rechtsregels die bijdragen aan het oplossen van een juridische casus
-oplossen van juridische casus, waaronder begrepen hanteren van een systematische aanpak bij het toepassen van rechtsregels op concrete gevallen

Communicatieve vaardigheden
-een gefundeerde en beargumenteerde positie innemen in een maatschappelijk, juridisch debat

Informatievaardigheden
-op een efficiënte manier juridische bronnen raadplegen en informatie verzamelen uit juridische (digitale) bibliotheken en databestanden, en de waarde, relevantie en kwaliteit van de informatie beoordelen
-op efficiënte wijze relevante ontwikkelingen bijhouden en kennis actualiseren

Climate Repair

Vakcode	A_HP006
Studiepunten	6.00
Periode	P4
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	I. Maas
Examinator	prof. dr. P.H. Pattberg
Betrokken Docenten	I. Maas, prof. dr. P.J.H. van Beukering
Onderwijsvormen	Seminar, Lecture

Doel vak

Do you want to develop your climate leadership? Are you ready to learn new transferrable skills to address the climate crisis? Do you want to learn from practitioners how to put climate action into practice? Then join this course on climate leadership: Climate Repair. This is done by providing scientific knowledge about the climate crisis, inspirational masterclasses of climate leaders in their field, and reflecting on leadership in interactive seminars. Moreover, you will also be challenged to enact climate leadership yourself in this course by working on a group project.

The aims of this course therefore are:

- Increase self-awareness through self-reflection
- Broadening scientific knowledge regarding the climate crisis
- Developing climate leader competencies
- Develop multi-disciplinary skills by collaborating in group assignments with fellow students from different disciplines
- Exploring creativeness combined with climate leadership

Inhoud vak

Tackling the climate crisis is one of the largest societal challenges today. While the science on climate change and mitigation and adaptation strategies are clear, the response from society is often slow for various reasons. To overcome this deadlock, we need climate leadership and this course is centred around that. Leadership has to be based on sound scientific knowledge and should bridge fundamental science and social sciences. However, you will not only learn about the scientific background regarding repairing our climate but also be offered a more holistic and interdisciplinary approach to leadership. This means that you will learn what it means to be a leader, with the scientific

knowledge and understanding behind the current global crises.

Climate or sustainability leadership is a type of leadership that aims to strive for (organizational) change whilst taking into account planetary and societal boundaries. Sustainability leadership is based on dialogue, inclusion, collaboration, and participation. This type of leadership requires particular skills and these include:

- Systems thinking. Addressing climate issues requires understanding complex structures and seeing the interconnections between social and natural systems.
- Collaboration. To effectively work on complex issues you need others to determine an effective approach. Sustainability leaders guide others in navigating this complexity by facilitating dialogue and collaborative sensemaking and decision-making processes.
- Self-reflection. Sustainability leaders are self-reflective professionals, who are aware of the values that drive them and align their work with these values.

This course has three types of lectures. These are Lectures, Masterclasses, and Seminars. The lectures aim to teach you the basics of the climate crisis, leadership, and social systems change. During the masterclasses, we will invite climate leaders from society and invite a scientific leader to provide a scientific perspective. The masterclasses cover a wide range of areas in which climate leadership can be expressed. This includes Non-Governmental Organizations (NGO’s), Start-ups, Law, Activism, Geo-engineering, Art, and Government. The seminars will have a more open format in which we will reflect on the masterclasses and work on the group assignment.

Aanvullende informatie onderwijsvormen

Combination of seminars, lectures, masterclasses and excursions

Toetsvorm

Group assignment, presentation, self-reflection

Vereiste voorkennis

No prior knowledge is required

Literatuur

Downloadable journal articles, reports and papers.

Cognitive Neuroscience

Vakcode	AB_1056
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. C.P.J. de Kock
Examinator	prof. dr. M.P. van den Heuvel
Betrokken Docenten	dr. S. van der Sluis, prof. dr. S. Spijker, dr. C.P.J. de Kock, dr. P. Rao MSc
Onderwijsvormen	Study Group, Computer lab, Lecture, Practical

Doel vak

Introduction to the field of cognitive neuroscience: understanding the biological mechanisms underlying cognitive processes such as learning and memory, discussing recent developments in the field with leading scientists, and acquiring knowledge on how the brain, and its different cell types, systems, and function.

Inhoud vak

In the first course of this Minor, you will learn the basics of cognitive neuroscience through a series of introductory lectures on brain function and (dysfunctional) cognitive behavior. More specifically, we will teach you the structure and function of the major building blocks of the brain, ranging from single cells to neuronal networks, and from emotion to motor control. We combine workshops and keynote lectures, delivered by renowned neuroscientists, to discuss recent advances in the field of learning and memory, brain plasticity, and brain disease (e.g., developmental disorders, Angelman syndrome, PTSD). Finally, you will learn about frontier brain imaging methods (e.g. MRI) and experience various technical approaches to measure the brain (e.g., histology) in hands-on practicals.

Aanvullende informatie onderwijsvormen

Teaching and learning activities will be scheduled on campus (by estimation) 2 times/week. Additional activities include individual assignments, group assignments, computer programming and self study.

Toetsvorm

Written exam & assignments

Vereiste voorkennis

This course is part of the minor Brain and Mind. University students need at least 90 ECTS to be eligible for (courses in) this minor. HBO students can follow (courses in) this minor if they have at least 120 ETCS.

Literatuur

Recent literature, to be announced at the start of the course.

Foundations of Behavioral Neuroscience
Carlson, Neil R.
(9th edition)

Exam material:
CH2, CH3, CH5, CH6 (pg. 136 - 146), CH7 & CH12

Aanvullende informatie doelgroep

Open to students from all educational backgrounds (e.g., exact, social, life and economic sciences) with an interest in the brain and mind.
Due to curriculum overlap, students from Biomedical Sciences and Health and Life Sciences from the VU cannot follow this course.

Overige informatie

Coordinators: Christiaan de Kock and Martijn van den Heuvel.
No special requirements to be met.
Part of minor Brain and Mind.
This minor course requires a minimum of 25 participants to take place.

Cognitive Psychology for AI

Vakcode	XB_0072
Studiepunten	6.00
Periode	P4
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. J.J. Snell
Examinator	dr. J.J. Snell
Betrokken Docenten	dr. J.J. Snell
Onderwijsvormen	Lecture

Doel vak

The goal of this course is for students to acquire a good understanding of the mind

and brain, and the various aspects of human cognitive functioning. You

will learn about perception, attention, memory, language processing,

decision-making, and about realms of cognition in which these elements

are combined. You will also learn to adopt the general mindset that

cognitive scientists need in order to unravel human cognitive

functioning. Finally, you will gain experience in conducting (small)

experiments, and reporting the results of those experiments. In sum, you will gain knowledge, learn to apply that knowledge, and learn to communicate insights at an academic level.

Inhoud vak

We will cover most realms of cognition, as discussed in the book by

Goldstein (neurons and neural mechanisms, perception, attention, memory,

language, learning, etc.). We will also discuss the potential

shortcomings of various lines of research (when might evidence be

regarded as conclusive?), and the characteristics of good theoretical

quality.

Aanvullende informatie onderwijsvormen

The course will mostly consist of plenary lectures (2 per week), and
twice during the course period an assignment that involves doing a small

online
experiment, inspecting your own data, and reporting that data.

Toetsvorm

70% of the grade: Exam

10% of the grade: Participation (quizzes, small assignments)

10% of the grade: Report 1 (in a group)

10% of the grade: Report 2 (in a group)

For the exam there is a resit opportunity; for the other components there is not. However, the first report is only included in the final grade if it doesn't have a negative impact on the result.

Literatuur

Bruce Goldstein & Johanna van Hooff: Cognitive Psychology

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence

Community-based Health Interventions

Vakcode	AB_1110
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	G. Tijsma MSc
Examinator	G. Tijsma MSc
Betrokken Docenten	dr. E.J. de Weger
Onderwijsvormen	Study Group, Lecture

Doel vak

Introduction

This course is especially suitable for students who are interested in the minor Global Health as you will use your newly acquired knowledge and apply it in a community engaged learning project that runs from period 1-3. Students will work in project teams on a real-life transdisciplinary global health challenge together with various local and/or global community partners. Students will cross disciplinary boundaries and interact with health professionals, patients, residents and others to gain an in-depth understanding of global heath problems, and to take cohesive and strategic actions to address these problems at a local level. Each course in the minor programme will incorporate a different sub-project, starting in the first two courses with a broad exploration and needs analysis on the topic, followed by a more thorough investigation on a specific challenge in the next two courses, and concluding n the final course with a proposal for a sustainable innovation. Training workshops are offered throughout the minor programme to support the development of specific skills such as transdisciplinary collaboration, reflection, social justice, global citizenship, dealing with diversity, creativity, leadership, and adaptability.

Course Objective

- To describe and explain the current status, history and theoretical underpinning of Community based Health Interventions (CbHIs).
- To describe and evaluate the different aspects of CbHIs (including design, implementation, monitoring and evaluation of CBHI).
- Identify organizational conditions for implementing CbHI (like community engagement, organizational learning and change, and attention for knowledge transfer and sustainability).

- Analyse strengths and opportunities of CbHIs, leading to a well-grounded advice for optimization.
- Discuss and analyse case studies of CbHIs in their context in a participatory way that is ethically sound.
- Describe and explain knowledge and understanding through verbal as well as written communication.

Inhoud vak

Numerous interventions are developed and implemented in the area of health, care, and prevention. Although there is often much attention for national or even international scale interventions, community-based health interventions (CbHIs) are a rapidly upcoming phenomenon. The underlying transition of perceiving health as an individual attribute to health as a result of complex social and local aspects is supporting the importance of CbHIs.

CbHIs are an innovative approach to severe and complex problems. In CbHIs, health is perceived as the result of interaction between individual and environmental aspects. Therefore, implementers of CbHIs take an interdisciplinary and participatory approach to public health issues. For instance, (self) management of diabetes, sexual health, mental health, and obesity, and the living environment are addressed by CbHI, but also community problems like social cohesion, loneliness of elderly or limited access to drinking water can be targeted. CbHIs are flexible and participatory in nature. As a result, they are easier to adapt to specific situations and are often designed in collaboration with the target group.

This course focuses on why CbHIs are essential for solving complex health and care issues and the types of interventions involved. We will take you through the history of CbHIs and the theoretical foundations of this strategy. In addition, we will give insights into aspects of design, implementation, monitoring and evaluation of CbHIs, taking into account appropriate attitudes, skills and knowledge to influence public health in a community setting. The ethical issues involved in community work are very important and issues such as stakeholder participation, sustainability and scaling-up of the intervention and its effects are discussed. Furthermore, the importance of learning from and adapting to emerging issues is discussed in relation to implementing CbHIs. We will explore the importance of learning capacity of the organisations that implement CbHIs, and their role as spiders in a web of multiple stakeholders that are involved with different perspectives, objectives

and goals.

In the practical part of all courses in the minor Global Health, including this one, you will use your newly acquired knowledge and apply it in a community engaged learning project. Students will work in project teams on a real-life transdisciplinary global health challenge together with various local and global community partners. Students will cross disciplinary boundaries and interact with health professionals, patients and others to gain an in-depth understanding of global health problems, and to take cohesive and strategic actions to address these problems at a local level. Each course in the minor program will incorporate a different sub-project, starting in the first two courses with a broad exploration and needs analysis on the topic, followed by a more thorough investigation on a specific challenge in the next two courses, and concluding in the final course with a proposal for a sustainable innovation. Training workshops are offered throughout the minor program to support the development of specific skills such as transdisciplinary collaboration, reflection, dealing with diversity, creativity, leadership and adaptability.

Aanvullende informatie onderwijsvormen

Lectures (18 hours), Self-study (65 hours), Work groups (12 hours),
Field visits (8 hours), Assignment (65 hours)

Toetsvorm

Individual assessment (50%), and group assignment (50%). All parts need
to be passed.

Literatuur

Book: Community Based Health Interventions: Principles and Application

by Sally Guttmacher, Patricia J. Kelly, Yumary Ruiz-Janecko

Articles on Canvas: e.g. Dahlgren, G., & Whitehead, M. (1991). Policies and strategies to promote social equity in health. Stockholm, Sweden: Institute for Future. *Studies*.

Aanvullende informatie doelgroep

Bachelor students from all fields of study are welcome to participate.

Overige informatie

Part of the minor Global Health.
This minor course requires a minimum of 20 participants to take place.

Aanbevolen voorkennis

We recommend that students have been enrolled in the courses Future challenges in global health and Drivers of change in global health.

Computational Intelligence

Vakcode	XB_0025
Studiepunten	6.00
Periode	P5
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. A. Yaman
Examinator	dr. A. Yaman
Betrokken Docenten	dr. A. Yaman, prof. dr. A.E. Eiben
Onderwijsvormen	Lecture, Practical

Doel vak

1. Knowledge and understanding of:
- Optimization techniques: Derivative-Free Optimization (Hill Climbing, Local Search), Gradient-based Optimization (Gradient Descent and Stochastic Gradient Descent)
- Evolutionary Algorithms
- Neural Networks (Fully-Connected Networks, Convolutional Networks)
- Sampling Methods: Metropolis-Hastings Algorithm, Simulated Annealing
- Reinforcement Learning: Q-learning
- Neuroevolution: Neural Architecture Search, NEAT.

2. Applying knowledge and understanding:
- How before mentioned optimization algorithms work and where to use them.
- How to formulate an evolutionary algorithm for a specific problem.
- How to analyze an optimization algorithm.
- What neural network fits best for given problem.

3. Making judgments:
- What optimization algorithm to use for given optimization problem (either Derivative-Free method or Gradient-based method).
- What layers of a neural network are suitable in a given problem.

4. Communication skills
- Presenting an analysis in a written form (a short report) for each assignment.

Inhoud vak

In the course Computational Intelligence, we will focus mainly on computational aspects of Artificial Intelligence, namely, optimization algorithms for solving learning problems. Specifically, we will consider problems that cannot be solved using information about gradient due to their combinatorial character or complexity of the objective function (e.g., non-differentiability, blackbox objective function). These problems pop up in computer science and AI, such as, identification of biological systems, task scheduling on chips, robotics, finding optimal architecture of neural networks. For this purpose, we will introduce different classes of algorithms that can be used to tackle these

problems, namely, hill climbing and local search, and evolutionary algorithms. Additionally, we explain sampling methods (Markov Chain Monte Carlo) and population-based sampling methods, and indicate how they are linked to evolutionary algorithms. In the second part of the course, we will discuss neural networks as current state-of-the-art modeling paradigm. We will present basic components of deep learning, such as, different layers (e.g., linear layers, convolutional layers, pooling layers, recurrent layers), non-linear activation functions (e.g., sigmoid, ReLU), and how to use them for specific problems. At the end of the course, we will touch upon alternative approaches to learning using Reinforcement Learning and Bayesian Optimization. We will conclude the course with a recently revived field of neuroevolution that aims for utilizing evolutionary algorithms in training neural networks.

Aanvullende informatie onderwijsvormen

Lectures and practical assignments.

Toetsvorm

The final grade is calculated based on the final exam (50 points) and 5 individual practical assignments (10 points each, 50 points in total). To pass the course, students are required to obtain at least 25 points from the final exam, and 55 points in total including all the points from the exam and assignments.

The exam can be retaken (a resit).

Solutions to practical assignments must be provided within given

deadlines. There is no resit option for practical assignments.

Literatuur

The literature will be made available on Canvas.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (track Intelligent Systems)

Aanbevolen voorkennis

Calculus
Algebra
Statistics
Programming in Python

Computational Thinking

Vakcode	X_400475
Studiepunten	3.00
Periode	P2
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. A. Bhulai
Examinator	dr. A. Bhulai
Betrokken Docenten	dr. A. Bhulai, L.H. Rekker MSc
Onderwijsvormen	Lecture, Practical

Doel vak

Knowledge and understanding:
at the end of the course, students will be familiar with basic knowledge of some solution strategies (e.g., guest and check, modeling, formulas,) and algorithms (search algorithms, sorting algorithms, and graph algorithms) to solve problems. Students will also have knowledge of

creating algorithms.

Applying knowledge and understanding:
students will be able to implement basic solutions strategies, search algorithms, sorting algorithms, and graph algorithms.

Making judgements:
students will be able to choose the right solution strategy and apply that to solve problems.

Communication skills:
students will be able to write a project report and give an oral presentation of their project to peers.

Learning skills:
students will be trained in analyzing problems, translating algorithms to flowchart, and writing pseudocodes to implement algorithms.

Inhoud vak

There are various strategies to solve everyday problems. Often a problem can be solved in different ways and there is not always a "best way". However, sometimes one way is more efficient than the other, or you find one approach easier or more pleasant than the other.
In this course, you will be acquainted with
different
solution strategies (such as modeling, formulas, guess and check) and algorithms (such as search algorithms, sorting algorithms and graph algorithms) to solve problems. You will learn to solve problems by reasoning and by using knowledge from other disciplines. In the practical sessions, you will use Python as a tool to resolve various problems using the
different
solution strategies and algorithms that have been discussed in the course. Since there are many ways to solve a problem, you will also start thinking about developing algorithms yourself. In this course, we encourage your problem solving and algorithmic thinking, as well as your creative and innovative skills. At the end of the course, you will work together with some other students in a group on a project. You will conclude the project with a short presentation.

Aanvullende informatie onderwijsvormen

(video) lectures, practical sessions, project, presentations, self-study

Toetsvorm

The final grade is based on the individual practicum assignments (45%), group project

assignment (40%), and two supervised quizzes on campus (15% together).

For all these three parts separately, the average grade should be at

least a 5.5 to pass the course.

During the project presentations, questions are asked that are taken into account during the grading of the project. This means that grades within a group may vary from person to person.

You CANNOT redo the practical assignments or the project if you have not

passed them or if you have not completed them. If the average grade of

the practical assignments or the project is less than 5.5, you fail the

course and must do the course again next year.

Literatuur

Syllabus and video lectures

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence - year 1

Overige informatie

This course revolves around self-study with video lectures in

combination with class sessions (once per week) for deepening

understanding and practical sessions (once per week).

Afwijkende intekenprocedure

For this course, please enroll for the module, via VUweb. The faculty

will enroll you for the other teaching methods.

Students will be placed in a practicum group in Canvas to

participate in the practical sessions.

Aanbevolen voorkennis

Basic knowledge of the Python programming language is recommended to

participate in this course.

A little mathematical understanding may be to your advantage.

Conflict and Peacebuilding: Local and Global Perspectives

Vakcode	S_CPB
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. M. Matelski
Examinator	dr. M. Matelski
Betrokken Docenten	dr. S. Sremac, dr. M. Matelski, prof. dr. D.A. Forster, dr. J.K. Maiyo
Onderwijsvormen	Seminar, Lecture, Written partial exam

Doel vak

The aim of this course is to introduce students to the causes and dynamics of conflict and the key approaches to conflict transformation, reconciliation and peacebuilding in (post-)conflict societies, with specific emphasis on the local level and with ample attention for individual circumstances and dispositions.

When finalizing the course, students will be able to:

- understand the main approaches and theories related to the root causes, local dynamics, transformation and

resolution of conflicts;

- understand the role and complexity of intersectional identities (e.g. religion, gender, class, ethnicity, sexuality, etc.) in conflict occurrence, transformation and resolution;

- apply multiple perspectives and insights from various disciplines to specific case studies in (post-)conflict societies.

Inhoud vak

Since the early twentieth century, the world has been shaken by war, ethnic cleansing, crimes against humanity and genocide, followed by transitional and transformative phases of conflict management, conflict resolution and peacebuilding. In order to understand the root causes of conflict and its local dynamics as well as conflict transformation and reconciliation efforts, it is necessary to focus on the everyday experiences, interests, and needs of the different actors involved in these processes on the micro-level. This course will therefore integrate local perspectives from various parts of the world in comparison to global trends and developments.

Given the changing nature of conflict, this course discusses and examines some of the key concepts and recent developments in relation to distribution and inequality, governance and human rights, as well as the influence of external interventions such as peacekeeping missions and development schemes, and the interplay between local and international justice mechanisms, which all influence the onset, course, and solutions available in situations of conflict.

The course will pay attention to these practices and initiatives in different national, cultural and political environments, drawing from examples in Africa, the Americas, (South)East Asia, Europe, and the Middle East. Based on various case studies presented by (guest) lecturers with expertise in fields such as migration, religion, ethnicity, gender and development, students will be presented with a comprehensive understanding of the root causes, dynamics, and transformation of conflict in various societies. Students will apply these insights in a group presentation and a written exam with short essay questions.

Aanvullende informatie onderwijsvormen

• Interactive (guest) lectures discussing various case studies.

• Working group sessions and group presentations.

• Study visit to an international tribunal and/or NGO in The Hague.

Toetsvorm

Individual exam

Group presentation

Literatuur

Selection of articles and book chapters (available via Canvas and VU library).

Aanvullende informatie doelgroep

Students in the minor Peace and Conflict Studies.

Overige informatie

This course has an exam in the last week of November and mandatory tutorials until 13 December. It is not possible to pass the course without taking part in the exam and the tutorials.

Aanbevolen voorkennis

Enrolment in 'Peace and Conflict: From Theory to Practice' is recommended.

Conversational AI

Vakcode	XB_0119
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. F. Ilievski
Examinator	dr. F. Ilievski
Betrokken Docenten	dr. F. Ilievski, prof. dr. P.T.J.M. Vossen
Onderwijsvormen	Lecture, Practical

Doel vak

Knowledge and understanding: at the end of the course, students will have a solid understanding of conversational AI concepts (e.g., dialogue acts), development and evaluation of dialogue systems, large language model (LLM) architectures and how LLMs can be deployed as conversational agents, challenges with LLMs as conversational agents and mitigation strategies, tool-augmented LLMs, grounding LLMs to external knowledge, learn the components of multi-agent and multi-modal agents, the role of LLMs in embodied environments, and understand considerations of pragmatics, storytelling, and efficiency.

Applying knowledge and understanding: students will be able to implement conversational AI using LLMs and test their system on various conversational tasks.

Making judgments: students will have a basic understanding of the

ethical and societal implications of the developments in conversational AI.

Communication skills: students will be able to write a scientific

report about an original research question in collaboration with a group of students.

Learning skills: students will be trained to develop conversational AI for multiple tasks and domains, develop an original and precise research question, and perform the corresponding experiments to answer their questions.

Inhoud vak

Foundational concepts of conversational AI, such as dialogue acts. Concepts and architectures of Large Language models as components of conversational AI. Understanding model limitations and designing mitigation strategies to address these limitations. LLMs with tool augmentation and grounding to external knowledge. Design and development of multi-modal and multi-agent LLM systems. Adapt and build conversational AI for specific target domains and applications.

Aanvullende informatie onderwijsvormen

Theoretical lectures and working group sessions (one of each per week)

Toetsvorm

Multiple choice exams on theory (50%) and assignments (50%).

Assessment details:

- Each of the three assignments must be at least 5/10.
- If a student scores below this grade, they can resubmit once based on feedback, a few days after the

feedback is given by the instruction team.
- Resubmissions for only one of the three assignments are allowed for each student.
- The assignment grade must be at least 5.5/10 on average over the three assignments.
- The exam grade must be at least 5.5/10.
- If a student scores below this grade, they can take a resit.

Literatuur

The literature list will be made available through Canvas

Aanbevolen voorkennis

Students are expected to have solid programming skills in Python and general understanding of AI concepts.

Creatief Schrijven I

Vakcode	L_NABALES103
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Nederlands
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	prof. dr. J.H.C. Bel
Examinator	prof. dr. J.H.C. Bel
Betrokken Docenten	dr. J. Grave, prof. dr. J.H.C. Bel
Onderwijsvormen	Werkcollege

Doel vak

Als je deze cursus hebt gevolgd kan je:

1) basale narratieve structuren en schrijftechnieken op het gebied van onder meer stijl, plot, scenes en personages, zoals o.a. beschreven en toegelicht door James Wood in Hoe fictie werkt, toepassen in korte creatieve schrijfoefeningen.

2) zaken als het vertelperspectief en de vrije indirecte rede zelf op creatieve wijze toepassen in een afgerond eigen verhaal.

3) op respectvolle wijze feedback geven op het creatieve werk van medestudenten.

4) met respect luisteren naar de feedback van medestudenten op zijn creatieve werk.

Inhoud vak

In het college Creatief schrijven verdiep je je in literaire technieken en leer je hoe je een verhaal schrijft. Als je zelf schrijft, begrijp je namelijk beter hoe het literaire mechaniek werkt (structuur, stijl, plot, genre, ruimte) en hoe je daarmee uitdrukking kunt geven aan de persoonlijke inzichten die je wilt overdragen in een verhaal. In een reeks colleges verdiep je je in de verschillende technieken die in fictionele teksten worden gebruikt. Dat gebeurt aan de hand van de opgegeven literatuur, door middel van oefeningen en door het zelf schrijven van een literaire tekst die elke week in omvang groeit.

Aanvullende informatie onderwijsvormen

Werkcolleges

Toetsvorm

• schriftelijk tentamen over James Wood, How fiction works/ Hoe fictie werkt (40%);

• het schrijven van een kort verhaal (60%).

Voor beide onderdelen moet minimaal een 5,5 worden behaald.

Literatuur

James Wood: *Hoe fictie werkt / How fiction works*(vertaling Arie Storm). Andere te lezen primaire en secundaire teksten worden voor aanvang van de cursus via Canvas bekendgemaakt.

Aanvullende informatie doelgroep

Het vak is bedoeld voor studenten van de BA-track Creatief schrijven Nederlands en voor de minoren Aan de slag met literatuur en De schrijfacademie. Andere belangstellenden kunnen bij de docent een aanvraag indienen of ze het college kunnen volgen.

Current Issues

Vakcode	R_Currissues
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	A.I. Carrozzini
Examinator	A.I. Carrozzini
Betrokken Docenten	A.I. Carrozzini, prof. dr. G.N. Cornelisse, prof. dr. G.T. Davies
Onderwijsvormen	Study Group, Lecture

Doel vak

• To learn to contextualise news about global issues within relevant

international and/or European legal frameworks and scholarly debates;

• To practice writing a research paper on a current issue within the scope of the minor;

• To practice peer review;

• To learn to cooperate with a fellow student;

Inhoud vak

The course is part of the minor Law and Global Justice. This course introduces students to selected topics within the scope of the minor which are of particular current importance or interest. Classes are interactive, involving some lectures, but also discussions and exercises. The aim is to help students understand the kinds of law and policy problems which are important at European and International level, and to critically evaluate the responses to these. This prepares the students for advanced courses at masters level, where they may engage with these problems in more detail. This course invites students to think critically and apply a combination of legal, social and historical perspectives and it develops both writing and oral skills.

Previous current issue topic areas include: family reunification policies; space law; climate engineering and climate litigation; integration policies; state surveillance; selling citizenship; fertility tourism and migrant smuggling.

Students will have to read and analyse academic literature and engage in active discussion of current issues, as well as formulating problems and questions in a short essay and/or presentations. Oral and writing analytic abilities are therefore the major skills advanced in this course.

Aanvullende informatie onderwijsvormen

- Interactive (guest-) lectures and seminars, in which discussion and problem-solving are involved; and

- Student conference, during which students present their short paper, moot court or debate.

Toetsvorm

- written exam, consisting of one essay question;

- participation in the course, consisting of the submission of mini-podcast on one of the topics of the course.

Literatuur

Preliminary reading lists will be announced on Canvas for a range of current topics.

Aanvullende informatie doelgroep

The course is part of the minor Law and Global Society. Apart from regular students, the course is also available for:

Students from other universities/faculties

Exchange students

Contractor (students who pay for one course)

Aanbevolen voorkennis

Exchange students - basics of EU law and integration, good command of English.

Data Analytics and Privacy

Vakcode	R_DAP
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	mr. T.H.A. Wisman
Examinator	dr. M.R. Leiser LLM
Betrokken Docenten	prof. mr. A.R. Lodder, S. De Conca, mr. T.H.A. Wisman, dr. M.R. Leiser LLM
Onderwijsvormen	Study Group, Lecture, Tutorial

Doel vak

Data Analytics and Privacy focuses on the role of European fundamental rights and legal principles in the regulation of data analytics, with a general focus on the right to privacy and data protection. After this course:

1. The student will identify and deconstruct the main legal implications concerning data protection and privacy, in particular with regard to the European legal framework.

2. The student will describe and apply the basics of European Union data protection law.

3. The student will know and understand the basics of data analytics in the context of privacy and data protection.

4. The student will outline and reflect on potential interferences with fundamental rights following data analytics.

Inhoud vak

Data analytics has the ability to pose significant challenges to privacy, human autonomy, and dignity. The ability to influence people in ways they can act against their own interests is what political scientists describe as power. Data analytics as an instrument to develop knowledge and facilitate the exercise of power is a phenomenon that requires regulation. Power in a constitutional democracy, whether exercised by the state or other parties, has to be subject to the law. Law offers the option to mediate between conflicting interests guiding toward a legal solution to a problem rather than one which is based on

brute force.

Privacy is a societal norm that is codified in Article 8 of the European Convention on Human Rights. EU data protection legislation in the form of the General Data Protection Regulation (GDPR) regulates the processing of personal data in EU member states. Privacy and data protection is a funny pair that have provoked many lawyers to write relentless essays on. Gutwirth and De Hert, two prominent privacy and data protection lawyers from Belgium, use an interesting distinction

between the two. They refer to privacy as an opacity instrument, because it protects a sphere in which the individual's behaviour is not being recorded. They refer to data protection as a transparency instrument, because it makes the exercise of power knowledgeable for those who are on the receiving end. Privacy and data protection laws offer rules that can be complied with by parties who want to use data analytics in their strategies and policies.

In this course, the basics of privacy and data protection law are explained in the context of data analytics. The idea is for students of computer science to understand the basics of the law and for students of law to understand the basics of data analytics. An objective is to understand the interaction between data analytics and privacy laws. An outcome is to understand how the laws of privacy and data protection apply to the development of computer systems, platforms, apps, software, etc.

Aanvullende informatie onderwijsvormen

Lectures, tutorials.

Toetsvorm

Exam - 35 Multiple choice questions worth 2 points each; 2 Open-ended essay style questions, each worth 15 points each.

Literatuur

Made available via Canvas.

Data Science: Visualization and Analytics in R

Vakcode	S_DSVAR
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. M.A.C.G. van der Velden
Examinator	dr. M.A.C.G. van der Velden
Betrokken Docenten	dr. M.A.C.G. van der Velden
Onderwijsvormen	Seminar, Computer lab

Doel vak

Upon completion of this course, the student is able to ...
• use various computational techniques in R: structuring digital data, visualization and systematic evaluation;
• critically reflect on the implications of the selection, structuring and manipulation of data for the outcome of their work;
• evaluate results critically and in a systematic manner;
• critically analyze other digital based research projects. They will be able to position their own work in the existing field of digital humanities and social analytics.
• collaborate in an advanced, interdisciplinary research groups;
• present their work in an academically convincing and ethical way for an interdisciplinary audience.

Inhoud vak

The explosion of digital information and increasing efforts to digitize existing information sources has produced a deluge of data, such as digitized historical news archives, literature, policy and legal documents, political debates and millions of social media messages by politicians, journalists, and citizens. Graphs and charts let you explore and learn about the structure of the information you have collected. Good data visualizations enable you to communicate your ideas and findings.
This course will offer analytical and practical training in digital visualization techniques using the open-source platform R. This course is placed in the broader scope of Digital Humanities and Social Analytics. In terms of critical reflection and skills this is a more advanced course within the Minor Digital Humanities and Social Analytics.
An important part of the classes will entail practical training in the visualization of data: what are the "right numbers" to present, how to present uncertainty in data, which ties in a network are important enough? The course will teach you how to transform data to a visual: from a basic graphical display to animated and BBC-worthy graphics (e.g. see
https://www.r-bloggers.com/create-data-visualizations-like-bbc-news-with
This course invites you to develop visuals from many data sources, such as numerical data, textual data, networked data, etc. At the end of the course you will be able to use attractive visualizations to present your research results in both oral and written communications.
After completion of this course you will possess knowledge of digital tools and opportunities of a field of research in order to continue to acquire computing skills andpursue further studies and / or a career that entails interdisciplinary collaboration, work with many types of data and media and involves high level critical and analytical skills.

Aanvullende informatie onderwijsvormen

Lectures and seminars

Toetsvorm

Individual and group assignments (40%), take-home exam (60%), both parts have to be evaluated with a sufficient grade to pass the course.

Literatuur

• Healy, K. (2018). Data visualization: a practical introduction. Princeton University Press. (online version freely available)
• Additional scientific articles and book chapters

Aanvullende informatie doelgroep

Students who take the University Minor ‘Digital Humanities and Social Analytics’. As long as there are available places, we welcome other students of all disciplines, including international exchange students. Please contact the coordinator in advance.

Overige informatie

This course is part of the minor Digital Humanities and Social Analytics.

Aanbevolen voorkennis

This course is designed for students who take the minor Digital Humanities and Social Analytics. For other students it would be helpful to familiarize with the basics of digital data in advance. Please contact the instructors for more information and advice.

Data Structures and Algorithms for AI

Vakcode	XB_0043
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. K. Tuma
Examinator	dr. K. Tuma
Betrokken Docenten	prof. dr. ir. F. Massacci, dr. K. Tuma, dr. F. Madeiral Delfim, L.H. Rekker MSc
Onderwijsvormen	Lecture, Seminar, Written partial exam

Doel vak

The goal of this course is to introduce Bachelor students enrolled in the target audience program to the basic concepts of data structures and algorithms. After taking this course, students will have gained (i) basic knowledge of data structures and algorithms, (ii) ability to analyse complexity of algorithms, and (iii) recognize trade-offs and choose an appropriate data structure and algorithm to solve a particular problem. Specifically, the student will have reached the following learning outcomes:

- Describe the basic data structures and algorithms that are used for common computational problems. [Dublin Descriptors]: Knowledge and understanding, Communication
- Explain the role of algorithm complexity (time and space) in addressing problems and make complexity judgements on simple programs. [Dublin Descriptors]: Applying knowledge and understanding, Communication
- Explain the rationale of the proposed solution and possibly apply appropriate techniques on a concrete problem instance. [Dublin Descriptors]: Applying knowledge and understanding, Communication
- Identify and review pros and cons of proposed techniques for solving problems. [Dublin Descriptors]: Knowledge and understanding, Lifelong learning skills

Inhoud vak

The content of the course is organised in weekly lectures, where the lecturer will cover the concepts of linear data structures (e.g., linked list, trees, graphs), basic notion of time and space complexity, which will be discussed on simple programs (e.g., sorting), and basic algorithms for traversing trees and graphs (e.g., search or decision trees). The introduced concepts will be accompanied by an explanation of the difference between solving the same problem by using a different method. More advanced topics will be covered towards the end of the course. The students will work on individual assignments during the weekly lab sessions which consist of practical programming challenges aimed to deepen their understanding of the topics covered during the lectures. The programming assignments will be automatically tested for correctness and efficiency, the latter being a determinant characteristic.

Aanvullende informatie onderwijsvormen

The course is based on the following teaching methods:

- Weekly lectures introducing new topics
- Weekly lab sessions where student TAs will supervise students working on practical exercises
- Assignments in the form of programing challenges including analytical tasks describing the implemented solution to an algorithmic problem

Toetsvorm

Assignments: 30%

Theory + Practicum A: 35%

Theory + Practicum B: 35%

Assignments:

Points will be assigned with the following grading scheme:

- 5% assessment of the first lab assignment (A1)

- 5% assessment of the second lab assignment (A2)
- 10% assessment of the third lab assignment (A3)
- 10% assessment of the fourth lab assignment (A4)

Theory + Practicum:

- Part A (35%): There is a first exam (in person) assessing the material covered in the first half of the lectures and the first two assignments. This examination takes place at the end of September.
- Part B (35%): There is a second exam (in person) assessing the material covered in the second half of the lectures and the third and fourth assignments. This examination takes place at the end of the course.

For multiple choice tests the university standard method of correction for the chances of random answers is applied.

Resit:

There is a single resit for the examinations Part A and Part B but not the assignments.

Passing requirement:

To pass the course, a student needs to pass (>5.5) on all components (assignments, theory, practicum)

Plagiarism:

The assignment submissions are automatically evaluated for plagiarism. If the submitted code (and not the boiler plate) is flagged for plagiarism the student will be reported to the Examination Board.

Literatuur

Roughgarden, Tim. Algorithms Illuminated (selected chapters of Part I, Part II and Part III). Soundlikeyourself publishing, 2017.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (Year 2)

Aanbevolen voorkennis

- Introduction to programming (Python) (X_401096). Good working knowledge of Python is strongly advised and if not present, the student should be prepared to put extra effort for programming in Python.
- Some background in discrete mathematics and calculus is advised (polynomial and exponentials functions, logarithmic function, function compositions, simple series, factorial calculus, etc.)

Data Structures and Algorithms for CS

Vakcode	XB_0091
Studiepunten	6.00
Periode	P2
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. A. Mehrabi
Examinator	dr. A. Mehrabi
Betrokken Docenten	dr. A. Mehrabi
Onderwijsvormen	Lecture, Seminar, Written partial exam

Doel vak

After taking this course, you:

- have basic knowledge of data structures and algorithms,

- know that different data structures are suitable for different computational problems, and

- are able to analyze the performance of algorithms, with regard to their time and space requirements.

The following list presents the course objectives according to Dublin Descriptors.

1. Learn the notations and the language of data structures and algorithms. (Knowledge and understanding.)

2. Learn some basic data structures including linear data structures (e.g., arrays, linked lists, queues, and stacks) and several efficient dictionary data structures (e.g., binary search trees, AVL trees, and hash tables). Also, familiarize with performances of the different data structures with regard to their query and storage requirements. (Knowledge and understanding.)

3. Learn algorithms for solving some basic problems including sorting, searching, and a few graph-related algorithms like BFS and DFS.

4. Be able to use the data structures and the algorithms learned from the course to solve an algorithmic problem. (Applying knowledge.)

5. Given an algorithmic problem, pick the most appropriate data structure in order to speed up a designed algorithm for the problem and analyze its performances. (Making judgements.)

6. Familiarize with few real-life problems that can be modeled as an algorithmic problem and can be thus studied using algorithms and data structures. (Communication.)

Inhoud vak

You will be acquainted with some basic linear data structures (e.g. arrays, lists, queues, and stacks), and several efficient dictionary data structures (e.g., binary search trees, AVL trees, hash tables). Although our lectures and discussions will be from a theoretical perspective, we will also discuss their implementation details whenever needed. Some fundamental algorithms will be discussed to solve basic problems: sorting, searching, and a few graph algorithms like BFS and DFS. It will be also explained how the time and space complexity of algorithms can be analyzed.

Aanvullende informatie onderwijsvormen

There will be two lectures and one exercise sessions per week. Attendance is not mandatory, but it is strongly recommended.

Toetsvorm

There will be one midterm exam (50%),and one final exam (50%). Roughly speaking, the midterm exam covers the first half of the course and the final exam covers the second half of the course. Your final grade is simply the sum of the grades you have obtained for the two exams. There is no minimum grade requirements for two exams in order to pass the course, except that their sum has to be at least 55.

The resit of the course covers the entire course and it replaces both the midterm exam and the final exam.

Vereiste voorkennis

Officially speaking, the course does not have any entry requirements, but the knowledge of the following topics is strongly advised in order to be able to follow the course.

i) calculus and in particular geometric and arithmetic series, and

ii) discrete mathematics, and in particular mathematical induction.

Literatuur

Introduction to Algorithms (3rd edition), by Thomas H. Cormen, Charles, E. Leiserson, Ronald L. Rivest, and Clifford Stein, MIT Press 2009.

Toelichting Canvas

We will use Canvas for our communications, posting announcements, publishing materials of the course (e.g., course slides, weekly exercises, and any other additional related materials).

Aanbevolen voorkennis

As mentioned earlier, a good knowledge in calculus and discrete math is strongly advised in order to be able to follow the course.

Data Wrangling

Vakcode	XB_0014
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. S. Bhulai
Examinator	prof. dr. S. Bhulai
Betrokken Docenten	prof. dr. S. Bhulai, M.J. Pellemans MSc, L. Wehbi
Onderwijsvormen	Seminar, Lecture

Doel vak

The course is geared toward getting data ready for its end purpose. After this course, the student should be able to:

1. acquire data from different online and offline sources,

2. understand how to clean and pre-process data,

3. transform data for analytics purposes,

4. perform feature engineering,

5. visualize data.

Inhoud vak

Data wrangling is the process of gathering data in its raw form and molding it into a form suitable for its end use. This course is about how to gather the available data and produce an output that is ready to be used. A number of common steps in the data wrangling process will be discussed, including acquiring, cleaning, shaping, and structuring the data, as well as feature engineering and visualization.

Aanvullende informatie onderwijsvormen

Lectures (6 x 2 hours) and Q&A sessions (3 x 2 hours).

Toetsvorm

The final grade is determined by hand-in assignments (30% of the final grade), a presentation, and a report (70% of the final grade). Both parts must be passed with a grade of at least 5.5. There are no resit options due to the nature of this course.

Literatuur

Slides

Aanvullende informatie doelgroep

3BA, 3LI, 3CS, 3IMM

Aanbevolen voorkennis

Programming experience in Python.

Databases

Vakcode	X_401008
Studiepunten	6.00
Periode	P5
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. J. Endrullis
Examinator	dr. J. Endrullis
Betrokken Docenten	dr. J. Endrullis
Onderwijsvormen	Lecture, Computer lab, Practical

Doel vak

The students will learn:

(a) to design and evaluate database schemas using Entity-Relationship diagrams, (Applying knowledge and understanding) (Making judgements)

(b) to understand the relational model (including integrity constraints such as key constraints and foreign key constraints), (Knowledge and understanding)

(c) to translate Entity-Relationship diagrams into relational schemas (and to enforce cardinality constraints by design), (Applying knowledge and understanding)

(d) to reason about good and bad database design using functional dependencies, (Making judgements)

(e) to apply normalization algorithms to transform a database schema into (various) normal forms, (Applying knowledge and understanding)

(f) to understand concurrency concepts like transactions, schedules and concurrency anomalies, and (Knowledge and understanding)

(g) to reason about different techniques for concurrency control (lock based concurrency control, optimistic concurrency control, multiversion concurrency control). (Making judgements)

Inhoud vak

The course is concerned with base principles and important aspects of relational databases. The course objective is to obtain a good knowledge and understanding of relational database systems. This includes the ability to develop good database models, and to query and update databases using SQL.

Aanvullende informatie onderwijsvormen

Lectures, exercise/practicum classes and individual homework.

Toetsvorm

The homework is mandatory for qualifying for the exam (70% of the homework points to qualify for the exam). In the case that at least 90% of the homework points are obtained, 0.5 bonus points are awarded for the final grade. At the end of the course, there is a final exam. The overall grade is the grade of the final exam plus the possibly 0.5 bonus point obtained for the homework. (The bonus is only added for students who pass the exam with a grade of at least 5.5.)

There is no resit opportunity for the homework.

Literatuur

Database Systems, The Complete Book, by: Hector Garcia-Molina & Jeffrey D. Ullman & Jennifer Widom. 2nd

edition.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 2)
Bachelor Business Analytics (year 2)
Bachelor Computer Science (year 2)

Aanbevolen voorkennis

Basic programming skills help.

Decolonizing Europe: History and Memory

Vakcode	L_GCBAGES200
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Nederlands / Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	drs. W.C. Manuhutu
Examinator	drs. W.C. Manuhutu
Betrokken Docenten	prof. dr. G.R. Jones, drs. W.C. Manuhutu
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

The course Decolonizing Europe. History and Memory has both historical and methodological learning objectives.

After the course, participants

1. understand the main approaches to the political history of the European nation-state and are able to situate leading historians in the historiographic debate on decolonization, postcolonialism and decoloniality;
2. understand how the course themes of decolonization and national histories in Europe play out in the institutional setting of museums and the construction of public and political discourse in present-day societies; and will be able to write a critical appraisal of museum exhibitions and collections and their role in European postcolonial society;
3. are able to critically review (in writing and speaking) a historical monograph, a work of fiction or a public/ political debate relevant to the course theme, and to develop an argued opinion about the issues at hand;
4. can reflect on their own ‘subject position’, and explore the course theme from various perspectives while acknowledging different experiences with respect to European postcolonial society.

Inhoud vak

The course focuses on the impact of European imperialism on the dynamics of nation-state formation with an emphasis on ‘postwar’ Europe. In 'Europe after Empire. Decolonization, Society and Culture' (first published in 2016) Elizabeth Buettner investigates the impact of decolonization on Europe. After 1945 all around the globe countries became independent, so what did that mean for Europe itself? How did these these events and developments shape politics and how are they remembered in present-day European societies? Therefore, the concept of multidirectional memory as developed by Michael Rothberg will be a point of departure in the course.

Students will come across key political developments that played a major role in the repositioning of Europe in the international arena after colonialism such as:

• the reordering of European nation-states after 1945 and the impact of the Cold War on decolonization;

• the changes within Europe and between Europe and the ‘Third World’ as the result of decolonization;

• the relationship between decolonization, the founding of the EU and gradual European integration and, simultaneously, the emergence of major ambiguities within nation-states concerning the concept of multicultural society.

. How public memory and representation of the colonial pasts and the post-colonial
presents is negotiated in present-day Europe.

The course investigates how these developments interacted and can be traced in political and cultural developments, with particular attention to a better understanding of colonialism as a history with a deep influence on notions of belonging, inclusion and exclusion and with respect to citizenship at the national and European level.

The colonial past not only ‘happened’ in the colonies, far away from Europe, but was also part of European history and still provides many different, contesting and opposing views and perspectives on the past and the making of contemporary society. Against the backdrop of a

political history, this course will discuss how historians,

philosophers, activists, politicians, writers, museum professionals and

others approach this history within a national, European or global frame

of reference.

Aanvullende informatie onderwijsvormen

The course will consist of weekly lectures and seminars in which the themes of the course will be discussed. Students will present readings and lead discussions on topics they have prepared in small groups.

A visit to a museum or a historical site is part of the course.

All lectures and seminars will be in English.

Toetsvorm

Class participation (10%); In class presentations and discussions

(15%); museum report (25%); final written assignment (50%)

Literatuur

The handbook used during this course is:

-Elizabeth Buettner, Europe after empire. Decolonization, society, and

culture (Cambridge 2015).

- Additional literature will be posted on Canvas.

Aanvullende informatie doelgroep

This course is an elective for BA 2 students History and International Studies; Students enrolled in the minor 'History'; exchange students.

Toelichting Canvas

All Information on the course schedule, additional literature, quizzes,

polls and assignments will be posted on Canvas

Aanbevolen voorkennis

Basic knowledge of modern history for those students outside the Geschiedenis and History and International Studies program

Designing Solutions for Global Sustainability

Vakcode	AB_1231
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M. Werthschulte
Examinator	dr. M. Werthschulte
Betrokken Docenten	dr. ir. M.G. van der Meij, dr. M. Werthschulte
Onderwijsvormen	Lecture, Seminar

Doel vak

This course aims to guide the student in designing solutions to complex challenges in global sustainability, within their own field of interest (e.g., energy, climate, water, waste, nature, food security, gender).

Inhoud vak

The course comprises interactive lectures and exercises and is assessed through an assignment. After this course students:

- Can meaningfully integrate People-Planet-Profit analyses into the design of innovative solutions for global sustainability challenges in a systematic and creative manner;
- Can characterize key components of the design creation process, namely analysis, synthesis and conceptualization;
- Identify and apply methods and processes to design sustainable solutions and communicate the results in a convincing and credible manner;
- Demonstrate a capacity to collaborate in interdisciplinary teams and contribute to a shared goal;
- Reflect on personal strength and weaknesses in developing and conceptualizing global sustainable solutions.

Aanvullende informatie onderwijsvormen

The course is organized in three thematic weeks and a fourth closing week, which follow the logic of a design process, sequentially addressing:

1. Design analysis and design specification (week 1)
2. Ideation: going through the creative design process of innovation, development, and actualization (week 2)
3. Conceptualisation and evaluation, user empathising, and visualisation for product / service pitching (week 3)
4. Completion: final presentation and review of the sustainability solution, and personal reflection (week 4).

Each week starts with a main lecture on Monday followed by two working sessions on Wednesday and Friday. The Wednesday session evolves mainly around Peer-to-Peer exercises while the Friday session will allow for more Peer-to-Teacher interaction. To accommodate the latter, two teachers will be present in the class-room on Friday. The working sessions and lectures will be highly interactive, for which there is a lot of space to work and reflect on your assignment.

Toetsvorm

The final course grade is completely based on an individual assignment, consisting of a pitch and a final report. To pass the course, students must obtain an overall grade of at least 5.5 (out of 10) and contribute to the compulsory activities. To calculate the overall grade, grades of the individual elements are averaged using the weights above.

Vereiste voorkennis

This course welcomes students of any background, as long as they have a strong affinity with sustainability challenges.

Literatuur

Week 1:

1. Tim Brown and Jocelyn Wyatt (2010). Design Thinking for Social Innovation. Development Outreach 12(1): 29 – 43 https://doi-org.vu-nl.idm.oclc.org/10.1596/1020-797X_12_1_29

2. Rikke Friis Dam and Yu Siang Teo (2019) 5 Stages in the Design Thinking Process. Interaction Design Foundation. https://www.interaction-design.org/literature/article/5-stages-in-the-de

3. Tassoul, M, & Buijs, J. (2007). Clustering: An Essential Step from Diverging to Converging. Creativity and innovation management 16(1): 16-26. DOI:10.1111/j.1467-8691.2007.00413.x. (focus on page 16 – 22) http://onlinelibrary.wiley.com/doi/10.1111/j.1467-8691.2007.00413.x/epdf

Week 2:

4. Tom Ritchey (2002 - revised 2013) General Morphological Analysis: A general method for non-quantified modelling. Swedish Morphological Society. 1-10. http://www.swemorph.com/pdf/gma.pdf

5. Hsiao, S. W., & Chou, J. R. (2004). A creativity-based design process for innovative product design*International journal of industrial ergonomics*, *34*(5), 421-443. https://doi.org/10.1016/j.ergon.2004.05.005

6. Tassoul, M, & Buijs, J. (2007). Clustering: An Essential Step from Diverging to Converging. Creativity and innovation management 16(1): 16-26. DOI:10.1111/j.1467-8691.2007.00413.x. (focus on page 22 – 26). http://onlinelibrary.wiley.com/doi/10.1111/j.1467-8691.2007.00413.x/epdf

Week 3:

7. Pruit, J., & Grudin, J. (2003). Persona's, practice and theory. DUX '03 Proceedings of the 2003 conference on

Designing for user experiences: 1-15. https://www.microsoft.com/en-us/research/wp-content/uploads/2017/03/prui

8. Chiara Diana, Elena Pacenti, Roberta Tassi (2009). Visual'tiles', Communication tools for (service) design, Conference Proceedings ServDes.2009; DeThinking Service; ReThinking Design; Oslo Norway 24-26 November 2009. http://www.ep.liu.se/ecp/059/006/ecp09059006.pdf

9. sSWOT: Eliot Metzger, Samantha Putt del Pino, Sally Prowitt, Jenna Goodward, Alexander Perera (2012). sSWOT: a sustainability swot user's guide. World Resources Institute (WRI). Washington DC. https://sustainabilitycasestudies.files.wordpress.com/2015/12/sustainabi

Aanvullende informatie doelgroep

This course is predominantly aimed at students that are following the ERM pre-master or the minor Sustainability: Global Challenges, Interdisciplinary Solutions. These students already worked on their personal "Grand Challenge" in the previous courses of this minor/pre-master and therefore have a head start in working on their solution. However, the course also welcomes students that did not follow these programs, although it may be more challenging for them as they need to catch up quickly in the first week of this condense course.

Overige informatie

When students from other universities want to register for this course, they may be asked to which programme this course/minor belongs. Kindly fill the BSc program "Aarde, Economie & Duurzaamheid".

Aanbevolen voorkennis

We strongly encourage students to take all courses of the minor "Sustainability, Global Challenges, Interdisciplinary Solutions", as a personal "Grand Challenge" comes back in different assignments of the five minor courses. Work in the preceding courses forms relevant input to the assignment of this course.

Development and Globalization: Anthropological Perspectives

Vakcode	S_DG
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. F.H. Eiró de Oliveira
Examinator	dr. F.H. Eiró de Oliveira
Betrokken Docenten	dr. F.H. Eiró de Oliveira
Onderwijsvormen	Lecture

Doel vak

The aim of this course is to introduce students to the interrelated themes of development and globalization and to help them gain insights into the issues of poverty, global inequality, governance and development from an anthropological perspective.

The course has the following learning goals.

The student has acquired knowledge and understanding of:

- development and globalization-related processes and their effects on power relations on the global, regional, national and local level.
- development and globalization-related processes and their effects on environment and natural resources, political relations and governance processes.
- anthropological analyses of development and globalization-related phenomena.
- different theoretical views on development as well as “development in practice” on an introductory level.

The student has acquired the competences to

- understand and reproduce knowledge related to the historical, sociocultural and political dimensions of international development and globalization and their role in shaping the contemporary world.

The student demonstrates

- a critical attitude towards the theory and practice of globalization and development.
- the ability to reflect on issues of global social and cultural inequality.

The student has acquired the skills to

- reproduce in written text knowledge of processes and phenomena related to globalization and development.

Inhoud vak

This course takes an anthropological perspective to analyse the interrelated themes of development and globalization. The historical development of capitalist economies in the “Global North” and the ongoing global restructuring of these economies have great impacts on economic and social processes in the “Global South.” State governments, supranational agencies, NGOs and multinational corporations all invest, or claim to invest, in the social equality and economic growth of so-called developing countries. Yet growth-oriented policies often have negative side effects, such as the reproducing of social inequalities, ecological degradation, cultural hegemony, economic exploitation, and the corrosion of local agency. In this course we untangle the politics of development taking place in the current era of global capitalism. Specifically, we analyze the interactions between (trans)national stakeholders and local populations, whereby we zoom in on the way those labelled 'poor' by stakeholders experience inequality and poverty. We also highlight potential and experienced gaps between intentions and outcomes of development policies and look at what anthropology can contribute to development debates and policies.

Aanvullende informatie onderwijsvormen

Lectures

Toetsvorm

Written assignment(s), including exam

Literatuur

To be announced in the course manual (on CANVAS)

Aanvullende informatie doelgroep

2nd year bachelor students in Cultural Anthropology and Development

Sociology;

Students in the minor Development and Global Challenges;

Students in the minor Anthropology;

The course is also open as an elective.

Digital Humanities and Social Analytics in Practice

Vakcode	XB_0015
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. V. de Boer
Examinator	dr. V. de Boer
Betrokken Docenten	dr. V. de Boer, dr. L. Stork
Onderwijsvormen	Seminar

Doel vak

- To gain knowledge of a specific Digital Humanities or Social analytics challenge and/or dataset.
- To apply the previously acquired skills and knowledge to a specific DH/SA challenge.
- To set up and execute a DH/SA project in a multi- and/or interdisciplinary team.
- To write a DH/SA research paper and reflect on the work done from one's own background.

Inhoud vak

In the Course "Digital Humanities in Practice", you will be able to

apply your Digital Humanities and Social Analytics knowledge and skills to a real-world DH challenge . We have teamed up with respected researchers and renowned institutes to present you with a number of possible projects.

Aanvullende informatie onderwijsvormen

Initial lectures.
Interactive working groups and feedback moments.
DH "internship" in small groups.

Toetsvorm

- Assessment of the quality of the work done in the project (30%).
- The final research paper describing the project, the research, and the results (30%).
- The presentations (20%), composed of the intermediate and final presentations (poster/video) scores, as well as the peer reviews by other course groups.
- A personal reflection (20%), listing the individual contribution of the student as well as the relation to previous courses (both within the minor and to other courses).

Vereiste voorkennis

A significant number of VU Digital Humanities Minor courses need to have been passed.

Aanvullende informatie doelgroep

Minor Digital Humanities

Double Burden of Disease

Vakcode	AB_1109
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. S.S. Rai
Examinator	dr. S.S. Rai
Betrokken Docenten	
Onderwijsvormen	Study Group, Lecture

Doel vak

• To understand key concepts, terminology and specific diseases of the double burden of disease
• To be able to explain the double burden of disease
• To understand the causes and implications of the double burden of disease in different contexts
• To gain insight into health interventions to address the double burden of disease
• To understand the complexities, consequences and possible interventions of the triple burden of disease
• To acquire skills on finding, reading and integrating relevant literature for a scientific essay
• To learn how to translate and communicate scientific results to the public, via online media

Inhoud vak

The course ‘Double Burden of Disease’ is a course within the ‘Global Health’ minor of the Bachelor programs Health Sciences, Biomedical Sciences and Health & Life (and other health-oriented Bachelor programs). The course is interdisciplinary in nature and is open to students from any discipline/bachelors program who are either taking the "Global Health minor" or interested in the topic of "Double or Triple burden of Diseases".

Low- and middle-income countries (LMICs) currently face a “double burden of disease”. This term was defined by the WHO as “first, the emerging epidemics of non-communicable diseases and injuries, which are becoming more prevalent in industrialized and developing countries alike, and second, some major infectious diseases which survived the 20th century – part of the unfinished health agenda” (WHO, 1999a). In essence, the health shift from communicable diseases (CDs) (e.g. hepatitis A, polio, measles, tetanus) to non-communicable/chronic diseases (NCDs) (e.g. cardiovascular diseases, cancers and diabetes), which health system experts had seen in high-income countries (HICs) after the second world war, is not happening in LMICs. Instead, people in LMICs now face the double burden of high rates of both NCDs and CDs. This is also described as double-disease burden, dual burden, joint burden, and combined burden.

In addition, recent conversations within the health and scientific field have highlighted the “triple burden” of communicable diseases, non-communicable diseases and additional factors such as mental health, injuries, socio-behavioural conditions, lifestyle changes, climate change, etc. This introduces additional complexity to the existing problems of the double burden of disease. Although this course focuses on the double burden of disease, we will also discuss these emerging triple burdens.

The beginning of the course presents an overview of communicable and non-communicable diseases as part of the conversation on the double and triple burden of disease. We will study key concepts and terminology as well as measurements related to the double burden of disease. The following lectures will address specific diseases as several examples of the double or triple burden of disease and how it affects people, professionals and health care systems. For example, HIV as an infectious and chronic condition, and mental health and related illnesses. Finally, we will discuss health system responses, interventions and programs to address the double (and triple) burden of

disease.

Important Note:

For students who are interested in the minor Global Health, we highly encourage you to either follow all 5 courses (periods1+2+3) within this minor, or atleast 4 courses (periods1+2)!

This is important because in the practical part of all courses in the minor Global Health, including this one, you will use your newly acquired knowledge and apply it in a community engaged learning project that runs from period 1 to 3. Students will work in project teams on a real-life transdisciplinary global health challenge together with various local and global community partners. Students will cross disciplinary boundaries and interact with health professionals, patients and others to gain an in-depth understanding of global health problems, and to take cohesive and strategic actions to address these problems at a local level. Each course in the minor program will incorporate a different sub-project, starting in the first two courses with a broad exploration and needs analysis on the topic, followed by a more thorough investigation on a specific challenge in the next two courses, and concluding in the final course with a proposal for a sustainable innovation. Training workshops are offered throughout the minor program to support the development of specific skills such as transdisciplinary collaboration, reflection, social justice, global citizenship, dealing with diversity, creativity, leadership and adaptability

Aanvullende informatie onderwijsvormen

- Lectures 18 hours (Introductory lectures and guest lectures are mandatory)
- Work groups 9 hours (Mandatory)
- Self study 130 hours
- Examination 3 hours

Toetsvorm

- Individual assessment (60%) [Written Exam, Essay]
- Group work (40%) [Group Blog, Blog Presentation]

Further details will be specified in the course guide upon start of the course.

All parts need to be passed (grade 5.5 or higher)

Literatuur

Information on selected reading materials will be made available on
Canvas

Aanvullende informatie doelgroep

This is an interdisciplinary course and is open to all disciplines/bachelor programs. Students from any discipline/program following the minor Global Health or other students with a background in health- or biomedical science or another relevant bachelor programme in the Beta sciences are welcomed.

Overige informatie

Guest lecturers will be invited to discuss fieldwork and research.

Aanbevolen voorkennis

We recommend students to have been enrolled in the minor courses Future
Challenges in Global Health and Drivers for Change in Global Health.

Drivers of Change in Global Health

Vakcode	AB_1108
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen

Vakcoördinator	B.G.J.S. Sonneveld
Examinator	B.G.J.S. Sonneveld
Betrokken Docenten	B.G.J.S. Sonneveld, drs. A.D.L. Mers, M. Gerber
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

• Acquire knowledge and understanding of important drivers of change and how they influence health and health care globally
• Be able to apply several data collection and analytic skills, using tools of demography, anthropology, sociology and epidemiology
• Critically analyze scientific publications, theories, hypotheses and arguments, and justify and present findings both orally and in writing

Inhoud vak

"Ours is a period of change - continual, multi-form, and multi-level - technical, scientific, economic and political." (Dan Smith, 2012)

The starting point of this course is that in every corner of our lives as individuals, communities and societies, there is change. These changes affect our health in many ways. We are for example confronted with new and emerging infectious diseases, an increase in non-communicable diseases, changing patient demand patterns and rising costs of health care. Changes in health and disease patterns and changes in health care are strongly influenced by the state of the world in terms of climate, demography, economic, politics, culture and technology. In addition, there is a trend of increasing globalization of the world, with a diversity of impacts on people in different parts of the worlds. The patterns and trends described above can be seen as drivers of change in global health as they bring along new problems and new opportunities regarding health and healthcare systems.

The course analyzes causality of the process of change and identifies subsequent responses that mitigate negative effects. We thereby aim to answer questions like: Can changes in health be attributed to a single driver
or are multiple factors at play? Is the impact of the driver generic or does
it target local conditions? How can we isolate the effect of the driver amidst
other factors? Indeed, identifying causality is a first step to support decision
makers and design successful policy interventions. Furthermore, people normally do not succumb to damaging effects of changes; instead affected groups respond with mitigating measures by developing instant coping mechanisms, for the short term, and adaptation measures that anticipate future expected shocks, for longer periods. The controversial situation that
short term planning of coping mechanisms undermines long term sustainable
future adaptation measures should, of course, be avoided.

This course consists of a series of lectures which provide insight into a number of important drivers of change in different parts of the world, such as urbanization, climate change, migration, technological development, and how they affect health and health care. Each lecture is accompanied by reading materials in the form of scientific articles and complementary background information from the book. Furthermore, the course consists of an assignment and project groups. For the assignment you conduct a case study into one specific driver of change in an assignment group with fellow students. You make use of scientific publications in different sub-assignments to come up with an evidence-based call for action with respect to the driver of choice. At the end of the course you will present your work orally and in the form of a written report together with your team. In the project group meetings

you work on your academic skills (data collection, reflection, analysis etc.) through various exercises related to the topic of this course.
During these meetings you also receive feedback on sub-assignments.

Aanvullende informatie onderwijsvormen

Lectures: 20 hours
Assignment (project group meetings + independent group work): 75 hours (12 + 63)
self study: 62,5
Exam: 2,5 hours

Toetsvorm

Written exam (50%) and group assignment (50%). Both the exam grade and the total group assignment grade should 5.5 or higher in order to pass the course.

The assignment includes a written report (40%) and a presentation (10%). The two assignment components (report and presentation) can compensate for each other if one part is insufficient.

Literatuur

- Course guide
- Assignment manual
- Selected scientific publications (announced on Canvas).
- Recommended: The penguin state of the world atlas, 2012 (ninth edition). Dan Smith. Penguin Books.

Aanvullende informatie doelgroep

This is a compulsory course in the bachelor minor global health

Dynamic Modelling for Human-Centered Systems

Vakcode	XB_0036
Studiepunten	6.00
Periode	P4
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M.C.A. Klein
Examinator	dr. M.C.A. Klein
Betrokken Docenten	dr. M.C.A. Klein, dr. N. Ullah
Onderwijsvormen	Lecture, Computer lab

Doel vak

At the end of the course, the student has knowledge and understanding of: the role that dynamic models can play in systems aiming to understand and support human behaviour; the steps required to develop a dynamic model; different representations and implementations that can be used for dynamic models.

The student is able to apply this knowledge and understanding to: build dynamic models based on a textual description of a process in a domain (e.g. psychology); integrate a dynamic model in an intelligent system to derive conclusions about the state of the environment and/or user; integrate a dynamic model in an intelligent system to decide on relevant actions.

The student is able to make judgments about: the appropriate representation for a dynamic model; the validity of a dynamic model based on simulation results; the ethical application of behaviour support systems.

The student has acquired communication skills to: report in a scientific and precise manner about the design and the evaluation of a dynamic model.

The student has acquired learning skills to: read and interpret semi-scientific texts from other domains (e.g. psychology, sociology).

Inhoud vak

Intelligent systems usually encode specific information about the context in which the system executes, e.g. the users, the physical environment and the task environment in which it is used. This information does not only cover environmental states, but also information on the various processes in the environment. Dynamic models are a useful way to encode these processes.

In this intense 7-week bachelor course, the students will learn how to develop dynamic models based on descriptions of processes related to human functioning. The course considers methodological aspects of

modelling, such as: the collection and specification of relevant knowledge, the specification of the model, the definition of simulation experiments and the validation of the results obtained from

the modelling process.

In addition, a methodology will be taught on how to use such dynamic models as basis for intelligent, human-centered systems that interprets and supports human behaviour. Based on realistic examples from psychology (where emotions or moods are modelled), bio-medicine (where physiological models of the body are used to measure intoxication), or sociology, it is shown how dynamic models of such a domain can be incorporated to derive conclusions about the current situation (assessment) or to decide on relevant actions (support).

During the course, student will do weekly assignments based on lectures and construct models. The models are implemented in Excel and later in Python.

Aanvullende informatie onderwijsvormen

Lectures and practical sessions.

Toetsvorm

Group assignments (in total 50% of the grade) and an individual final exam (50% of the grade). Both elements should be graded with at least a 5.5.

There is a resit for the exam. At most one assignment can be redone if the average of the assignments is below 5.5.

Literatuur

A reader is available via Canvas.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 1)

Aanbevolen voorkennis

Python programming.

Economic Challenges for Minor Students

Vakcode	E_MUE_ECMS
Studiepunten	6.00
Periode	P1
Vakniveau	100

Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. R.I. Luttens
Examinator	dr. R.I. Luttens
Betrokken Docenten	dr. R.I. Luttens, prof. dr. H.L.F. de Groot
Onderwijsvormen	Study Group, Lecture

Doel vak

This course provides an overview of the development of economic thought as well as an insight into the relevance of economic theories for important current economic issues.

After successfully completing this course, the student can:

Academic Skills:

- explain the economic way of thinking;
- identify fundamental questions in economics;
- place economics as a discipline in relation to other social sciences.

Bridging Theory and Practice - Knowledge:

- describe the most important economic concepts;
- explain the role of different institutions in the economic process;
- identify the main differences and similarities between various economic schools;
- appoint the different fields of study of the program.

Bridging Theory and Practice - Application:

- understand various forms of market and government failure;
- analyze current policy proposals from an economic perspective.

Broadening Your Horizon:

- name the most important works of prominent thinkers in economic science;
- situate economic theories in a social, political and historical context.

Self Awareness:

- define their own position and attitude with regards to different economic schools of thought;
- constructively engage with positions and attitudes that differ from their own.

Inhoud vak

Economic theories do not come out of thin air. Crucial for this course is the insight that economists, both in the present and in the past, respond to the social, political and economic circumstances of their time. Their theories and policy proposals are not always in agreement with each other, which gives rise to the emergence of different paradigms and competing schools of thought. Studying the development of economic thought in its historical context is an excellent way to appreciate the richness of economics as a scientific tradition.

Each lecture starts with the discussion of an important period in economic history, starting from classical antiquity up to modern times. Each lecture ends with the discussion of a current 'economic challenge'. The following topics will be discussed: climate change, markets and morality, Brexit, limits on growth, aging, sweatshops, market failure, influence of advertising, the Euro crisis, Covid-19 pandemic, robotisation, nuclear threat and big data.

The relevance of the theory to the contemporary debate is always demonstrated. Along the way, we introduce the most important works of leading thinkers in economics such as Adam Smith, Karl Marx, Alfred Marshall and John Maynard Keynes. The fact that their ideas about the correct way of research and policy-making differed greatly from each other underlines the philosophy of this course that tackling important social questions from different angles promotes the understanding of what economics is all about.

Aanvullende informatie onderwijsvormen

Lectures, tutorials, and Q&A lecture at the end of the course

Toetsvorm

Exam and weekly tutorial assignments

Literatuur

- Robert L. Heilbroner, The Worldly Philosophers: The Lives, Times And Ideas Of The Great Economic Thinkers, Touchstone 7th Revised edition (August 1999)
- Selection of original texts via Canvas
- Lecture slides via Canvas
- Glossary via Canvas

Educatieve Minor Didactiek 1

Vakcode	O_EMDID1
Studiepunten	6.00
Periode	P1
Vakniveau	400
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	C.L. Geraedts
Examinator	drs. L.J. van Well-van Grootheest
Betrokken Docenten	prof. dr. M. Meeter, dr. O.P.M. van Buuren, drs. A.J.C. Monquil, drs. L.A. van der Bruggen, W. Maas, drs. J.K.W. Riksen, drs. H.R. Goudsmit, drs. Y.G. Meindersma, dr. I. Pauw, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. W. Jongejan, drs. L.J. van Well-van Grootheest, dr. H.B. Westbroek, C.L. Geraedts, dr. A.A. Kaal, J. Quartel MA, R.B. van der Vos, R.M.C. Canisius, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, drs. E.D. van Noort, drs. S.M. van den Broek MA MSc., dr. S.M. Calor, prof. dr. ir. F.F.J. Hermans, drs. N.H. Ypenburg
Onderwijsvormen	Hoorcollege, Werkgroep

Doel vak

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Inhoud vak

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio’s vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio’s werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Aanvullende informatie onderwijsvormen

Hoorcolleges, werkcolleges en zelfstudiemodules

Toetsvorm

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literatuur

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). *Inventing ourselves. The secret life of the teenage brain*. London: Transworld Publishers Ltd.

- Ebbens, S. & Ettekoven, S. (2020). *Effectief leren*. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). *Gedragsproblemen in de klas in het voortgezet onderwijs*. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). *Lessen in orde*. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Aanvullende informatie doelgroep

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Educatieve Minor Didactiek 2

Vakcode	O_EMDID2
Studiepunten	9.00
Periode	P2+3
Vakniveau	400
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	C.L. Geraedts
Examinator	drs. L.J. van Well-van Grootheest
Betrokken Docenten	drs. N.H. Ypenburg, drs. E.D. van Noort, dr. O.P.M. van Buuren, prof. dr. M. Meeter, dr. I. Pauw, drs. J.K.W. Riksen, drs. H.R. Goudsmit, drs. Y.G. Meindersma, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. W. Jongejan, drs. L.J. van Well-van Grootheest, dr. H.B. Westbroek, C.L. Geraedts, dr. A.A. Kaal, drs. A.J.C. Monquil, drs. L.A. van der Bruggen, W. Maas, drs. J. Quartel MA, J. Quartel MA, dr. B. de Vries, R.M.C. Canisius, R.B. van der Vos, drs. E.D. van Noort, drs. S.M. van den Broek MA MSc., dr. S.M. Calor, prof. dr. ir. F.F.J. Hermans, prof. dr. ir. F.F.J. Hermans
Onderwijsvormen	Hoorcollege, Werkgroep

Doel vak

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Inhoud vak

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen

toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema's centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Aanvullende informatie onderwijsvormen

Hoorcolleges, werkcolleges en zelfstudiemodules

Toetsvorm

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Literatuur

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Aanvullende informatie doelgroep

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

Overige informatie

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Educatieve Minor Praktijk 1

Vakcode	O_EMPRAK1
Studiepunten	6.00
Periode	P1
Vakniveau	400
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	drs. Y.G. Meindersma
Examinator	drs. Y.G. Meindersma
Betrokken Docenten	drs. E.D. van Noort, dr. O.P.M. van Buuren, dr. A. Handelzalts, drs. J.K.W. Riksen, drs. H.R. Goudsmit, drs. Y.G. Meindersma, dr. I. Pauw, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. L.J. van Well-van Grootheest, dr. H.B. Westbroek, C.L. Geraedts, dr. A.A. Kaal, drs. A.J.C. Monquil, drs. L.A. van der Bruggen, W. Maas, drs. N.H. Ypenburg, R.B. van der Vos, R.M.C. Canisius, J. Quartel MA, drs. E.D. van Noort, dr. S.M. Calor, drs. S.M. van den Broek MA MSc., prof. dr. ir. F.F.J. Hermans
Onderwijsvormen	Werkcollege

Doel vak

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen vmbo-tl en/of onderbouw havo/vwo.

Inhoud vak

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage omvat minimaal 48 klassencontacturen in vmbo- tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Aanvullende informatie onderwijsvormen

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Toetsvorm

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Educatieve Minor Praktijk 2

Vakcode	O_EMPRAK2
Studiepunten	9.00
Periode	P2+3
Vakniveau	400
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	drs. Y.G. Meindersma
Examinator	drs. Y.G. Meindersma
Betrokken Docenten	drs. E.D. van Noort, dr. O.P.M. van Buuren, drs. A.J.C. Monquil, drs. L.A. van der Bruggen, W. Maas, drs. N.H. Ypenburg, drs. J.K.W. Riksen, drs. H.R. Goudsmit, drs. Y.G. Meindersma, dr. I. Pauw, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. L.J. van Well-van Grootheest, dr. H.B. Westbroek, C.L. Geraedts, dr. A.A. Kaal, dr. A. Handelzalts, R.M.C. Canisius, R.B. van der Vos, J. Quartel MA, dr. B. de Vries, drs. E.D. van Noort, drs. S.M. van den Broek MA MSc., dr. S.M. Calor, prof. dr. ir. F.F.J. Hermans
Onderwijsvormen	Werkcollege

Doel vak

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen vmbo-tl en/of onderbouw havo/vwo.

Inhoud vak

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 2 van de stage omvat minimaal 72 klassencontacturen in vmbo- tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Tijdens de klassencontacturen observeert de student lessen, verzorgt de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten te formuleren en te bespreken.

Aanvullende informatie onderwijsvormen

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Toetsvorm

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met zijn werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Vereiste voorkennis

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Educatieve Minor, Peergroep

Vakcode	O_EMPEERGR
Studiepunten	0.00
Periode	P1+2+3
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	drs. L.J. van Well-van Grootheest
Examinator	drs. L.J. van Well-van Grootheest
Betrokken Docenten	
Onderwijsvormen	Werkcollege, Werkgroep

Doel vak

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen - de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- student kan leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Inhoud vak

In de peergroup staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de

peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Aanvullende informatie onderwijsvormen

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Toetsvorm

Onderdelen uit de peergroup worden getoetst in de startproef en de basisproef, en komen terug in de praktijk.

Literatuur

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Aanvullende informatie doelgroep

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Afwijkende intekenprocedure

Hoewel Peergroup onderdeel uitmaakt van de vakken Didactiek 1 en 2 moeten studenten apart voor de peergroupwerkgroepen ingeschreven staan.

Emerging Technologies for E-Business and Online Commerce

Vakcode	E_MFEO_ETEOC
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. R. Mousavi Baygi
Examinator	dr. R. Mousavi Baygi
Betrokken Docenten	dr. R. Mousavi Baygi, L.A. Falcke
Onderwijsvormen	Lecture, Seminar

Doel vak

Academic & research skills: Be able to critically assess the potential of emerging digital technologies for value creation in today's economy.

Knowledge: Knowledge of the technological, organizational, processual, and economic aspects of the digital technology.

Bridging Theory and Practice: Ability to translate and apply theoretical knowledge into practice. After successfully completing this course you will be able to apply the knowledge you obtain about this domain in analyzing business cases related to emerging technologies in e-business and online commerce.

Professional social skills: Ability to work effectively in teams and communicate on project findings.

Self-awareness: Ability to reflect about the learning process itself, which enables students to take responsibility for their learning and knowledge creation.

Inhoud vak

The digital age brings about many changes that challenge our understanding of how firms operate and create value. This course thus focuses on the properties and potentials of emerging digital technologies for value creation for businesses. It explores properties of digital technology, different types of digital innovations, different models of organizing and creating value from digital innovation, as well as their implications for organizations and societies. More specifically, in this course, we will focus on three key issues: First we will review some major “emerging” digital technologies. Then we explore platforms and ecosystems as two important models for organizing & enabling digital innovation. Finally, we dive into the broader implications of emerging technologies for organizations and society, including their unintended consequences. In a course-long group project, students will devise platform- or ecosystem-based digital initiatives that leverage an emerging technology to create value for a legacy firm or a start-up.

Aanvullende informatie onderwijsvormen

- Lectures
- Tutorials

Toetsvorm

- Group Project
- Individual Assignments

Vereiste voorkennis

This course is part of the minor E-business and Online Commerce. Students should at least be familiar with the content of 'Introduction to E-business and Online Commerce' and 'Consumer Science for Online Commerce'

Literatuur

Various resources that will be made available via Perusall.

Aanvullende informatie doelgroep

This minor can be followed by all SBE bachelor students. In addition, advanced bachelor students (third year) from other faculties are welcome to join. Particularly those with in an interest in Business and Organization Studies, Digital Technology, Entrepreneurship, Economics. It is especially interesting for:

- Future managers who want to understand how digital technologies can be implemented to transform existing business
- Entrepreneurs that want explore and exploit the opportunities of digital technologies
- Future consultants in digital technology, strategic business consultants, or government policy consultants

Overige informatie

This course is part of the minor E-business and Online Commerce and Digital Business Innovation.

Toelichting Canvas

Students are expected to continually check Canvas and any Canvas-generated emails.

Aanbevolen voorkennis

Courses in period 1 and 2 of the Minor E-business and Online Commerce

English Language Test

Vakcode	VU_ELT
Studiepunten	0.00
Periode	P1
Vakniveau	0
Onderwijstaal	Engels
Faculteit	
Vakcoördinator	
Examinator	

Betrokken Docenten	
Onderwijsvormen	

Doel vak

At VU Amsterdam, having a strong command of the language of your study programme is crucial to your academic success. Therefore, all first-year Bachelor students must take a Dutch or English language proficiency test at the beginning of their studies. The required test depends on the language of instruction for your Bachelor's programme.

See also: Language Proficiency Test - Vrije Universiteit Amsterdam (vu.nl)

Inhoud vak

The test consists of multiplechoice and fill-in-the-blank questions on the following topics:

- Grammar
- Spelling and punctuation
- Structuring
- Vocabulary/usage
- Phrasing
- Reading comprehension
- Pronunciation
- Classroom English

Aanvullende informatie onderwijsvormen

The language proficiency test will take place in September in the DigiTenT at the VU.

Toetsvorm

The language proficiency test is a digital test consisting of 200 closed questions.

Overige informatie

The VU English Language Test is linked to a first-year course. The final grade for this course is awarded only after passing the English Language Test. Students who do not obtain a satisfactory result for the test have to take the refresher course offered by the Taalloket.

For more information on the VU English Language Test, please see also:Language Proficiency Test - Vrije Universiteit Amsterdam (vu.nl)

Afwijkende intekenprocedure

You do not need to register for the test. The Educational Office of your faculty will ensure your registration for the language proficiency test. Once registered, you can find the test's date and time on your personal timetable. If you have not been registered by 8 September, please contact the student desk. You can also check the timetable of your programme for the test schedule.

Ensure you know your VUnet ID and corresponding password, as you will need these details to log in to the language test. Tip: change your password to something you can easily remember.

Environment and Development

Vakcode	S_ED
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	drs. W.A.M. Tuijp
Examinator	drs. W.A.M. Tuijp
Betrokken Docenten	drs. W.A.M. Tuijp, dr. J.K. Maiyo
Onderwijsvormen	Study Group, Excursion, Lecture

Doel vak

The aim of the Environment & Development course is to enable you as a student to examine and critically reflect on the relationships between economic and social development, and the environment.

After completion of the course you are able to:

• recognise and describe the current and potential impacts of the major international environmental concerns;

• recognise, analyse and explain the complexities of environmental issues related to development at a global level;

• distinguish, analyse and explain different perspectives on environmental problems and possible solutions;

• recognise organisational and governance issues related to environment and development and be able to think through their consequences for actual practices.

Moreover, as a student you learn to find, analyse and critically reflect on scientific literature within a relevant topic and become a project team member in a research environment, presenting and discussing the researched topic to an audience.

Inhoud vak

What do we mean by the concepts of environment and development and how are the two related? What are the causes and consequences of global environmental change? How is the global community dealing with ecological problems? How can the world adequately feed more than 9 Billion people by 2050? Is sustainable development, with its notions of environmental 'friendliness', really achievable? During the course we will address these and other critical and societally relevant questions.

Sustainable economic development and sustainable environmental management are two key challenges facing the world today. The relationship between development problems and environmental issues is complex and transboundary – it cuts across the globe. The way we, as a society, allocate financial, human and natural resources directly influences how we manage local, national and global environments. The Environment & Development course is about understanding the world we live in and questioning the sustainability of our choices as individuals and as members of society.

The E&D course takes a multi- and interdisciplinary perspective and looks both at: (1) Global Issues - such as environment-trade-poverty links; and (2) Local Issues - focusing on land degradation, deforestation, water scarcity and how these have an impact on human development. The course comprises a series of lectures by both VU staff as well as guest lecturers from academia, NGOs and the private sector.

By means of illustrated case studies from all over the world, students learn to appreciate the complexity and interlinked nature of environmental and development issues at both global and local levels.
Furthermore, we encourage students to develop their own (critical) opinion regarding current (and potential) international environmental challenges and their impacts.

Aanvullende informatie onderwijsvormen

Mandatory (guest) lectures, working group assignments and exercises, field trip, tutorials.

Toetsvorm

A mix of group assignments (presentations) and individual assignments (research paper), as well as peer reviewed assignments (trip report).

Literatuur

Each week, a selection of articles will be posted on Canvas to be studied in advance, check the course manual also (on Canvas).
Students will discuss the literature via weekly discussions.
For background reading on the key concepts in this class we will use "Paths to a Green World: the political economy of the global environment" by Jennifer Clapp and Peter Dauvergne (2011) 2nd edition. Cambridge - MIT Press.

Aanvullende informatie doelgroep

The E&D course is open to third year VU bachelor students, including exchange students, who join the full Minor Development & Global Challenges.

Overige informatie

Comments from former students on the group assignments:

"We learned a lot from each other, our ideas, strengths and beliefs as well as how difficult it can be to address a topic that not many studies have covered so far, and how important it is to narrow down and focus more on specific topics to get relevant results".

"What we learned most from this assignment was that everybody has their own strengths and weaknesses and that each person's strengths can be used to the fullest extent to create an overall comprehensive and informative presentation".

"The group project was demanding and difficult at times, but it taught us a lot and made us realize how lucky we were to be in a good group with members who all put a lot of effort to finish a project that we were

happy with".

"We learned a lot from this group project, especially about our subject

on food loss: how important it is to address the complicated and complex

problem that food loss is to our planet and society. We found our topic

very interesting and would have liked to delve deeper into the subject

if we had more time".

Aanbevolen voorkennis

Interest in environmental issues and social questions

Equational Programming

Vakcode	X_401011
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. F. van Raamsdonk
Examinator	dr. F. van Raamsdonk
Betrokken Docenten	dr. F. van Raamsdonk
Onderwijsvormen	Lecture, Seminar, Practical, Written partial exam

Doel vak

The objective of the course Equational Programming is to obtain a basic understanding of the foundations of functional programming via lambda calculus and equational reasoning and to acquire basic functional programming skills.

For lambda calculus, at the end of the course, the student will:

Knowledge and understanding:

- know alpha-conversion, lambda terms, beta-reduction, reduction sequences, reduction graphs,

- know fixed points,

- know weak and strong normalization and confluence,

- know several reduction strategies,

- know the encoding of several data types,

- know simple types and be familiar with its typing system;

Applying knowledge and understanding:

- be able to give reduction sequences and reduction graphs,

- be able to reason about fixed points,

- be able to reason about weak and strong normalization and uniqueness of normal forms,

- be able to apply a reduction strategy,

- be able to encode elementary data types and recursive functions,

- be able to give elementary typing derivations;

Making judgements:

- be able to analyze reduction graphs,

- be able to reason about encodings.

For the equational specification part, at the end of the course, the student will

Knowledge and understanding:

- know equational specifications,

- know algebras, models, initial models,

- know the notions of syntactic derivability and semantic consequence, know Birkhoff's theorem,

- know the notion of initial specification,

- know homomorphisms and isomorphisms between Sigma-algebras;

Applying knowledge and understanding:

- be able to give a derivation,

- be able to define an (initial) specification,

- be able to give an (initial) model,

- be able to give or show non-existence of an isomorphism;

Making judgements:

- be able to analyze with respect to being a model, containing junk or confusion, being an initial model

- be able to reflect on specifications and their relation with functional programs;

Communication:

- be able to write proofs and derivations;

For the functional programming part, at the end of the course, the student will

Knowledge and understanding:

- be familiar with the basics of the functional programming language Haskell: pattern matching, recursive functions, algebraic data types, and the basics of monads;

Applying knowledge and understanding:

- be able to write small Haskell programs with pattern matching, and recursion,

- be able to deal with partial functions using monads.

Learning skills:

- have acquired flexibility in using the functional style of programming.

Inhoud vak

In the practical work, we use the functional programming language Haskell. We practice with the basics such as lists, recursion,

data-types, and a bit of monads.

The theoretical part is concerned with the foundations of functional programming in the form of lambda calculus and equational specifications. We study untyped lambda calculus reduction theory and expressive power: beta reduction, reduction strategies, confluence, encoding of

data types, fixed-point combinators, and recursive functions. In addition, we study the lambda-calculus with simple types, its typing

system and a type inference algorithm, and possibly strong normalization of simply-typed lambda-calculus. In the equational specifications part, we study the syntax and semantics of equational systems, and we work towards the results that all initial models are equal up to isomorphism and that the term model is an initial model.

Aanvullende informatie onderwijsvormen

We have two on campus lectures per week for the duration of seven weeks.

Twelve lectures discuss material and two lectures are meant for questions and wrap-up.

We have one on campus exercise class per week for the duration of seven weeks.

The lectures are concerned with the theoretical part, so lambda calculus and equational specifications.

In addition, we have one online Q&A afternoon per week for the functional programming part, for the duration of six weeks.

Toetsvorm

There are three obligatory programming assignments with grade pass/fail.

All three must be pass in order to pass the course.

A fail grade for an assignment can during the course be improved to a pass grade using the feedback. After the course, there is only a resit possibility for the programming part for those students who already actively participated (as in: handed in all assignments) and have a pass grade for the exam, in consultation with the teacher.

There is a written closed book midterm exam in week 4 of the course about the lambda calculus part that counts for 45%.

There is a written closed book final exam in week 8 of the course about the equational specifications part and a bit of functional programming

that counts for 55%.

Both parts should have grade at least 5.0 in order to pass the course.

In addition, the calculated final grade must be at least 5.5 to pass the course.

The resit is either about one of the two parts, or about both parts.

Literatuur

Course notes, slides, and exercise sheets provided via Canvas.

We further refer to various online material about Haskell for example the Haskell wiki.

Aanvullende informatie doelgroep

Bachelor Econometrics and Operations Research (year 3)

Minor Applied Mathematics

Minor Deep Programming

Toelichting Canvas

We use Canvas for the schedule of the course, and to make available the course notes and the slides for the lectures, the exercises for the exercise classes, and the programming assignments.

We use Canvas further to provide feedback to the obligatory programming assignments, and to the non-obligatory homework assignments, to give the grade for the midterm, and for announcements.

Aanbevolen voorkennis

For the theoretical part, some familiarity with formal reasoning is recommended (as for example taught in the course Logic and Modelling).

Ethical AI

Vakcode	XB_0117
Studiepunten	3.00
Periode	P4
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. G. Löhr
Examinator	dr. G. Löhr
Betrokken Docenten	dr. E.M.A.L. Beauxis Aussalet, dr. G. Löhr
Onderwijsvormen	Lecture

Doel vak

Knowledge and insight: Has demonstrable knowledge and understanding of a discipline that builds on and exceeds the level achieved in secondary education; usually operates at a level where, with the support of specialised manuals, there are some aspects that require knowledge of the latest developments in the field of expertise. – Summarize key ideas from primary texts in the analytic philosophy of AI.

Applying knowledge and insight: Is able to apply his knowledge and insight in such a way that it shows a professional approach to his work or profession, and also has competencies for drawing up and deepening arguments, and for solving problems in the field of expertise. – Place contemporary debates on artificial intelligence

within its proper historical and philosophical contexts. – Apply key philosophical theories relevant to assessing the ethical relevance of cutting-edge AI technologies.

Judgement: Student is able to collect and interpret relevant data (usually in the field of expertise) with the aim of forming an opinion that is partly based on weighing relevant social, scientific or ethical aspects.

Communication: Is able to communicate information, ideas and solutions to an audience consisting of specialists or non-specialists. Learning skills Has the necessary learning skills to enter into a follow-up study that presupposes a high level of autonomy. – The student can clearly structure and express own philosophical views in writing and speech.

Learning skills: Has the necessary learning skills to enter into a follow-up study that presupposes a high level of autonomy. – Relativize and contextualize potential threats posed by AI but also identify threats that might be overlooked.

Inhoud vak

Everyone seems to be talking about AI these days. While some present AI as the savior of humanity, others claim that it will soon become conscious and inevitably destroy us. In this course, we will provide the philosophical background that will allow you to contextualize the most important academic and non-academic debates about AI. You will be able to see for yourself which debates are] exaggerated, and which debates should be paid more attention to. We will start with some basic ethical issues (responsible AI, moral agency of AI). Those discussions will soon motivate more theoretical or foundational topics from the philosophy of mind and language. Some of these topics are old, but recent progress in AI research (e.g., transformer models like GPT or LaMDA) also raises new philosophical issues.

Aanvullende informatie onderwijsvormen

Interactive lectures, Supervised group work, tutorials, and exercises

Toetsvorm

Essay 70%, 2500-3000 (strict maximum word limit, excluding bibliography, avoid footnotes, in-text citation, APA)

Individual Presentation 20% (graded), 15 min max (I will time this).

Participation 10% (participation in discussions, and paper proposal, not graded),

Ethics

Vakcode	W_BA_ETEN
Studiepunten	6.00
Periode	P2
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. L. Bastian
Examinator	dr. L. Bastian
Betrokken Docenten	dr. L. Bastian
Onderwijsvormen	Lecture, Seminar

Doel vak

Students can:

1. Remember and explain the most influential theories within the field of normative ethics, including utilitarianism, deontology, and virtue ethics
2. Remember and explain the most influential theories within the field of meta-ethics, including relativism, moral

realism, skepticism and theories of moral responsibility
3. Analyse and evaluate the relative advantages and disadvantages of these theories
4. Apply these theories in normative ethics to analyze contemporary moral problems
5. Compose a case study based on this analysis

Inhoud vak

Ethics is a branch of philosophy that focuses on questions, such as

- “In virtue of what are actions right or wrong (morally obligatory, morally permissible, or morally impermissible)?”
- “What makes a certain state of affairs good or bad?”
- “What constitutes a good life?”

In this course, we will critically explore different theories that have been developed which offer answers to these questions. Along the way spend

time examining how these ethical theories apply to contemporary moral issues. We also discuss several epistemic, metaphysical and semantic questions, such as whether we can know moral claims, whether they are objectively rather than relatively true, and what their semantic features are.

Aanvullende informatie onderwijsvormen

The course is six weeks long, with 2 lectures per week, and one workgroup every two week.

Toetsvorm

The final grade for 'Ethics' is determined by a weighted average of the following components:

- Reading assignments: 10% (assessment of learning objective 3)
- Case-study (group assignment): 30% (assessment of learning objectives 4 & 5)
- Final exam (knowledge questions and short essay questions): 60% (assessment of learning objectives 1-3)

In order to pass the course, a passing grade needs to be achieved in both the case-study and the final exam.

Aanvullende informatie doelgroep

BA1-students Philosophy; students in the Minor Philosophy.

Toelichting Canvas

Students must check Canvas regularly: all updates, changes and recent information will be communicated there.

EU Governance in an International Context

Vakcode	S_EUGIC
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. O. Terzi MSc
Examinator	dr. O. Terzi MSc
Betrokken Docenten	dr. O. Terzi MSc
Onderwijsvormen	Lecture

Doel vak

Learning outcomes:

Knowledge and understanding – The student has acquired knowledge and understanding of:

(1) the history of European integration and the main theories trying to understand and explain the process of European unification;

(2) the institutional structure of and politics within European Union and how it interacts with member states;

(3) key aspects of EU policy making in the internal and external arena.

Skills – The student is able to:

(4) recognise the key historical milestones of European integration and the current challenges the EU faces currently when it comes to its institutional and political make-up;

(5) apply a variety of theoretical approaches to understand and explain past and current developments.

Attitude – The student displays:

(6) critical reflection upon the European integration process, the EU's institutions and policies as well as the way democracy functions at the EU level.

Inhoud vak

This course familiarizes students to the way the EU operates, its institutional architecture, its history, and its modes of decision-making. The course highlights how EU decision-making affects domestic politics, whilst at the same time is situated in a broader, international context. It will introduce students to the key theoretical debates about the European integration process (neo-functionalism, liberal intergovernmentalism, etc.), and it will pay special attention to the main characteristics of EU decision-making.

The theoretical and practical insights will then be applied in a number of selected policy domains involve both policy making within the EU (e.g. monetary integration, regulation of the single market, making of common foreign and security policy) and outside the EU, that is to say the EU's engagement in the global realm (e.g. enlargement, trade, security).

Aanvullende informatie onderwijsvormen

Lectures.

Toetsvorm

Final exam.

Literatuur

Cini M. & Perez-Solorzano Borragan, N. (2022). *European Union Politics*, Oxford: Oxford University Press.

Additional articles to be announced in the course manual (see CANVAS).

Aanvullende informatie doelgroep

Second-year Bachelor students in Political Science (all tracks and/or specializations)

Second-year Bachelor students in Bestuurs- en Organisatieswetenschappen, specialization Besturen

European Cultural History

Vakcode	L_GABAMKD103
Studiepunten	6.00
Periode	P1

Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. J.O.W. Singerton
Examinator	dr. J.O.W. Singerton
Betrokken Docenten	dr. J.O.W. Singerton, dr. P.H. Moser
Onderwijsvormen	Lecture, Seminar

Doel vak

This course aims to provide MKDA students a framework for better understanding developments in European cultural history. At the end of this course, students are expected to:
1. have basic knowledge of European history from Antiquity to the present;
2. have a level of knowledge of and insight into historical developments and current contexts pertaining to visual and material culture of the Western culture area (from the past and up to today's globalizing world)
3. have a level of knowledge of historical periodization, needed to situate European cultural and political developments in their context of emergence.

Inhoud vak

The long distance covered by humanity has resulted in the world we are in today. Throughout the past, people and movements have worked towards offering the world a different face. Without close study of that past, much of the present world cannot be understood. This course presents a concise overview of the major episodes in European cultural history, from classical Antiquity and the Middle Ages to early modern, modern and contemporary history, thereby highlighting the links with art and culture.

Aanvullende informatie onderwijsvormen

The lectures of this course are offered in English. Students following the Dutch (i.e. bilingual) track may in some cases submit their written assignments in either English or Dutch. Exceptions are multiple-choice tests, assignments with peer reviewers across language tracks, and group assignments (across language tracks).

Toetsvorm

Written exam (70%), assignments (30%).

Connection between learning objectives and assessment:
All three learning objectives will be assessed in both the exam and the assignments

Literatuur

Will be announced in the syllabus/Canvas.

Aanvullende informatie doelgroep

This course is compulsory for all BA MKDA students in their first year.

It is not open as an elective.

Overige informatie

For attendance rules, see Teaching and Examination Regulations. Students should be aware that regular attendance is necessary to pass the course.

Evolutionary Psychology

Vakcode	P_BEVOLPS
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	prof. dr. M. van Vugt
Examinator	prof. dr. M. van Vugt
Betrokken Docenten	prof. dr. M. van Vugt, S.E. Hudspith
Onderwijsvormen	Hoorcollege

Doel vak

De cursus zal studenten kennis laten maken met - en hen vertrouwd maken - met de belangrijkste theorieën, concepten en onderzoeksresultaten in de opkomende wetenschappelijke discipline van de evolutionaire psychologie. Het hoofddoel van de cursus is om studenten inzicht te geven in waarom mensen zich gedragen zoals ze doen door ze te bekijken door een evolutionaire lens (zoals een bioloog naar een bepaalde diersoort zou kijken). Centraal in deze cursus staat de vraag of bepaalde menselijke eigenschappen, eigenschappen en gedragingen het product zijn van geëvolueerde psychologische mechanismen die onze voorouders in staat stelden om verschillende problemen met betrekking tot hun overleving en voortplanting op te lossen. Voorbeelden zijn psychologische aanpassingen voor het vinden van een partner, voor (groot)ouderschap, altruïsme, agressie, werken, leiden, status en zelfverdediging.

Tijdens de cursus krijg je inzicht in enkele basisvragen over de menselijke natuur:

-Waar evolueerde de mens?

-Wat vertellen genen over ons gedrag?

-Wat verklaart onze relatief grote hersenomvang?

-Hoe verschillen mannen en vrouwen in hun seksuele strategieën?

-Waarom geven we de voorkeur aan bepaalde voedingsmiddelen boven andere?

-Wat is de evolutionaire logica achter oorlogsvoering?

-Is religie een aanpassing of bijproduct?

-Hoe werkt culturele evolutie?

-Waarom praten, zingen, dansen, spelen en kijken we naar sport?

Inhoud vak

Deze cursus traint studenten in het verklaren van menselijke psychologie en gedrag vanuit een evolutionair perspectief door gebruik te maken van inzichten uit de evolutionaire psychologie, biologie en antropologie. Tijdens

de cursus zullen we evolutionaire psychologie gebruiken om onderwerpen als menselijke seksualiteit, agressie, samenwerking, ouderschap, religie en cultuur (onder andere) te begrijpen. De cursus bestaat uit frontale hoorcolleges, opdrachten en een eindexamen.

Aanvullende informatie onderwijsvormen

Colleges en essay-opdrachten

Toetsvorm

multiple choice examen en twee essay-opdrachten

Vereiste voorkennis

Nee

Literatuur

Buss, D. M. (2019). Evolutionary Psychology: The New Science of the Mind (sixth edition). Routledge. Plus relevant scientific articles as conveyed by the teacher through Canvas.

Aanvullende informatie doelgroep

VU students

Overige informatie

50 studenten max van VU minor Psychology and Brain

Afwijkende intekenprocedure

NA

Toelichting Canvas

Op Canvas (vanaf Augustus)

Aanbevolen voorkennis

Basiskennis psychologie en kennis van statistiek en methoden in psychologisch onderzoek is aanbevolen

Extended Research Questions in Bioinformatics

Vakcode	XB_0074
Studiepunten	6.00
Periode	P1+2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. H. Mouhib
Examinator	dr. H. Mouhib
Betrokken Docenten	dr. ir. K.A. Feenstra, dr. Q. Peng MSc
Onderwijsvormen	Study Group, Computer lab, Lecture

Doel vak

This course is the follow-up to Research Questions in Bioinformatics.

You may choose one of the research topics of the Bioinformatics group to

study. In the process, you will get a chance to interview a researcher

about a scientific publication. You will be introduced to how research

is conducted in the Bioinformatics section. Finally, you will explain

the chosen topic to your fellow students in a visual representation and in a

presentation.

After following this course the student;

- Knows a specific example of research in the Bioinformatics field.

[Knowledge and understanding]

- Understands the knowledge the research is built upon and what the long

term goal of this line of research is. [Knowledge and understanding]

- Is able to ask relevant questions about a research paper or a

presentation. [Communication]

- Is able to give a clear presentation of a bioinformatics method and

its results. [Communication]

- Is able to describe the contents of a bioinformatics scientific

publication in their own words. [Communication]

- Can write purposeful feedback on a written assignment of a peer.

[Making judgements]

Inhoud vak

Research within Bioinformatics focuses on developing (computational) methods to understand biological experiments, and to predict biological function. During this project, you will discover why Bioinformatics is of great importance for (medical) research.

Aanvullende informatie onderwijsvormen

In this course, you'll engage in four parts related to your chosen subject, each taking approximately one week:

1. Research Insight: Select a recent paper from our group members to gain an understanding of our ongoing research. The focus is on comprehending the research question rather than conducting the work itself.
2. Comprehensive Exploration: Delve deeper into the subject by analyzing two or three additional papers. Compare how different researchers approach the research question presented in these papers.
3. Visual representation: Craft a visual representation of your subject (infographic/ poster/ concept map) where you provide context for the research and break down the research question in an accessible manner.
4. Peer Review and Presentation: Engage in peer review by evaluating and providing feedback on fellow students' work. Additionally, present your findings to the student group.

Toetsvorm

• [~50%] Scientific review article in 1000 words, containing the

research question, briefly formulated, a review of the consulted

literature, and three figures: schematic summary of research method and

at least one with results / graph.

• [pass/fail] assessment of articles of the other students (peer

review).

• [~50%] presentation (15 minutes) of your poster, with discussion and questions (15 minutes).

More detailed grading and compensation rules will be posted on Canvas.

In case of a failing grade for the popular scientific article this can

be redone after the course.

Literatuur

• Course material on Canvas
• Selected articles for study and presentation.

Aanvullende informatie doelgroep

This course is only open to (pre-master / bachelor) students in the minor Bioinformatics and Systems Biology

Overige informatie

This course runs in periods 1 and 2. The final presentations will be given in week 5 of Period 2.

Filmmaking in practice: pre-production, production and post-production

Vakcode	R_FIP
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	dr. S. Stolk
Examinator	dr. S. Stolk
Betrokken Docenten	prof. dr. W.G. Werner, dr. J.H. Roeland, dr. E. van Roekel, dr. S. Stolk
Onderwijsvormen	Study Group, Lecture

Doel vak

During the course, students create a research-based short documentary collaboratively. Upon completion of the course, students 1) master basic technical skills in documentary filmmaking (pre-production, production and post-production) and are able to create (as a team) a short documentary about a self-chosen societal topic that touches on the limits of articulation and representation, and 2) are able to reflect on the production of the short documentary, in relation to the notion of truth and knowledge claims in academic productions.

Inhoud vak

Under the guidance of a teacher and an experienced filmmaker, students learn the basic skills of academic filmmaking, including 1) pre-production skills: writing a research and film proposal, methods, ethics and data management, planning and organization, access to the field, storyboarding, and synopsis. 2) production skills: camera work, audio recording, use of light and tripod, creating footage (a- and b-roll), archiving. 3) post-production skills: basic control editing software programs, editing techniques, creating a narrative, color and sound correction, creating a logline, tagline, and trailer.

Aanvullende informatie onderwijsvormen

Interactive lectures; film screenings; and group discussions.

Toetsvorm

A short documentary based on research and a reflection report. In the report students 1) contemplate on the cinematographic and narrative choices made in the documentary and research activities, 2) students explain their approach to truth telling and the way this approach has informed their documentary. 3) Students reflect on the academic value of the knowledge they produced in the documentary.

Vereiste voorkennis

The four courses in the minor Visual Evidence:

1. Introduction course: Learning how to look

2. Showed trials: Truth telling in documentary film and court

3. AudioVisual Anthropology (AVA)

4. Seeing is believing

Literatuur

See course manual published on Canvas.

Aanvullende informatie doelgroep

The course is only open to students in the minor Visual Evidence.

Overige informatie

The course is part of the minor ‘Visual Evidence’.

Afwijkende intekenprocedure

Enrolment through emailing the minor coordinator Sofia Stolk via s.stolk2@vu.nl

Toelichting Canvas

Canvas is the main communication platform of the course.

Five O’Clock Neurosciences (Honours)

Vakcode	A_HP003
Studiepunten	6.00
Periode	P5
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. N.A. Goriounova
Examinator	prof. dr. H.D. Mansvelder
Betrokken Docenten	dr. K. Linkenkaer Hansen, dr. M.M. Karnani, dr. C.P.J. de Kock, dr. R. Min
Onderwijsvormen	Study Group

Doel vak

At the end of the course you should be able to:
1. Explain and discuss current theories on the biological basis of such phenomena as consciousness and its altered states, free will, moral behaviour and creativity
2. Distinguish between speculation based on personal opinion and arguments based on the scientific evidence
3. Form your own ideas based on big topics in neuroscience and support it by examples from current research
4. Think in terms of future perspectives in research and create your own ideas on research questions and possible experiments
5. Give an oral presentation for which you selected the key concepts from a popular science book and relevant information from neuroscience research
6. Formulate critical questions and engage in discussion

Inhoud vak

What are the neural circuits that make some people more creative than

others? How do we feel in our brain? How are we able to experience the

emotions of other people? What is consciousness? Do we have free will?

In our culture, we are used to think of the mind, heart and soul as

something separate from the body and brain. However, decades of neuroscience research have shown that our ability to feel the emotions of the self and understand the emotions of others, to be creative or alter our states of consciousness are inseparable from the workings of the brain and its intimate connection with the whole body. Are you curious about how modern neuroscience uses its arsenal of techniques to tackle the neural mechanisms of uniquely human traits such as consciousness, emotions, empathy and creativity? Well, then this course is for you!

Five O'clock Neurosciences aims to actively engage you in exploring the unknown territory of neuroscience and its big questions. You will not learn neatly packed answers out of the classic textbooks but will be challenged to ask your own questions and search for answers. This is why the course is designed in a unique way and uses nonstandard approach and study material. Instead of textbooks, you will read the books of famous neuroscientists and primary research papers, but also use TED talks and Sci-Fi films to investigate how modern neuroscience approaches the study of big questions in philosophy in psychology. The material is organized around several themes where you will present and discuss on such topics as free will, consciousness, creativity, aging, and human intelligence.

Aanvullende informatie onderwijsvormen

- Short lectures (14 hours)
- Student presentations (40)
- Discussion assignments: each student formulates two questions/ discussion points and a research question with an idea for an experiment (14)
- Active debate (14)
- Essay (60)
- Self-study (26)

Toetsvorm

- Presentation (40%)
- Discussion assignment (pass/fail)
- Essay (60%)

Literatuur

- book to be selected by the student (explanation of this point happens during the first meeting)
- scientific papers

Aanvullende informatie doelgroep

This course is intended for second and third year bachelor students that

have been selected for the VU honours programme.

Overige informatie

Students are expected to actively participate in oral presentations and

discussions. The topics discussed in the course provide an excellent

preview and preparation to the Research Master of Neurosciences

(https://masters.vu.nl/en/programmes/neurosciences/).

Toelichting Canvas

The course only runs every other year, from March until the end of May

on uneven years (2025, 2027, 2029 etc). Updates on the exact dates of the

course will be posted on Canvas before the start of the course.

Aanbevolen voorkennis

A first year's bachelor background in life sciences is recommended, but
not absolutely required.

Formal Logic in AI

Vakcode	XB_0116
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. V.N. Stebletsova
Examinator	dr. V.N. Stebletsova
Betrokken Docenten	dr. V.N. Stebletsova, dr. P.S.H. Koopmann
Onderwijsvormen	Seminar, Lecture

Doel vak

Knowledge and understanding: students will be acquainted with central logic-based

formalisms for symbolic AI such as modal logics and description logics.

Applying knowledge and understanding: students will know syntax and semantics of different modal and description logics,

how to reason in these logics, aspects of expressivity, and be able to determine which

logic is suitable for a particular application. They will be able to formalise

real-life knowledge and processes involving terminology, time, and strategic reasoning using appropriate logics.

Making judgements: students will be able to reason about and assess the

application decisions of different modal and description logics, and be able to determine which logic is suited for which application scenario.

Learning skills: students will be trained in writing specifications for properties of process systems based on

concepts of knowledge, time, strategic reasoning using a particular language and verifying those properties by model checking.

Inhoud vak

During the course we discuss logics that are relevant for symbolic AI:

— modal logics, in particular, Basic Modal logic, Temporal logic, Epistemic logic, some multi-modal logics most applicable to specification and verification of correctness properties of artificial Intelligence systems;

— description logics; in particular the description logic ALC and some of its relevant extensions

— connection between modal logics and description logics;

— issues relevant to understanding capacities of logics, such as deductive reasoning, model checking and expressiveness

Aanvullende informatie onderwijsvormen

During 7 weeks, 2 lectures (both 1.5 hours) and 1 exercise class (2 hours) each week.

There is no mandatory attendance for any lectures, well mandatory presence during exercise classes.

Toetsvorm

Assignments 40% (individual work).

An exam 60% (individual work)

There is a resit for the exam, no resit is possible for

practical assignments.

Literatuur

A reader for the modal logic part.

F. Baader, I. Horrocks, C. Lutz, U. Sattler: An Introduction to Description Logic. Cambridge University Press. 2017. ISBN 9781139025355

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 3)

Aanbevolen voorkennis

Basic knowledge of logic (propositional logic and first-order logic)

Foundations of Business Administration

Vakcode	E_MUB_FBA
Studiepunten	6.00
Periode	P1

Vakniveau	100
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. V. Duplat
Examinator	dr. V. Duplat
Betrokken Docenten	dr. V. Duplat
Onderwijsvormen	Study Group, Lecture

Doel vak

Students learn to apprehend real-world business situations by applying specific theoretical perspectives or using related analytic tools.

In particular, after following the course students:

• Have an advanced understanding of the traditional and emerging theoretical frameworks and concepts developed for studying organizations (foundational knowledge),

• Are able to adopt theoretical frameworks and apply tools to real-world situations and organizations (application),

• Are able to report, expose and defend their analyses and business recommendations, both verbally (report) and orally (presentation) (critical, creative and practical thinking), and

• Are able to work in small teams and properly allocate tasks among team

members under time pressure (Consulting project).

Therefore, besides the content-centered objectives and goals, the course has for objectives to impact students' critical thinking, creativity in their problem solving approach. Critical thinking in which students analyze and evaluate; creative thinking in which students imagine and create; practical thinking in which students solve problems and make decisions. By the end of the course, students have had the opportunity to approach business situations or contexts they are familiar with by using concepts and frameworks aimed at understanding those situations and contexts and innovating them.

Inhoud vak

New approaches to business and management constantly emerge. The course on the Foundations of Business Administration provides insights on traditional and new business approaches, while adopting an even-handed appreciation for theory and practice. The course familiarizes the students with the three main theoretical perspectives on organizations - Modern, Symbolic and Post-modern perspectives - and presents analytical tools rooted in those perspectives. It puts the students in a situation where those tools must be used for providing ideas to be explored by a

real-world firm.

These lectures are organized in five parts: (1) introduction of the three perspectives and their assumptions over time; (2) interdependency between organizations and their environment; (3) organizational social structure and organizational culture; (4) technology and physical structure of organizations; and (5) organizational power, control and conflict. Throughout the lectures, each perspective, concept and analytical tool is presented by referring to real-world and current business situations.

Aanvullende informatie onderwijsvormen

The course combines lectures and tutorials. The tutorials involve groups of 30 to 35 students. These latter tutorials combine case studies, round-table discussions about recent academic, business and press articles, role-games and making videos. A key objective is for you to develop your own personal synthesis and approach for identifying and addressing problems that managers face in organizations. Needless to say, preparing before attending classes gives you the best way to practice in lieu of actually designing and managing organizations. Throughout the tutorials, students will apply the theoretical frameworks and analytical tools introduced in the lectures to real-world organizations and situations.

Attendance to both lectures and tutorials is highly recommended to pass the course!

Toetsvorm

- Two group assignments under the form of a Consulting project (oral presentation, video-making, interviews and written reports) (team assessment)

- Individual quizzes or mini-essays on a weekly basis (individual assessment)

- Final written exam (individual assessment).

Vereiste voorkennis

NA

Literatuur

- Murphy, Willmott and Daft (2020, 4th edition) “Organization theory and design: an international perspective” is used as a main reference.

- Selection of business and managerial articles that will be posted on Canvas.

Overige informatie

Taking the other P1 course in the minor is strongly recommended!

Aanbevolen voorkennis

P1 course "Business Professionals"

From Protein to Cell

Vakcode	AB_1052
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. D. Bald
Examinator	dr. D. Bald
Betrokken Docenten	dr. D. Bald, K. van Rossum
Onderwijsvormen	Lecture, Practical

Doel vak

Study fundamental properties of proteins in a concerted theoretical and practical approach. We will cover principles and applications of investigating proteins as well as antibiotics targeting proteins. In the laboratory practicals selected features of key proteins will be studied. The design of experiments to address a certain open question and understanding of experimental protocols are an important aspects of this course.

Inhoud vak

The course consists of a mixture of lectures, practicals (in groups), and individual study. We will cover concepts/methods/techniques that you can

use to study a broad range of relevant questions, e.g: how to study a protein

and how to design an experimental strategy.

Aanvullende informatie onderwijsvormen

Lectures: 8 h, practicals: 20 h, individual study: 140 h.

Toetsvorm

Written reports on the performed experiments (100%).

Literatuur

Lecture slides and experimental protocols. Any biochemistry textbook can be used for repetition.

Aanvullende informatie doelgroep

Students in the Minor Biomolecular Sciences.

Overige informatie

This course is scheduled in the second half (weeks 5-8) of period 2.

A lab coat is mandatory for the lab practical lab work. Prior

participation in the course Experimental Cell Biology I is expected.

As this course mainly consists of practical exercises the examination is

offered only once per study year.

Aanbevolen voorkennis

Participation in Experimental Cell Biology I.

Future Challenges in Global Health

Vakcode	AB_1042
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	drs. E.A. Schrammeijer MSc
Examinator	drs. E.A. Schrammeijer MSc
Betrokken Docenten	drs. E.A. Schrammeijer MSc, M. Gerber, drs. A.D.L. Mers
Onderwijsvormen	Study Group, Lecture

Doel vak

• The student can list the five future challenges in global health and classify examples of health problems under each of these challenges.
• The student can explain how new developments in health and life sciences interact with global health challenges.
• The student can examine different policies developed at both the national and international level to address global health challenges.
• The student can compare policy-making processes between countries and the different visions that exist on policy.
• The student can apply certain methods, such as causal analysis and document review to perform a policy analysis within a team of students, on a real-world global health challenge.
• The student can develop a policy brief on the basis of different sources of information (policy documents, scientific publications, grey

literature), justify and present a synthesis of his/her findings verbally and in written form.

Inhoud vak

The course ‘Future Challenges in Global Health’ is the first course within the minor ‘Biomedical and Health Interventions’ as well as the minor 'Global Health' of the Bachelor’s programs Health Sciences, Biomedical Sciences and Health & Life (and other health-oriented Bachelors programs).

The world of biomedical and healthcare interventions is in constant flux – new and emerging infectious diseases, changing disease patterns, demographic changes, rising costs of health care; all of which add complexity to the already considerable challenges. At the same time, some innovative answers to these challenges have emerged, such as novel pharmaceuticals, neurotechnologies, gene therapy, e-Health and m-Health (e.g. using a smart phone as a heart rate monitor or as a tool in losing weight or enhancing physical activity), and field test kits replacing entire laboratories. This begs the question: How can we make these answers fit the challenges, which are constantly emerging? History reveals a number of health interventions, which have shown to not be that effective, as well as a numerous unintended consequences (for example how does an anti-malaria campaign lead to collapsing roofs and to cats being parachuted over Borneo?). This course explores how we can learn from these experiences, and use the evidence on effective biomedical and health care interventions to develop better health policies.

The beginning of the course provides you with an overview of both current and future challenges, in addition to scientific advancements in global health. We will also study how various countries and organizations, like the WHO, UN, and EU have addressed and dealt with these challenges, and why their policies have (not) been effective. We will use policy models to analyse real-world problems, solutions and policies (e.g. intervention programs). Addressing challenges in global health means both having a thorough knowledge of the health problem and potential interventions and perhaps more crucially, to understand the policy process and gain insight into how interventions can be effectively put into practice.

A prominent example of this is the following. We know there is an ideal intervention to prevent the spread of HIV/AIDS: a condom. Yet, the disease still spreads, and this is not due to a lack of knowledge. This leaves us with lingering questions such as when is scientific knowledge important? Where do politics come in? Do we need to involve more people in setting up health interventions? Do we need to work in public-private partnerships?

Important Note:

For students who are interested in the minor Global Health, we highly encourage you to either follow all 5 courses (periods1+2+3) within this minor, or at least 4 courses (periods1+2)!

This is important because in the practical part of all courses in the minor Global Health, including this one, you will use your newly acquired knowledge and apply it in a community engaged learning project that runs from period 1 to 3. Students will work in project teams on a real-life transdisciplinary global health challenge together with various local and global community partners. Students will cross disciplinary boundaries and interact with health professionals, patients and others to gain an in-depth understanding of global health problems, and to take cohesive and strategic actions to address these problems at a local level. Each course in the minor program will incorporate a different sub-project, starting in the first two courses with a broad exploration and needs analysis on the topic, followed by a more thorough investigation on a specific challenge in the next two courses, and concluding in the final course with a proposal for a sustainable innovation. Training workshops are offered throughout the minor program to support the development of specific skills such as transdisciplinary collaboration, reflection, social justice, global citizenship, dealing with diversity, creativity, leadership and adaptability.

Aanvullende informatie onderwijsvormen

Lectures (21 hrs), training workshop (10 hrs), working groups (16 hrs), self study (110 hrs), exam (3 hrs)

Toetsvorm

Exam(s) (individual assessment) (50%) and assignment (group assessment) (50%). Both parts need to be passed.

Vereiste voorkennis

The course is open to students from all fields of study.

Literatuur

Selected materials are made available through Canvas.

Aanvullende informatie doelgroep

Course for students within the minor Global health and the minor Biomedical and health interventions.

Overige informatie

Part of the minor Global health and the minor Biomedical and health interventions. This minor course requires a minimum of 25 participants to take place.

Global Development and Prosperity

Vakcode	AB_1275
Studiepunten	6.00
Periode	P2

Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M. Schaafsma
Examinator	dr. M. Schaafsma
Betrokken Docenten	dr. M. Schaafsma
Onderwijsvormen	Study Group, Lecture

Doel vak

The aim of the course is to provide students with a profound understanding of how contemporary global challenges are linked to socio-economic systems and how transformations in these systems can deliver just and sustainable prosperity.

By the end of this course, students will be able to:

- Explain existing, dominant measures and visions of development and economics and their positive and negative social and environmental effects, including inequality and climate change;

- Describe alternative measures of prosperity, and discuss their potential for transformations towards sustainable futures;

- Examine alternative visions of prosperity and achieving just and sustainable futures;

- Illustrate transformations towards just and sustainable futures through case study examples.

Inhoud vak

Over the last centuries, humankind has seen tremendous improvements in health, education and income, but at the same time this drive for economic growth and consumerist lifestyles has caused serious and sometimes irreparable damages to global biodiversity, climate systems and habitats, as well as major socio-economic inequalities and injustices.

The course first explores the history of such growth paradigms. Has emphasis on GDP growth equally benefited different world regions and people’s quality of life? We will discuss how growth-oriented visions of development have been globalized, and how they relate to current global environmental and social challenges.

This leads us to the question: What does GDP as an indicator tell us? What does it mean to live a good life? How do you measure if someone lives a good life? We review some of the dominant measures of prosperity and the assumptions on which they rest, as well as alternative measures of prosperity, including happiness, quality of life and capabilities. Does it matter which metric we use?

Next, we explore alternative ideas and visions of prosperity for the planet and respond to these global challenges. What would a prosperous society focused on long-term well-being look like, and how do we get there? How can citizens and consumers, farmers, or academics contribute to such a society? We will discuss the underlying assumptions of some of the existing frameworks for sustainable development as well as more radical proposals for transition and transformation towards prosperity. Using case studies of suggested transitions, we will critically evaluate whether such transformations will be sustainable and just, and how different stakeholders react to, and benefit from, these transformations.

Aanvullende informatie onderwijsvormen

The course consists of lectures, guest lectures, and seminars for discussions and group activities. The course runs for 7 weeks during period 2. Each week, there will be a lecture addressing the main material (2 hours), a guest lecture by VU researchers or external speakers illustrating the topic of the week (1 hour), and a seminar to explore the topic further (2 hours).

Some of the seminars will be dedicated to the development of the argument map (see below). This is an

opportunity to get feedback on the content and format of the argument map from peers and lecturers. Some lectures will be pre-recorded. Attendance in all live sessions is mandatory, but will be checked irregularly. We assume students are sufficiently serious and will show up at every class, barring unforeseen circumstances. Attending class is likely to increase your grade so attending classes is in your own interest.

The course corresponds to 6 ECTS, equivalent to 168 hours of study time. Students can expect the following:

- Attending/viewing lectures: 35 hours
- Mandatory readings: 45 hours
- Preparing for seminars / group discussions: 38 hours
- Developing argument map: 50 hours
- TOTAL: 168 hours

Toetsvorm

The overall mark for this course is based on two separate grades:

- Argument map, accompanied by an annotated bibliography and 2 peer-reviews (50%)
- Closed-book written exam (50%)

Students who take part in the minor Sustainability, Global Challenges, Interdisciplinary Solutions are recommended to develop their argument map for their topic/case selected in course 1.

Further details on the assignments will be provided in the course manual.

Vereiste voorkennis

Students do not require a specific disciplinary background, but affinity with the natural environment and social-economic development is desirable.

Literatuur

Each week, a selection of articles will be recommended to be studied in advance of the lectures and seminars.

Aanvullende informatie doelgroep

This course is part of the minor "Sustainability: Global Challenges, Interdisciplinary Solutions". We strongly encourage students to take all courses of the minor.

This course is suitable for students with an interest in the link between the natural environment and social-economic development.

This course can also be taken as:

• Extra-curricular course for VU-students

• Extra-curricular course for non-VU-students.

Toelichting Canvas

The course manual, as well as further course materials (e.g. lecture slides, reading materials), and course announcements will be circulated through Canvas.

Aanbevolen voorkennis

We strongly encourage students to take all courses of the minor "Sustainability, Global Challenges, Interdisciplinary Solutions", as a personal "Grand Challenge" comes back in different assignments of the five minor courses. Work in the preceding courses forms relevant input to the assignment of this course.

Global English

Vakcode	L_ETBAETK209
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. L.M. Rupp
Examinator	dr. L.M. Rupp
Betrokken Docenten	dr. L.M. Rupp
Onderwijsvormen	Lecture, Seminar

Doel vak

1. Knowledge

You are able to describe factors in the spread of English in the world

and the ways in which varieties of English have evolved world-wide.

You are also able to describe some of the social consequences of Global

English (e.g. language loss, social inequality).

2. Skills

Within theories of language variation and change, you are able to

analyse some of the major linguistic features of varieties of English around the world.

You are also able to assess social issues in society that involve the English language, such as

linguistic discrimination and implmented language policies.

3. Attitude

You are able to present a well-informed perspective of the nature of

different Englishes and the impact of the global spread of English on

people in societies around the world.

4. Communication

You are able to report on topics in Global English in short oral

presentations and in writing.

5. Competence

You are able to identify people’s needs regarding the English language,

including your own, and you are able to decide on the best way to act

upon it.

Inhoud vak

In the lectures, we will examine the spread of the English

language around the world. We will begin with regions in the world where

English is spoken as a first language (England, the Celtic countries,

the U.S., Australia, etc.). We will then move on to regions where

English is spoken as a second language (Africa and Asia), and from there

to regions where English is used as a foreign language or a lingua

franca (e.g. The Netherlands, Europe, China, and so

on).

We will explore different issues in the global spread of

English. These include: linguistic issues (variation in English, World

Englishes), social issues (language and identity, language and power,

language death, linguistic exclusion and discrimination), and literary

concerns (the use of English in postcolonial literature).

Some of the lectures are presented by guest lecturers from Scotland,

Australia, Uganda and Japan.

In the seminar, we will address questions that have arisen from the

lectures or the reading and we will discuss course work.

Aanvullende informatie onderwijsvormen

Lectures (2 hours per week) and seminars (2 hours per week)

Toetsvorm

Exam (100%, individual grade).
To successfully complete the course, you must obtain a grade of ≥5,5 for the exam and have submitted mandatory course work.
There is an attendance requirement of 80% for both the lectures and the seminars.

Vereiste voorkennis

Students must have passed Academic English CIS-L&S Grammar (L_EABAALG103) and Academic English CIS-L&S Writing (L_EABAALG104) and they must have participated in English: International Communication (L_ETBACIW102).

Exchange students should have taken equivalent courses.

Students in the Minor 'English' are exempted from these entry requirements. They need to contact the Onderwijsbureau of the Faculty of Humanities (FGW) to register for the course (onderwijsbureau.fgw@vu.nl).

Literatuur

Schneider, E.W. 2020. English Around The World. Cambridge: Cambridge University Press. Second edition.
Other literature and materials will be made available in Canvas.

Aanvullende informatie doelgroep

CIS students, Literature & Society students, students in the Minor 'English', History students, premaster- and exchange students.

Afwijkende intekenprocedure

This course has seminars, but you cannot self-register for a seminar in VU-net. If you want to know the days/times at which the seminars have been scheduled, go to rooster.vu.nl; click the 'Options' button at the right-hand side of the course title and select ‘Choose groups’.

A couple of weeks before the start of the course, your course coordinator will invite you to sign up for a seminar via Canvas.

Global Political Economy

Vakcode	S_GPE
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	prof. dr. E.B. van Apeldoorn
Examinator	prof. dr. E.B. van Apeldoorn
Betrokken Docenten	prof. dr. E.B. van Apeldoorn, I.T. Liu
Onderwijsvormen	Lecture

Doel vak

Learning outcomes:

Knowledge and understanding – The student has acquired knowledge and understanding of:

(1) the contemporary global political economy, in particular how the contradictory process of globalization reshapes the relationship between states and markets;

(2) rival concepts and theories within International Political Economy and their application to issues in contemporary global political economy.

Inhoud vak

This course offers students an introduction to the subject of International Political Economy (IPE). Throughout, the course will be guided by the question to which extent, and how, the current process of globalization is changing the relationship between states and markets, between public regulation and the private economy, between state and capital. Traditionally IPE studies the relationship between ‘the economic’ and ‘political’ within the interaction of – patterns of co- operation and conflict between – national states. If anything, the global financial and economic crisis of 2008 and beyond has made clear that this state-centric perspective is no longer adequate.

At the same time the crisis has also shown that states, although apparently vulnerable in the face of global market forces, are also crucial when it comes to protecting the workings of global capitalism. This shows that indeed the relationship between states and markets is not a one-way street. In other words, politics and policies are shaped by the interests and activities of transnational (market) actors and by economic globalization but the latter is also driven by politics, and shaped (indeed enabled) by the policy choices that states make. It is from this perspective that this course will examine the various approaches within international political economy; the historical evolution of the global political economy; the globalization of production and the role of transnational corporations; the international monetary system and the globalization of finance; the global financial crisis and the eurozone crisis; the political economy of development; the rise of China and other emerging powers, and the political economy of energy and the environment.

Aanvullende informatie onderwijsvormen

Lectures.

Toetsvorm

Final exam.

Literatuur

Balaam, D.N. & Dillman, B. (Eds). (2019). *Introduction to International Political Economy*. Pearson New International Edition (Seventh edition). Harlow: Pearson Education.

Aanvullende informatie doelgroep

Second-year Bachelor students in Political Science, track Global Politics and specialization Mondiale Politiek.

Aanbevolen voorkennis

Some introductory-level knowledge of political science and International Relations as well as of basic (macro-)economics is recommended but relevant concepts will also be explained in class.

Governance and Regulation of Emerging Technologies

Vakcode	R_GRET
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	P.B. Cardozo Solano LLM
Examinator	P.B. Cardozo Solano LLM
Betrokken Docenten	prof. mr. A.R. Lodder, I. Bratu, P.B. Cardozo Solano LLM
Onderwijsvormen	Study Group, Tutorial, Lecture

Doel vak

The aim of this course is to explore various ways to regulate and govern

societal changes caused by new technological developments. After this

course the students understand the various regulatory and

governance instruments, such as laws, regulation via technology,

self-regulation, standardisation, and how and when to apply these to new

technologies, including so-called disruptive technologies.

Specifically, the course fulfills the following learning outcomes:

- The students will familiarize themselves with recent technological advancements, will learn about existing technological developments and their impact in our increasingly digital societies;

- The students will gain understanding of ethical dimensions of modern technologies, setting the scene for legal analyses;

- The students will gain understanding of various laws, regulations and governance mechanisms applicable to various technologies in multiple sectors and domains;

- The students will demonstrate the ability to work with peers by acknowledging and supporting an inclusive and diverse environment;

- The students will demonstrate ability to clearly formulate legal arguments verbally and in writing and to resolve complex legal problems from multiple perspectives.

Inhoud vak

This introductory course of the Minor Technology, Law and Ethics offers an introduction into and an overview of the ways technology can or should be regulated. Important general concepts to be discussed are the economy (market powers), the law (regulation and case law), social

conventions and ethics, and the architecture (e.g. the software).

The course examines the fundamentals of regulation from legal and economic perspectives. It explores the range of regulatory tools available and considers their advantages and disadvantages through practical examples. As the course progresses, it examines approaches to the regulation of technology, beginning with Lawrence Lessig's framework, extending to network communitarianism and gatekeeper theory, and exploring nudge theory and behavioral economics. Students will analyze these regulatory perspectives and approaches and discuss their strengths and weaknesses in the context of emerging technologies.

Moreover, the course provides insight into the impact of regulation on key issues for the technology sector, including monopoly control, the promotion of competition, the concept of disruptive technologies, quality standards, and navigating the increasingly complex interplay between regulation, innovation, and the protection of fundamental rights.

Aanvullende informatie onderwijsvormen

Lectures and tutorials.

Toetsvorm

Written exam (individual; digital exam on campus or take-home via Canvas).

Literatuur

Materials will be made available via Canvas.

Aanvullende informatie doelgroep

Apart from regular students, the course is also available for:
Students from other universities/faculties
Contractor (students who pay for one course)

Governance of Global Sustainability

Vakcode	AB_1229
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. J. Dell'Angelo
Examinator	dr. J. Dell'Angelo
Betrokken Docenten	dr. J. Dell'Angelo
Onderwijsvormen	Lecture, Seminar

Doel vak

The main objective of this course is for you to develop an understanding of the different modes and multidimensional challenges of sustainability governance and the critical ways in which it is possible to think about decisions that have impact on the environment. Collectively as a class, and individually, you will develop knowledge in the politics of the environment by engaging with core scientific literature and discussing and tackling real world socio-environmental problems. By the end of the course, you will have gained an understanding of the complexity of the political dimension of the environment and the decisions that society deals with that impact sustainability. You will develop theoretical knowledge that will give you the ability to think critically and question mainstream societal decisions, politics and policies about the environment.

By the end of this course you will be able to:
 Understand and describe different critical principles and theories about multilevel governance of sustainability and the politics and policies of Sustainable Development
 Apply different theories to think critically about and different sustainability problems, by analyzing them from diverse and diverging perspectives
 Characterize different modes of sustainability governance and identify their strengths and limitations in relation to different types of socio-environmental problems
 Formulate your own analysis and suggestions regarding societal decisions that impact local and global sustainable development

Inhoud vak

Inhoud vak

Humanity deals with complex socio-environmental problems that impact the wellbeing of current and future generations. Nature and ecosystems globally are threatened and under suffering increasing deterioration. It is important to develop critical knowledge, understanding and perspectives about socio-environmental problems and potential solutions. Governance plays a key role in dealing with global sustainability issues. In this course we will study alternative theories about the sustainability challenges that will allow you to think of socio-environmental problems critically and we will learn about the different modes of governance that play a role in addressing issues of sustainability and sustainable development. We will deal with the political dimension of the environment and learn about how different theoretical perspective challenge the dominant paradigms. We will discuss and debate key sustainable development and socio-environmental problems such as the FEW nexus, climate change, water challenges, environmental conflicts etc.

Aanvullende informatie onderwijsvormen

We will mix: frontal lectures, seminars, discussions and group activities.

The structure of the course consists of a mix of lectures and group discussions, activities and assignments. The group activities in class will include simulations and team assignments. Readings will be assigned for
each session and it will be essential to be well prepared in order to successfully participate and address key questions that we will tackle.

Participation: Due to the complexity of the issues addressed and the concentrated length of the course I highly recommend attending every session. Of course, if you will have serious family or health reasons you can count on my help for making up for your absence and catching up. The amount of the material that we will go through and the very short time to prepare between the end of classes and the final exam will not allow you to be successful in this course if you procrastinate reading and if you don’t participate.

I will lead group discussions on key questions related to the readings. To make the most of our time together, I will call on students directly to address specific question during the group discussion. I will apply the “Socratic method” and challenge you to defend your opinions. (This is a method that might seem a little intimidating but is particularly useful to develop the capacity to debate publicly and defend your opinions – in line with the learning goals of the course.)

Preparation advice: I can’t stress enough the importance to hit the ground and start running from day one, actively participate and not fall behind. A critical aspect is that you make an effort to study the assigned readings before class. If you don’t you will see pages adding up quickly and you not be as able to engage productively in discussions and have less information and tools to understand the lectures. I look forward to work with you, so please also take advantage of office hours.

Toetsvorm

Assessment is based on exam, presentations and written

group assignments

Literatuur

For each week, a selection of articles will be made to be studied in advance.

Overige informatie

We strongly recommend that students interested in this course take all courses of the Minor Sustainability: Global Challenges Interdisciplinary Solutions

Aanbevolen voorkennis

Interest in sustainability issues and social and environmental justice and politics

Grand Challenges for Sustainability

Vakcode	E_MFS_GCS
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. ir. J.H. Ansink
Examinator	dr. ir. J.H. Ansink
Betrokken Docenten	C. Leisher, prof. dr. P.J.H. van Beukering, dr. ir. J.H. Ansink
Onderwijsvormen	Study Group, Lecture

Doel vak

The objective of this course is to characterize a path toward sustainable development and to identify the Grand Challenges the world faces in making the transition from the business-as-usual path to the sustainable development path.

ACADEMIC SKILLS / KNOWLEDGE

- The biophysics behind global environmental problems such as climate change and biodiversity loss;
- The importance of the 17 Sustainable Development Goals for achieving sustainable development;
- The drivers of economic growth and development and why the management of natural resources cannot be left to the free market;
- The importance of good governance for achieving sustainable development;
- Whether the government can and should intervene to obtain sustainable development and combat poverty, climate change, biodiversity loss, and resource depletion;
- The role of urban areas, in which more than half of the world population currently lives, for achieving sustainable development.

RESEARCH SKILLS / QUANTITATIVE SKILLS

After successfully completing this course, you will be acquainted with theoretical and empirical methods for studying economic growth, the effects of market failures, the management of natural resources, and the effects of different policy interventions.

BRIDGING THEORY AND PRACTICE

After successfully completing this course, you will be able to select, frame and qualitatively describe a challenge in the context of sustainability that is urgent, multidisciplinary, and linked to your personal interest. You will also be able to present a well-founded research proposal in an online pitch that is embedded in the academic literature.

SOCIAL SKILLS

After successfully completing this course, you will be able to present and actively discuss themes relevant to this course.

BROADENING YOUR HORIZON

After successfully completing this course, you will be able to explain:

- The interactions of the world economy, global society, and the natural environment that are important for sustainable development;
- Why sustainable development calls for socially inclusive and environmentally sustainable economic growth.

Inhoud vak

Sustainable development is the central challenge of our time. Today, the Earth is inhabited by more than 8 billion people. This is 10 times more than at the start of the Industrial Revolution in the 18th century. Every year, these people produce more than 100 trillion US dollars of output or 200 times more than at the start of the Industrial Revolution. Both population and output are projected to keep growing in the 21st century. Although global economic development has brought widespread prosperity, more than 700 million people still live in extreme poverty, and human activities may plunge the world into a gigantic environmental crisis caused by climate change and biodiversity loss. We live in the Anthropocene where human activity has become a dominant influence on the Earth's climate and natural environment. To eradicate poverty and prevent environmental catastrophes, a transition needs to be made from the business-as-usual path to the sustainable development path. Making this transition requires good governance by governments, citizens, and businesses.

The course is organized around the Sustainable Development Goals as adopted by the UN in 2015. The first week will start with a general introduction that sketches several important sustainability issues, illustrated by empirical evidence. During the course, we pay attention to the scientific as well as to the economic and societal dimensions of the identified challenges for sustainability. Furthermore, both analytical side (i.e., how to make sense of the interactions of the economy, society and the environment?) and the normative or ethical side (i.e., what should be the objectives of a well-functioning society?) of sustainable development will be discussed during the course.

The topics that will be dealt with during the course are:

1. Growth and development and links to capital accumulation and technological change;
2. Ending global poverty, improving education, and providing basic health care;
3. Management of natural resources and planetary boundaries;
4. Climate change including climate science and environmental policies;
5. Biodiversity and land-use change;
6. Global governance.

Aanvullende informatie onderwijsvormen

Lectures (with interactive elements); Tutorials (including presentation and discussion sessions); MOOC (to do at home prior to the lectures and tutorials)

Toetsvorm

Written exam – Individual assessment

Interim Assignments – Individual and group assessment

Literatuur

Sachs, Jeffrey D., The Age of Sustainable Development, 2015, Columbia University Press, New York.

https://cup.columbia.edu/book/the-age-of-sustainable-development/9780231173155

Collection of articles.

Great Minds I

Vakcode	W_BA_MND1
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen

Vakcoördinator	dr. L.W. Lee
Examinator	prof. dr. M. Martijn
Betrokken Docenten	prof. dr. M. Martijn, dr. L.W. Lee
Onderwijsvormen	Lecture

Doel vak

The objective of this course is to acquire critical understanding of

some of the highlights of the ancient and medieval philosophy time periods. After this

course you will have

1) knowledge of some of key thinkers from philosophy in the ancient and medieval philosophy time periods;

2) understanding of the main questions this philosophy seeks to answer.

After this course you are able to

3) interpret philosophical texts from the ancient and medieval philosophy time periods;

4) use and apply some of the core concepts of philosophy;

5) outline some strands of development in the ancient and medieval philosophy time periods in your

own words.

Inhoud vak

This course largely focuses on Western philosophy from 6th century BCE to 15th century CE. It introduces you to Ancient and Medieval philosophy, by studying the theories of a number of well-known thinkers: Plato, Aristotle, Augustine, and Thomas Aquinas, while also attending to the theories of those who are less well-known, such as Mengzi [Mencius], Laozi, Ibn Rushd [Averroes], and Christine de Pizan. We will focus on the relation between world, knowledge and goodness (i.e., metaphysics, epistemology, and ethics).

Aanvullende informatie onderwijsvormen

Interactive lecture; seminar; text analysis

Toetsvorm

Weekly assignments (pass/fail, assessment of learning objective 3, 4);
final exam (100%, assessment of learning objectives 1, 2, 4, 5). The
assignments need to be sufficient (i.e. 'pass'), the final exam
determines the grade.

Literatuur

A selection of primary sources, to be distributed through Canvas.

Aanvullende informatie doelgroep

Minor students Philosophy

Great Minds II

Vakcode	W_BA_MND2
Studiepunten	6.00
Periode	P2+3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. C.H. Krijnen

Examinator	dr. C.H. Krijnen
Betrokken Docenten	dr. C.H. Krijnen
Onderwijsvormen	Lecture

Doel vak

The student acquires the ability:
1. To recognize and interpret the fundamental questions within modern and contemporary philosophy;
2. To identify and explain the basic ideas and concepts of some important philosophers from the 17th-20th century;
3. to make connections, and to differentiate between these concepts and ideas.
4. to read, analyze, and interpret primary philosophical texts;
5. to further develop argumentative skills, orally and in writing.

Inhoud vak

In this course the most important thinkers from the philosophical

history of the 17th until the 20th century will be discussed. It concerns philosophers who have

had an ineffaceable influence on philosophical thinking in general and

thinking about science and culture in particular. The following

philosophers are discussed in Modern philosophy: Descartes, Hume, Kant,

Hegel. Concerning Contemporary Philosophy, the course is presently under construction. Four philosophers will be addressed. Info on the program will follow on Canvas. Presently, Nietzsche, James, Peirce, Wittgenstein, Heidegger, Horkheimer/Adorno, Marcuse, Arendt, Foucault, Deleuze belong to the possible candidates.

Aanvullende informatie onderwijsvormen

Lectures to introduce the philosophical movement and philosopher's work; text reading seminars in which a text(fragment) of the philosopher is discussed. Attendance of 80% of the text reading seminars in the classes on Modern Philosophy and in the classes on Contemporary Philosophy is mandatory.

Toetsvorm

- Assigments about the primary texts (complete/incomplete) (assessment

of learning objectives 1, 2, 3, 4, 5);

- Midterm test on Modern Philosophy with essay questions (50%)

(assessment of learning objectives 1, 2, 3, 4, 5);

- Final test on Contemporary Philosophy with essay questions

(50%)(assessment of learning objectives 1, 2, 3, 4, 5) .

- There is a mandatory attendance of 80% in the lectures in part I and

80% in the lectures in part II in order to achieve learning objectives 4

and 5.

Literatuur

The literature list will be published at Canvas two weeks before the start of the course.

Aanvullende informatie doelgroep

The course is part of the minor Philosophy.

Toelichting Canvas

Two weeks before the start of the course, a detailed program will be made available through Canvas.

Aanbevolen voorkennis

Great Minds I

Hadith-wetenschappen

Vakcode	G_BATRSAL026
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit Religie en Theologie
Vakcoördinator	dr. Y. Ellethy
Examinator	dr. Y. Ellethy
Betrokken Docenten	dr. Y. Ellethy
Onderwijsvormen	Hoorcollege

Doel vak

Na afronding van deze module:
- kan de student de terminologie op het gebied van de hadithwetenschappen benoemen, definiëren, uitleggen, analyseren, en toepassen;
- kan de student vergelijkingen maken tussen westerse historische methodologie en Hadithmethodologie, de hadithwetenschappen integreren in en confronteren met bestaande historische theorieën en eigen theoretische positiebepaling verwoorden;
- kan de student de niet-islamitische en islamitische kritiek tegen de autoriteit van de Soenna en de betrouwbaarheid van de Hadithmethodologie weergeven, bediscussiëren en hierover argumenteren;
- kan de student uitleg geven over de belangrijkste concepten van de principes van de hadithwetenschappen [uṣūl al-ḥadīth];
- kan de student de inhoud van een aantal in het college behandelde Koran- en Hadithteksten weergeven en deze teksten analyseren en uitleggen volgens de methode van de Koran- en hadithwetenschappen blijkend uit presentaties en werkcollege discussies;
- kan de student in hoofdlijnen iets vertellen over de hedendaagse discussies en problematiek van de hadith in de moderne tijd, het belang van dit vakgebied uiteenzetten in een bredere maatschappelijke of wetenschappelijke context.
- kan de student de belangrijke Soenna-hadithliteratuur benoemen en hierover uitleg geven en literatuuronderzoek doen blijkend uit schrijfopdrachten en presentaties;
- kan de student de hadiths classificeren en toeschrijven aan een bepaalde autoriteit en deze classificeren;
- kan de student enkele hadithteksten vanuit een eigen perspectief toelichten, verwoorden, in verband brengen met andere posities, en het belang hiervan uiteenzetten in een bredere maatschappelijke en wetenschappelijke context;
- kan de student hermeneutische competenties gebruiken, achtergrondinformatie bij actuele kwesties aanleveren en de opgedane kennis in link met de praktijk toepassen.

Inhoud vak

Inhoud vak

Hadith-wetenschappen is een vervolg op en verdieping van Inleiding in de Koran en Soenna. De inhoud wordt verdiept met meer aandacht voor: hadithwetenschappen/methodologie, terminologie van de hadithwetenschappen usul/mustalah al-hadith, en de hedendaagse discussies over de autoriteit van de Soenna. Het gaat dus om: geschiedenis van usul al-hadith, classificatie van de hadiths, analyse van isnâd/sanad en matn, relatie tussen Koran en Soenna , deconstructie en beoordeling van een sanad, leeswijze van een sanad, criteria van betrouwbaarheid van een overlevering/overleveraar, aanvaardbaarheid en onaanvaardbaarheid van een hadith, aantasting van een isnâd of matn, hadith commentaar (sharh) en methodes van takhrij van een hadith.

Aanvullende informatie onderwijsvormen

Hoor- en werkcolleges met schriftelijke opdrachten en tussentijdse papers. Een klassieke bron-tekst (matn) van de hadithmethodologie wordt uitgelegd en ganalyseerd; een aantal relevante artikelen, boekhoofdstukken en hadith teksten worden behandeld. Er wordt aandacht gegeven aan de interactieve deelname van de studenten. Vragen worden aan het begin van het college beantwoord en besproken. Aan het eind van de serie hoorcolleges wordt een werkstuk gepresenteerd, in werkgroep besproken en beoordeeld.

Toetsvorm

Schriftelijk tentamen (80%)

Schrijfopdracht (20%)

Literatuur

Verplicht:
- Azami, M., Studies in Hadith Methodology and Literature, Indianapolis: American Trust publications, 1977
- Brown J., "The rules of Matn criticism:There are no rules", Islamic Law and Society 19 (2012), pp. 356-396 (Canvas)
- Ibn Al-Salah, An Introduction to the Science of the Hadith, trans. Dickinson E., Reading: Garnet Publishing Ltd, 2006.

Aanbevolen:
- Kamali M., A Textbook of Hadith Studies, Leicestershire: The Islamic Foundation, 2009
- Maloush T., Early Hadith Literature and the Theory of Ignaz Goldziher, Phd thesis, University of Edinburgh, 2000
- Siddiqi, M., Hadith for Beginners, New Delhi : Goodword Books, 2000 (VU Bibliotheek)

Overige informatie

Voor deze module geldt een aanwezigheidsplicht van 80%.

Afwijkende intekenprocedure

Je kunt je alleen intekenen voor dit vak als je bent ingetekend op de minor Arabisch en bronnen van de Islam of de minor Islam, of als je de track Islam (onderdeel van de bachelor Theology and Religious Studies) of de track Vrije Keuze (onderdeel van de bachelor Theology and Religious Studies) volgt.

Aanbevolen voorkennis

- Inleiding in de Koran Soenna (G_BATRSPC123) [Note: n.v.t. voor minor-studenten]

- Arabisch IV (G_BATRSPC202) [Note: n.v.t. voor minor-studenten]

- Geschiedenis van de Islam tot 1800 (G_BATRSPC106) [Note: n.v.t. voor minor-studenten]

Health Economics

Vakcode	E_MFHC_HEC
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	prof. dr. W. Janssens
Examinator	prof. dr. W. Janssens
Betrokken Docenten	dr. T.J. Galama, prof. dr. W. Janssens
Onderwijsvormen	Study Group, Lecture

Doel vak

In this course students learn to:
• Apply micro-economic theory to the study of health policy and healthcare
financing;
• Understand and address the drivers of individual health-related behavior;
• Explain the origins of inequalities in health outcomes over the life cycle, and within and between countries;
• Formulate economic requirements for effective healthcare systems;
• Identify healthcare market failures and explain how/why governments may intervene;
• Evaluate the impact and effectiveness of healthcare policies using economic theory;
• Perform longitudinal analyses of developments in healthcare;
• Collect and synthesize empirical evidence to communicate on matters of healthcare.

Inhoud vak

The central theme of this course is the economic optimization of healthcare systems, investigating both the demand- and the supply-side. This course provides students the necessary economic knowledge to analyze healthcare policies and individual health-related behavior, and to examine the economic effects of healthcare policies and interventions.
The following topics will be addressed:
• healthcare systems' main objectives;
• market imperfections and principal-agent relationships in healthcare;
• methods to evaluate the economic performance of healthcare policies;
• health financing and health insurance;
• benefit-incidence analyses of healthcare programs;
• socioeconomic and life cycle inequalities in health outcomes;
• prevention, treatment and (socioeconomic) impact of infectious diseases (such as COVID19, HIV/AIDS, malaria);
• the growing burden of non-communicable diseases;
• individual behavior and lifestyle choices that affect health outcomes;
• the link between mental health, depression and socioeconomic outcomes.

Aanvullende informatie onderwijsvormen

- Two lectures per week (on-campus)
- One tutorial meeting per week (on-campus)

Toetsvorm

- Written exam (individual assessment)
- Weekly written tutorial assignments (group assessment)

Vereiste voorkennis

None.

Literatuur

- Selection of chapters from Bhattacharya, J., T. Hyde and T. Tu (2014), "Health Economics", Palgrave McMillan.
- Papers, as listed in the course manual and published on Canvas.

Aanbevolen voorkennis

None.

History of AI

Vakcode	XB_0113
Studiepunten	3.00
Periode	P5
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. D.J. Beckers
Examinator	dr. D.J. Beckers
Betrokken Docenten	dr. D.J. Beckers
Onderwijsvormen	Study Group, Lecture, Seminar

Doel vak

Students acquire knowledge and understanding about the history of computing / AI from various perspectives: computing as a scientific goal, computing as a government (administrative or military) objective / ideal, computing as an economic enterprise (Knowledge and understanding).

Students can apply that knowledge and understanding to describe the meanings digital culture has or may have for various people in contemporary society (Applying knowledge and understanding).

Students are able to communicate their knowledge and understanding in an effective way, and from various perspectives - thereby they will be better equipped to reflect on their subject of study (Communication) (Making judgements)

Inhoud vak

Various subjects from the history of computing will be treated. Several highlights will be discussed and placed within the social context of its time. By discussing these highlights from several points of view the history of computing / AI will serve as a way to illustrate the various roles of computing / AI in society.

Aanvullende informatie onderwijsvormen

Lectures, and (non mandatory) workgroups and assignments. Optional quizzes helping to study; extracurricular activities for more in-depth info.

Toetsvorm

Written exam; assignments allow students to skip a specifically designated 1 point question in the exam, if the four best

assignments are graded with an average score of 6 or better.

Resit possible for the exam. In principle resubmission of assignments is not allowed.

Literatuur

Martin Campbell-Kelly a.o., *Computer: A history of the information machine* (fourth edition, 2023). Students are required to read chapters 4-15.

Aanvullende informatie doelgroep

Bachelor AI (year 1)

Overige informatie

More information with the course coordinator: W&N room A652, d.j.beckers@vu.nl.

Toelichting Canvas

On Canvas (in the "Modules" tab) will be announced which chapters you're supposed to prepare for the next lecture. There are optional extra's as well.

Honours Research Project 1

Vakcode	X_HP003
Studiepunten	6.00
Periode	Ac. Year (sept)
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. ir. A. Iosup
Examinator	prof. dr. ir. A. Iosup
Betrokken Docenten	prof. dr. ir. A. Iosup
Onderwijsvormen	

Doel vak

If you are a talented and motivated student, the VU Honours Programme offers you, through the research project of the Honours Programme, the opportunity to invest more in your own area of study and to broaden your horizons.

For this research project, you will focus on a high-quality, scientific research project that is unique for each student. You will learn scientific knowledge and skills, learn through embedding in one of our research groups how the field of your choice works at a professional level, and, depending on your project, learn to relate to design, analysis, engineering, ethics, and other processes specific to the field.

The research project is split into three main modules -- this is the first module, focusing on general research skills and on defining a scientific research project.

By the end of completing these modules, you will learn the skills and knowledge corresponding to the level of a starting, independent researcher with a high potential for further development. The successful project may end with a high-quality publication, good to present in a prestigious scientific venue or to publish in a top journal in the field.

Notice that the learning outcomes of this research project cover all Dublin Descriptors (Knowledge and understanding, Applying knowledge and understanding, Making judgements, Communication, Lifelong learning skills).

Inhoud vak

In this first module of the research project, you will learn the general methods for conducting scientific research, with a focus on Computer

Science. The course will focus on the skills and knowledge specific to the field where the student will develop a scientific research project, and on defining a scientific research project.

Aanvullende informatie onderwijsvormen

The primary actor of this module is the student, who will spend much time in individual and group study.

Students will jointly participate, with a fixed and useful frequency decided by the supervisor, in a seminar, workshop, or face-to-face meeting organized by the supervisor on various topics relevant to the research project and/or to the development of students into starting, independent researchers. Students will also formulate and ask questions about the process and the content of the project.

To support individual development, this module is also designed to match the professional work of the research group where the student is embedded. To this end, the main pedagogic technique here combines the “learning by doing” of the research project with the immersion of the student in a real-world, high-quality research group. Students will learn about the operational challenges and approaches taken by these professional researchers.

Toetsvorm

Oral presentation and a written report. This course is Pass/Fail, and students need to pass both the presentation and the report assessment to Pass the course.

The written report should aim to identify at least: (1) the context, (2) the problem, (3) the main research question(s), (4) the research approach, (5) a preliminary, but concrete and feasible, plan.

To limit the burden on both student and supervisor for research projects that span across multiple modules of the Honours Research Project (that is, the student continues to modules 2 and 3), the written report can take the form of a problem statement, and later be combined into a final report of the project across all modules.

Literatuur

Depending on the project, the supervisor can recommend various items of literature or guide the student in conducting a literature survey.

Aanvullende informatie doelgroep

Students admitted to the Honours Programme CS/AI.

Overige informatie

Students can continue a research project beyond the 18 ECs typically offered for the three modules of the Honours Research Project. The HP

does not limit the maximum number of ECs, it just specifies the mandatory minimum. For example, students can choose to deepen their

research project so they can give more meaningful answers to an important problem, or to earn additional credits by training for and

then attending various conferences and seminars where they present their work to the national and international community.

Afwijkende intekenprocedure

Before starting a research project the student will agree individually with a senior staff member (UD, UHD, HGL), who will act as the main

supervisor, on the general topic of the project (e.g., systems security, distributed systems, high-performance computing, artificial intelligence, bioinformatics). The topic of the project can be proposed by either the student or the senior staff member, but it must be agreed upon by both parties. To register for this course, contact the HP

Coordinator, Selma Zaghow, at s.m.a.zaghow@vu.nl.

Toelichting Canvas

This course uses Canvas to form a knowledge community between students.

Honours Research Project 2

Vakcode	X_HP004
Studiepunten	6.00
Periode	Ac. Year (sept)
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. ir. A. Iosup
Examinator	prof. dr. ir. A. Iosup
Betrokken Docenten	prof. dr. ir. A. Iosup
Onderwijsvormen	

Doel vak

If you are a talented and motivated student, the VU Honours Programme offers you, through the research project of the Honours Programme, the opportunity to invest more in your area of study and to broaden your horizons.

For this research project, you will focus on a high-quality, scientific research project that is unique for each student. You will learn scientific knowledge and skills, learn through embedding in one of our research groups how the field of your choice works at a professional

level, and, depending on your project, learn to relate to design, analysis, engineering, ethics, and other processes specific to the field.

The research project is split into three main modules -- this is the second module, focusing on developing a scientific research project.

By the end of completing these modules, you will learn the skills and knowledge corresponding to the level of a starting, independent researcher with a high potential for further development. The successful project may end with a high-quality publication, good enough to be presented at a prestigious scientific venue or to be published in a top journal in the field.

Notice that the learning outcomes of this research project cover all Dublin Descriptors (Knowledge and understanding, Applying knowledge and understanding, Making judgements, Communication, Lifelong learning skills).

Inhoud vak

In this second module of the research project, you will learn how to develop a scientific research project, in a particular field of Computer

Science.

Aanvullende informatie onderwijsvormen

The primary actor of this module is the student, who will spend much time in individual and group study. Developing the research project will require significant conversations with the research group and especially the supervisor. The student should plan and then spend significant periods of time with only light supervision or even working independently.

Students will jointly participate, with a fixed and useful frequency decided by the supervisor, in a seminar, workshop, or face-to-face meeting organized by the supervisor on various topics relevant to the research project and/or to the development of students into starting, independent researchers. Students will also formulate and ask questions about the process and the content of the project.

To support individual development, this module is also designed to match the professional work of the research group where the student is embedded. To this end, the main pedagogic technique here combines the “learning by doing” of the research project with the immersion of the student in a real-world, high-quality research group. Students will learn about the operational challenges and approaches taken by these professional researchers.

Toetsvorm

Oral presentation and a written report. This course is Pass/Fail, and students need to pass both the presentation and the report assessment to

Pass the course.

The written report should aim to identify at least: (1) the context, (2) the problem, (3) the main research question(s), (4) the research approach, (5) a preliminary, but likely feasible, answer or solution.

To limit the burden on both student and supervisor for research projects that span across multiple modules of the Honours Research Project (that is, the student continues to modules 3), this written report can later be combined into a final report of the project across all modules.

Vereiste voorkennis

Completion of module 1 of the Honours Research Project.

Literatuur

Depending on the project, the supervisor can recommend various items of literature or guide the student in conducting a literature survey.

Aanvullende informatie doelgroep

Students admitted to the Honours Programme CS/AI.

Overige informatie

Students can continue a research project beyond the 18 ECs typically offered for the three modules of the Honours Research Project. The HP does not limit the maximum number of ECs, it just specifies the mandatory minimum. For example, students can choose to deepen their research project so they can give more meaningful answers to an important problem, or to earn additional credits by training for and then attending various conferences and seminars where they present their work to the national and international community.

Afwijkende intekenprocedure

Before starting a research project the student will agree individually with a senior staff member (UD, UHD, HGL),

who will act as the main

supervisor, on the general topic of the project (e.g., systems security, distributed systems, high-performance computing, artificial intelligence, bioinformatics) The topic of the project can be proposed by either the student or the senior staff member, but it must be agreed upon by both parties. To register for this course, contact the HP

Coordinator, Selma Zaghow, at s.m.a.zaghow@vu.nl.

Toelichting Canvas

This course uses Canvas to form a knowledge community between students.

Honours Research Project 3

Vakcode	X_HP006
Studiepunten	6.00
Periode	Ac. Year (sept)
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. ir. A. Iosup
Examinator	prof. dr. ir. A. Iosup
Betrokken Docenten	prof. dr. ir. A. Iosup
Onderwijsvormen	

Doel vak

If you are a talented and motivated student, the VU Honours Programme offers you, through the research project of the Honours Programme, the opportunity to invest more in your area of study and to broaden your horizons.

For this research project, you will focus on a high-quality, scientific research project that is unique for each student. You will learn scientific knowledge and skills, learn through embedding in one of our research groups how the field of your choice works at a professional level, and, depending on your project, learn to relate to design, analysis, engineering, ethical, and other processes specific to the field.

The research project is split into three main modules -- this is the third module, focusing on deepening and completing a scientific research

project. By the end of completing these modules, you will learn the skills and knowledge corresponding to the level of a starting, independent

researcher with a high potential for further development. The successful project may end with a high-quality publication, good to present in a prestigious scientific venue or to publish in a top journal in the field.

Notice that the learning outcomes of this research project cover all Dublin Descriptors (Knowledge and understanding, Applying knowledge and understanding, Making judgements, Communication, Lifelong learning skills).

Inhoud vak

In this third module of the research project, you will learn how to deepen and complete a scientific research project, in a particular field of Computer Science.

Aanvullende informatie onderwijsvormen

The primary actor of this module is the student, who will spend much time in individual and group study. Deepening the research project will require significant conversations with the research community and especially the supervisor. The student should plan and then spend

significant periods with only lightly supervision or even independently.

Students will jointly participate, with a fixed and useful frequency decided by the supervisor, in a seminar, workshop, or face-to-face meeting organized by the supervisor on various topics relevant to the research project and/or to the development of students into starting, independent researchers. Students will also formulate and ask questions about the process and the content of the project.

For individual development, this module aims to match the professional work of the research group where the student is embedded. To this end, the primary educational technique here combines the “learning by doing” of the research project with the immersion of the student in a real-world, high-quality research group. Students will learn about the operational challenges and approaches taken by these professional researchers.

For outstanding projects that can lead to high-quality publications, the supervisor will guide the student in the editorial process typical of high-quality science.

Toetsvorm

Oral presentation and a written report. This course is Pass/Fail, and students need to pass both the presentation and the report assessment to Pass the course.

The written report should aim to identify at least: (1) the context, (2) the problem, (3) the main research question(s), (4) the research

approach, (5) the answer or solution, (6) evidence supporting the answer or solution, (7) an analysis of the main contribution of the research project.

To limit the burden on both student and supervisor for research projects that span across multiple modules of the Honours Research Project, this written report can be combined into a final report of the project across all modules.

For outstanding projects, the report can be written in the form of a scientific publication and submitted to a top venue. For these cases, the project should also aim to publish free open-source software and FAIR data, plus a reproducibility summary (e.g., following the ACM or the IEEE formulations leading to trusted artefacts).

Vereiste voorkennis

Completion of modules 1 and 2 of the Honours Research Project.

Literatuur

Depending on the project, the supervisor can recommend various items of literature or guide the student in conducting a literature survey.

Aanvullende informatie doelgroep

Students admitted to the Honours Programme CS/AI/IS.

Overige informatie

Students can continue a research project beyond the 18 ECs typically offered for the three modules of the Honours Research Project. The HP does not limit the maximum number of ECs, it just specifies the mandatory minimum. For example, students can choose to deepen their research project so they can give more meaningful answers to an important problem, or to earn additional credits by training for and then attending various conferences and seminars where they present their work to the national and international community.

Afwijkende intekenprocedure

Before starting a research project the student will agree individually with a senior staff member (UD, UHD, HGL),

who will act as the main

supervisor, on the general topic of the project (e.g., systems security, distributed systems, high-performance computing, artificial intelligence, bioinformatics) The topic of the project can be proposed by either the student or the senior staff member, but it must be agreed upon by both parties. To register for this course, contact the HP

Coordinator, Selma Zaghow, at s.m.a.zaghow@vu.nl.

Toelichting Canvas

This course uses Canvas to form a knowledge community between students.

HP Research Community and Practice

Vakcode	X_HP010
Studiepunten	6.00
Periode	Ac. Year (sept)
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. S. Lusini
Examinator	prof. dr. ir. A. Iosup
Betrokken Docenten	dr. S. Lusini, prof. dr. ir. A. Iosup, dr. A. Yaman
Onderwijsvormen	Lecture

Doel vak

By taking this course, you will be able to:

- Understand the research process, including research methods and practical aspects. (Knowledge and understanding)

- Develop your own research project, including a selection of research methods and approaches relevant to your project. (Knowledge and understanding, Applying knowledge and understanding, Communication, Making judgements, Lifelong learning skills)

- Be able to present your own research project, in writing and through an oral presentation. (Communication)

Inhoud vak

The VU BSc Honours Programme promises to provide students with a deeper understanding and experience in the curricular aspects of their Bachelor. One of the most important topics, which currently is included only implicitly in the study programme, is the research process---the philosophy, theories, methods, and body of knowledge associated with the scientific method of our field.

This course aims to introduce the principles and practice of the research process in the fields of computer science and artificial intelligence.

Aanvullende informatie onderwijsvormen

Lectures with Socratic engagement in online sessions, self-study and self-reflection, guided report-writing. There are 16 sessions spread over the entire academic year, with about 2x meetings per month. Exact timings to be announced at the start of the course.
Students and course professors meet at mid-term to discuss progress and draft of the research portfolio.
Students and course professors meet at the end, for a one-day seminar where all students present and discuss the research process.

Toetsvorm

1. Students formulate one **research proposal**, preferably linked with the HP research project. The formulation delivers one report, for which we offer a template students can use.
2. Students formulate a **research portfolio**, by self-reflecting on the main topics of the course. The portfolio

must cover at least 60% of the topics, each addressed in a short overview (suggested: 1-page recto and verso).
3. Students formulate a **final report**, which also includes the preliminary reports from points 1 and 2. This report is due at the end of thecourse. The portfolio should be delivered at mid-term (the first part, for feedback) and at the end (the final version).
4. Students prepare a **presentation of their research proposal and final report**.

Relatively to point 1, which presents the start of the research project, this report also includes a reflection on the entire duration and final results of the research project. Relatively to point 2, this report also includes an analysis and self-reflection about how the entire course helped with the research project. The final report cannot add more than 6 pages of new material, excluding references and the title page.

Mandatory parts:

Students have to complete the detailed requirements for each of the assessment points 1, 2, and 3.

All assignments, as well as the course, are graded as pass or fail.

Literatuur

No textbook. The course provides: slide decks via Canvas. There is a list of recommended readings, including several books.

Aanvullende informatie doelgroep

This course is only open for students enrolled in the BSc Honours Programme AI/CS of the Beta Faculty.

Afwijkende intekenprocedure

To register for this course, contact the HP Coordinator, Selma Zaghow, at s.m.a.zaghow@vu.nl.

Toelichting Canvas

This course uses Canvas to form a knowledge community between students.

Human Rights and Citizenship

Vakcode	R_HumRC
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	dr. K.J. Swider
Examinator	dr. K.J. Swider
Betrokken Docenten	prof. dr. G.R. Jones, dr. K.J. Swider
Onderwijsvormen	Study Group, Tutorial

Doel vak

After successfully completing this course you will be able to:

- Understand how the concept of citizenship operates in national and international legal systems;
- Gain an insight into the role of citizenship policies in the formation of statehood in (post-)colonial and state succession contexts;
- Understand international and regional legal norms on statelessness and multiple nationalities;
- Understand the hierarchical function and racialized instability of citizenship and rights in colonial and postcolonial contexts;
- Apply the knowledge of core concepts and theories acquired in this course to analyse your chosen citizenship regime;

- Gain a comparative perspective on how various states across the globe regulate acquisition and loss of their citizenships;
- Critically evaluate the relationship between the status of citizenship and the concept of human rights;
- Critically engage with the concept of ‘integration’ and analyse the assimilationist shift of mandatory integration measures.

The course contributes to achieving the following end objectives of the Bachelor degree:

2&3 - Interconnectedness between different legal orders: national, European, international; 4 - understanding of legal terminology in English; 5 - understanding that the law develops and manifests itself in social contexts; 7 - reading, understanding and analysing legal sources; 9 - critical reflection on laws from historical and philosophical perspectives; 10 - reflection on one's own position in society where the relevant laws function; 16 - conducting research into a legal problem and writing down the findings; 21 - finding legal sources.

The course also contributes to the objectives of the Law in Action by asking students to apply the theoretical knowledge acquired to a concrete citizenship context of their choosing, reflecting on the political, societal, historical and anthropological relevance of the legal category of citizenship worldwide.

Inhoud vak

Citizenship is an important concept for national and international legal systems, but also a core notion in political science, sociology, and anthropology, as well as a salient political issue in many states. This course offers a cross-disciplinary insight into the seemingly omnipresent yet poorly defined phenomenon of citizenship. Together we will attempt to understand core legal terminology surrounding citizenship, and engage with national and international laws that regulate how citizenship ought to be acquired and lost by individuals. We will look at how states assert their statehood and sovereignty through the law and politics of citizenship in different contexts of state formation, including state succession and (post-)colonialism. Finally, we will study the nexus between citizenship, migration and human rights through a critical look at integration policies, as well as the hierarchical function and racialised instabilities of citizenship. We approach the topic with awareness of how we are situated within it, and reflect on how our own relation with our (lack of) citizenship(s) shape our perspectives on the topic.

At the start of the course each student chooses any citizenship regime they want to explore, and applies the knowledge acquired in the course to analyse their chosen regime. In class we exchange findings about different citizenship regimes and gain a comparative perspective from each other. At the end of the course each student puts together a Portfolio based on the work they have done during the course, which forms the basis for the assessment.

Aanvullende informatie onderwijsvormen

Weekly lectures, obligatory weekly assignments.

Toetsvorm

The course is graded on the basis of a written Portfolio.

Submission of compulsory weekly assignments is required for passing the course.

The format for the re-examination is determined based on the number of students needing re-examination. It may take the form of an oral re-examination, or a written assignment equivalent to the Portfolio.

Vereiste voorkennis

Successful completion of university-level courses at 2nd year Bachelor level, and ability to engage with interdisciplinary academic literature, is essential. Legal skills, such as ability to read and understand national and international legal documents and judgments, is desirable. Background in either law, anthropology, sociology or political science is desirable.

Literatuur

Will be announced on Canvas.

Aanvullende informatie doelgroep

Apart from law students of the VU, the course is also available for:

- Students from other universities/faculties
- Exchange students
- Contractor (students who pay for one course)

Human Rights and the Border

Vakcode	R_HumRB
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	prof. dr. mr. B. de Hart
Examinator	prof. dr. mr. B. de Hart
Betrokken Docenten	mr. dr. Y. Arbaoui, prof. dr. mr. B. de Hart
Onderwijsvormen	Study Group, Tutorial

Doel vak

The overall aim of the course is to increase knowledge of the law concerning borders and the understanding of the changing meanings of borders. The course invites students to explore borders from a (global) legal and sociological perspective.The specific learning aims of this course are:

1. To develop some insight into the coherence and conflicts between relevant domestic, European and international law;
2. To understand the ways in which the law develops and manifests itself in a social context;
3. To read, understand and analyze legal and theoretical texts and arguments, including case law and legislation;
4. To take a well-founded and substantiated position in a social-legal debate.
5. To understand the different categories of ‘migrants’ created by the law and the attaching differences with regard to the right to cross borders and the sanctioning of illegal border crossing;
6. To identify relevant domestic, European and international law and deal with conflicts among them;
7. To critically reflect on legislation and case-law concerning borders, practices concerning borders as well as on the human dimension of borders.

Inhoud vak

The operation of borders and border control in practice may differ

greatly from how it may be understood to operate in theory. In this course, the knowledge of the law on borders will be connected to societal reality. The course explores the convergence and discontinuities between physical borders and social identities, and the emerging tensions between the nation state and migrants' enjoyment of their fundamental human rights. It provides an opportunity to study which laws regulate people's mobility, creating different categories of 'migrants'. The course covers the issue of asylum at the border and other forms of how human rights provisions can generate migration rights. In addition, the course addresses current issues such as border deaths, forced return of migrants, smart borders and racialization of borders.

Aanvullende informatie onderwijsvormen

The course basically consists of (interactive) lectures. In addition, field excursions (or online alternatives) may take place, enabling students to learn how borders work in practice.

Toetsvorm

The course will be concluded with a written exam and/or a paper.

Literatuur

Will be announced on Canvas.

Aanvullende informatie doelgroep

This course is open to students of various disciplines who have completed their first year of their Bachelor program, including exchange students.

Overige informatie

This course is part of the minor 'Law and Global Society: Internet,

Migration and Climate Change' and of the minor 'Migration Studies'.

Human-Computer Interaction

Vakcode	XB_0013
Studiepunten	6.00
Periode	P5
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. ing. M. Alimardani
Examinator	dr. ing. M. Alimardani
Betrokken Docenten	prof. dr. J.R. van Ossenbruggen, L.H. Rekker MSc, dr. ing. M. Alimardani
Onderwijsvormen	Lecture, Practical

Doel vak

At the end of this course, students will be able to:

Knowledge and understanding:

1. Describe the fundamental principles of user experience design

2. Describe fundamental input and output techniques for human-computer interaction

Applying knowledge and understanding:

3. Gather requirements for designing interactive systems

4. Design a human-agent interaction to answer a scientifically grounded research question.

5. Apply low-fidelity and high-fidelity prototyping methods in HCI.

Making judgements:

6. Conduct user studies to empirically evaluate the UX and usability of their interaction design

Communication:

7. Communicate design solutions and evaluations on an academic level

Lifelong learning skills:

8. Work in teams to achieve a common goal

Inhoud vak

In many application domains, AI systems must collect input from humans or collaborate with them to enhance their performance and decision-making processes. This course will introduce students in the AI program to the existing theories and methods for the design and evaluation of interactions between humans and [intelligent, socially interactive, or immersive] computing systems. The aim is to give students an understanding of the concepts of usability, user experience (UX), and user-centered design, and to equip them with empirical research skills for prototyping and iterative testing of such interactive computing systems.

The course is divided into two parts, each achieving different learning goals;

In the **theoretical part of the course (lectures)**, students will learn about the core topics in HCI such as the lifecycle of interaction design, human factors, interaction elements, and common research methods in HCI studies.

In the **practical part of the course**, students will acquire hands-on research experience where they collaborate in groups to design, conduct, and report on an HCI experiment with human participants.

Aanvullende informatie onderwijsvormen

This course is taught on campus. The lectures will not be recorded.

Attendance in the practical sessions of the course is mandatory.

Toetsvorm

The final course grade is composed of two assessment components:

- Written Exam (Individual - 50% of the final grade)

- Research project report (Group work - 50% of the final grade)

There is also the opportunity for students to collect bonus points by taking the in-class quizzes that are randomly given in some lectures without prior notification.

The final grade will be calculated

Final Course Grade = 0.5 x Individual Exam Grade + 0.5 x Group Project Report Grade + Bonus points (if any is collected by the student from in-class quizzes)

To pass the course, students must pass both the individual exam and the research project report (for each component a grade >=5.5 is required).

Please note:

- The group must submit one report and all group members will receive the same grade.
- Resit opportunities will be provided for both the individual exam and the group project report.
- The in-class quizzes are not mandatory and hence a resit opportunity will not be provided.

Literatuur

MacKenzie, I. Scott. *Human-Computer Interaction : An Empirical Research Perspective*, Elsevier Science & Technology, 2013. *ProQuest Ebook Central*, https://ebookcentral.proquest.com/lib/vunl/detail.action?docID=1110719.

(Students have free access to the online version of the book through their VUnetID)

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence

Toelichting Canvas

All details of the course syllabus will be published on the Canvas page.

Aanbevolen voorkennis

Knowledge of programming, statistics, qualitative and quantitative research methods

Identity, Diversity and Inclusion

Vakcode	S_IDI
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. I.E. Waldring
Examinator	dr. I.E. Waldring
Betrokken Docenten	dr. M.C. de Regt, dr. I.E. Waldring
Onderwijsvormen	Lecture

Doel vak

This course is designed to introduce students to the various issues

concerning diversity and inclusion in an increasingly globalizing world.

The course focuses in particular on contemporary issues concerning

processes of inclusion and exclusion in the Dutch/European context.

The central questions in this course are:

1. How and why are identities based on ethnicity, gender, class and

sexuality constructed by both insiders and outsiders?

2. How do (groups within) European/Dutch societies respond to diversity?

3. What are the relevant mechanisms of inclusion or exclusion?

4. How should we contextualize current debates and practices related to

inclusion/exclusion processes in relation to Dutch/European historical developments?

Learning outcomes:

Knowledge and understanding - The student has acquired knowledge and understanding of:
(1) the relevant forms and dimensions of social identities;
(2) theories of identity construction, inclusion and exclusion;
(3) the questions, debates and policies on diversity in contemporary Western societies, and the differences between societies thereof;
(4) the challenges of contemporary developments - such as globalization and individualization - on contemporary forms of diversity.

Application - The student has acquired the competences to:
(5) apply acquired knowledge in the analysis of contemporary forms of diversity.

Making judgements - The student is able to demonstrate:
(6) a critical stance in contemporary debates over identity, diversity and inclusion.

Inhoud vak

Identity issues have become very prominent in our globalizing world. While migration is often presented as one of the main causes of the increasing emphasis on identity, other developments, such as those related to (cultural) globalization and economic transformations, have had a strong impact as well. In addition to ethnic and religious diversity, gender inequalities, class differences and issues related to sexual diversity have changed The Netherlands and other European societies. Ethnicity, gender, class and sexuality are markers of identity, but have also become axes of inclusion and exclusion in contemporary European societies.

This course discusses how ethnic diversity intersects with other forms of diversity. While historical constructions of the nation were already gendered, in contemporary discourses on national identity, gender and sexuality have become more prominent as markers of

national inclusion and exclusion. Or, as in Europe ethnic diversity

largely coincides with class distinctions, how does this affect feelings

of belonging and inclusion?

Various identity aspects will be studied in relation to possible

reactions to experienced threats to identity and/or social exclusion. The course will

also zoom in on cases of local conflict and on contemporary debates related

to the subject, like the one on feminist solidarity in an age of diversity.

Aanvullende informatie onderwijsvormen

Lectures.

Toetsvorm

Digital exam.

Literatuur

Various articles including texts by Cooper (2015), Crul (2016), Young (1990), Woodward (2003), and others (see CANVAS).

Aanvullende informatie doelgroep

2nd year bachelorstudents in Cultural Anthropology and Development Sociology.

Students in the Minor Sociology and the Minor Gender & Diversity.

Also open as an elective course for Exchange Students.

Imagining the Dutch: themes in Dutch History

Vakcode	L_GCBAALG003
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Nederlands / Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	drs. W.C. Manuhutu
Examinator	dr. P.H. Moser
Betrokken Docenten	drs. W.C. Manuhutu, dr. P.H. Moser, prof. dr. P.J.E.M. van Dam, dr. H.M.E.P. Kuijpers
Onderwijsvormen	Hoorcollege

Doel vak

The main goal of this course is to familiarize diverse group of students

coming from different countries and educational backgrounds with some

core aspects of the history of the Netherlands and of the way the

Netherlands is imagined.

Students will also learn how to connect Dutch history with a number of

present-day academic as well as public debates on issues such as

national identity, tolerance, environment, colonialism, World War II,

democracy and multiculturalism.

They will learn how to critically connect these themes to different representations of Dutch history in the media, in museums and the internet.
As this course is open to non-history students, it will also introduce several concepts of history as an academic discipline, by discussing subfields of history such as political history, colonial history, environmental history and national history.
After taking this course, students will be able to recognize normative thinking in academic literature and in the work of historians, to read and summarize academic writing, to use scientific literature in debates and to critically addresses historical information and images.

Inhoud vak

Often, the perception of the Netherlands at home and abroad is ridden with stereotypes. It is a country of cheese, herring and coffeeshops. Canals, clogs, windmills and a very liberal attitude to life choices are also elements in this imagery. The struggle against water is yet another recurring theme in the way the Dutch are perceived and see themselves.

The course will cover a period of over 500 years from 1500 until the present, giving an overview of the multiple histories of the Netherlands. Besides written texts, audio-visual sources will also be employed in order to illuminate the themes that will be discussed. Breaking away from the Spanish Habsburg Empire in the late sixteenth century, the Dutch Republic became a world power in the course of the seventeenth century. As a republic it was an exception in an age of kingdoms and empires. The seventeenth century was marked by economic, scientific and cultural success. The foundations for a large colonial empire were laid in the seventeenth century. This period has been coined the ‘Golden Age’, a term that has come under scrutiny and debate in recent years. In the course the emergence of the Dutch Republic and the way the ‘Golden Age’ came into being and is remembered will be discussed. Several characteristics that are considered ‘typically Dutch’ such as a pragmatic, entrepreneurial mentality are linked to this period The merchant and the minister are two other images that are often used in characterizing the Dutch. As a country dependent on trade, international relations have always been of the utmost importance. As a small nation the Netherlands also has a tradition in international law

as well as a tendency to imagine itself as holding the moral high-ground

and a guide to the world.

During the course among others the following themes will be discussed:

-The emergence of the Netherlands as an independent political nation and

its development as a Republic in the seventeenth century

-Notions of tolerance and religious diversity

-The economic, scientific and cultural developments in the Netherlands (

Rembrandt, Hugio Grotius)

-The Dutch and water: a maritime power at sea, managing water at home

-The Dutch as a colonial power and a post-colonial society

-The transition from a Republic to a Kingdom

- Pillarrized consensus democracy and present-day Dutch politics

- The Dutch and the World Wars

- Migration, diversity and race In the Netherlands

Discussion among students about the content of the lectures and the

course literature is part of this course, which is specifically designed

to connect history with contemporary issues. Quizzes and polls will be

employed to stimulate class engagement.

Aanvullende informatie onderwijsvormen

In period 1,each week there will be two lectures All the thematic lectures will be given by various lectures, primarily from the History Department of the VU. Presence in class is mandatory. Students who have a reason to be absent have to inform the course coordinator in time of their absence, which is limited to a maximum of two times, after which the student can be penalized with exclusion from participating in the exams.

Toetsvorm

There is one written assignments (40%), as well as a written exam

(60%).

Literatuur

Handbook: Kennedy, J. C. (2017). A concise history of the Netherlands. Cambridge University Press.

Students will also be asked to read additional articles or book chapters related to each theme, which will be posted on Canvas in due time.

Overige informatie

This course will be provided two times during the academic year in

period 1 (L_GCBAALG003) and in periods 4&5 (L_GCBAALG004).

All the lectures will be in English. Students who follow a Dutch language program can make their assignments in Dutch.

Impact of Algorithms on Human Lives

Vakcode	X_HP015
Studiepunten	6.00
Periode	
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. A. Bhulai
Examinator	dr. A. Bhulai
Betrokken Docenten	dr. A. Bhulai
Onderwijsvormen	Lecture

Doel vak

After completing the course “Impact of algorithms on human lives”

- you can explain what an algorithm is in basic terms,
- you can distinguish differences between algorithms, artificial intelligence, machine learning, and deep learning,
- you can describe the advantages and disadvantages of algorithms,
- you can explain how AI technologies work in basic terms,
- you can explain what impact algorithms and technologies have on our lives and how they influence us, and
- you will be able to better judge the benefits or potential problems associated with these technologies.

Inhoud vak

Today we live in an algorithmic society; algorithms are all around us. Throughout the day, we use different technologies and algorithms through the many devices that have become essential in our lives, such as computers, smartphones, and smartwatches. Many of them make our lives a lot easier. But are they all good for us?

In this course, you will get an explanation in layman's terms, without programming and complex mathematical formulas how these technologies work and how they affect our lives. We will discuss the basic concepts so that by understanding the basics, we can demystify these technologies and understand how they impact our daily life. This will allow us to conduct a well-informed and expert debate about under what circumstances these influences are positive and when negative. Understanding these basic concepts, for example, will also help us to better judge a news story about one of these technologies, whether it is right or wrong, and what benefits or potential problems these technologies bring.

Aanvullende informatie onderwijsvormen

(guest) lectures, discussions, in-class activities, debate, self-study

Toetsvorm

The course concludes with a debate and a reflection report that is graded with a pass or fail.

Aanvullende informatie doelgroep

This course is for students taking the honours program.

Overige informatie

1) A laptop computer may be useful in this course.

2) The course is taught only in odd-numbered academic years.

Toelichting Canvas

For this course, we use Canvas.

Information Management

Vakcode	X_401085
Studiepunten	6.00

Periode	P6
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. I.G. Gerostathopoulos
Examinator	dr. I.G. Gerostathopoulos
Betrokken Docenten	dr. I.G. Gerostathopoulos, A.S. Iyer
Onderwijsvormen	Lecture, Seminar, Written partial exam

Doel vak

Through this course, students will study the topic of information management in breadth and in depth. After taking this course, the student will be able to:
1. Explain the strategic relevance of information systems for modern organizations; (Knowledge and understanding) (Applying knowledge and understanding)
2. Analyze the level of competition within an industry and determine is implications for an information systems strategy; (Applying knowledge and understanding)
3. Understand the impact of the advent of the internet on the management of information within and between organizations; (Knowledge and understanding)
4. Identify the various phases in the development of an information system; (Applying knowledge and understanding)
5. Model simple and moderately complex business processes with a formal modeling technique (Petri nets); (Making judgements) (Applying knowledge and understanding)
6. Analyze process models with respect to various behavioral properties. (Lifelong learning skills)

Inhoud vak

No organization can do without information systems. For some organizations, such systems are even of strategic relevance, as they offer a clear competitive advantage. Think, for example, of how Amazon has become such a dominant retailer or how an organization as Uber has conquered the taxi market.

This course explains the relevance and use of information systems in modern organizations. We will briefly sketch how the role of information systems has developed over the years to reach its current ubiquitous level. Special attention is devoted to the rise of the internet and its impact on traditional organizations, as well as the emergence of new types of (cloud-based) organizations.

Reasoning from the organizational importance of information systems, we will look into the way information systems are developed such that organizations can achieve their objectives. We will pay considerable attention to an important phase in information system development, namely how we analyze and model business processes. For this purpose, we will rely on the use of classical Petri nets.

This course will approach the topic of information management in breadth and in depth. Breadth is achieved by giving an overview of all relevant topics in the area of information management; depth is attained by introducing students to a powerful, formal modeling technique that they will learn to master in the context of organizational analysis.

Aanvullende informatie onderwijsvormen

This course consists of lectures and practicals.
Attendance is not mandatory but highly encouraged.

Toetsvorm

Intermediate exam (individual, 30%) and final project (group-based, 70%).

Resit option: resit of Intermediate exam, re-submission of the final project.

Literatuur

1. "Business Information Management: Improving Performance using Information Systems",

by Dave Chaffey and Steve Wood. ISBN: 9780273686552.

2) "Modeling Business Processes", Wil van der Aalst and Christian Stahl.

ISBN: 9780262015387 (print), 9780262296465 (eBook).

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 1)

Information Retrieval

Vakcode	X_400435
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. R.M. Siebes
Examinator	dr. R.M. Siebes
Betrokken Docenten	dr. R.M. Siebes
Onderwijsvormen	Lecture, Seminar, Computer lab, Written partial exam

Doel vak

At the end of this course, students will:

- understand how search engines and other information retrieval systems work (knowledge and understanding);

- understand the general principles and methods of information retrieval systems (knowledge and understanding);

- have acquired some basic skills and experiences in programming important aspects of information retrieval systems (apply knowledge and

understanding).

Inhoud vak

This course covers the core aspects of information retrieval and search engines, including indexing, Boolean retrieval, the different types of

queries, query execution, the vector space model, web crawling, networks, link analysis, PageRank, classification, and clustering.

Aanvullende informatie onderwijsvormen

Lectures, which are video-recorded, and practical sessions. The lecture recordings are made available through Canvas afterward. Physical presence in the lectures is recommended but not mandatory. The practical sessions provide help for the assignments and are optional.

Toetsvorm

Six programming assignments in Python (5% of final grade each) and a final exam in the form of a multiple-choice test (70% of final grade). The final exam must be completed with a passing grade to pass the course. There is no resit for the programming assignments.

Literatuur

"Introduction to Information Retrieval" by Manning, Raghavan and Schütze. The book can be found online: http://nlp.stanford.edu/IR-book/

Aanvullende informatie doelgroep

Minor Applied Mathematics

Minor Business Analytics & Data Science

Minor Bioinformatics & Systems Biology

Bachelor Econometrics and Operations Research

Aanbevolen voorkennis

Basic programming skills are expected (in Python). Students without. previous programming experience should reserve additional time in the

weeks before the course starts to learn the basics of Python programming on their own.

Inleiding in de Koran en Soenna

Vakcode	G_BATRSPC123
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Nederlands
Faculteit	Faculteit Religie en Theologie
Vakcoördinator	dr. Y. Ellethy
Examinator	dr. Y. Ellethy
Betrokken Docenten	dr. Y. Ellethy
Onderwijsvormen	Hoorcollege

Doel vak

Na afronding van deze module:

- kent de student op hoofdlijnen de ontstaansgeschiedenis, de indeling

en

de thematiek van de Koran en de Hadith;

- kan de student beknopt uitleg geven over visies op de geschiedenis van

de Goddelijke openbaring in het algemeen en de openbaring van de Koran aan de profeet Mohammed in het bijzonder;

- kent de student de ontstaansgeschiedenis, de verzameling en de

verspreiding van de Koranische tekst in hoofdlijnen;

- kent de student de westerse discussies en kritiek i.v.m de historische

ontwikkeling van de tekst van de Koran en kan hierop reageren op

een wetenschappelijke manier;

- kent de student de algemene kenmerken, inhoud, stijl en historische

context van de Koran in hoofdlijnen;

- heeft de student fundamentele kennis omtrent de terminologie van de

Koranwetenschappen (en basiskennis van de Soenna en Hadith

terminologie);

- kent de student de geschiedenis en de ontwikkeling van de Koran- en

Hadithwetenschappen en de betreffende klassieke en moderne literatuur in

grote lijnen;

- kan de student een werkstuk van enkele pagina’s schrijven over de

positie van de Koran en de Soenna binnen de Islam.

Inhoud vak

In deze cursus (met meer focus op de Koranwetenschappen) worden gezaghebbende visies op de geschiedenis van de openbaring, de verzameling en de ontstaansgeschiedenis van de Korantekst, de betreffende kritiek, en de belangrijkste kernpunten en terminologie binnen de Koranwetenschappen ʿulūm al-Qur'ān behandeld. De student krijgt ook basiskennis van de positie van de Soenna binnen de Islam, het ontstaan en de ontwikkeling van de Hadith wetenschappen, terminologie en klassieke literatuur. In de module Hadith-wetenschappen zal meer nadruk worden gelegd op de Soenna en Hadithmethodologie.

Aanvullende informatie onderwijsvormen

Hoor- en werkcolleges met schriftelijke opdrachten en tussentijdse papers. Er wordt aandacht gegeven aan de interactieve deelname van de studenten. Vragen worden aan het begin van het college besproken. In aansluiting op elk hoorcollege-onderdeel wordt een werkstuk geschreven en in werkgroepen besproken en beoordeeld.

Toetsvorm

Schriftelijk tentamen (80%)
Schrijfopdracht (20%)

Literatuur

Verplichte literatuur:

- Al-A'ẓami, M., The History of the Quranic Text from Revelation to

Compilation, Leicester: UK Islamic Academy, 2003.

- Leemhuis, F. "Koran"; "Soenna", in J. Waardenburg (ed.), Islam: Norm,

Ideaal en Werkelijkheid. Houten: Fibula, 20005 , pp. 54-74; 75-79.

(Canvas).

- Ljamai, A., Inleiding tot de Studie van de Koran, Zoetermeer: Meinema,

2005, hoofdstukken 1, 2, 3 en 4 t/m p. 71.

-Motzki, H., The Collection of the Qur'an: A Reconsideration of Western Views in Light

of Recent Methodological Developments, Der Islam 78 (2001), pp. 1-34.

- Watt, M. (et al.), Bells Inleiding tot de Koran, Utrecht: de Ploeg,

1986, hoofdstukken 1 en 2 t/m p. 39.

- Nederlandse Koranvertaling.

Aanbevolen literatuur:

- Ali, M., Sirat Al-Nabi and the Orientalists, Madinah: King Fahd

complex for the Printing of the Quran, 1997 (Section IV: Receipt of Way).

- Hamidullah, M., An Introduction to the Conservation of Hadith in the

Light of the Sahifah of Hammam ibn Munabbih, Kuala Lumpur: Islamic Book

Trust, 2003.

- Verdere literatuur wordt voor aanvang van het college bekend gemaakt

via

Canvas.

Aanvullende informatie doelgroep

Deze module maakt onderdeel uit van de bachelor Theology and Religious Studies - Islam, de universiteitsminor Islam en de minor Arabisch en Bronnen van de Islam.

Overige informatie

Aanwezigheid 80%

Afwijkende intekenprocedure

Je kunt je alleen intekenen voor dit vak als je bent ingetekend op de minor Arabisch en bronnen van de Islam of de minor Islam, of als je de track Islam (onderdeel van de bachelor Theology and Religious Studies) of de track Vrije Keuze (onderdeel van de bachelor Theology and Religious Studies) volgt.

Inleiding Inspanningsfysiologie

Vakcode	B_IF
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. J.J. de Koning
Examinator	dr. J.J. de Koning
Betrokken Docenten	dr. J.J. de Koning
Onderwijsvormen	Practicum, Werkcollege, Deeltentamen schriftelijk, Hoorcollege

Doel vak

Doel van dit vak is het verkrijgen van kennis van de bouw en werking van organen en orgaansystemen die een rol spelen bij het bewegen en de energiehuishouding. Na afloop van de cursus kan de student de belangrijkste onderdelen van deze organen en orgaansystemen benoemen, de bouw en werking van deze onderdelen benoemen en de werkingsmechanismen beschrijven. Daarnaast kan de student deze kennis toepassen tijdens het meten van verschillende variabelen tijdens rust en inspanning. Ten slotte is de student in staat de uitkomsten van de metingen te interpreteren en te verwerken.

Inhoud vak

Tijdens de cursus wordt, na een inleiding, de bouw en de werking van de voor het bewegen belangrijkste fysiologische systemen, zoals de bloedsomloop, de ademhaling, de energievoorziening, de spieren en de hormoonhuishouding besproken. Daarbij wordt zowel het functioneren tijdens rust als tijdens fysieke inspanning besproken.
Bij de practica wordt de theoretische kennis verder uitgebreid en toegepast bij het registreren van de stofwisseling in rust, het ECG en de bloeddruk, de verschillende longvolumina en ademhalingsparameters, de hartfrequentie tijdens fysieke inspanning, het lichamelijk prestatievermogen, het dagelijkse energieverbruik en de dagelijkse voedselopname.

Aanvullende informatie onderwijsvormen

Aanvullende informatie onderwijsvormen

De cursus is opgebouwd rond 6 thema's (Meten is weten; Zuurstof van atmosfeer naar bloed; Zuurstofcirulatie in het lichaam; Zuurstofgebruik in de spier; Energie voor de spieren; Limitaties van het systeem en training). Ieder thema heeft een themadocument met daarin de leerdoelen van ieder thema. Ieder thema heeft een aantal 'Meet Ups', welke dienen ter verduidelijking van de leerstof. Deze Meet Ups zijn gestructureerd rond vragen van studenten ('Flipped Classroom Approach') en behulpzaam voor de
bestudering van de grote hoeveelheid leerstof. Door middel van vraagstukken en opdrachten wordt verder inzicht in de leerstof verkregen. Daarnaast
volgt iedere student een aantal werkgroepen en practica. De practica zijn verplicht en dienen ter aanvulling op de leerstof en bieden bovendien de gelegenheid om de kennis van de leerstof toe te passen en te verdiepen. Voorwaarde voor deelname aan het practicum is dat de student voor elke bijeenkomst steeds de betreffende stof in het boek en de cursushandleiding bestudeerd heeft. Na elke practicumbijeenkomst wordt het practicum door iedere student uitgewerkt aan de hand van opdrachten (inhoud en tijdstip van inleveren volgens de richtlijnen in de cursushandleiding). Inschrijving voor de practica gaat via Canvas.

40 uur/ Meet Ups
12 uur/ practica en werkgroepen
20 uur / uitwerking practicum
3 uur / tussentoets
3 uur/ eindtoets
90 uur / zelfstudie

Toetsvorm

De tentamenstof beslaat de hoofdstukken van het boek ("Exercise Physiology: nutrition, energy, and human performance") zoals beschreven in de zes thema's, de themadocumenten en besproken tijdens de Meet Ups, de diverse practica
en de studiehandleiding.
Het tentamencijfer zal bestaan uit een gemiddelde van de eerste en tweede deeltoets. De eerste deeltoets (thema 1-3) wordt halverwege de cursus gehouden en de tweede deeltoets (thema 4-6) aan het einde va de cursus. Beide toetsen worden schriftelijk afgenomen en bestaan uit meerkeuze en open vragen.
Om een cijfer voor het tentamen te krijgen dient men aan de practicumverplichtingen te hebben voldaan. Deze verplichtingen zijn: alle practicumbijeenkomsten (actief) volgen en de bijbehorende opdrachten (voldoende) maken.

Literatuur

De verplichte literatuur bestaat uit:

- W.D. McArdle, F.I. Katch, V.L. Katch: Exercise Physiology: nutrition,

energy, and human performance , 9th edition (2022). International

edition.

- De cursushandleiding en (thema)documenten via Canvas

Overige informatie

De practica zijn verplicht. Het tentamencijfer is alleen geldig als alle practicumbijeenkomsten zijn gevolgd en de betreffende opdrachten zijn ingeleverd. Bij het eventuele missen van een practicumbijeenkomst of opdracht met een geldige reden dient zo spoedig mogelijk contact opgenomen te worden met de practicumbegeleiders voor het plannen van een inhaalbijeenkomst.

Afwijkende intekenprocedure

De indeling van werkgroepen/practica/tutorgroepen etc. vindt plaats via Canvas.

Toelichting Canvas

De beschrijving van de zes thema's (themadocumenten) zijn te vinden op Canvas.

Inleiding Psychologie (UM)

Vakcode	P_UINLPSY
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. W. Donk
Examinator	dr. W. Donk
Betrokken Docenten	dr. W. Donk
Onderwijsvormen	Hoorcollege

Doel vak

-Kennis en begrip hebben van het vakgebied psychologie
-Het kunnen toepassen van deze kennis zoals blijkt uit de vaardigheid basisvragen te kunnen beantwoorden op het gebied van de psychologie

Inhoud vak

Het vak geeft een overzicht van de psychologie. Wat is de genetische en biologische basis van gedrag? Wat is bewustzijn? Hoe zien we, leren we, onthouden we en denken we? Waarom gedragen we ons zoals we doen? Naast deze fundamentele vragen zullen o.a. ook de volgende onderwerpen aan bod komen: intelligentie, sociale psychologie, de ontwikkeling, persoonlijkheidsleer, psychopathologie en psychologische behandelmethoden.

Aanvullende informatie onderwijsvormen

-14 hoorcolleges Contacturen: 168 (24 college, 3 examen, 141 zelfstudie)

Toetsvorm

- Multiple choice tentamen

Literatuur

Phelps, E.A., Berkman, E.T., & Gazzaniga, M.S. (2022). Psychological Science (7th edition). Norton.

Overige informatie

De cursus wordt Engelstalig aangeboden.

Intelligent Systems

Vakcode	XB_0031
Studiepunten	6.00
Periode	P2
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen

Vakcoördinator	dr. K.S. Schlobach

Examinator	dr. K.S. Schlobach
Betrokken Docenten	dr. K.S. Schlobach, dr. N. Ullah
Onderwijsvormen	Seminar, Lecture, Practical, Written partial exam

Doel vak

Knowledge and understanding: at the end of the course, the students will be familiar with basic knowledge of some of the core aspects of AI: state-space representations, search, adversarial search, logic, automated reasoning, reasoning with uncertainty and vagueness and machine learning.

Applying knowledge and understanding: students will be able to implement basic (adversarial) search algorithms, as well as knowledge based and adaptive methods to build Intelligent Agents.

Making judgements: the students will have a basic understanding of the ethical and societal implications of the developements in AI.

Communication skills: students will be able to write a scientific reports about an original research question in a group of students.

Learning skills: students will be trained in acquiring a set of complex AI related topics in a restricted period of time, come up with an original research question and perform the necessary (empirical) research.

Inhoud vak

The course will provide an introduction to some of the basic concepts of Artificial Intelligence, such as search, adversarial search, knowledge representation and machine learning.

Aanvullende informatie onderwijsvormen

2 lectures of 2 hours per week.
Working groups to practice the theoretical material (2 hours)
Practical groups to apply the acquired knowledge (flexible, on demand)

Toetsvorm

The grade will be determined via two (digital) exams (35%+35%), and a number of practical assignments (30% in total). All three components have to been completed successfully to pass the course. There will be a (single) resit exam, combining both partial results, but no resits for the practical assignments.

Literatuur

Chapters from "AI a Modern Approach", Russell and Norwig, provided as a part of a reader.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 1)

Aanbevolen voorkennis

Some basic Programming in Python, as tought in the the first part of the Introduction to Programming course in P1.

International Relations and Global Governance

Vakcode	S_IRGG
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. J. Hogan
Examinator	dr. J. Hogan
Betrokken Docenten	dr. J. Hogan
Onderwijsvormen	Lecture

Doel vak

Learning outcomes:

Knowledge and understanding - The student has acquired knowledge and understanding of:

(1) the rival theories and approaches within the discipline of International Relations (IR) and their meta-theoretical foundations, such as Realism, Liberalism, Constructivism, Marxism, Postcolonialism Poststructuralism and Feminism;

(2) the most important characteristics of these different theoretical perspectives and are able to relate them to key thinkers within these approaches;

(3) how and why interstate relations and global governance - within the context of ongoing globalization - affects nation states and societies, and how different actors, such as states, governments, international organisations, multinational corporations (MNCs), and NGOs, collaborate and conflict within these transnational processes;

(4) contemporary world politics and pertinent themes such as democracy, security, the rise of emerging powers, gender, neo-colonialism, the global political economy and (under)development, and can apply and contrast the different theoretical perspectives of IR (e.g., Realism, Liberalism, Constructivism, Marxism, see above) to case studies and questions related to these contemporary issues.

Skills – The student is able to:

(5) apply their knowledge of the conceptual and theoretical toolbox of IR, and of contemporary developments, to issues such as war, humanitarian intervention, democracy, development cooperation, international negotiations and international political economy.

Attitude – The student displays:

(6) an ability to critically reflect upon the rival theoretical approaches in the discipline of IR.

Inhoud vak

This course offers a comprehensive overview of the discipline and subject of International Relations (IR) and its main concepts and theories and approaches. Throughout, the course will be guided by the question to what extent, and how, the current process of globalization is changing the nature and content of world politics, approaching this question from the various competing theoretical perspectives that IR has to offer.

Traditionally, the object of study for IR has been the conflict between and co-operation of sovereign states. This model is, however, increasingly regarded as outdated inasmuch as more and more non-state actors such as multinationals, NGOs and transnational social movements appear to play a prominent role in world politics. In addition, we can also observe transnational forms of regulation through international organizations and emerging structures of what is called ‘global governance’.

The question has been raised whether in the face of these processes of globalization and transnationalisation,

states have lost the sovereignty that used to be the basis of the international system. On the other hand, there are still many instances where state power is very visible. Indeed, recent geopolitical developments and events, such as the rise of China or Russia's invasion of Ukraine, have highlighted that classical themes of interstate rivalry and international security may not have lost their relevance.

In this course the focus will be on seeking to understand these issues from various theoretical lenses. Students will learn how different perspectives highlight different structures and different actors and processes, and how a deeper knowledge of these theories and their main concepts allows for a deeper understanding of the richness of IR and its relation to the rest of the social sciences, and of the complexity of today's globalized world politics.

Aanvullende informatie onderwijsvormen

Lectures

Toetsvorm

Election observation report (pass/fail); International negotiation simulation; Final exam

Literatuur

Baylis, J., Smith, S., & Owens, P. (Eds.). (2023). *The Globalization of World Politics: An Introduction to International Relations (9th edition)*. Oxford University Press.

Various journal articles, all available through the VU library website or Google Scholar.

Aanvullende informatie doelgroep

Second-year Bachelor students in Political Science, track Global Politics and specialization Mondiale Politiek.

Students in the Minor Political Science

Internet Governance

Vakcode	R_InternGov
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	mr. T.H.A. Wisman
Examinator	mr. T.H.A. Wisman
Betrokken Docenten	mr. T.H.A. Wisman
Onderwijsvormen	Study Group, Lecture

Doel vak

At the end of this course students:

• Understand the basics of the Internet and the basics of governance;

• Understand what Internet governance is, both in the broad and the narrow sense, are able to explain how they relate and are capable of applying them to real-world phenomena;

• Understand and are able to apply the modalities of Lessig;

• Understand and are able to apply the models of Solum;

• Know and understand European law and case law regarding topics such as privacy, freedom of expression and copyright, and are able to apply to this Lessig's modalities and Solum's models;

• Are able to analyse text (policy documents, law, case law) and reflect on the relation between technology (internet), law and power;

• Are able to reflect on internet governance phenomena and the underlying dynamics of regulation.

Inhoud vak

In the first half of this course the focus is on models and modalities which are used to analyse issues of internet governance. These can be used to determine which parties are involved (the question WHO => stakeholders) and how they (attempt to) promote or defend their interests (the question HOW => the things that regulate/modalities/models) in internet governance issues. The answers to these questions are not always clear-cut. Interests can intertwine and Machiavellian politics may be lurking beneath everyday facades.

Armed with these analytical models the emphasis of the second half of this course will shift to applying them to different phenomena. This half of the course deals with specific legal subjects and the concomitant case law of the Court of Justice of the European Union. In this second half Solum's models of internet governance will be used together with Lessig's modalities to establish how stakeholders seek to pursue their interests and how this impacts various fundamental rights, such as the right to freedom of expression and the right to privacy. Students will attain insight in the opaque processes which can spur lawmaking and critically reflect on why some laws are in action and the inaction of other laws.

This course is a required part of the minor Law and Global Society. The transnational nature of the internet in conjunction with its technical features, make internet governance a suitable area for students to verse themselves in matters of transnational law, power and technology. This course has no prerequisites.

Aanvullende informatie onderwijsvormen

Lectures and tutorials.

Toetsvorm

Exam

Vereiste voorkennis

None.

Literatuur

Amongst others:
L. Lessig, Code and Other Laws of Cyberspace (Basic Books, New York 2006)
L.B. Solum, Models of Internet Governance

Material will be made available on Canvas before the start of the course.

Aanvullende informatie doelgroep

Apart from regular students, the course is also available for:
Students from other universities/faculties
Exchange students
Contractor (students who pay for one course)

Internship Minor Business Administration - New Ways of doing Business

Vakcode	E_MUB_IMBA
Studiepunten	12.00
Periode	P1+2, P2+3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. V. Duplat

Examinator	dr. V. Duplat
Betrokken Docenten	
Onderwijsvormen	

Doel vak

Academic & research skills

• Ability to examine and understand problems from different

perspectives;

• Ability to put forward well-founded, substantiated points of view, both in spoken and written format;

• Ability to apply acquired knowledge to other problems and in other contexts.

• Ability to translate practically relevant problems into (academically) relevant research questions;

• Ability to design and execute a project using a systematic, analytical approach in a real business environment (of profit or not-for-profit organizations).

Bridging theory and practice - knowledge

• Have specialized, in-depth knowledge and insights about the minor theme;

Bridging theory and practice - practice

• Ability to make connections between theories, models, and concepts of that specific minor theme / discipline.

• Ability to apply theoretical knowledge in a specific organizational context;

• Ability to formulate relevant recommendations for practice based on your knowledge acquired;

Social skills

• Have a better understanding of roles and needs of different types of stakeholders that you need to interact with as a professional;

• Ability to work well in a team and reflect on your own role in the team.

Broadening your horizon

• Have a better understanding of what the expectations of the academic and professional field are in terms of knowledge and skills needed;

• Have awareness of the various career opportunities the field offers.

Self-awareness

• Ability to reflect on your own responsibilities as well as others;

• Ability to reflect on your personal development;

• Ability to receive and are able to deal with feedback from others.

Inhoud vak

Increasingly organizations and maybe even your future employer are looking for experience as well as academic credentials. The School recommends doing an internship, because it is an excellent way to apply the knowledge and (academic) skills which you acquired during your studies. Your most important learning goal as a student-intern is to familiarize yourself with professional and market-related skills in a real and new organizational environment.

With the job market becoming increasingly competitive, gaining relevant experience will give you a good start into your professional career.

Companies offer a wide range of internships in various disciplines. What is crucial in obtaining approval for your internship and eventually obtaining your study credits, is that there is a clearly defined project that allows you to fulfill the learning objectives. Also, the project needs to allow for an individual assessment.

Finally, note that in order to obtain your internship credits, your

internship has to be pre-approved by the minor coordinator and supervised by a School member that is assigned to you by either the minor coordinator.

Aanvullende informatie onderwijsvormen

On-site Internship

Toetsvorm

Written report – Individual assessment

Literatuur

Literature relevant to the theme of the minor and internship should be used to develop a solution to the problem that is investigated with the internship project.

Overige informatie

You have to arrange your internship in either in period 1 and 2, or in period 2 and 3. To prepare for it you are warmly invited to contact Career Services at the Faculty of Business and Economics. They can help you find an internship and get the best out of it.

Internship Minor Managing Digital Innovation

Vakcode	E_MUM_IMDI
Studiepunten	12.00
Periode	P1+2, P2+3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	K.C. Cepa
Examinator	K.C. Cepa
Betrokken Docenten	
Onderwijsvormen	

Doel vak

Academic & research skills

- Ability to examine and understand problems from different perspectives;
- Ability to put forward well-founded, substantiated points of view, both in spoken and written format;
- Ability to apply acquired knowledge to other problems and in other contexts.
- Ability to translate practically relevant problems into (academically) relevant research questions;
- Ability to design and execute a project using a systematic, analytical approach in a real business environment (of profit or not-for-profit organizations).

Bridging theory and practice - knowledge

- Have specialized, in-depth knowledge and insights about the minor theme

Bridging theory and practice - practice

- Ability to make connections between theories, models, and concepts of that specific minor theme / discipline.
- Ability to apply theoretical knowledge in a specific organizational context;
- Ability to formulate relevant recommendations for practice based on your knowledge acquired.

Social skills

- Have a better understanding of roles and needs of different types of stakeholders that you need to interact with as a professional;
- Ability to work well in a team and reflect on your own role in the team.

Broadening your horizon

- Have a better understanding of what the expectations of the academic and professional field are in terms of knowledge and skills needed;
- Have awareness of the various career opportunities the field offers.

Self-awareness

- Ability to reflect on your own responsibilities as well as others;
- Ability to reflect on your personal development;
- Ability to receive and are able to deal with feedback from others.

Inhoud vak

Increasingly organizations and maybe even your future employer are looking for experience as well as academic credentials. The School recommends doing an internship, because it is an excellent way to apply the knowledge and (academic) skills which you acquired during your studies. Your most important learning goal as a student-intern is to familiarize yourself with professional and market-related skills in a real and new organizational environment. With the job market becoming increasingly competitive, gaining relevant experience will give you a good start into your professional career.

Companies offer a wide range of internships in various disciplines. What is crucial in obtaining approval for your internship and eventually obtaining your study credits, is that there is a clearly defined project that allows you to fulfill the learning objectives. Also, the project needs to allow for an individual assessment.

Finally, note that in order to obtain your internship credits, your internship has to be pre-approved by the minor coordinator and supervised by a School member that is assigned to you by either the minor coordinator.

Aanvullende informatie onderwijsvormen

On-site internship

Toetsvorm

Written report – Individual assessment

Literatuur

Literature relevant to the theme of the minor and internship should be used to develop a solution to the problem that is investigated with the internship project.

Aanvullende informatie doelgroep

IMPORTANT

- Subscription to the internship through VU.nl is not possible.
- CONTACT THE MINOR COORDINATOR as soon as you have an INITIAL proposal for the internship. Approval of the minor coordinator is essential in order to be able to do a minor internship.

- The general internship manual will be available through VU.nl (including more details on a time plan and practical matters). CAREFULLY READ THE MANUAL ON VU.nl (go to Services > Degree programme > Internship, or Serviceplein > Opleidingsprogrmma > Stage). The manual will provide more insights in what is exactly expected in terms of your internship proposal, the concrete requirements, and the related time line of activities.
- After completing the internship the subscription to the course as well as the registration of the result will be done by the back office.

Interpreting Information in Text by Humans and Machines

Vakcode	L_PABAALG005
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. E. Maks
Examinator	dr. E. Maks
Betrokken Docenten	dr. E. Maks
Onderwijsvormen	Seminar, Lecture

Doel vak

In this course, students are trained in the computational analysis of texts. In
particular, we explore the process of identifying and annotating information in historic and contemporaneous texts such as novels, lyrics, letters, newspaper articles, movie scripts, blogs and other social media texts using manual and automatic methods.

By the end of the course :
* students are able to formulate a humanities or social analytics research question that can be answered by the information found in a (set of) texts of their choice
* students are able to make the relevant information explicit by carrying out a linguistic annotations task.
* students are able to build and apply basic text mining techniques (in particular sentiment and emotion analysis) to find the relevant information automatically.
* students are able to reflect on the results of the automatic analysis by performing a qualitative and quantitative error analysis.
* students are able to present their findings in a research paper.

Inhoud vak

This module addresses the process of systematic text analysis through (1) human annotation and (2) automatic analysis using text mining techniques.

Human annotations make information that is implicit in data explicit, allowing researchers to explore their data, identify patterns and answer various research questions in a methodologically sound way. It also requires the use of some type of interpretation model and it results in an analysis that can be compared across annotators. The degree to which annotators agree or disagree (the so-called Inter Annotator Agreement) tells us something about the reproducibility of the interpretation process, the matureness of theoretical notions and the criteria used to apply them to real data.
Text mining techniques can be used to automatically find the same or similar information in text. Some of these techniques are off the shelf software, but most of them need to
be built or fine-tuned to carry out a specific task. How do these techniques work? Can a machine do better than humans? Is it possible to use the automatic annotations to extract useful informations from the text and to answer research questions?

Aanvullende informatie onderwijsvormen

Lectures, Work groups (2 * 1,5 hrs each week)

Toetsvorm

Weekly assignments and a final research paper.

Aanvullende informatie doelgroep

Students in the VU Minor Digital Humanities and Social Analytics.
NB. If you want to follow this course while not being enrolled in this minor, please contact the coordinator.

Overige informatie

attendance is mandatory

Aanbevolen voorkennis

Programming Skills (Python)

Introduction to Artificial Intelligence

Vakcode	X_401089
Studiepunten	3.00
Periode	P1
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. I. Tiddi
Examinator	dr. I. Tiddi
Betrokken Docenten	dr. M.C.A. Klein, dr. V.N. Stebletsova, dr. I. Tiddi, dr. K.S. Schlobach, dr. V. de Boer, B. Postma, prof. dr. F.A.H. van Harmelen, dr. K. da Silva Miras de Araujo, dr. N. Ullah, B. Borsboom, dr. A. Yaman, dr. A.J. Lisowska MBA
Onderwijsvormen	Meeting, Lecture, Seminar

Doel vak

After completing this course, students will have:

- a basic understanding of a number of subfields of AI; (Knowledge and understanding)

- a good understanding of the academic skills required in studying AI; (Applying knowledge and understanding)

- some basic skills in writing a project proposal (Communication)

- a solid understanding of how the degree program in AI is structured. (Lifelong learning skills)

- some experience with reflecting on the role of AI in society (Making judgements)

Inhoud vak

The course combines 3 types of lectures:

- general lectures to survey the richness of the area of Artificial Intelligence (AI)
- methodology lectures on how to conduct research in AI ;
- individual working groups sessions where both academic skills and the individual AI topics are discussed.

In addition, students will work in groups on small projects to find, analyse and solve (theoretically) interesting AI problems using material gathered from the lectures.

The goal of the course is to provide a broad introduction to AI and to provide students with the basic academic skills and knowledge that are necessary for studying AI at the Bachelor's level.

Aanvullende informatie onderwijsvormen

- General Lectures on subfields of AI
- General Lectures on the methodology for AI research
- Working groups
- 1 session on how to deal with diversity in the classroom

Toetsvorm

Assessment is based on:

- a group assignment (about 60%)
- individual contributions consisting in written reflections about some of the lectures (20%)
- written MCQ exam about topics discussed in class (20%).

For students who failed, there will be an opportunity to resit the submissions (either resubmit reflections or improve the group report).

Literatuur

As part of the group project, a literature search will be required on a self-chosen topic.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 1)

Introduction to Digital Humanities and Social Analytics

Vakcode	L_AABAALG076
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. L. Viola
Examinator	dr. L. Viola
Betrokken Docenten	dr. H.M.E.P. Kuijpers, dr. L. Viola
Onderwijsvormen	Seminar, Lecture

Doel vak

Upon successful completion of this course, students will be able to:

- Understand and articulate the key issues and debates in digital humanities, especially concerning cultural heritage, digital methods, and the reliability of digital research;
- Critically evaluate digital tools, data, and methodologies used in DH and Social Analytics research;
- Apply digital tools to analyze humanities and social data, acknowledging the potential for bias and error;
- Demonstrate an ability to manage digital projects and create a digital humanities workflow, from data acquisition to dissemination of results;
- Collaborate effectively in interdisciplinary teams to conduct digital humanities research.

Inhoud vak

This course gets students acquainted with Digital Humanities (DH), a field that explores the intersection of digital technology and the humanities, and Social Analytics, a field that interprets digital interactions and relationships of people, topics, ideas, and content. The course provides students with the theory, tools, and methods needed to analyze and interpret digital data of interest to the humanities and social sciences. It focuses on three main themes:

1. Debates in Digital Humanities and Social Analytics
2. Computation & the Humanities: components and workflows

3. Digital methods

Students will engage in critical discussions and hands-on projects that expose them to the practical challenges and opportunities of building a digital humanities workflow as well as theoretical questions within the field of Digital Humanities and Social Analytics.

In this course, students' diversity is a resource that gives them a chance to learn from and with each other. Students mostly work in groups; in this way, they develop understanding of the value of collaboration in a DH research setting whilst becoming acquainted with DH professional practices where collaboration is the norm. This is added to the intrinsic highly multi-disciplinary nature of the Minor which can be chosen by students from any discipline.

Aanvullende informatie onderwijsvormen

Classes will consist of a combination of bi-weekly lectures, discussion, interdisciplinary group work and hands-on practicals. Students are required to attend at least 80% of the classes. Students who fail to do so without a valid reason will be excluded from the course.

Toetsvorm

Project portfolio (70%) and engagement in discussion and practical activities (30%)

Vereiste voorkennis

Students from all disciplines are welcome; no prior knowledge is required.

Literatuur

Readings will be announced through CANVAS

Aanvullende informatie doelgroep

Students who take the University Minor 'Digital Humanities and Social Analytics'. As long as there are available places, we welcome other students of all disciplines, including international exchange students.

Introduction to Digital Innovation

Vakcode	E_MUM_IDI
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. J. van Angeren
Examinator	dr. J. van Angeren
Betrokken Docenten	dr. J. van Angeren, dr. N.B. Schirrmacher
Onderwijsvormen	Study Group, Lecture

Doel vak

ACADEMIC SKILLS:

Develop the ability to understand and critically examine theories and

concepts relating to digital innovation, in order to:

- Understand and examine theories and concepts relating to digital innovation;
- examine the relationship between technological development and digital innovation;
- reflect on the implications of digital innovation across different levels.

KNOWLEDGE:

Develop a fundamental understanding of the domain of digital innovation,

and the main trends and issues that permeate it. More specifically,

students will gain knowledge concerning the following:

- Understand what digital innovation is, what different types of digital innovation exist, what the fundamental characteristics of digital innovation are, and how digital innovation relates to novel technological developments (e.g., artificial intelligence, blockchain, internet of things, metaverses);
- describe the architecture of digital innovation and its fundamental building blocks;
- explain the impact of digital innovation on work, organizations, markets, and society.

BRIDGING THEORY AND PRACTICE:

Develop the ability to use the elementary characteristics and theories

of digital innovation to explain the fast-paced digital transformation

of organizations and industries. Specifically:

- apply theoretical knowledge about digital innovation to a real-life case
- conceive of and develop prototypes for digital tools.

SOCIAL SKILLS:

Students gain the following social and collaborate skills:

- Ability to work in a multidisciplinary teams on a real-life business case;
- present project results orally and in writing.

SELF-AWARENESS

In terms of self-awareness, students gain the following skills:

- Take responsibility for their own learning;
- open up to and deal with feedback from others.

Inhoud vak

This course introduces students to the domain of digital innovation. Digital innovation can refer to innovations in technology (i.e., the development of new technologies), innovations applying technology (i.e., the development of new products, services, processes, or business models facilitated by digital technologies), and innovation with technology (i.e., the changing innovation process). We focus on developing a comprehensive, interdisciplinary understanding of digital innovation. We define digital innovation and related terms, conceptualize the main drivers of digital innovation (i.e., digital technology), and discuss digital innovation's distinct characteristics. We pay special attention to the sociotechnical nature of digital innovation, by emphasizing the importance of the context within which digital innovations are set and used. The course also focuses on developing a better understanding of the typical architecture of digital innovation. After discussing business model innovations as a special, prevalent case and/or consequence of digital innovation, the last part of the course focuses on the implications of digital innovation for work, organizations, the structure of markets and industries, and society.

Students will also put their newly acquired knowledge of digital innovation into practice. In teams, they will be working on a real-life business case. They will develop a minimal viable product (MVP) that addresses the case at hand. The assignment will simulate the process of a typical digital consultancy assignment. Student teams will pitch their digital solutions to a designated jury of experts during a dedicated session at the end of the course.

Aanvullende informatie onderwijsvormen

Lectures

Case tutorials

Toetsvorm

Final exam - Individual assessment

Case assignment - Group assessment

Literatuur

The readings for this course consist of academic- and practitioner-oriented research papers. The (links to) required readings will be made available via Canvas.

Aanvullende informatie doelgroep

This course is the introductory module to the minor on Managing Digital Innovation, but the course is also open to students from outside of this minor, with the exception of students in (International) Business Administration at Vrije Universiteit Amsterdam.

Introduction to Programming (PYTHON)

Vakcode	X_401096
Studiepunten	6.00
Periode	P2
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. A.J. van der Ploeg MSc
Examinator	dr. A.J. van der Ploeg MSc
Betrokken Docenten	J. Veltman, dr. A.J. van der Ploeg MSc
Onderwijsvormen	Lecture, Practical, Written partial exam

Doel vak

The goal of this course is to teach students to solve problems using structured programming.

Learning Python is actually a side effect that happens because the programming language to practice structured programming happens to be Python.

Knowledge and understanding: At the end of the course, students will be familiar with basic programming concepts, such as types, variables, expressions, statements, loop, methods.

Applying knowledge and understanding: Students will be able to program small to medium-sized programs when given a natural language description of the problem.

Communication skills: Students will be able to write code such that it can easily be understood by others.

Learning skills: The students will learn how to read library documentation.

Making judgments: Students will be able to judge whether a given program solves a certain problem.

Inhoud vak

During this course, students learn to write programs in Python using types (int, boolean, float, list and str), expressions, assignment statements, if-statements, iterations (while- and for-statement), They also learn standard functions, module math, as well as how to make functions, perform I/O, make classes and use objects.

Attendance at the practical may be mandatory depending on if you did not already solve the exercises for that week.

Aanvullende informatie onderwijsvormen

Lectures and practicals.

Toetsvorm

There is an exam that determines the end grade of the course. To make

the exam grade valid, the student is required to hand in individual

ungraded programming assignments. There will be a resit exam.

Literatuur

An on line book is used (How to Think Like a Computer Scientist,
Learning with Python 3, by Jeffrey Elkner, Allen B. Downey,
and Chris Meyers) see URL:
http://openbookproject.net/thinkcs/python/english3e/index.html

Aanvullende informatie doelgroep

Minor Bioinformatics and Systems Biology
Bachelor Biology
Bachelor Biomedical Sciences
Master Biomolecular Sciences
Master Bioinformatics and Systems Biology

Overige informatie

This course cannot be followed by students in the Bachelor Artificial
Intelligence. They follow the similar course called "Introduction to
Python programming" (XB_0082).

Afwijkende intekenprocedure

Choosing a TA and scheduling will be done in the beginning of the first
week of period 2. There will be 2 hours of practical for module 1 and 6
hours practical for the module 2 - 6.

Introduction to Psychology and its Methods

Vakcode	XB_0069
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. D.G. Özkan MSc
Examinator	dr. D.G. Özkan MSc
Betrokken Docenten	dr. D.G. Özkan MSc
Onderwijsvormen	Hoorcollege

Doel vak

-Kennis en begrip hebben van het vakgebied psychologie en haar methoden van onderzoek

-Het kunnen toepassen van deze kennis zoals blijkt uit de vaardigheid basisvragen te kunnen beantwoorden op het gebied van de psychologie en haar methoden van onderzoek

Inhoud vak

Het vak bestaat uit twee delen. In het eerste deel krijgt de student een overzicht van de psychologie met een nadruk op de biologische cognitieve kanten van het vak. Wat is de genetische en biologische basis van gedrag? Wat is bewustzijn? Hoe zien we, leren we, onthouden we en denken we? Het begrip intelligentie zal ook aan bod

komen. In het tweede deel worden onderzoeksmethoden besproken met de bedoeling de student in staat te stellen onderzoek in methodologisch opzicht te evalueren. Aan de orde komen: wetenschapstheorie, logische redeneringen, beschrijvend, correlationeel en experimenteel onderzoek, empirische cyclus, meetschalen, betrouwbaarheid, validiteit, steekproef en populatie, onafhankelijke, afhankelijke en controle variabelen, contaminerende variabelen, random error, binnen-proefpersonen en tussen-proefpersonen designs, factoriële, kleine-N en quasi-experimentele designs.

Aanvullende informatie onderwijsvormen

- 8 hoorcollege over psychologie - 6 hoorcollege over methodologie

Toetsvorm

Multiple choice tentamen

Optie voor hertentamen

Literatuur

- Phelps, E.A., Berkman, E.T., & Gazzaniga, M.S. (2022). Psychological Science (7th edition). Norton.

- Morling, B. (2020). Research Methods in Psychology: Evaluating a World of Information (4th edition). Norton.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence

Overige informatie

De cursus wordt Engelstalig aangeboden.

Introduction to Python for Humanities and Social Sciences

Vakcode	L_AABAALG075
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. L.G. de Passos Morgado da Costa BA
Examinator	dr. L.G. de Passos Morgado da Costa BA
Betrokken Docenten	dr. L.G. de Passos Morgado da Costa BA, dr. P.J.M. Sommerauer
Onderwijsvormen	Seminar

Doel vak

This course is meant to introduce you to the basics of the Python programming language. There is a lot to discover about Python and programming in general, and you will probably learn something new every day if you continue programming after this course. In the master-level version of the course, our goal for you is to become an independent programmer who is able to find solutions to new problems.

Knowledge and comprehension: in the first half of the course, you will:

- Become familiar with the basics of python syntax and the basic data types in python
- Learn how to use basic python tools to work with and manipulate the basic data types in the Python Standard Library

Application: In the second half of the course, you will:

- Learn how to apply your knowledge to deal with different file types (e.g., plain text, CSV/TSV, JSON)

- Learn how to use external libraries for text analysis
- Learn how to explain your code and results to others by documenting it so it can be shared

Analysis: Throughout the entire course, you will develop the following skills:

- Learn how to analyze (and debug) your own code and other people's code
- Learn how to divide a problem into small and manageable sub-problems
- Learn how to structure your code around small, self-contained, and understandable modules

Synthesis: At the end of the course, you will:

- Be able to follow a suggested structure to work on small programming projects
- Be ready to learn how to work with new libraries we have not covered in the course
- Be able to continue to develop as a programmer in an independent and self-guided manner

We will focus on readability and understandability so that you will be able to share your code and results with others, and re-use your code in the future. This is a practical course, in which you will get a lot of hands-on experience. Due to the nature of this course, active participation is required.

Inhoud vak

During this course, you will learn how to analyze text data using the Python programming language. No programming knowledge is required; we believe that anyone can learn how to program.

You will learn how to extract information from text corpora; deal with different file types (plain text, CSV, JSON). We will focus on readability and understandability of your code so that you will be able to share it with others, and reuse your code in the future.

Aanvullende informatie onderwijsvormen

The course is organized in blocks. Blocks typically follow this routine: - Lecture 1: introduction of concepts in the form of an (interactive) lecture. Students are expected to have worked through preparatory exercises and ask questions in class. - Lecture 2: The lecturer focuses on dedicating time to more difficult concepts from the block. If there is enough time, students can already start working on the assignment. Students will have the opportunity to ask for help and clarification. The deadline for an assignment is typically soon after this lecture. - Lecture 3: this lecture is a feedback session. The lecturer provides general feedback about the submitted assignments.

Toetsvorm

Assessment has three components:

1) Mid-term Exam (20%): This exam tests basic concepts concepts in blocks 1 and 2. These concepts are the foundation of blocks 3 and 4. The exam will take place in class, at the end of block 2. To pass this course, you need a passing grade (at least 5.5).

1) Two graded individual assignments (40%): The assignments are designed to practice your programming and problem-solving skills. Moreover, they allow us to keep track of your progress, and identify topics that require more attention in class. To pass this course, each assignments needs a passing grade (at least 5.5).

2) Final exam (40%): The final exam is designed to test your knowledge of Python. To pass this course, you need a passing grade (at least 5.5).

Literatuur

To be announced on Canvas. All materials are freely available online (https://github.com/cltl/python-for-text-analysis/tree/master).

Aanvullende informatie doelgroep

All bachelor students who want to get acquainted with programming in Python.

Overige informatie

Students need a working computer (or laptop) with access to the command line to complete this course. Any working laptop is sufficient. Note that tablets, phones and netbooks are usually not sufficient.

Aanbevolen voorkennis

There are no prerequisites to take this course but an interest in text analysis is recommended.

Introduction to Python Programming

Vakcode	XB_0082
Studiepunten	6.00
Periode	P1+2
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M. Verano Merino MSc
Examinator	dr. M. Verano Merino MSc
Betrokken Docenten	dr. M. Verano Merino MSc, B. Postma
Onderwijsvormen	Lecture, Practical, Written partial exam

Doel vak

This course aims to teach students how to solve problems by means of a computer program.

Learning Python is actually a side effect that happens because the programming language used to solve problems during the course is Python. However, the knowledge acquired by the students during the course is transferable to other programming languages like Java or C.

Knowledge and understanding: at the end of the course, students will be familiar with basic programming concepts, such as data types, data structures, variables, expressions, control flow statements, methods, and classes.

Applying knowledge and understanding: Students will be able to write and read small to medium-sized programs when given a natural language description of the problem.

Communication skills: Students will be able to read code written by others and write code such that others can understand it.

Learning skills: Students will learn how to read technical documentation (e.g., programming language and library references) and how to use programming as a tool for solving problems.

Making judgments: Students will be able to judge whether a given program solves or not a certain problem.

Inhoud vak

This course will equip students with a wide range of programming skills. They will learn how to write programs for solving problems using different data types (e.g., int, boolean, float, string, and custom types), data structures (e.g., lists, sets, dictionaries, tuples, and matrices), expressions, control flow statements (e.g., conditionals, and loops), and more advance programming techniques such as comprehensions. They will also gain a thorough understanding of standard functions, learn how to define functions, perform I/O tasks, and grasp the basics of Object-Oriented Programming.

Aanvullende informatie onderwijsvormen

There will be 1 lecture per week, as well as 1.5 hours of practical session per week. There is also a significant amount of programming exercises to practice the different programming concepts taught during the course.

Toetsvorm

A mandatory midterm (25%) and final exam (65%) determine the course's final grade. Students must obtain at least a 5.0 on the midterm exam to be able to take the final exam.

Moreover, there are several programming assignments (10%) that are non-resit. This means completing them on time and to the best of your abilities is important.

There will be a resit exam, and **only** students from the previous year (2023-2024)**can** transfer grades for assignments. Possible bonus points are not transferable to the resit exam.

Literatuur

The book used in this course is Think Python: How to Think Like a Computer Scientist, Allen B. Downey, Green Tea Press. The book is available online at: https://greenteapress.com/wp/think-python-2e/.

Aanvullende informatie doelgroep

Bachelor students of Artificial Intelligence (year 1)

Overige informatie

You will be allocated to a TA in the first week of period 1; this also determines the scheduling of the practical sessions.

Introduction to Reinforcement Learning

Vakcode	XB_0118
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. S. Yu
Examinator	dr. S. Yu
Betrokken Docenten	dr. S. Yu
Onderwijsvormen	Lecture, Practical

Doel vak

At the end of the course, students will:

1. Understand and explain the fundamental concepts of reinforcement learning (RL), such as Markov decision process, value functions, etc. (Knowledge and understanding)
2. Knowledge on different RL algorithms. (Knowledge and understanding)
3. Characterize and differentiate between different RL algorithms (Temporal-Difference learning, Monte Carlo, SARSA, Q-learning, Policy Gradients, etc.). (Applying knowledge and understanding; Making judgments)
4. Model a RL agent for sequential decision-making problems. (Applying knowledge and understanding)
5. Implement RL algorithms in standard benchmarks. (Applying knowledge and understanding).

Inhoud vak

Reinforcement Learning is one of the three main paradigms in machine learning, alongside supervised learning and unsupervised learning. It is a rapidly advancing field in artificial intelligence, focusing on how an agent can learn to maximize cumulative rewards while interacting with a complex and uncertain environment. Reinforcement Learning has diverse applications, including games, robotics, and general decision-making. This course offers a comprehensive introduction to the fundamental concepts of Reinforcement Learning. Topics will include key algorithms such as Monte Carlo, Temporal-Difference learning, SARSA, Q-learning, and Policy Gradients, among

others.

Aanvullende informatie onderwijsvormen

Lectures (two per week) and working groups.

Toetsvorm

There will be a written exam which is supported by lectures (counting for 40%) and a group assignment (60%) of the final grade. There will be a resit exam, but NO resit for the group assignment.

Literatuur

Sutton, Richard S., and Andrew G. Barto. *Reinforcement learning: An introduction.* MIT press, 2018.

Aanvullende informatie doelgroep

BSc Artificial Intelligence

Introduction to the minor Visual evidence: learning how to look

Vakcode	R_IMVE
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	prof. dr. W.G. Werner
Examinator	prof. dr. W.G. Werner
Betrokken Docenten	prof. dr. W.G. Werner, dr. J.H. Roeland, dr. E. van Roekel
Onderwijsvormen	Study Group, Lecture

Doel vak

In this course, we explore what the practice of looking at film entails for the comprehension of knowledge productions in different disciplines. We examine and compare multiple practices of visual knowledge productions between different fields of study (e.g., law, anthropology, and theology), taking documentary film as our point of departure. We ask questions such as: while watching a documentary film, what is it exactly that you are looking at? What makes us convinced or carried away by a visual story? How do we make sense of what we hear or see?

In the course, we examine in detail the multiple ways in which documentary films craft and substantiate their claims to truths through audiovisual means. And we ask what happens when filmmakers deliberately break these disciplinary rules. In so doing, the course objectives center on the cultivation of audiovisual literacy and critical reflection on how we consume and interpret documentary film in our professional and personal lives.

Inhoud vak

During the course, we watch and analyze (together and individually) multiple genres of audiovisual storytelling around one current societal issue and compare the different representational practices. In this way, we will recognize the different motivations, aesthetics, and functions of visual evidence inside and outside academia. Concretely, we will zoom in on the representations of current global migration in artistic documentary, essay film, reportage, virtual reality, and audiovisual evidence in court.

Aanvullende informatie onderwijsvormen

Interactive lectures; film screenings; and group discussion.

Toetsvorm

Short papers comparing different genres and their respective claims to truth and knowledge.

Vereiste voorkennis

Not required

Literatuur

see course manual as will be published on canvas.

Aanvullende informatie doelgroep

Apart from regular students, the course is also available for:

Students from other universities/faculties

Contractor (students who pay for one course).

Overige informatie

The course is part of the minor ‘Visual Evidence’. The course is mandatory for students who follow the minor. The course is open for everyone.

Toelichting Canvas

Canvas is the main communication platform of the course.

Islam en Europese cultuur

Vakcode	G_BATRSAL005
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit Religie en Theologie
Vakcoördinator	dr. W. Boender
Examinator	dr. W. Boender
Betrokken Docenten	dr. W. Boender
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

Na afronding van deze module:

-heeft de student een grondig inzicht verworven in de geschiedenis van islam en moslims in Europa;

-heeft de student kennis opgedaan van beeldvorming over islam en moslims in Europa in verleden en heden;

-heeft de student analytische vaardigheden verworven om deze geschiedenis kritisch te bestuderen;

-heeft de student inzicht verworven in hedendaagse sociaal-religieuze vraagstukken in Europese context;

-heeft de student inzicht verworven in manieren waarop islamitische jurisprudentie (fiqh) wordt uitgewerkt en toegepast in de context van Europa.

Inhoud vak

Maurits Berger, de auteur van het tekstboek dat we in deze cursus gaan doornemen, stelt dat "the topic of 'Islam in Europe' is more than a story told by stringing together historical events. It is a story about us, Europeans, whether Muslim or non-Muslim, of native or foreign origin, and how we struggle with our past and present in the continent called Europe, how we shape our identity and, particularly, the uneasy relationship between who we are and who we want to be" (2014:17). Om deze 'story' te begrijpen, gaan we in deze cursus diep in op de geschiedenis van engagement, vanaf het vroegste begin tot aan de meest recente tijd. We volgen enerzijds de verhaallijn van Berger, die vooral het perspectief vanuit Europa op islam en moslims belicht. Anderzijds gaan we in op verschillende manieren waarop moslims islam vormgeven in de Europese context. We belichten daartoe o.a. de ontwikkeling van *fiqh al-aqalliyāt* (islamitische jurisprudentie voor minderheden), institutionele mogelijkheden binnen de scheiding tussen religie en staat voor bijvoorbeeld ritueel slachten, en we bespreken enkele studies over 'identity politics'.

Aanvullende informatie onderwijsvormen

Voor de colleges (maandag en woensdag) geldt een aanwezigheidsplicht van 80%.

Toetsvorm

Wekelijkse verwerkingsopdrachten (voldaan)

Tussenopdracht t.b.v. essay-paper (30%)

Eindopdracht: essay-paper (70%)

Literatuur

Aanschaffen: Maurits S. Berger, 2014,*A Brief History of Islam in Europe. Thirteen Centuries of Creed, Conflict and Coexistence*, Leiden: Leiden University Press

Aanvullende literatuur wordt aangegeven op Canvas.

Aanvullende informatie doelgroep

De module is een verplicht onderdeel van de Bachelor TRS profiel Islam en van de universiteitsminor Islam. Het is tevens toegankelijk als keuzevak.

Aanbevolen voorkennis

Er is geen voorkennis vereist.

Islamitische ethiek

Vakcode	G_BATRSAL049
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit Religie en Theologie
Vakcoördinator	dr. ing. H. Quadir
Examinator	dr. ing. H. Quadir
Betrokken Docenten	dr. ing. H. Quadir
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

Na afronding van deze module:

- heeft de student kennis over hoe islamitische ethiek wordt gedefinieerd en hoe het zich verhoudt tot het concept van akhlāq (morele waarden) en ãdāb (manieren);

- Heeft de student kennis hoe islamitische ethiek gegrond is in juridische methodologie (uṣūl al-fiqh) en de maqāṣid al-sharīʿa (doeleinden van de sharia);

- beheerst de student de centrale islamitische terminologie m.b.t. islamitische ethiek;

- heeft de student zicht op hoe islamitisch ethiek zich verhoudt tot geestelijke verzorging, hedendaagse politiek en openbaar beleid;

- heeft de student zicht op de spanningen die vernieuwingen in het islamitisch ethisch redeneren in de moderniteit veroorzaken.

Inhoud vak

In deze module wordt nagegaan hoe islamitische ethiek wordt gedefinieerd en welke implicaties en implementaties het heeft tot:

- Uṣūl al-fiqh en de maqāṣid al-sharīʿa.

-Biomedische ethische kwesties en uitdagende vraagstukken in geestelijke verzorging.

-Economische en financiële kwesties.

-Vrijheid van meningsuiting in de islam en liberale democratie.

-Mensenrechten.

Ethiek is een denksysteem of theorie dat probeert antwoord te geven op wat het juiste handelen is. In deze module zullen we nagaan hoe dit in de islam is geregeld. Hiertoe zullen we eerst onderzoeken hoe islamitische ethiek is gedefinieerd en gegrond in de Schriftuurlijke bronnen van de islam en ook andere erkende bronnen.

In de colleges zullen diverse artikelen en hoofdstukken worden besproken die ingaan biomedische kwesties zoals abortus en palliatieve zorg. Ook zullen diverse maatschappelijk en politieke ethische kwesties de revue passeren.

Aanvullende informatie onderwijsvormen

Werkcolleges gedurende drie weken. In de vierde week maken de studenten het tentamen en maken zij de schrijfopdracht. De student bereidt ieder college voor d.m.v. het lezen van verplichte literatuur. Er geldt een aanwezigheidsplicht.

Toetsvorm

Schrijfopdracht (20%) (minimum cijfer 5,5)

Presentatie (10%) (minimum cijfer 5,5)

Tentamen (70%) (minimum cijfer 5,5)

[Note: De deeltoetsen moeten ieder met minimaal 5,5 behaald zijn; indien dit niet het geval is, dan dient dit onderdeel herkanst te worden.]

Literatuur

- Draz, M. A., The Moral World of the Qur'an, transl. D. Robinson and R. Masterton, London: I.B. Tauris, 2008.
- Mohiuddin, A., Mehrunisha Suleman, M., Shoaib Rasheed, S. & Aasim I. Padela, A.I., When can Muslims withdraw or withhold life support? A narrative review of Islamic juridical rulings, Global Bioethics, 31:1, 29-46, 2020. DOI: 10.1080/11287462.2020.1736243
- Kevin, R.A., “Islamic Law as Islamic Ethics.” The Journal of Religious Ethics 186–203. 1983.
- Sachedina, A., Islamic Ethics. Fundamental Aspects of Human Conduct, New York: Oxford University Press, 2022.

Aanvullende informatie doelgroep

Deze module maakt deel uit van het traject Islam van de Bachelor Theology and Religious Studies en van de universiteitsminor Islam.

Islamitische theologie/Kalam

Vakcode	G_BATRSAL027
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit Religie en Theologie
Vakcoördinator	dr. M. Ajouaou
Examinator	dr. M. Ajouaou
Betrokken Docenten	dr. M. Ajouaou
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

Na afronding van deze module:

- kan de student het ontstaan en de ontwikkeling van Kalām scholen

(Islamitische

theologie) beschrijven;

- kan de student tenminste 5 discussiepunten onder de belangrijkste

moslimtheologen

uit de klassieke periode (700-1200) van islamitische theologie/ Kalām

weergeven;

- kan de student de algemene principes van de islam en de principes van

de

mu‘tazila en de repliek van Ash‘ariyya benoemen en uitleggen;

- kan de student het verschil tussen islamitische scholen weergeven;

- kan de student de theologische uitgangspunten van verschillende oude

en nieuwe

islamitische stromingen vergelijken en analyseren;

- leert de student aan de hand van casuïstiek methodieken (kennen) om de

toepassing van Kalām te vergemakkelijken;

- is de student vertrouwd met het brede spectrum en de verschillende

dimensies (leer, praxis, zingeving, ethiek. ideologie) van de

islamitische religiositeit en de plaats van Kalām daarin;

- kan de student daarnaast het voorgaande in de huidige context

plaatsen.

Inhoud vak

Dit vak behandelt centrale thema's uit de islamitische theologie/kalam: de definities van de islamitische godsdienst, betekenis van de islamitische geloofsbelijdenis, enkele belangrijke islamitische begrippen, de zes geloofspunten en de zuilen van de islam, het begrip God en goddelijke eigenschappen, profeetschap in de islam, hiernamaals, de predestinatieleer en de vrije wil, goed en kwaad (ethiek), het begrip ‘majeure zonden’ en de meetlat van geloof en secularisatie. In de module komen de belangrijkste principes van de pre-Kalām scholen aan de orde (Khawāridj en Mardji’a, Quadarriya en Djabriyya) en Kalām scholen (Mu‘tazila, Asj’ariyya en Maturdiyya). De studenten passen de opgedane inzichten toe op de hedendaagse religiositeit van moslims. Dit geschiedt middels casuïstiek, opdrachten en discussie in werkgroepen.

Aanvullende informatie onderwijsvormen

Hoorcolleges en werkcolleges

Toetsvorm

Schriftelijk tentamen (50%)
Tussentijdse opdrachten, waaronder een presentatie (50%)

Literatuur

- Winter, Tim (ed.). Classical Islamic Theology. Cambridge: Cambridge University Press, 2008. [Note: Deze bron is te raadplegen via de

universiteitsbibliotheek.]
- Ajouaou, Mohamed. Wie is moslim? Geloof en Secularisatie onder Westerse Moslims. Zoetermeer: Meinema, 2014. [Note: Deze bron is te raadplegen via Canvas.]
- Syllabus.

Aanvullende informatie doelgroep

Alle doelgroepen

Key Strategies in Disability and Neuropathy

Vakcode	AB_1045
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	B.S. Mengistu
Examinator	B.S. Mengistu
Betrokken Docenten	dr. R.M.H. Peters, dr. M.M. Waltz
Onderwijsvormen	Study Group, Lecture

Doel vak

• To define the concept of disability according to different models/philosophical ideas/worldviews (e.g., how do people experience disability?)
• To identify different causes and consequences of impairment, using neuropathy as a model
• To compare and identify strengths and weaknesses of interventions and innovations that i) aim to prevent impairment and disability, ii) focus on care and rehabilitation and iii) aim for inclusion.
• To familiarize yourself with important debates around human rights, ethics, participation, innovations in the field of disability and neuropathy
• To reflect on your own worldview related to disability
• To practice research skills during the i) problem-based learning sessions, including: formulating research objectives, searching for relevant literature, summarising the literature, presenting key findings, participating in scientific discussions; and ii) Community Service Learning project/Photo essay, including: working in partnership (e.g., with disabled people), formulating lines of argumentation.

Inhoud vak

All over the world, people with impairments and illnesses experience activity limitations and have difficulties participating in society due to barriers, including physical barriers, legal and policy barriers, and psychological and social barriers, such as stigma. When people with an impairment or illness face barriers they cannot easily overcome, they experience disablement. The scale of this problem is huge. The World Health Survey and the Global Burden of Disease estimate state the prevalence of disability as 15.6% and 19.4%, respectively (WHO, 2011). This course looks at issues surrounding disability. We will review relevant theories and models related to disability, including the medical model, social model and ICF model. During the course you will

investigate questions such as ‘What is disability?,’ ‘What is normal?,’ ‘How do different worldviews influence how people see disability?,’ and ‘What is my own worldview?’.

Disorders that affect the nervous system (neuropathies) are responsible for many forms of impairment. Neuropathies that are important in this course include diabetes, leprosy, intellectual disability, autism, mental illness, spina bifida and cerebral palsy. It is important to understand that there are differences of opinion about these conditions: for example, many people consider autism and some forms of mental functioning that can be labelled as mental illness to be variant, but normal, forms of neurological function, while others consider them impairments that should be eliminated.

Different interventions and technologies, and various health and support policies and practices, have been developed to address health issues and remove the social and environmental barriers that affect disabled people. During the 20th and 21st centuries, developments and innovations in health and life sciences have resulted in an exponential growth in scientific knowledge about people, society and the environment. The idea that we know who we are seems to increase, but is this truly the case? For example, what does having a disability mean for a person’s identity, and what does the existence of disability mean for our dominant image of human nature, or the drive for standardization or even perfection? Innovations bring forth possibilities for new interventions and technological gadgets (e.g. bionic prosthetics, cochlear implants, microchips that enhance intelligence), but how do we select and use these? Who decides what is appropriate for whom?

In this course you will learn to reflect on various theories related to disability, and think critically about strategies to address the medical and social problems faced by people diagnosed with a neuropathy.

Note:

In the practical part of all courses in the minor Global Health, including this one, you will use your newly acquired knowledge and apply it in a community engaged learning project. Students will work in project teams on a real-life transdisciplinary global health challenge together with various local and global community partners. Students will cross disciplinary boundaries and interact with health professionals, patients and others to gain an in-depth understanding of global health problems, and to take cohesive and strategic actions to address these problems at a local level. Each course in the minor program will incorporate a different sub-project, starting in the first two courses

with a broad exploration and needs analysis on the topic, followed by a more thorough investigation on a specific challenge in the next two courses, and concluding in the final course with a proposal for a sustainable innovation. Training workshops are offered throughout the minor program to support the development of specific skills such as transdisciplinary collaboration, reflection, dealing with diversity, creativity, leadership and adaptability.

Aanvullende informatie onderwijsvormen

-Lectures (20 hours)

-Work groups (24 hours)

-Community Service Learning project (6 hours)

-Self-study (remaining hours)

Toetsvorm

- Individual exam (60%)
- Community Service Learning project (30%)
- Participation in workgroups (problem-based learning) (10%)

All three parts need to be passed individually.

Literatuur

There are ~10 compulsory readings for this course and some suggested readings (an overview of these readings will be provided on Canvas).

Aanvullende informatie doelgroep

Bachelor students from Biomedical Sciences, Health & Life, Health Sciences, bachelor programs in the natural sciences, and similar bachelor programs that participate in the minor Biomedical and Health Interventions or in the minor Global Health. Students from other programmes are welcome.

Knowledge and Data

Vakcode	X_400083
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. V. de Boer
Examinator	dr. V. de Boer
Betrokken Docenten	dr. J. Chen, dr. V. de Boer, B.B. Kruit, drs. W.X. Wilcke MSc
Onderwijsvormen	Computer lab, Lecture, Seminar

Doel vak

The objective of the Knowledge and Data course is to make students

acquainted with methods and technologies used for expressing knowledge

and data, in particular on the Web. By the end of this course, students

will have built an intelligent web application that queries and reasons

over integrated knowledge from various sources obtained from the Web.

All of this will be based on formal logic theory.

Knowledge and understanding: at the end of the course, students will be

familiar with:

- Theory of Data, Information and Knowledge
- Predictable inferencing and formal systems
- Linked Data and Knowledge Graphs
- Semantic Web technology stack (RDF, RDFS, OWL)
- Ontology Engineering
- Knowledge-driven Data Science

Application of Knowledge and Insights: students will be able to:

- Represent knowledge and data in various formalisms (RDF, RDFS, OWL)
- Implement basic (RDFS) reasoning,
- Develop advanced knowledge models in RDFS and OWL
- Work with SPARQL for querying (distributed) knowledge graphs
- Integrate acquired knowledge in an intelligent semantic data driven application.

Judgement: Students will be able to assess the value of available

datasets and ontologies for web applications, and to choose the

appropriate technology for a specific application.

Communication: Students are able to write a report about a developed

application.

Learning skills: The skill to acquire and apply knowledge and skills

about fundamental knowledge representation concepts as well as

state-of-the art technology, both individually as in a group context.

Inhoud vak

In this course, we study formalisms that are useful and necessary to represent knowledge and data, in particular when these knowledge and data are to be reused, e.g. published and consumed on the Web. We introduce the concept of Knowledge Graphs, the technologies and representation formats (RDF, RDFS, OWL) for expressing semantics and linked data in a web-accessible format, use the SPARQL query language to query over this data. We finally build a data science application that uses integrated data for some intelligent task.

Even though content on the web is generally produced from structured data sources (databases), its representation is in a form that is meant for human consumption. Linked Data allows to scale the walls of this siloed information space, by reusing identifiers and vocabularies across these datasets, and presenting that information in a way that is appropriate for machine consumption.

Aanvullende informatie onderwijsvormen

The course consists of lectures where theory is discussed and Working group sessions in which exercises related to the theory are discussed

Students will work on individual practical assignments in the first half of the course and will be supported in computer practicals. Students will also collaborate in groups for a final project assignment.

Toetsvorm

The final grade will be determined by three components that each count for 1/3 of the final grade:

- Weekly individual practical assignments (in total 5 assignments of which the results are averaged)
- A final group project, assessed on the basis of a final document, application and (video) presentation
- An exam testing the theory

Each component should be passed (5.5 or higher) in order to pass the course. There will be a resit option for the exam, the practical assignments and project can not be re-taken.

Weekly intermediary quizzes will be used to provide formative feedback.

Literatuur

We will provide the (online) reading material through Canvas.

Recommended literature: A Semantic Web Primer (3rd edition) Grigoris Antoniou, Paul Groth, Frank van Harmelen and Rinke Hoekstra, MIT Press, September 2012

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence year 2

Aanbevolen voorkennis

- Basic programming (Python)
- Logic (Basic propositional and predicate logic)

Law and Ethics of Reproductive Technologies

Vakcode	R_LERT
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	prof. dr. mr. B.C. van Beers
Examinator	prof. dr. mr. B.C. van Beers
Betrokken Docenten	prof. dr. mr. B.C. van Beers
Onderwijsvormen	Seminar, Lecture

Doel vak

This interdisciplinary course explores the bioethical, biolegal and biopolitical dilemmas that are raised by technological developments at the intersection of reproductive medicine and genetics.

This course will enable the student to critically reflect upon legal and ethical dimensions of current public debates on the regulation of assisted reproductive technologies. This course will teach the student to come to an understanding of the key concepts and categories within legal regulation of reproductive technologies, and to connect these with various normative ethical theories.

Through an examination of the existing legal frameworks surrounding reproductive and genetic technologies from the perspectives of law and bioethics against the background of ongoing contemporary political and societal discussions, the student will be trained to integrate ethical reasoning, daily practices and legal rules and regulations into a normative evaluation of these technologies. In this process the student will be encouraged to take a legally and ethically argued position in scientific debates on current developments in the field of assisted reproductive technologies through written and oral presentations of a legal and philosophical nature.

Inhoud vak

Technologies at the intersection of reproductive medicine and genetics

offer news ways of creating human life. These technologies make it possible to assemble, genetically screen, choose and, possibly, even design one’s future children. How can societies decide who may access these technologies to create which kinds of children? Which rights, whose rights and which public values should be taken into account within the regulation of this complex field? And what are the legal and ethical limits to these currently emerging forms of 'liberal eugenics'?

The general focus in this course will be on the role and meaning of human rights and human dignity for the regulation of assisted reproductive technologies.

Topics in this course include:

- law and ethics of prenatal testing
- selective reproduction and ‘designer babies’
- reproductive markets and reproductive tourism
- reproductive rights
- gestational and commercial surrogacy
- wrongful life claims
- the welfare of future children
- sperm and egg cell donation/selling
- eugenics and human enhancement
- the status of embryos and gametes

As students in this course are invited to discuss recent reproductive technologies and trends against the background of legal, ethical and societal approaches, this course fits in very well with the VU law school's focus on 'law in action'.

Aanvullende informatie onderwijsvormen

This 3 week course will be taught through interactive tutorials and lectures.

Toetsvorm

Written open-book exam.

Literatuur

All literature will be made available on Canvas, and will include legal and philosophical academic literature, legal and political documents, policy reports, news articles and audiovisual materials.

Aanvullende informatie doelgroep

Because this course is also part of a university minor (Technology, Law

and Ethics), it is open to students from various academic backgrounds.

Apart from regular students, the course is also available for:
Students from other universities/faculties
Exchange students
Contractor (students who pay for one course)

Aanbevolen voorkennis

No special knowledge of law, philosophy or bioethics is required to be able to participate in this course. A basic knowledge of human rights and a keen interest in the contemporary dilemmas surrounding reproductive technologies are a plus.

Linear Algebra and Calculus

Vakcode	XB_0114
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	E. Sandin Vidal
Examinator	E. Sandin Vidal
Betrokken Docenten	E. Sandin Vidal
Onderwijsvormen	Seminar, Lecture, Practical

Doel vak

Upon completion of the course, students will:

- Demonstrate a solid comprehension of fundamental concepts in Linear Algebra and Calculus, including vectors, matrices, linear systems, differentiation, and integration (Knowledge and Understanding).
- Apply mathematical techniques to solve basic exercises and problems across different levels of complexity within Linear Algebra and Calculus, such as solving linear systems or diagonalization, showcasing proficiency in computations and problem-solving strategies (Applying Knowledge and Understanding).
- Engage in rigorous mathematical reasoning by providing proofs of statements and theorems, and by explaining underlying concepts and principles (Applying Knowledge and Understanding; Making Judgments).
- Develop the ability to articulate mathematical solutions and explanations effectively, engaging clear communication of mathematical concepts (Communication Skills).
- Cultivate learning skills necessary for further study in mathematics and AI, including the ability to independently explore advanced topics, identify areas for improvement, and pursue self-directed learning opportunities (Learning Skills).

Inhoud vak

Calculus:

- Limits
- Derivatives
- Partial derivatives
- Gradients
- Integrals

Linear Algebra:

- Linear systems of equations
- Row reduction
- Linear maps
- Matrix and vectors operations
- Vector spaces
- Determinants
- Eigenvalues, eigenvectors and diagonalization
- Inner product, orthogonalization and Gram Schmidt

Aanvullende informatie onderwijsvormen

Theory lectures and exercise tutorials each week.

Toetsvorm

At least one final exam at the end of the course. Possibility of a midterm or other assessments.

Logic and Sets for AI

Vakcode	XB_0083
Studiepunten	6.00
Periode	P4
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. K.S. Schlobach
Examinator	dr. K.S. Schlobach
Betrokken Docenten	prof. dr. R.C.A.M. van der Vorst, dr. P.S.H. Koopmann, dr. K.S. Schlobach
Onderwijsvormen	Lecture, Seminar, Written partial exam

Doel vak

After completing this course, the student can:

1. express logical statements in propositional and predicate logic

(Knowledge and understanding) (Applying knowledge and understanding)

2. reason about the meaning of such formulas through truth tables and

models (Applying knowledge and understanding) (Making judgements)

3. argue formally whether one formula implies another one, or that they

are equivalent (Applying knowledge and understanding) (Making

judgements)

4. reduce a propositional formula to disjunctive or conjunctive normal

form (Applying knowledge and understanding)

5. apply reasoning algorithms on propositional formulas (Applying

knowledge and understanding)

Furthermore, the student is able to:

6. reason about set constructions through Venn diagrams and the algebra

of sets (Applying knowledge and understanding) (Making judgements)

7. construct and interpret formal, graphic, and matrix representations

of sets, relations and functions

8. determine and argue whether (Applying knowledge and understanding)

(Making judgements):

a. a relation is reflexive, transitive, symmetric or antisymmetric.

b. a relation is an ordering relation, equivalence relation, or a

function

c. a function is injective or surjective

9. construct and interpret compositions of relations (or functions) and their inverses (Applying knowledge and understanding) (Making judgements)

10. construct a proof by mathematical induction (Applying knowledge and understanding)

Inhoud vak

The sets part of the course starts by introducing the concepts of sets, Venn diagrams, product sets and relations. The student then learns the main characteristics and properties of three particular types of relation: ordering relations, equivalence relations and functions. The sets part concludes with a study of the principle of mathematical induction.

The logic part focuses in the first place on propositional logic: truth tables, boolean operators, functional completeness, logical puzzles, SAT-solving. In addition the student will
learn the meaning and use formulas of predicate logic, to express mathematical properties and sentences from natural language.

Aanvullende informatie onderwijsvormen

Every week, there is one 2-hour lecture and one 2-hour tutorial for the logic part of the course, and one 2-hour lecture and one 2-hour tutorial for the sets part of the course.

Toetsvorm

A written midterm exam (40% of the grade) and a written final exam (60% of the grade). For both the midterm and the final exam, at least a 5.0 must be achieved (and the overall mark must be at least 5.5.). The resit exam covers all material of the course.
It is not possible to resit only the midterm exam or only the final exam of the course.

Literatuur

All course materials are provided via Canvas

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 1)

Logistics Analysis

Vakcode	X_401084
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen

Vakcoördinator	dr. ir. S. Dabia
Examinator	dr. ir. S. Dabia
Betrokken Docenten	dr. ir. S. Dabia, ir. S.J.G. Miltenburg
Onderwijsvormen	Lecture, Seminar, Practical

Doel vak

The course Logistics Analysis is an exciting course that will challenge you in various ways. By taking Logistics as a point of departure, we bring together several perspectives and analyze business problems faced by logistics companies. Taking a logistics perspective will stimulate you to think about organizations in a different way, bringing together knowledge from different fields and realizing that this creates challenges and conflicts that managers need to deal with. You will learn to systematically describe logistical systems, and identify problems that emerge in these systems. Moreover, this course offers you a number of tools that allow you to analyze logistical systems, optimize them, (re)design them and assess the consequences of suggested improvements. Important topics such as production management, inventory management, and maintenance management are addressed, which are essential, hands-on tools any logistics professional should be able to work with.

Course Objectives:

The overall objective of the course Logistics Analysis is to furnish you with knowledge and skills to analyze logistics problems in a variety of industries including manufacturing, transportation and services. More specifically, we expect that upon completion of the course you are able to:

- Understand and explain (the importance of) logistical systems and their complexity;

- Analyze, model, and optimize real-life logistical systems by applying quantitative tools;

- Interpret and discuss the use of these tools in specific logistics situations as well as their general strengths and weaknesses;

- Write a clear management report containing useful recommendations for the management of a logistical company;

- Clearly report your work: analysis, modelling, solution approach, computational results, and managerial insights.

Inhoud vak

The course Logistics Analysis consists of a set of modules, each module covers a specific topic. The covered modules are:

Module 0: Introduction to Logistics

Module 1: Forecasting

Module 2: Production Management

Module 3: Inventory Management

Module 4: Maintenance Management

Module 5: Transportation Management

Aanvullende informatie onderwijsvormen

Lectures and tutorials

Toetsvorm

Group assignments (50% of the final grade)

Written examination (50% of the final grade)

Both the exam grade and the group assignment grade must be at least 5.0. Furthermore, to pass the course, the final grade must be at least 5.5 (6 rounded).

Literatuur

Provided via Canvas

Aanvullende informatie doelgroep

3BA

Aanbevolen voorkennis

English writing

Working with excel

This is a quantitative course. Hence, the following pre-knowledge is**highly required:**

Basic calculus and algebra (e.g. integration and differentiation of functions with a single or multiple variables etc)

Basic probability theory (e.g. basic probability distributions, conditional probability etc)

Basic stochastic processes (e.g. basic Markov chains)

Basic linear programming theory

NOTE: Please reconsider your decision taking the course if you don't possess the above-mentioned pre-knowledge.

Machine Learning

Vakcode	X_400154
Studiepunten	6.00
Periode	P4
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. P. Bloem
Examinator	dr. P. Bloem
Betrokken Docenten	dr. P. Bloem, dr. ir. M. Mohammadi
Onderwijsvormen	Lecture, Seminar

Doel vak

Upon completion of this course, students will:

- be acquainted with the dominant concepts of machine learning methods, including some theoretical background. (Knowledge and understanding)

- acquire knowledge of established machine learning techniques such as linear models, neural networks, decision trees and ensemble methods

(Knowledge and understanding)

- learn some statistical techniques to assess and validate machine learning results. (Apply knowledge and understanding, make judgments)

Inhoud vak

Machine learning is the discipline that studies how to build computer programs that learn how to behave from examples, rather than following explicit instructions. It is a subfield of artificial intelligence that intersects with statistics, cognitive science, information theory, and probability theory.

Recently, machine learning has become increasingly important for the design of search engines, robots, and sensor systems, and for the processing of large scientific data sets. Other applications include handwriting or

speech recognition, image classification, medical diagnosis, stock market analysis and bioinformatics.

The course covers a wide variety of machine learning techniques, but puts particular emphasis on gradient descent optimization, backpropagation, neural networks and deep learning. Some discussion on the broader social impact of machine learning technology is included.

Aanvullende informatie onderwijsvormen

The course consists of pre-recorded videos, interactive lecture/QA sessions (two per week) and optional homework assignments discussed in working groups (one per week). The practical assignment is supported by small exercises to help with the relevant technologies, and informal presentations at project groups (one per week).

There is no mandatory attendance for any lectures or workgroups, except that one member of each group must be present at the weekly project group. A large amount of the material is freely available at https://mlvu.github.io The course is taught in English.

Toetsvorm

The course assessment consists of two parts: an examination and a practical assignment. The examination consists of a standard exam and four online quizzes. The examination and practical assignment both comprise 50% of the final

grade. To pass the course, the examination grade should be at least 5.5, the practical assignment grade should be at least 4.5 and the average should be at least 5.5.

The examination is made individually, and the practical assignment is made in groups. There is a resit for the exam, no resit is possible for the practical assignment.

Literatuur

There is no textbook. Some reading material will be provided digitally.

Aanvullende informatie doelgroep

This course is only open to the following programs. If you are not a student of these programs, registration is not available.

It is currently part of the curriculumn of:

Bachelor Artificial Intelligence

Bachelor Business Analytics

Bachelor Computer Science

Premaster Artificial Intelligence

Premaster Business Analytics

It is available as an elective to students of:

Bachelor Mathematics

Bachelor Medische Natuurwetenschappen

If you are in the MSc Bioinformatics, you need to register for*Scientific Machine Learning* (XM_0138), which shares some aspects with this course.

Afwijkende intekenprocedure

Workgroup registration will be done through Canvas in the first week of the course. Only a registration for the main course itself is required to take part in all parts of the course.

Aanbevolen voorkennis

We require that students have some prior experience with linear algebra, calculus (limited to differentiation), probability theory and statistics. An overview and explanation of the required preliminaries can be found at https://mlvu.github.io/preliminaries/

A basic understanding will suffice and we will take some time to go over the basics again. Feel free to register if you have no experience with any of these, but expect to put in a little extra effort in the first weeks (using the materials provided).

Programming experience in python is also required.

Mechanics and Thermodynamics in the Cell

Vakcode	X_422589
Studiepunten	6.00
Periode	P2
Vakniveau	400
Onderwijstaal	Nederlands
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. D. Dulin
Examinator	dr. D. Dulin
Betrokken Docenten	dr. D. Dulin
Onderwijsvormen	Deeltentamen schriftelijk, Werkcollege

Doel vak

Below is provided a point-to-point list of the learning goals of the class that can be used as guidelines for the exams. At the end of this course, the students will:

- know the basic concepts of physics (in particular mechanics, statistical physics and thermodynamics), physical chemistry and mathematics to describe quantitatively biology on the cellular and molecular level.
- use these basic concepts to derive quantitative and predictive models that give a deeper insight into living systems.
- implement and compare different models to analyze quantitatively experimental data obtained from state-of-the-art microscopy approaches and formulate new biological models.

Inhoud vak

1. Basic concepts in thermodynamics and statistical physics to describe biological process such as protein-ligand binding, and polymer conformation.
2. Basic concepts in mechanics to describe how biological beam responds to stress and strain. Combining statistical physics and mechanics to describe semi-flexible polymer such as DNA.
3. Basic concepts of hydrodynamics to understand how liquid flows in the channels, and how dimensions impact the nature of the flow and the motion of objects in liquid.
4. Derivation of the law of diffusion from a macroscopic and microscopic perspective and its application to describe small molecule motion, protein-ligand association kinetics, and molecular motors translocation.

Aanvullende informatie onderwijsvormen

Lectures (4h per week) & Tutorials (2h per week)

Toetsvorm

Written exam (60% of final grade) + written tests (2x, 10% of final grade each) on part of the course contents + tutorial assessments (20% of final grade).

Literatuur

Phillips, R., Kondev, J., and Theriot, J., Physical Biology of the

Cell. 2nd Edition New York: Garland Science, 2012 (ISBN 0815344503). Chapter 1 to 13 (except 9 and 11).

(1st edition is also fine)

Aanvullende informatie doelgroep

3N, 3MNW mi-BB, 3WN, 3S

Overige informatie

This course is part of the Minors Biomedische Beeldvorming and

Bioinformatics and Systems Biology.Mathematics: Calculus & Mathematische

Methoden (or comparable)

Physics: basics of mechanics & thermodynamics

Aanbevolen voorkennis

Mathematics: Calculus & Mathematische Methoden (or comparable)
Physics: basics of mechanics & thermodynamics

Metabolomics

Vakcode	AB_1291
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. P.E.G. Leonards
Examinator	prof. dr. P.E.G. Leonards
Betrokken Docenten	prof. dr. P.E.G. Leonards, dr. L.A. Baumann
Onderwijsvormen	Study Group, Lecture, Seminar, Practical, Written partial exam

Doel vak

Molecular biology studies the functioning of cells and organisms at the molecular level, and is applied in many fields. Metabolomics is a rapidly emerging field in molecular biology, and is one of the so-called omics technologies (genomics, proteomics, metabolomics). Metabolomics measures small molecules – the metabolites – in the cell or organism with the aim of determining the health status. But it can also determine the influence of the environment on health, for example through diet or exposure to toxic substances. Examples of metabolites are amino acids, ATP, sugars, neurotransmitters, lipids, etc.

Metabolomics aims to measure the complete set of metabolites (the metabolome) in organisms and is widely used in the medical, toxicological, and biological field. Metabolites function in a network and are transported to the desired destination via the streets as in a city. Metabolites have a major influence on the phenotype, behavior but also on the development of organisms.

This course aims to provide insight into the principles of metabolism in organisms, how metabolomics can be used in practice, how metabolites can be determined in organisms, and how the large amount of data can be handled, processed, visualized and statistically analysed (Big data analysis). The course has a practical approach and consists of lectures followed by a metabolomics experiment (practical). The practical is an

important part of the course in which the student will conduct an experiment with zebrafish eggs and determine metabolites in the zebrafish embryos. Another important part is the processing of the large amount of data generated by using data visualization and statistical tools (introduction to processing of Big Data).

The following topics are covered:
• What is metabolomics
• What are metabolic networks
• Basic principles and workflows for determining and processing metabolite data
• Methods for the analysis of metabolites
• Applications of metabolomics
• Setting up a metabolomics experiment
• Processing and analysis of large data (visualization, metabolite networks, statistical analysis)

The practical is an important part of the course to give the student a better insight into the various aspects of metabolomics. The student carries out a metabolomics experiment with zebrafish eggs, whereby the development from egg to embryo is monitored. The metabolites (lipids and neurotransmitters) are determined, and the data is processed, among other things by visualizing the data, creating metabolic networks and statistically processing the data.

The learning objectives at the end of the course are:
• The student has knowledge and insight into the topics covered from metabolomics in cells and organisms. The interactions between metabolites, proteins and genes.
• The student has insight into the methods of measuring metabolites and the challenges of studying the metabolome.
• The student gains experience with chemical analysis methods and the analysis of large data sets.
• The student has practical experience in setting up and conducting a metabolomics experiment.
• The student can process and analyse collected experimental data and can do the interpretation.

Inhoud vak

The course provides an introduction to metabolomics and will discuss and use the tools and techniques (practical) needed to conduct a metabolomics study. Various applications of metabolomics in biology, medical and toxicological fields will be discussed. A focus point in the course is the performance of a metabolomics experiment in which you will work with eggs of zebrafish. You follow the development from egg to embryo, determine lipids and neurotransmitters in the embryos, and process and analyse the large amount of data.

In this course a large number of fundamental and practical topics from metabolism and metabolomics are covered such as:
• Metabolic Networks
• Metabolomics (principles, workflows in metabolomics, applications)
• Introduction to chemical analysis methods (chromatography, mass spectrometry)
• Introduction to data processing (including visualization and Big data analysis)

In the practical, the students will apply the material from the lectures via a metabolomics experiment with zebrafish to understand the principles of metabolic networks, the use of analytical equipment to determine metabolites, and the processing and analysis of large data sets.

Aanvullende informatie onderwijsvormen

Lectures (18 hours)

Metabolomics practical (28 hours)

Data processing (20 hours)

Presentation (16 hours)

Practicum report (8 hours)

Self-study (78 hours)

Toetsvorm

Examination (50% of the final grade, average grade at least 5.5).

Presentation (25% of the final grade, average grade at least 5.5).

Practicum report (25% of the final grade, average grade at least 5.5).

Attending practicals and working groups must be sufficient.

Literatuur

Background literature, not needed.

Metabolome analysis : an introduction. Silas G. Villas-Bôas, et al.

2007. ISBN-13: 978-0-471-74344-6. Chapters 1-5.

Aanvullende informatie doelgroep

Part of the minor Biomolecular Sciences.

Mind and Machine

Vakcode	AB_1060
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. L.N. Cornelisse
Examinator	dr. L.N. Cornelisse
Betrokken Docenten	dr. L.N. Cornelisse, dr. K. Linkenkaer Hansen
Onderwijsvormen	Study Group, Excursion, Computer lab, Lecture

Doel vak

To introduce the students to the basic principles of brain modelling, artificial intelligence, and brain computer interfacing, discussing the practical applications as well as the ethical, moral, and philosophical aspects.

Specifically, at the end of the course the student should be able to:
1. explain the meaning of key concepts treated in the course and to give examples of where key concepts are already applied (services or products).
2. describe most commonly used forms of, as well as recent trends in, brain modeling, AI and BCI as discussed in the course.
3. reproduce the underlying principles of brain modelling, AI and BCI at the level discussed in the course.
4. develop, present and defend a business proposal, i.e., an idea for a product or service that exploits the technologies and concepts presented in the course.
5. formulate a scientifically informed opinion about the ethical aspects of AI and BCI.

Inhoud vak

What is intelligence and what is unique about human intelligence? People have always been fascinated with the idea to create intelligent computers and robots and to integrate computers in the brain to enhance its performance. In recent years these technologies have become so advanced that they become more and more present in our daily live, from the personal assistant in your smart phone, to smart software in self-driving cars, to portable EEG headsets. Many see this as the start of a new era where smart machines will be completely integrated in our society, taking over many tasks and services now still done by humans. More dystopian views on the integration of human and machine are shown in science fiction films and series like Terminator 3 and Black Mirror. This has led to the realization that intelligence is not unique to humans but can exist in machines, and forces us to reflect on whether computers could ever reach or surpass human level intelligence, or merge with biological brains. , and if the brain can be of inspiration to improve Artificial Intelligence (AI).

In this course, the basic principles of brain inspired artificial intelligence, realistic computer simulations of the brain, and brain-computer interfacing (BCI) will be discussed. To investigate one of
these topics in more detail students work in groups to start their own virtual start-up company and prepare a business proposal in which they describe a new commercial application of artificial intelligence or brain computer interfacing. The business project is presented to peers and a reviewer during a poster session at the end of the course. In addition, students will gain hands-on experience in computer practicals and an EEG-based neurofeedback competition, and discuss the ethical, moral, and philosophical aspects of artificial intelligence and brain-computer-interfacing.

Disclamer: this course is aimed at students without a background in computer science or AI. It will cover general concepts of AI and BCI, and their impact on society, rather than an in-depth discussion of all technological details.

Aanvullende informatie onderwijsvormen

Lectures 35 hrs
Practicals 8 hrs
Business project 60 hrs

Toetsvorm

Exam 50%
Business project 40%
Discussion 10%

For all three sub-grades a grade ≥ 5.5 must be obtained. It is not possible to compensate.

Vereiste voorkennis

This course is part of the minor Brain and Mind. University students need at least 90 ECTS to be eligible for (courses in) this minor. HBO students can follow (courses in) this minor if they have at least 120 ETCS.

Literatuur

To be decided

Aanvullende informatie doelgroep

All students with an interest in the computational abilities of the brain and brain-inspired technology, that do not have a background in

Artificial Intelligence or computer science.
Due to curriculum overlap, students from Biomedical Sciences and Health and Life Sciences from the VU cannot follow this course.

Overige informatie

Part of minor Brain and Mind.
This minor course requires a minimum of 25 participants to take place.
Central Academic Skills:
Creative thinking to apply learned knowledge in the design and development of a product or service in the field of mind and machine.
Reflecting on acquired knowledge during the course to discuss ethical and philosophical implications.

Aanbevolen voorkennis

Two years of study at bachelor's level. Basic knowledge of neurobiology.

Minor English: English in my own Discipline

Vakcode	L_ETBAALG008
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. G.A. Dreschler
Examinator	dr. G.A. Dreschler
Betrokken Docenten	dr. G.A. Dreschler
Onderwijsvormen	Seminar

Doel vak

You gain insight into features of the English language as used in various text types which are common in your own academic discipline and the professional domains associated with your specialization. After successfully completing this course you will have (i) knowledge of the different types of texts written in your own discipline, either on an academic level or in the professional domain; (ii) insight into linguistic features related to structure, formality and stance in one or two text types in your discipline; and (iii) knowledge of several types of analysis and methods used in genre analysis and corpus linguistics. You will be able to apply these methods independently to a selection of texts from your discipline and use the knowledge gained from these analyses in your own writing.

Inhoud vak

In the first couple of sessions, we will discuss different methods of analysis used in linguistics for analyzing characteristics of texts and apply them to texts. You will then choose one of these methods and apply this in an analysis of a collection of English texts in your own discipline, present the preliminary results of your analysis, and write a final research article in which you report on the analysis, following conventions from linguistic papers.

Aanvullende informatie onderwijsvormen

2 seminars of 2 hours per week in weeks 1 - 3.

Toetsvorm

The grade for this course will be based on the final report (after rewriting) (75%); and the grade for the presentation (25%). In order to pass the course, you need a 5.5 minimum for each component.

Literatuur

Materials will be made available or listed on Canvas.

Aanvullende informatie doelgroep

The Minor Engels/Minor in English as a whole is aimed at bachelor and premaster students across the university who want to improve their written English in an academic context. The Minor is not open for students in the BA programme CIW who are following the track in English Language and Communication Studies.

Overige informatie

Attendance is obligatory (80% minimum). If you miss more classes, you may still be allowed to finish the course if you submit an alternative assignment, but only if you've attended 70% of the classes.

Aanbevolen voorkennis

This course is only available as part of the Minor Engels/Minor in English. Students must have completed Writing 2 before embarking on this course.

Minor English: Grammar and Writing 1

Vakcode	L_ETBAALG007
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	A.J. Gambrel
Examinator	A.J. Gambrel
Betrokken Docenten	dr. G.A. Dreschler, A.J. Gambrel, dr. A. Martonfi
Onderwijsvormen	Study Group, Lecture, Instruction course, Seminar

Doel vak

After successfully completing this course you will have knowledge of and insight into

- the language which typifies academic writing in English and English requirements of text structure, as well as into how these aspects are different from other languages.

-the most important aspects of English grammar, particularly those which typically cause students problems when writing formal English.

-the strengths and weaknesses of your own English writing skills.

After successfully completing this course you will be able to

-write and edit well-structured text in clear, cohesive and formal academic English.

- develop the strengths and reduce the weaknesses in your own writing.

Inhoud vak

The course consists of a writing and a grammar component. In the writing component of the course the emphasis is on (a) identifying the paragraph structures, sentence structures and kind of language used across a range of academic texts in all kinds of disciplines, and (b) getting to grips with the basic problems involved in writing good, formal English (e.g. differences between English and Dutch/other languages, the essentials of English punctuation, formal style). The grammar component consists of a practical introduction to basic aspects of the grammar of contemporary English, with special attention on the problems that students typically have when writing formal English.

Aanvullende informatie onderwijsvormen

For the writing component: 1 hr per week lecture; 2 hrs per week seminar.
For the grammar component: 1 hr per week lecture; 2 hrs per week seminar.

Toetsvorm

(i) a series of writing assignments on a subject related to the student's own discipline (50%);

(ii) a multiple choice computer test on grammar (50%).

In order to pass the course, you need a 5.5 minimum for each component.

Literatuur

For the writing component: Hannay, M. & J.L. Mackenzie (2017). Effective Writing in English. 3rd edition. Bussum: Coutinho.
Materials for the grammar component will be announced at a later stage.
Additional materials will be made available on Canvas.

Aanvullende informatie doelgroep

Bachelor students across the university who want to improve their written English in an academic context; the course is not open for students who have done academic English in their academic core. The course is part of the Minor Engels/Minor in English but can also be followed separately.

Overige informatie

Attendance is obligatory (80% minimum). If you miss more classes, you may still be allowed to finish the course if you submit an alternative assignment, but only if you've attended 70% of the classes.

Note that this is an English writing course rather than simply a writing skills course. The assumption is that participants have already successfully completed an academic skills course in their first or

second year of university study.

Aanbevolen voorkennis

At least one year of university study, including experience in writing

academic text; premaster students may also follow this course as long as

they have completed an academic skills course.

Minor English: Pronunciation and Presentation

Vakcode	L_EABAALG006
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. L.M. Rupp
Examinator	dr. L.M. Rupp
Betrokken Docenten	A.J. Gambrel, dr. L.M. Rupp
Onderwijsvormen	Study Group, Seminar, Lecture

Doel vak

Regarding pronunciation, you will be able to describe the most common

difficulties in English pronunciation. You will also be able to describe

the effects of particular pronunciation features on the intelligibility

and credibility of English accents, as well as the importance of

pronunciation features for identity construction in English. You will be able to able what you have learnt in analyses of your own English accent and the English accents of other speakers.

Regarding presentation, you will be able to strategically apply structural elements, prosodic

features and conversational patterns in such a way that they will help

you convey a clear message.

By the end of the course, you will be able to adapt pronunciation

features in the English accent that you wish to develop and you are able

to confidently give an oral presentation.

Inhoud vak

In the pronunciation module:

1) We will set goals for the English accent that you wish to develop, both professionally and at a personal level.

2) We will look into he most common difficulties in English pronunciation

and the effects that these difficulties may (inadvertently) have on the

intelligibility and credibility of your accent. We will also take

account of features that seem part of your identity.

2) You will conduct exercises, practise ear-training and be given tools that help you

analyse English pronunciation features.

4) You will work on pronunciation features in your own English accent accordingly.

As far as the presentation component is concerned:

1) Students give presentations on academic topics. During these presentations they will show awareness and some mastery over the following elements:

- structure that conveys an academic idea and its relevance and adheres to a time limit
- prosodic features (intonation, voice, articulation, etc.)
- body language (posture, eye-contact, etc.)
- persuasive and conversational patterns (repetition, rhetorical questions, etc.)

2) Students engage in useful ways with their peers, offering feedback and working collaboratively

3) Students demonstrate awareness of their strengths and weakness with regards to the topics and skills addressed in the course.

Aanvullende informatie onderwijsvormen

Pronunciation: weekly lectures (1,5 hour) and seminars (1,5 hours).
Presentation: weekly seminars (3 hours).

Toetsvorm

Two recordings of your own English pronunciation (50%) and a

presentation in English on an academic subject (50%). In order to

successfully complete the course students must score a grade of ≥5.5 for

each of the two components.

80% attendance and submission of the all of the course work by the set

deadlines is conditional for being awarded a grade for the recordings

and the presentation.

Literatuur

Literature and other materials will be made available in Canvas.
The pronunciation module is accompanied by online audio and video material (https://english-pronunciation.weebly.com).

Aanvullende informatie doelgroep

Students across the university who wish to work on their English pronunciation and presentation skills.

Minor English: Writing 2

Vakcode	L_ETBAALG005
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. G.A. Dreschler
Examinator	dr. G.A. Dreschler
Betrokken Docenten	dr. G.A. Dreschler, J.K. Branum
Onderwijsvormen	Seminar, Lecture

Doel vak

After successful completion of the course students will feel confident that they can write a bachelor dissertation in English and embark on a Master's degree where English is the language of tuition. Specifically, the course aims to help students in:

• getting more practice in writing formal, academic English;
• developing reading skills which will allow them to note linguistic and structural features of relevant academic text types in their own discipline;
• gaining insight into how specific linguistic structures can contribute to text coherence and text cohesion;
• acquiring greater knowledge of the stylistic and rhetorical aspects of written formal texts;
• getting greater insight into the strengths and weaknesses of their English writing skills, and knowledge of how to further develop strengths and reduce weaknesses.

Inhoud vak

For this course we focus on your position as a writer in the academic world, i.e. as someone who is engaged in academic discourse. This means that you need to be aware of appropriate structures at sentence level as well as at text level, of ways of using language to refer to other writers, and of ways of using academic language effectively. The emphasis in this course is on (a) gaining more insight into the language and style of your own academic discipline, (b) improving coherence, compactness and readability, and (c) expanding your grammatical repertoire.

Aanvullende informatie onderwijsvormen

2 hrs per week lecture; 2 hrs per week seminar.

Toetsvorm

A series of shorter writing assignments (50% in total) and a final paper of 2000 words on a

subject related to your study (50% of the mark). In order to pass the

course, you need a 5.5 minimum for each component

Literatuur

Hannay, M. & J.L. Mackenzie (2009). Effective Writing in English. 2nd edition. Bussum: Coutinho.
Separate materials available via Canvas.

Aanvullende informatie doelgroep

Bachelor and premaster students across the university who want to

improve their written English in an academic context, with the exception

of students of CIS who are following the specialization in English

Language and Communication Studies, students of Literature & Society.

Overige informatie

Attendance is obligatory (80% minimum). If you miss more classes, you may still be allowed to finish the course if you submit an alternative assignment, but only if you've attended 70% of the classes.

Aanbevolen voorkennis

Students must have either (a) completed an introductory academic English

course earlier in their university studies or (b) already completed

Minor English: Grammar and Writing 1.

Minor Seminar in Political Science

Vakcode	S_MSPS
Studiepunten	6.00
Periode	P1+2+3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	drs. B. Slijper
Examinator	drs. B. Slijper
Betrokken Docenten	drs. B. Slijper
Onderwijsvormen	Study Group

Doel vak

Learning outcomes;

A. Knowledge and understanding - The student has acquired knowledge and understanding of:

(1) the various branches of political science as a discipline, and that of (normative) political theory in particular;

(2) the approaches towards normative questions in political theory;

(3) a number of key topics in (normative) political theory, and the positions and arguments taken on these topics.

B. Skills - The student is able to:

(4) apply normative political theories to concrete contemporary contested issues;

(5) write a short, theoretically embedded and informed position paper on a contemporary contentious issue.

C. Attitude(s) - The student displays:

(6) the willingness to critically (self-)reflect upon the normative dimension of contemporary contested issues.

Inhoud vak

Certainly in the European tradition, political science has been a broad, thematic “discipline” from its inception. In addition to political science as an empirical social science, there is also political science in the tradition of the behavioral sciences such as economics (political economy) and psychology (political psychology), as well as in that of the arts and humanities such as law (constitutional law), history (political history) and philosophy and ethics (political philosophy or political theory).

In the four other courses in this Minor Political Science, the student will predominantly be familiarized with political science as an empirical social (and/or behavioral) science. There, the student will learn how to adequately describe and explain political phenomena.

In this Minor Seminar, we will however engage with normative political theory. The central questions will therefore not be descriptive or explanatory, but normative and evaluative, such as:

- Is democracy the prefered model for the governance of societies? And why (not)?
- How to understand the idea of individual freedom? And what -if at all- should be the limits of that freedom?
- How to understand the idea of human equality? And how should that be translated into policies?
- What is a just and fair distribution of resources, such as income and wealth?
- Is it possible the think in terms just and unjust wars? And if so: what determines the difference?

Students will read some seminal texts from political theorists on these topics, supported by additional introductory

texts. Moreover, the students are invited and stimulated to actively engage with these topics, position themselves within the debate(s), and develop and enhance their skills to clarify and defend their position.

Aanvullende informatie onderwijsvormen

Seminars.

Please note that attendance is mandatory.

Toetsvorm

Participation (complete/incomplete and conditional for a passing grade)

Group presentation and discussion (40%)

FInal position paper (60%).

Vereiste voorkennis

Being signed up for S_IRGG and S_SPC

Literatuur

To be announced in the course manual (see CANVAS)

Multatuli multicultureel

Vakcode	L_AABAALG083
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	prof. dr. J.H.C. Bel
Examinator	prof. dr. J.H.C. Bel
Betrokken Docenten	prof. dr. J.H.C. Bel
Onderwijsvormen	Werkcollege

Doel vak

• Studenten verwerven inzicht in de vele aspecten van één enkele literaire tekst;

• Studenten leren een storyline te ontwerpen voor het Multatuli-museum;

• Studenten zijn in staat in een recensie een onderbouwd oordeel te geven over een literaire tekst;

• Studenten leren een podcast te maken over de Max Havelaar.

Inhoud vak

Na een inleiding over het kolonialisme en de (post)koloniale theorie, wordt één enkele literaire (post)koloniale tekst centraal gesteld, Multatuli's Max Havelaar. Deze roman wordt vervolgens vanuit verschillende perspectieven bekeken, namelijk in literair-historische, receptie-historische, historische, juridische, medische en boek-historische context. Ook wordt het Nachleben van de tekst bestudeerd in de vorm van toneel- en filmbewerkingen. Een excursie naar het Multatuli-museum, het Tropenmuseum en/of het Museum voor Volkenkunde in Leiden maakt deel uit van het programma. Een bekende radiopresentator geeft een gastcollege over het maken van een goed radioprogramma, ter voorbereiding op de eindopdracht (podcast).

Aanvullende informatie onderwijsvormen

werkcollege (2 uur per week).

Toetsvorm

• Tussentijdse opdrachten - mondeling en schriftelijk, waaronder een storyline - (moeten voldoende zijn);

• Recensie over de centrale roman (60%);

• Als eindopdracht maken de studenten een podcast waarbij zij verschillende personages uit de roman interviewen (40%).

Voor alle onderdelen moet minimaal een 5,5 worden behaald.

Literatuur

De te lezen primaire en secundaire teksten worden voor aanvang van de cursus via Canvas bekendgemaakt. De Max Havelaar van Multatuli is verplichte literatuur.

Aanvullende informatie doelgroep

Dit vak is onderdeel van de minor 'De Schrijfacademie'. Deze minor staat open voor alle studenten die hun eerste studiejaar hebben behaald, van binnen en van buiten de VU.

Overige informatie

Deze cursus is een verplicht onderdeel van de minor ‘De Schrijfacademie’.

Multi-Agent Systems

Vakcode	XB_0024
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. K.V. Hindriks
Examinator	prof. dr. K.V. Hindriks
Betrokken Docenten	prof. dr. K.V. Hindriks, dr. M.E.U. Ligthart
Onderwijsvormen	Study Group, Lecture

Doel vak

The course has the following topic and goals.
1. Agents and Multi-Agent Systems
Agent, multi-agent system, cognitive state, beliefs, actions, communication, coordination. The student is able to define the elementary concepts, and to apply their relevant aspects in the design of programs. (Applying knowledge and understanding)
2. Agent-Environment Interaction
Action, percept. The student is capable of analyzing the environment in which the agent operates, and to identify the actions and relevant percepts that are available. The student is able to use the classification, actions and percepts as the basis for the design of an agent. (Applying knowledge and understanding)
3. Agent and Multi-Agent Program
Action rules, modules, MAS file, launch rules. The student is able to describe the relevant programming constructs. In addition, the student is able to apply the constructs to write a multi-agent and agent programs. (Applying knowledge and understanding)
4. Basic Concepts Prolog
Facts, rules, clause, queries, rule-based reasoning. The student is able to define the basic concepts in Prolog and describe the relationship between these concepts. Based on these concepts the student is able to solve simple problems in Prolog. (Applying knowledge and understanding) (Making judgements)
5. Prolog Programming
Negation as failure, cut, recursion, lists. The student is able to apply constructs and these techniques to write a Prolog program. The student is able to solve problems by using a combination of negation as failure,

recursion and the use of lists. In particular, the student is able to implement some search algorithms in Prolog. (Applying knowledge and understanding)
6. Reasoning in Logic Programming
Unification, backtracking, depth-first search, linear search, backward chaining. The student is able to explain the computational model of logic and reasoning in Prolog and to use these concepts. Simple tasks with unification of terms can be made by the student. The student is able to construct a derivation of a unification (resolution). (Applying knowledge and understanding) (Communication)
7. Develop a MAS
The student is able to build a relatively simple multi-agent system. Concepts relating to systems of rational agents are introduced to make complex decisions. (Applying knowledge and understanding) (Lifelong learning skills)

Inhoud vak

AI (Artificial Intelligence) techniques that are discussed in this course are knowledge representation and reasoning techniques, and multi-agent technology. Students are taught how to develop a multi-agent system that uses knowledge representation to reason about the environment in which the multi-agent system operates.

Aanvullende informatie onderwijsvormen

Lectures, lab sessions (with computer/ laptops). Attendance at labs is obligatory.

Toetsvorm

The course is assessed through an exam and a practical part.

The final grade is a weighted average of the individual exam grade (75%) and the

grade of the practical part (25%).

Practical part: Students get 3x a practical assignment that must be made

in pairs. The assignments are graded with a mark. The grade for the

practical part is the average of the marks for the assignments. There

must be a minimum of 5 on average obtained for the assignments to pass

the practical part.

The exam consists of open or multiple-choice questions about all the

material of the course. To pass the exam the grade must be at least a 5.

Both the practical part and exam must at least be a 5 but the final grade should minimally be a 5.5 to pass the course.

There will be a fourth bonus assignment that can be completed for a full

bonus point on the course end grade.

There is no second chance for the practical assignments.

Literatuur

Materials: Slides. Learn Prolog Now! (Blackburn et al.). A cognitive agent programming guide. (all available online)

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence
Bachelor Business Anatlytics

Nation and Migration

Vakcode	S_NM
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	E. Liberati
Examinator	E. Liberati
Betrokken Docenten	
Onderwijsvormen	Lecture

Doel vak

This course introduces students to the study of international migration and how the phenomena, perceptions, and policies towards migration reflect and shape the governance and self-perception of nation-states.

Learning outcomes

Knowledge and understanding. Students have acquired knowledge and understanding of:
(1) the key theories, concepts and methods for the study of migration in the social sciences.

Application. Students have acquired the competences to:
(2) apply these to analyse migration dynamics in selected case studies.

Attitude. Students are able to demonstrate:
(3) a critical perspective on current events surrounding migration.

Inhoud vak

Today’s debates on migration are often inseparable from nations’ perceptions of themselves and each other. This course discusses the relevance of migration in today’s global world, particularly in relation to identity concerns (diasporas, transnationalism, nationalism, multicultural societies), development (migration and development) and international political issues (migration governance and ethnography of the state). The course introduces students to major theories to understand migration, but privileges the adoption of constructivist approaches. It invites students to look at migration from the perspectives of people engaging in migration directly, of people

encountering migrants as new neighbours, or of people tasked with the function of controlling and governing migration. Through this perspective, students engage with processes of community building and belonging, and with the power struggles associated with migration. They acquire a thorough theoretical knowledge and critical view of these phenomena and key concepts that can help understand them. How do migrants construct their identities on the move? How do transnational communities and diasporas develop? How do national communities respond to migration and deal with the diversity that derives from it? How is migration governed and controlled by state apparatuses in migrant countries of origin and destination? How do the bureaucrats and professionals dealing with migration translate migration policies into everyday practices? What are the implications of migration for development and social transformation in both origin and destination societies? We will also consider how the relationship between nation and migration is affected by perceptions of gender and sexuality.

Aanvullende informatie onderwijsvormen

The course will be a combination of lectures and group work. Students should come prepared to discuss readings assigned for each class and answer questions on them. Students will discuss questions in groups and then report back to the plenary.

Toetsvorm

Written assignments and exam(s).

Literatuur

To be announced in the course manual (see CANVAS).

Aanvullende informatie doelgroep

2nd year bachelor students in Cultural Anthropology and Development Sociology;

Students in the Minor Anthropology and in the minor Development & Global Challenges.

Nature versus Nurture

Vakcode	AB_1057
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. R.M. Brouwer
Examinator	dr. S. van der Sluis
Betrokken Docenten	dr. S. van der Sluis, dr. R.M. Brouwer
Onderwijsvormen	Study Group, Computer lab, Lecture, Practical

Doel vak

Students learn how individual differences in human complex behavior can be explained. In this course, the emphasis is strongly placed on recent developments in human genetic research.

Inhoud vak

Humans vary in their behavior and in their vulnerability to disease. In the

Nature versus Nurture course, the influence of genes and of environmental factors on

individual differences in human behavior will be discussed. The course starts with an historical overview of the nature-versus-nurture debate, and describes how twin- and family studies paved the way for current genetic research. The course specifically focuses on the rapid developments in the field of complex trait genetics, that allow researchers to search the entire genome for parts of the DNA that are predictive of human behavior and disease. We will examine how the research design and definition of phenotype influences the outcome and interpretation of these studies. We will discuss recent large-scale gene-finding studies and discuss how these translate to state-of-the-art neuroscience research, and how they inspire studies on personalized treatments in medical practice. In addition, ethical aspects of current genetic research will be discussed.

Aanvullende informatie onderwijsvormen

Lectures, discussion groups, presentations.

Toetsvorm

The final grade of Nature versus Nurture is based on weekly assignments, a group presentation on an ethical aspect of genetic research, and a written exam (multiple choice and open questions). The course Nature versus Nurture is successfully completed if both the exam grade and the final aggregate grade are 5.5 or higher.

Vereiste voorkennis

This course is part of the minor Brain and Mind. University students need at least 90 ECTS to be eligible for (courses in) this minor. HBO students can follow (courses in) this minor if they have at least 120 ETCS.

Literatuur

Lecture notes, scientific papers, TBA during course

Aanvullende informatie doelgroep

This course is part of the minor "Brain and Mind". This minor is accessible to third year BSc students from alpha, beta, and gamma sciences (e.g., Sociology, Psychology, Economics, Law, Artificial Intelligence, computer science, criminology) and students from Life sciences (e.g., Biology, Physics, Chemistry, Medicine, Movement Science, Nutrition) with a broad interest in neuroscience. Students of Biomedical Sciences and Health and Life Sciences as well as students who plan to pursue a career in Neuroscience cannot follow this minor but are referred to the morespecialized minor "Biomolecular/Neurosciences".

Aanbevolen voorkennis

Broad interest in brain, behavior, psychology, genetics and neuroscience

Neuro- en Revalidatiepsychologie

Vakcode	B_NEURREVPSY
Studiepunten	6.00
Periode	P3
Vakniveau	200
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. A. Ledebt
Examinator	dr. A. Ledebt
Betrokken Docenten	dr. A. Ledebt, dr. J.F. Stins
Onderwijsvormen	Hoorcollege, Hoorcollege, Werkgroep

Doel vak

Na afloop van de cursus heeft de student een goed beeld van hoe neuropsychologische kennis, theorieën en onderzoeksmethoden kunnen bijdragen aan het revalidatietraject van mensen met hersenletsel (met name CVA). Ook zal de student in staat zijn wetenschappelijke artikelen over dit onderwerp kritisch te lezen en vragen over deze artikelen te beantwoorden en te presenteren aan medestudenten.

Inhoud vak

Neuropsychologie onderzoekt de relatie tussen hersenen en gedrag, en in het bijzonder de gevolgen van hersenschade op verschillende gedragsniveaus. Neuropsychologische stoornissen kunnen cognitief van aard zijn (bv geheugen, spraak), emotioneel (bv de stressreacties) of gedragsmatig (bv vermoeidheid en motivatie). Het vakgebied van de revalidatiepsychologie belicht de talrijke psychologische processen die inherent zijn aan de revalidatie van mensen met een lichamelijke en/of neurocognitieve aandoening. Al deze psychologische facetten hebben grote invloed op de kans van slagen van het traject van (neuro)revalidatie. De cursus beoogt een brug te slaan tussen enerzijds de theoretische inzichten uit de neuro- en revalidatiepsychologie, en anderzijds de uitdagingen die optreden in de praktijk van de revalidatie. De nadruk zal liggen op hersenschade na CVA (wat abrupt optreedt) en niet op neurodegeneratieve aandoeningen zoals dementie en de ziekte van Parkinson. De bewegingswetenschapper die werkzaam zal zijn in de revalidatie (bv als behandelaar of wetenschappelijk onderzoeker) zal na afloop van deze cursus uitstekend in staat zijn om psychologische en neuropsychologische kennis en instrumenten in te zetten in dit werkveld.

De cursus is opgedeeld in 3 onderwerpen, die elk in een afzonderlijke cursorische week worden besproken, als volgt: (1) psychologische factoren in de (neuro)revalidatie, (2) cognitieve stoornissen in de (neuro)revalidatie, en (3) motorisch (her)leren na beroerte.

Aanvullende informatie onderwijsvormen

De cursus bestaat uit hoorcolleges en werkcolleges.

Toetsvorm

Het eindcijfer bestaat uit de score behaald op het schriftelijk tentamen (70%) en de punten behaald tijdens de opdrachten (30%). Schriftelijk tentamen met open-eindvragen en/of gesloten vragen.

Literatuur

Losse artikelen. De literatuurlijst en de Cursushandleiding worden tzt online bekendgemaakt.

Overige informatie

Voor de werkcolleges geldt een aanwezigheidsplicht. Werkcolleges worden

voorafgegaan door een aantal opdrachten (bv vragen over literatuur) die op Canvas worden geplaatst. Alle studenten (werkzaam in groepjes) moeten opdrachten maken en inleveren. Tijdens de werkcolleges moeten studenten hun antwoord presenteren voor de groep.

Afwijkende intekenprocedure

Wegens de werkcolleges en bijhorende opdrachten is een maximum van 100 deelnemers mogelijk.

Toelichting Canvas

Studenten werken gedurende de cursus in tweetallen (soms drietallen) aan de groepsopdrachten. Studenten mogen zelf aan het begin van de cursus groepjes vormen via Canvas. Studenten die geen partner hebben worden door de docenten toegewezen aan een groep.

Aanbevolen voorkennis

Van de deelnemers wordt verwacht dat zij globaal kennis hebben van neuroanatomie en neurofysiologie zoals bijvoorbeeld behandeld in het boek “Neurowetenschappen, een overzicht” van Ben van Cranenburgh.
De cursus sluit wel aan bij een aantal cursussen die eerder in de Bachelor Bewegingswetenschappen worden gegeven, te weten: Psychologie (BA1); Motorisch leren en ontwikkelen (BA2); Neurowetenschappen (BA2); Revalidatie (BA3).

Neuropsychology (UM)

Vakcode	P_UNEUPSY
Studiepunten	6.00
Periode	P3
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. D.A. Crafa MSc
Examinator	dr. D.A. Crafa MSc
Betrokken Docenten	dr. D.A. Crafa MSc
Onderwijsvormen	Study Group, Lecture

Doel vak

To receive knowledge and insight on an introductory level on the consequences of brain damage on the behaviour of people. Also to be introduced to the field of the clinical neuropsychologist.

Inhoud vak

Using our knowledge of the working of the brain we discuss the principles of neuropsychology. Neurological and psychiatric disorders as agnosis, aphasia, and dementia. On the basis of patient examples that correspond as closely as possible to the literature, the neuropsychological diagnostics, among others, are illustrated.

Aanvullende informatie onderwijsvormen

Lectures

Toetsvorm

Exam (mc questions)

Vereiste voorkennis

Introduction to PSY and Biological PSY are required for this course.

Literatuur

Kolb, B. and I.Q. Whishaw, (2021, 8th Edition). Fundamentals of Human

Neuropsychology. Worth Publishers.

This edition of the textbook differs significantly from past editions and cannot be substituted.

Aanbevolen voorkennis

This is a UM course, but other students who meet the entry requirements may also enroll.

This course requires basic knowledge of psychology, but its contents may be too basic for psychology or medical students.

New Ways of Working

Vakcode	E_MUM_NWW
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. A. Sergeeva
Examinator	dr. A. Sergeeva
Betrokken Docenten	dr. A. Sergeeva
Onderwijsvormen	Study Group, Lecture

Doel vak

After completing the course, students will:
- Understand how the properties of digital technologies require, as well as enable,
new approaches to working and organizing
- Have knowledge of relevant theories of how working, coordinating, and managing
in these new environments is different from traditional workplaces and critically reflect upon the underlying assumptions of these theories
- Understand the interplay between technology and work practices and be able
to analyze and demonstrate that interplay
- Be able to apply academic insights to analyze and develop solutions for real
life problems

Inhoud vak

“New ways of working” refers to new practices through which knowledge workers collaborate in increasingly distributed and flexible organizational contexts enabled by digital technologies. In this course we elaborate on the nature of these new practices and critically examine the role of technology in various aspects of work and organizing. The lectures are structured in such a way that we first

discuss the academic understanding of what constitutes “work” and “technology” in the modern world. We then move on to critically reflect on how particular aspects of work are being challenged by digital transformation. The main themes discussed in this course revolve around 1) the role of physical space and co-location for organizing 2) the role of knowledge in organizing 3) the nature of occupations and occupational boundaries.
In general, during the course, we debate such questions as “How does automating work impact existing occupational relations in the workplace?” We learn to approach such questions critically, equipped with theory and practical experience. In order to gain more in-depth understanding of one of the new work forms, you also engage in a team project, where you select one specific trend in the world of work to study in depth. You examine this trend through the lens of the theory and show that you are able to analyze it in an academic way. In such a way, you learn to apply the basic concepts to examine one particular emerging digital phenomenon in the world of work, and develop recommendations for companies operating in the digital era.

Aanvullende informatie onderwijsvormen

The course will consist of a combination of interactive lectures, student-led sessions and tutorials. The lectures aim at developing your academic skills, and providing you with new knowledge. The lectures are based on the literature and discuss the concepts that will be later used in the group sessions and your specific project. Since the lectures are interactive, you are expected to attend to the lectures having read the literature and contribute actively with your answers to questions.
In the student sessions we will use a variety of interactive formats to examine the course topics in
more depth.

Aim of the student sessions is to (a) facilitate a deeper understanding of how the literature relates to contemporary problems in the world of work (b) critically reflect on the literature applying it to particular cases (c) provide you a more hands-on experience of new ways of working enabled by digital technologies. You will have to present, compare and contrast, criticize and apply the readings to various real cases.
Student sessions also help you to prepare for the final exam and perform a team project.

The tutorials are aimed to assist you in completing a team project. Two tutorials will be dedicated to
the presentations of your interim results, while other sessions will be used to provide you with feedback on your progress towards fulfilling the project.

Toetsvorm

The overall grade will consist of your grade received for the final individual exam (60%), a group project (40%) as well as in-class quizzes (pass / fail) that will assess your learning progress and class participation.

In order to pass the course, you need a minimum of 5,5 as a total course grade. In addition, the grade for each assignment and an exam must be at least (not lower than) 5.00.
Assessment matrix in the course manual provides an overview of the learning objectives assessed in each of the different parts.

The exam will consist of open questions that focus on your understanding and application of these concepts discussed during the course and sometimes will require you to provide examples to illustrate your understanding of the material. Exam questions can relate to the course readings and discussion in class (lectures and tutorials).

Vereiste voorkennis

None

Literatuur

A selection of readings (mostly academic papers, but also book chapters and thoughtful business magazine articles) will be made available befor the start of the course.

Oral History & Biography

Vakcode	L_AABAGES208
Studiepunten	6.00
Periode	P3
Vakniveau	200
Onderwijstaal	Nederlands / Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. N.F.F. Karrouche
Examinator	dr. N.F.F. Karrouche
Betrokken Docenten	dr. N.F.F. Karrouche, dr. D.G. Hondius
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

Aan het einde van de cursus zijn de studenten in staat om:

1. De kenmerken van oral history en biografie als bronnen en methoden van historisch onderzoek te begrijpen en hun waarde voor historisch onderzoek te beoordelen.
2. Een oral history interview voor te bereiden, af te nemen en te interpreteren, en op te slaan in een digitaal archief voor hergebruik.
3. Kritisch te reflecteren op oral history en de biografische methode en een eigen positie te onderbouwen en in te nemen in (gevoelige) academische debatten over deze bronnen en methoden, inclusief hun digitale hergebruik.

Inhoud vak

Dit is een intensieve methodologische cursus over oral history onderzoek en de biografische methode. In een reeks werkcolleges leer je hoe je mondelinge geschiedenissen en biografische bronnen kunt beoordelen, interpreteren en hergebruiken voor historisch onderzoek. Je leert ook oral history interviews af te nemen door samen met je medestudenten een collectie over een vooraf bepaald onderwerp samen te stellen.

Aanvullende informatie onderwijsvormen

Werkcolleges, 4 tot 6 uur per week. Aanwezigheid is verplicht.

Toetsvorm

- Participatie
- Oral history interview
- Essay

Vereiste voorkennis

Studenten moeten het eerste jaar van hun studie hebben afgerond.

Literatuur

Lynn Abrams, Oral History Theory (2016) en een selectie artikelen. De volledige leeslijst wordt bekendgemaakt bij aanvang van de cursus.

Aanvullende informatie doelgroep

Deze cursus staat open voor BA 2 studenten Geschiedenis en History & International Studies en studenten in de minor History. Andere studenten van Geesteswetenschappen en Sociale Wetenschappen zijn welkom, maar moeten al hun eerstejaars vakken hebben afgerond en moeten eerst contact opnemen met de coördinator om toegelaten te worden tot de cursus.

Overige informatie

Dit vak maakt deel uit van de minor History.

Toelichting Canvas

Lesmateriaal, inclusief handouts en powerpoints, wordt na elk seminar online beschikbaar gemaakt op Canvas. Het handboek en de gebruikte literatuur zijn digitaal beschikbaar via de universiteitsbibliotheek.

Aanbevolen voorkennis

Studenten moeten eerstejaarsvakken academische vaardigheden succesvol hebben afgerond.

Peace and Conflict: From Theory to Practice

Vakcode	S_PC
Studiepunten	6.00
Periode	P1+2+3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. M.R. Luitjens
Examinator	dr. M.R. Luitjens
Betrokken Docenten	dr. M.R. Luitjens
Onderwijsvormen	Lecture

Doel vak

The primary objective of the course ‘Peace and Conflict: From Theory to Practice’ is to familiarise students with the practical context of issues of peace and conflict.

When finalising the course, students will:

- have knowledge and understanding of the practical context regarding peacebuilding as experienced by practitioners;
- be able to apply their knowledge, understanding and problem-solving abilities to specific cases within the broader multi/interdisciplinary and multilevel context of peace and conflict;
- be able to critically assess peacebuilding attempts (in the broadest form), using their knowledge base built throughout the lectures;
- be able to communicate the knowledge and rationale that underpin their conclusions to an audience of specialists and non-specialists in a clear and unambiguous manner;
- be able to analyse, synthesise and co-create potential solutions to complex problems in the domain of peace and conflict;
- be able to give and receive constructive feedback to and from their peers on a variety of products, including oral presentations and written work, and use this feedback to revise and improve their work;
- be able to work in multi-disciplinary student teams and to contribute to the functioning and productivity of the group by sharing insights from their own disciplinary background.

Inhoud vak

Since the end of the First World War, the world has seen various waves of peacebuilding practices. During this course, these waves will be critically analysed and as such the complexity of peacebuilding practices will be presented. The approach that professionals adopt to deal with such complexities is central to the course ‘Peace and Conflict: From Theory to Practice’. Whereas the other courses in the minor ‘Peace and Conflict Studies’ deal with the philosophy and ethics of political violence, criminological and psychological approaches to violent behaviour, international security, and intersectional perspectives on peacebuilding and reconciliation, this module adopts a crucial focus of practical engagement. This course familiarizes students with a genealogy of peacebuilding, demonstrating the ways in which peacebuilding practices have altered over the course of the last centuries. This provides a backbone, and as such a critical analytical framework, to understanding the environment that practitioners in the peace and conflict sector work in. It thereby aims to familiarise students with the very practical complexities that governments, inter-governmental structures (e.g., European Union, United Nations), non-governmental organisations (NGOs), or Civil Society Organizations (CSOs) face when working in the context of peace and conflict.

Through a series of lectures and guest lectures by practitioners, and by working together in multidisciplinary student teams, students in this course are invited to combine theoretical knowledge with insight into these practical challenges, and to apply what they have learned throughout the Minor ‘Peace and Conflict Studies’ to complex, real-world case studies of international responses to (political) violence and armed conflict. Working in this setting will provide the students with a critical lens to understanding the complex and challenging work in peace and conflict settings.

Aanvullende informatie onderwijsvormen

Weekly lectures in P1, P2 and P3.

Toetsvorm

Exam (40%); Final (video) group project (35%); Guest-lecture assignment (25%); Film-assignment (pass/fail).

Students need to obtain a 4 minimum on each of the assignments.

Vereiste voorkennis

Students must have earned 90 credits, including 60 of the first year.

Being registered for 'Political Violence and the Human Condition' and 'Philosophy and Ethics of Political Violence: Peace, War and Terrorism' is required.

Aanvullende informatie doelgroep

This course is intended for students in the minor Peace and Conflict Studies. The course is also open as an elective course to students from any Dutch university or other institution of higher education, and exchange students.

Overige informatie

The minor is taught in English.

Philosophy and Neuroethics

Vakcode	W_BA_PNEU
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. C.B. Ranalli
Examinator	dr. C.B. Ranalli
Betrokken Docenten	L.C. de Bruin, dr. P. Robichaud, dr. S. Coninx, dr. G. Löhr, dr. C.B. Ranalli
Onderwijsvormen	Study Group, Lecture

Doel vak

Students are introduced to the most important concepts and positions in philosophical debates on the impact of neuroscience and neurotechnologies on individuals and society.

Lerning Objectives

1. You can describe, explain, and reflect on the main philosophical positions and concepts regarding the mind-body problem and identify what they imply for empirical research in psychology and neuroscience;

2. You are able to describe, explain, and reflect on philosophical positions and concepts regarding science and

objectivity and identify how they affect contemporary discussions on the impact of neuroscientific research and neurotechnologies;

3. You can describe, explain, and reflect on the most important ethical frameworks and identify how they affect contemporary discussions on the impact of neuroscientific research and neurotechnologies;

4. You can describe, explain, and reflect on current topics in neuroethics and identify their relevance for society (e.g., neuro-enhancement);

5. You are able to give an assessment of the ethical and epistemological implications of advancing developments in artificial intelligence research both on an individual and societal level (e.g., AI and autonomy; information bubbles and echo chambers);

6. You can describe, explain, and reflect on the main philosophical positions on free will and identify what they imply for neuroscientific research as well as our self-understanding as humans.

Inhoud vak

Neuroscientific research and emerging neuro-technologies give rise to new social, legal, epistemological, and ethical issues. In this course, students are familiarized with key concepts, positions, and arguments in philosophical debates on these topics, including (1) common approaches to the mind-body problem and their implications for scientific research; (2) philosophical underpinnings of contemporary debates on science and objectivity; (3) central ethical frameworks and their applications; (4) current topics in neuroethics and their relevance for society;(5) implications of certain advancing developments in artificial intelligence research, and (6) the main philosophical debates on free will and their potential significance.

Aanvullende informatie onderwijsvormen

(Interactive) lectures

Toetsvorm

Written exam (100%) (Learning Objectives 1 - 6)

Literatuur

Course manual and literature will be published a few weeks ahead of the beginning of the course on Canvas.

Aanvullende informatie doelgroep

This course is part of the Universiteitsminor Technology, Law and Ethics.

Aanbevolen voorkennis

Philosophy requires active reading and thinking. It's helpful to read How to read philosophy before embarking on this course.

Philosophy and the Ethics of Political Violence: Peace, War and Terrorism

Vakcode	S_PEV
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. A.F. Pacheco Lozano

Examinator	prof. dr. G.J. Buijs
Betrokken Docenten	prof. dr. G.J. Buijs, prof. dr. F. Enns
Onderwijsvormen	Seminar, Written partial exam

Doel vak

The aim of this course is to introduce students to the main philosophical and ethical concepts on violence and non-violence, war and peace (-building), and the phenomenon of terrorism – in the context of a globalized world. Special attention will be given to religiously motivated violence and the potential role of religion in peace-building, including a postcolonial lens.

Learning Objectives

When finalizing the course, students will have knowledge and understanding of

- The prospects and problems of the main political-philosophical and ethical concepts of war and peace;
- The prospects and problems of violent and non-violent peacebuilding concepts;
- The historical, cultural and economic contexts in which certain approaches concerning war and peace have emerged and have been applied;
- The prospects and problems of religion as one of the main contributors to violence (terrorism) as well as to non-violent peace (-building) in a globalized world.
- Becoming aware of the continued challenges of colonial mindsets and structures.

Inhoud vak

Philosophy Part

As far as we go back in history, we find traces of human violence. Does this mean that human nature is violent per se? Or has violence other sources and causes? This type of questions has occupied philosophers, especially in modern times, the times that are characterized both by a continuation of the high level of violence that we encounter throughout human history and by an increasing moral denouncement of violence. So, in philosophy, key questions that are raised with increasing intensity are: what is violence, how does it occur and can we overcome violence and enter into a state of non-violence or even ‘eternal peace’? Is violence ‘selective’, i.e. does it affect certain groups more than others, e.g. women? There is also an increasing focus on the aftermath of conflict and oppression in history. How can we deal with a violent past? How can ‘restorative justice’ ever be just?

Ethics Part

For centuries, in the Western World the concepts of war and peace have been developed and discussed in the field of philosophy linked to theology, due to the fact of the corpus christianum (the medieval concept of a unity of church and state). The just-war-theory is the predominant model of reasoning in this tradition, challenged only by some religious minorities who pronounced non-violence as the moral obligation within Christian ethics. During the Enlightenment period, this societal unity of political and religious powers begins to fall apart, due to new ways of thinking and reasoning. This has lead the (European) societies into violent (freedom-) struggles within, resulting in a clear separation of “church and state”. This paved the way to secular states on the one hand and religious plurality on the other. Yet, it also supported colonization.

Current phenomena like forms of terrorism, “New Wars” as well as the “Renaissance of the Just War theory” demonstrate, that moral reasoning of religious communities still plays a major role in orienting people of faith – and implicitly also people of no faith – in their ethical judgements. This is not only true for some ethical dilemmas (such as collective self-defense, emergency assistance for populations at risk or violent struggles for political liberty and independence) but also for concepts of non-violent resistance, peace-building, and reconciliation (see Mahatma Gandhi, Martin Luther King, Nelson Mandela, World Council of Churches etc.). – In times of economic globalization, cultural diversity, and religious plurality the discourse on war, (non-) violence and peace (-building) finds itself in rapidly changing contexts – and new forums of analysis and engagement, including postcolonial studies.

Aanvullende informatie onderwijsvormen

Seminar-style with presentations and discussions.

Attendance mandatory (80%)

Toetsvorm

For the Ethics part: Group assignments and final exam (50%)

For the Philosophy part: Reading Reports and final exam (50%)

Literatuur

The Ashgate Research Companion to Religion and Conflict Resolution, ed. by Lee Marsden (2012).

Additional articles will be provided.

Aanvullende informatie doelgroep

Bachelor students in the minor Peace and Conflict Studies.

The course is also open as an elective course.

Overige informatie

Week 1-3 PHILOSOPHY (Buijs)

Week 4-6 ETHICS (Enns & Pacheco Lozano)

Philosophy of Science

Vakcode	W_BA_SCIE
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. J.S. de Boer
Examinator	dr. J.S. de Boer
Betrokken Docenten	dr. J.S. de Boer
Onderwijsvormen	Lecture

Doel vak

1. The student has achieved a broad knowledge of central debates in

contemporary philosophy of science.

2. The student understands the key concepts and arguments employed in

these debates.

3. The student is able to present his/her own well-argued views

regarding the nature of scientific knowledge and its role in today's

culture and society.

4. The student is able to apply the general philosophical analyses to concrete cases, and can explain the relevance of these analyses to science practioners.

Inhoud vak

This course will address key-subjects in philosophy of science:

- Classical philosophy of science;

- Historical and sociological research in science;

- Realism and the nature of scientific explanation;

- Science and values;

- Feminist's perspectives on science.

- Philosophy of science issuues in various disciplines

-Students also learn how to assess the value of resources according to the purpose they serve

Aanvullende informatie onderwijsvormen

Lectures, assignments, debates

Toetsvorm

Exam with MPC questions and open questions (60)%: objectives 1, 2 and 4

Writing assignment (40%): objectives 3 and 4.

Quizzes: pass/fail.

Passing the quizzes is prerequisite for taking the exam.

There is a minimum requirement of 5.0 for both the

exam and the writing assignment.

Literatuur

Kent W. Staley, An Introduction to the Philosophy of Science (Cambridge University Press, 2014). Other required readings will be announced through Canvas.

Aanvullende informatie doelgroep

BA2-students Philosophy; students in the Minor Philosophy.

Aanbevolen voorkennis

Recommended is a general familiarity with systematic or historical problems from philosophy.

Political Philosophy

Vakcode	W_BA_PP
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	prof. dr. G.J. Buijs
Examinator	prof. dr. G.J. Buijs
Betrokken Docenten	prof. dr. G.J. Buijs
Onderwijsvormen	Lecture

Doel vak

At the end of this course, the student has achieved the following goals:
Knowledge and understanding:
1. Knowledge and understanding of the contemporary political and political-philosophical landscape: you are able to recognize the most important viewpoints/political ideologies within contemporary political philosophy and the contemporary political landscape, you can indicate the core convictions of each of them and how each ideology relates to the other ones.

2. Philosophical analysis: you can identify and understand the

underlying presuppositions of the various schools of thought and critically analyze these presuppositions.

3. Getting acquainted with "famous texts": for each movement you can indicate why a particular text fragment from the tradition of political philosophy is typical or foundational for a particular political-philosophical movement.

4. Analysis of current ‘hot topics’ in the public-political sphere: you will study a number of current social themes and learn to recognize how different schools of thought think about these issues. You will also learn to formulate and argue your own point of view, in an interdisciplinary context with students from other VU-faculties.

Skills:
5. Learning to identify key points in the study material: every week you write a short reflection on readings.

6. Debating skills: a round of debates is held every week in which we try to determine the 'solving potential' of a particular school of thought for an urgent social problem. You will make a brief analysis of this problem on the basis of internet, newspapers and magazine articles and statistical data and you will learn to introduce the results of your analysis briefly and (hopefully) convincingly in a debate.

7. Dialogue and reflection: In the VU-wide Broader Mind Course associated with this course you learn to approach personal and social themes from various disciplinary perspectives, to reflect on them and to discuss these themes with students from very different disciplinary and cultural backgrounds.

Inhoud vak

Political philosophy or political theory (often used interchangeably) deals with hot topics in today’s society. Together we explore the ideas and visions that people use when designing, redesigning or assessing the way their society is organized. In this course you get to know the most important past and contemporary ideologies within the modern political spectrum, such as liberalism, socialism, feminism, ecological thinking. We also look - particularly during the lectures - at the historical-cultural background of these ideologies: where do they come from, what are the core ideas?

And we practice as well public debates, and reflecting on your own opinions, together with people from very different backgrounds. Listening is as important as talking. This is partly done in the lectures, partly in the university-wide Broader Mind Course, which is associated with this course.

We always try to do this in an informed, academic way: arguments are key. But in the selection of arguments political ideologies are important maps for people. In the past, from Plato, but certainly in modern times, all kinds of ideas and visions have been formed that inspire people and on which people rely on in dealing with issues in society.

The course has two parts. The main part will be offered in Period 1. Next to this you will participate, with your philosophical background acquired in Period 1, in the university-wide offered Broader Mind Course, which is organized in the spring (period 4 and 5, in total 40 hours), in which you will meet students from all faculties of the VU and together work on and discuss themes that focus on personal development and societal challenges (and the relations between those). Through a smart mix of online assignments and carefully scheduled meet-ups is the Broader MInd Course organized in such a way that it can be very well combined with other courses.

In the main course, we will study poltical ideologies and apply these already to current societal issues as well as to some themes that will be discussed as well in the Broader Mind Course. What do the various political ideologies have to say on inequality, on climate change, on abortion rights, on drugs use, on education? You may self choose the topic that you want to explore and discuss. them. Discussions will take place as 'talkshows'.

Aanvullende informatie onderwijsvormen

In Period 1: Watching videoclips on the main materials, class discussions and collective readings, debates/talkshows. The Broader Mind part consists of 'blended learning': online assignments, to be fullfilled at a time of your own choosing, and offline meetups in interdisciplinary groups of 25 students, working together on experience-based learning assignments. In total the Broader MInd part will take about 40 hours, over 4 months, so about 2,5 hours per week, no activities in exam periods.

If for certain reasons it really is not possible to take the Broader Mind Course in the Spring, there is a special Autumn group that can take this course simultaneously the current couse in the first semester.

Toetsvorm

- Written assignments and participation in talkshows (both: pass/ fail), meant to practice goals 4,5, 6
- Written exam that will determine your grade (Goals 1,2,3)
- Broader Mind certificate which will validate your grade (achieving goals 4,6,7)

Literatuur

Key reading is Andrew Heywood, Political Ideologies. An Introduction. London: Palgrave Macmillan (7th edition) - almost the entire book.

- fragments of "classical" texts or source texts of a certain ideology. (on Canvas)

Aanvullende informatie doelgroep

BA2-students Philosophy, track English

Overige informatie

If you want to have a preview of the Broader Mind Course, see

https://vu.nl/en/student/extra-challenge/broader-mind-course

Aanbevolen voorkennis

It may help when you have already taken the introductory Ethics Course and the course on social & cultural philosophy but this is not a formal entry requirement.

Political Violence and the Human Condition

Vakcode	S_PVHC
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. M. Weerdesteijn MSc
Examinator	dr. M. Weerdesteijn MSc
Betrokken Docenten	W.E. Born, dr. M. Weerdesteijn MSc
Onderwijsvormen	Seminar, Lecture

Doel vak

This course aims to provide students with knowledge and understanding of the perpetrators and victims of political violence. It will cover the reasons why, and the processes and mechanisms through which, people get involved in political violence, as well as the impact this has on victims and their communities.

When finalizing the course, students will have knowledge and understanding of:

- The driving forces of violent behaviour on an individual and group level;

- The psychological foundations of violent behaviour;

- The adverse psychosocial and intergenerational consequences of violence for victims and communities;

-Effective strategies to hold perpetrators accountable and reduce the adverse psychological and psychosocial consequences of violence for victims and communities;

- The prospects and challenges of an interdisciplinary approach to violent behaviour that combines, amongst other disciplines, psychology and criminology.

In addition, students will be able to:

- Apply theories of multiple disciplines to understand the causes and consequences of violence as well as the effectiveness of post-conflict interventions;

- Critically evaluate the relevance of different theories from multiple disciplines to help understand contemporary cases of political violence and their aftermath;

-Critically appraise and integrate the literature on psychological and psychosocial interventions as well as transitional justice mechanisms in the aftermath of political violence.

Inhoud vak

Mass atrocities are frequently perpetrated during wars and they have a devastating effect on the victims and their communities. The perpetrators and the victims of this violence have been studied from numerous disciplines including, but not limited to, criminology, clinical psychology, social psychology and history. Studies across these different disciplines have focused on elements and processes that influence the perpetrator and contribute to mass violence. In addition, extensive scientific literature exists on the consequences of mass violence for the exposed society, community and the individual, and how

individuals, communities and countries may deal with the past.

In this course, these perspectives will be integrated to provide an overview of the reasons why, and the processes through which, individuals perpetrate mass atrocities. The hypothesis that these individuals are ordinary people in extraordinary circumstances will be discussed by analysing theories as well as case studies. Furthermore, the appropriateness of individual accountability for these collective manifestations of political violence will be discussed, as well as

potential alternatives. A second central focus of the course will be the psychological and psychosocial consequences of political violence and war-related trauma for its victims; the affected individuals, communities and societies. In this part of the course we will also focus on how to interfere with the development of such adverse consequences, on an individual and societal level.

Aanvullende informatie onderwijsvormen

Lectures and seminars

Toetsvorm

Written assignments

Vereiste voorkennis

Being registered for 'Philosophy and Ethics of Political Violence: Peace, War and Terrorism' is required.

Literatuur

Selected articles and book chapters that are available on-line

Overige informatie

This course is only available for students who are also enrolled in the course Philosophy and the Ethics of Political Violence: Peace, War and Terrorism

Principles of Bioinformatics

Vakcode	X_401094
Studiepunten	6.00
Periode	P3
Vakniveau	300

Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. E. Bosdriesz
Examinator	dr. E. Bosdriesz
Betrokken Docenten	dr. ir. K.A. Feenstra, prof. dr. J. Heringa, dr. E. Bosdriesz, dr. J.M. Urtasun Elizari, dr. Q. Peng MSc
Onderwijsvormen	Lecture, Computer lab, Practical

Doel vak

Are you interested in bioinformatics? Would you like to know how huge amounts of data can be analyzed in order to discover new biology? Would you like to solve open questions in scientific research? This course is open for any Bachelor student in a Science Degree (including Biology or Biochemistry).

Principles of Bioinformatics is the concluding course for the minor Bioinformatics and Systems Biology. The lectures provide a comprehensive overview of key topics in the field. In practical sessions, you'll work on a bioinformatics benchmark, gaining hands-on experience with current algorithms and uncovering areas for further advancement in this rapidly evolving research field

Learning goals

The student:

• is aware of the major issues, methodology, data resources, and available algorithms in bioinformatics. [Knowledge and understanding]

• can collaborate on a project in a group of diverse backgrounds. [Communication]

• gains hands-on experience in scripting and handling basic mathematical equations as a means of solving bioinformatics problems. [Applying and understanding]

• Ask questions about something that is unclear. [Communication]

• Knows what biological databases contain and how to use them. [Applying and understanding]

• Understands and can explain what a bioinformatics methods benchmark is and why it is relevant. [Knowledge and understanding]

Inhoud vak

The following topics are covered:

• Evolutionairy profiles

• Data resources in for the Life Sciences, including the Gene Ontology Database (GO)

• Homology Searching (BLAST / PSI-BLAST)

• Genomics, Massively Parallel Sequencing (MPS)

• Benchmarking

• Computational Analysis of Genome Sequencing (Genome Assembly)

• Differential gene expression

• Clustering, machine learning and pattern recognition

There are practical sessions that aim to show you existing solutions within the field of Bioinformatics. In the practicals you will use existing databases and (web-server) solutions to solve biological problems. You will also use python scripts to automate queries to databases and web servers to investigate the value of current Bioinformatics Algorithms.

Practical work:

One individual assignment:

• Homology searching

One group assignment in Python including:

• Running (PSI-)BLAST in Python, using the GO database, creating a ROC plot

Aanvullende informatie onderwijsvormen

• 6 lectures together with 'Introduction to Bioinformatics' (2 hours per lecture)

• 2 lectures specific to 'Principles of Bioinformatics'

• 6 group practical sessions (two hour sessions), partially supervised.

• 1 small assignment of which one together with 'Introduction to Bioinformatics'.

Toetsvorm

• [pass/fail] One individual assignment

• [~10%] One individual assignment

• [~40%] Group assignment consisting of three graded parts

• [~50%] Digital exam to assess: lecture topics

You are allowed to resit one part of the course if that makes you able to pass the course. The resit will be either oral or written after the end of the course.

Literatuur

• Course material (slides, scientific papers) on canvas

Aanvullende informatie doelgroep

This course is available for students in the minor 'Bioinformatics and Systems Biology' and students from the major Biomedical Mathematics.

Overige informatie

This course is part of the Minor Bioinformatics and Systems Biology. Most of the lectures of this course will be given together with the BSc course "Introduction to Bioinformatics".

Aanbevolen voorkennis

Programming in Python, Molecular Biology.

An interest in programming and biological problems.

Probability and Statistics

Vakcode	XB_0115
Studiepunten	6.00
Periode	P4
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. R. Hindriks
Examinator	dr. R. Hindriks

Betrokken Docenten	dr. R. Hindriks
Onderwijsvormen	Lecture, Seminar, Practical

Doel vak

After this course the student should be able to

- understand the setup of a probability space, events, probability and conditional probability (knowledge and insight);

- use the basic rules of probability theory (sum rule, product rule, law of total probability, Bayes' rule) (applying knowledge and insight);

- understand what a random variable is, what its distribution is, both for discrete and continuous random variables (this includes the concept of probability mass function, probability density function, and cumulative distribution function)

(knowledge and insight);

- compute the expectation and variance of a random variable (applying knowledge and insight);

- compute the expectation and variance of a sum of independent random variables (applying knowledge and insight);

- derive discrete probability distributions for probabilistic experiments associated to the Bernoulli, Binomial, Negative Binomial and Poisson distribution (applying knowledge and insight);

- compute probabilities for continuous random variables, specifically the uniform, exponential, and normal distribution (applying knowledge and insight);

- perform calculations based on the law of large numbers and central limit theorem (applying knowledge and insight);

- reproduce the definition of an estimator, carry out elementary computations involving point estimators (applying knowledge and insight);

- understand what confidence intervals are and be able to derive them in simple settings using a pivot (applying knowledge and insight);

- understand the setup of classical hypothesis testing (critical region, p-value, type of errors) and apply this to simple hypothesis testing problems (knowledge and insight);

- derive likelihood ratio tests in simple settings (applying knowledge and insight);

- perform basic computations with the statistical software package R (judgement, learning skills);

- carry out statistical analyses using the software package R (judgement, learning skills);.

Inhoud vak

From the book by From Pishro-Nik:

Chapter 1: Basic concepts

Chapter 3: Discrete random variables, Bernoulli, Binomial, Negative Binomial, Poisson

Chapter 4: Continuous random variables; sections 4.0, 4.1 and from 4.2 uniform, exponential and normal random variables

Chapter 7: Law of large numbers and central limit theorem; sections 7.0 and 7.1

Chapter 8: Classical statistics (point estimation, interval estimation, hypothesis testing); sections 8.1, 8.2, 8.3 and 8.4

Aanvullende informatie onderwijsvormen

6 weeks with 4 hours lecturing each week. Exercise classes. Two R assignments, to be done in groups of 2.

Attendance to lectures and classes is not mandatory but strongly recommended.

Toetsvorm

The exam counts for 80%. Each of the assignments counts for 10%. To pass the course, the (unrounded) grade of the exam needs to be at least 5. The exam has a resit, the assignments do not.

Literatuur

Hossein Pishro-Nik, Introduction to Probability, Statistics and Random Processes. The book can be ordered but is also freely available on www.probabilitycourse.com.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 2)

Aanbevolen voorkennis

Elementary set theory

Calculus

Project Collective Intelligence

Vakcode	XB_0026
Studiepunten	6.00
Periode	P6
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. ir. E. Ferrante
Examinator	dr. ir. E. Ferrante
Betrokken Docenten	dr. ir. E. Ferrante, dr. A. Yaman
Onderwijsvormen	Lecture

Doel vak

The main objectives of this course are:
1) To understand the notion of collective intelligence and natural and artificial systems, and the difference between modeling them at the macroscopic versus the microscopic scale,
(Knowledge and understanding)
2) To gain insight into open research issues related to collective intelligence and collective dynamics,
(Knowledge and understanding)
3) To learn the use of simulators to conduct empirical research in this field. (Apply Knowledge and understanding) (Make judgements) (Lifelong learning skills)

Inhoud vak

The whole is greater than the sum of its parts. Or isn't it?
This course is concerned with collective intelligence and collective dynamics.
We will elaborate on the differences between modeling these systems at the microscopic versus the macroscopic scale, discuss the philosophical as well as the technical aspects of such modeling, and learn to conduct experiments with virtual agents that interact in virtual worlds.
Collective Intelligence concepts will be also positioned and studied in relationship with Artificial Life and Swarm Robotics.

Aanvullende informatie onderwijsvormen

The course combines literature study and moderated discussions with experimental work in computer simulations. Students will work in small

groups.

Toetsvorm

Students will work in teams of three, four, or five (depending on total student numbers).

They will subject to a compulsory evaluation every week that involves a presentation and demonstration with simulation videos. Answering questions during the group presentation and discussion is part of the assessment

Grading is based on:

1) Quality of the presentation and degree of achievement of the task:

a. Flocking: the agents need to move together in a group and in a common direction. This task is mainly meant as a formative training for the simulation environment (0% of the final grade - only a formative grade will be given, assessed at the end of Week 1)

b. Aggregation: the agents should aggregate all in one shelter among those available in the environment. (20% of the final grade, assessed at the end of Week 2)

c. Collective dynamics simulation: A collective dynamics linked to a specific scenario will be given to students to simulate and study. Examples of this from past editions of the course are: covid spread simulation, prey/predator dynamics. This task runs for the final two weeks of the course (40% of the final grade, the first 20% assessed at the end of Week 3 and the other 20% at the end of Week 4)

2) short report of the approach and experimental results (40% of the final grade)

The resit options are the following. In case one of the evaluations listed under point 1) is missed (b or a component of c), the students have the chance to submit a video containing a presentation and a live demo. In this case, the grade of the specific component (b, or a part of c) will be capped to 55%

In case of late submission of the report, a penalty of 10% on the grade of the report component will be applied for each 24 hours of delay. A maximum of 5 days delay is allowed (corresponding to a penalty of 50% on the grade of the report component).

Literatuur

Introductory papers on swarm robotics and collective dynamics, e.g. the review here https://link.springer.com/article/10.1007/s11721-012-0075-2

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (Year 2)

Afwijkende intekenprocedure

Students take this course in groups. Creating groups sufficiently early is of paramount importance given the tight schedule of the course. Therefore, students are requested to check Canvas announcements up to one week before the official start of the course, and execute any instructions about group formation contained in the announcement.

Aanbevolen voorkennis

Coding skills in Pyhton are necessary.
Students are required to download, install, and use simulation software.
Students should also have the will to switch to a more research-oriented mindset (the course is not about following instructions, but proposing research problems and approaches to solve them, with a strong motivation).

Project Conversational Agents

Vakcode	XB_0101
Studiepunten	6.00
Periode	P3
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	prof. dr. K.V. Hindriks
Examinator	prof. dr. K.V. Hindriks
Betrokken Docenten	prof. dr. K.V. Hindriks
Onderwijsvormen	Lecture, Seminar

Doel vak

1. Develop a conversational agent by developing a simple ontology for a knowledge base for the cooking domain which enables the agent to reason about recipes (Applying knowledge and understanding)

2. Apply and understand conversation design patterns (Applying knowledge and understanding)

3. Design and implement visual support for different stages in the conversation (Applying knowledge and understanding)

4. Evaluate a prototype conversational agent for the cooking domain (Making judgments)

5. Plan and organize work together in a project group using appropriate tools (Applying knowledge and understanding)

Inhoud vak

Students learn how to develop a cognitive dialogue agent that uses Google's DialogFlow and builds on the natural conversation framework of Robert Moore. Groups will collaborate todevelop a conversational agent for the cooking domain, with a focus on the recipe selection task in this domain. They will design and implement domain knowledge, conversational patterns, natural language understanding and visuals. The agent will be a voice assistant with visual support that supports the user to find a suitable cooking recipe.

Aanvullende informatie onderwijsvormen

Lectures, lab sessions (with computer/ laptops). Attendance at labs is obligatory. Students will work in groups of 6.

Toetsvorm

The grade is based on the quality of the basic conversational agent (30%), the quality and originality of extensions to the basic agent (30%) and the written report (30%). In addition, students are required to complete an individual quiz that targets their knowledge on the use of Dialogflow, Conversation patterns, implementing visuals with Prolog and Bootstrap, and evaluating conversational agents. The quiz will be published on canvas for a limited duration during the final lecture, to ensure that it tests their knowledge of these subjects, and will count for 10% towards the grade: a pass will add 1 full point to the grade, whereas a fail will add nothing.

There are no resit options for the project work (including the report).

Literatuur

Conversational UX Design: A Practitioner's Guide to the Natural

Conversation Framework

Robert J Moore en Raphael Arar | 29 mei 2019

ISBN-10: 1450363024

ISBN-13: 978-1450363020

A cognitive agent programming guide (online)

Learn Prolog Now! (online)

Github tutorial (online)

Bootstrap tutorial (online)

Dialogflow tutorial (online)

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence

Aanbevolen voorkennis

You should have completed Multi-Agent Systems (XB_0024)

Project Intelligent Systems

Vakcode	X_401076
Studiepunten	6.00
Periode	P3
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M. Cochez
Examinator	dr. M. Cochez
Betrokken Docenten	dr. M. Cochez
Onderwijsvormen	Lecture, Practical, Written partial exam

Doel vak

The course teaches students:

1) how to develop an intelligent agent (bot) based on standard Intelligent Systems technology, and extensions thereof (Knowledge and understanding) (Apply knowledge and understanding);

2) basic research and experimental analysis skills through the analysis of how effectively the developed software

performs in a controlled scientific experiment. (Making judgments) (Communication)

3) how to report on their research findings. (Communication)

Inhoud vak

In the course Project AI, a broad variety of techniques from the area of Artificial Intelligence are applied, in particular those from the course Intelligent Systems like adversarial search, knowledge representation and machine learning.

The course focuses on the development and scientific analysis of methods for rational agents that perform effectively in a card game (the Austrian trick game Schnapsen). The outcome is a number of intelligent game-bots, and a thorough evaluation of the impact of parameters on their performance. The results of the experiment are written in a scientific report. This experience provides students with seminal skills that will be applied throughout the AI Bachelor study, and through subsequent studies and career activities. This course provides insights into how proper documentation of an experiment can ensure that:

- results are formulated objectively;

- developed scientific reports are well-specified and contain complete

experiment metrics.

Aanvullende informatie onderwijsvormen

There will be a few introductory lectures (refreshing relevant knowledge and introducing the practical), followed by sessions during which we discuss research methods and the academic process. The practical work will be conducted in 3-person groups, but there is also a personal evaluation component. The groups will be self-directed, and will require to consult the course assistants during practical sessions.

Toetsvorm

The course is assessed based on 3 parts.

First, the final group report, which contains the results of your work. There will be a combination of self and peer reviewing to individually balance grades within a group.

Second, the individual test, where the students show that they can individually code an intelligent bot, create a research setup and analyze results, and report on research findings. Besides, the student will reflect on their work and show their ability to generalize their findings. This test is done in a controlled environment at the university.

Third, the peer review. In this task, students provide peer reviews for other groups. The decision on whether you pass this task is made based on the quality of the reviews you write for other groups, not the reviews you receive.

To pass the course, you will need to pass each of the 3 parts. For the group report and individual test, you need a grade of 5.5 or higher. For the peer review, you must get a pass (this component is graded pass/fail). Then, your final grade is computed as a weighted sum of the group report (60%) and individual test (40%).

Not all aspects of every task will be graded (random check). For example, there are multiple peer reviews, but we might only grade some of them at random to determine your grade.

There is no resit option for this course.

Literatuur

The literature required for this course will be distributed via Canvas.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 1)

Overige informatie

As this course is a project-oriented course, the student groups are expected to be working on this project individually for most of the time.

However, some of the meetings are mandatory to ensure a good progress of each group.

Aanbevolen voorkennis

The knowledge taught in the course Intelligent Systems and Introduction

to Programming are required to succeed in the course.

Project Science and Tech-based Entrepreneurship

Vakcode	A_HP007
Studiepunten	6.00
Periode	
Vakniveau	400
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M.L. Blankesteijn
Examinator	dr. M.L. Blankesteijn
Betrokken Docenten	dr. M.L. Blankesteijn
Onderwijsvormen	Lecture, Seminar

Doel vak

Develop entrepreneurial competencies:

- Dealing with uncertain situation by
- o Taking initiative
- o Be courageous
- o Act responsible
- ● Takes quick and informed decisions
- ● Be creative and innovative
- ● Be oriented towards result and output
- ● Develop Entrepreneurial leadership
- ● Be cooperative
- ● Be constructive
- ● Develop autonomous/independent attitude
- ● Feedback: giving and receiving
- ● Develop Communication skills:
- o Pitching
- o Writing report

Inhoud vak

This course is an introduction to “Science-based Entrepreneurship”.

To be a successful science and tech entrepreneur, one needs sound

disciplinary knowledge of a practical scientific domain, combined with

knowledge of and hands-on experience with innovation processes and

entrepreneurship. In science-based entrepreneurship both natural and

social scientific principles are combined in a process of

entrepreneurial engagement to support valorization of ideas.

In this course we enable you, as a student, to initiate and/or contribute on a fundamental, practical level to such processes. You take the freedom and responsibility to make the project both a challenge of translating theory into practice and an experience in personal entrepreneurial development. You build on your disciplinary academic background to independently design and test a science-based entrepreneurial idea. You come up with this idea yourself, or, you approach a science-based (start-up) company with whom you collaborate to develop the idea (for example a start-up in the Demonstrator lab at the VU). In executing the project you (of course) get support from the teachers, entrepreneurial role models and fellow students. You choose a topic. You work on developing the topic into an idea. You design strategies on how you are going to test the idea. Through this process you gain insight into how a particular technological innovation or the application of innovative knowledge from a specific fundamental research domain in your disciplinary field of research creates commercial and societal value.

Aanvullende informatie onderwijsvormen

Lectures, pitching, group work.

Toetsvorm

Individual and group assignments

Literatuur

Blankesteijn, M., et al. (2020). " Science-based entrepreneurship education as a means for university-industry technology transfer." International Journal for Entrepreneurship and Management. International Journal of Management Education 18(3): 100425.
Perusso, A., et al. (2020). "The contribution of reflective learning to experiential learning in business education." Assessment & Evaluation in Higher Education 45(7): 1001-1015.
Thomassen, M. L., Middleton, K. W., Ramsgaard, M. B., Neergaard, H., & Warren, L. (2019). Conceptualizing context in entrepreneurship education: a literature review. International Journal of Entrepreneurial Behavior & Research.
Kuratko, D. F., & Morris, M. H. (2018). Corporate entrepreneurship: A critical challenge for educators and researchers. Entrepreneurship Education and Pedagogy, 1(1), 42-60.
Krueger, N. F. (2009). The microfoundations of entrepreneurial learning and… education: the experiential essence of entrepreneurial cognition. Handbook of university-wide entrepreneurship education, 35-59.

Afwijkende intekenprocedure

This course is offered every other year. The next time it will be offered is in the academic year 2025-2026.

Project Situated AI

Vakcode	XB_0121
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M.C.A. Klein
Examinator	dr. M.C.A. Klein
Betrokken Docenten	dr. M.C.A. Klein
Onderwijsvormen	Practical

Doel vak

The aim of the course Project Situated AI is to let students apply the different concepts that have been taught in the courses during period 1 and period 2 of the minor in a semi-realistic application (Applying knowledge and insight).

Students will learn to make decisions about suitability of different techniques for specific scenarios (Judgement).

Students will learn to integrate different techniques into a demonstrator and to explain the working and rationale behind the demonstrator both orally as in written form (Communication).

Inhoud vak

This course concludes the minor Situated AI. It is a full-time, 4 weeks course, in which students will work in groups on several practical assignments related to the other courses of the minor. One week will be devoted to building a robot application, another to the implementation of a conversational AI system, another week to the implementation of a reinforcement algorithm, and one to the application of logical reasoning. The different components will be combined in one application scenario, for which the students will build an demonstrator. Students will write a report about their demo and the choices made.

Aanvullende informatie onderwijsvormen

Workshops, tutorials, project work.

Toetsvorm

Each of the weekly assignments will be graded. In addition, the final demonstration together with the accompanying report will be graded. All components count for 20% of the final grade.

The individual contribution to the project will be assessed via an interview. The individual contribution to the project will be assessed via an interview. In case that unequal contribution to the group project becomes apparent, an individual grade can be adapted.

There is no option for a resit of the assignments.

Aanvullende informatie doelgroep

Bachelor students Artificial Intelligence registered for the minor Situated AI.

Aanbevolen voorkennis

This course can only be followed if all other courses of the minor Situated AI have been followed as well.

Project Socially Aware Computing

Vakcode	XB_0042
Studiepunten	6.00
Periode	P3
Vakniveau	200
Onderwijstaal	Engels

Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	C. Gerritsen
Examinator	C. Gerritsen
Betrokken Docenten	C. Gerritsen, dr. N. Ullah
Onderwijsvormen	Lecture, Seminar

Doel vak

After having completed this course students are able to:

· Perform a study following a full research cycle: setting up an experiment (conceptualisation,

formalisation), performing the experiment (simulation), analysing and interpreting its results

(applying knowledge and understanding; making judgements)

· Apply and integrate knowledge about programming, modelling, statistics, conducting

research and academic writing to actual societal challenges

(applying knowledge and understanding; making judgements)

· Build a multi agent-based model in the modelling environment Netlogo

(applying knowledge and understanding)

· Report the results of the project in written form

(communication; lifelong learning skills)

Inhoud vak

Within this course students learn to address a societal challenge involving social interaction by means of agent-based simulation. They learn to identify relevant concepts and relations, formalise these in a dedicated environment (Netlogo), conduct simulations to test different hypothetical scenarios and analyse their results.

They do this by working individually on assignments and a project in a chosen domain. They report their findings in written form.

Aanvullende informatie onderwijsvormen

This project course is given by means of active working group meetings. These meetings are mandatory. The course is divided into two parts. During the first week the students will work on assignments to learn how to work with Netlogo. If they pass these assignments they are allowed to start their project. They will work on their project during the next three weeks.

During the first two weeks the students have 20 hours of work group sessions per week. During the third week there will be 12 hours of work group sessions. In the final week of the course (week 4) the students can schedule an individual meeting with a member of the teaching team. Attendance during the work group sessions is mandatory.

Toetsvorm

Pass/fail individual assignments that need to be passed to start the project

Individual project report (95%) + self-reflection (5%)

We only offer a resit for the project. If you fail the assignments you will have to take the course again next academic year.

This course consists of individual assignments and an individual project. Students need to work on the assignments in class. Their work is checked by the teacher or TA. The meetings are mandatory. The students also work on their project in the mandatory meetings.

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (year 2)

Aanbevolen voorkennis

Knowledge on programming (Introduction to Python Programming) and statistics (Statistical Methods) is highly recommended.

Religions and Gender

Vakcode	G_BATRSAL054
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit Religie en Theologie
Vakcoördinator	dr. L. Minnema
Examinator	dr. L. Minnema
Betrokken Docenten	prof. dr. C.A.M. van den Berg, dr. L. Minnema
Onderwijsvormen	Seminar, Lecture

Doel vak

After finishing this course, the student is able to:

- describe analytically how certain aspects of gender have been or become an issue in religions;

- articulate major parallels and differences between a number of religious traditions in their approaches to gender issues in the past and present;

- describe both religious insider views and academic outsider views;

- link historical patterns to present-day debates and phenomena.

Inhoud vak

This course addresses the important question of the relationship of contemporary and past religions (Buddhism, Christianity, Hinduism, Islam, Shinto, Greek and Babylonian religions) to gender. Gender issues related to male and female models in religious narratives, historical shifts in the religious status of women, mother goddesses and female power, and religious views of homosexuality will be addressed across cultures and religions. The central focus will be to discover underlying patterns of similarity and difference. Moreover, in two afternoon sessions and one morning session a link will be made between historical patterns in various traditions, and contemporary interpretations of and discussions on

religions and gender.

Six entire morning sessions plus two shorter afternoon sessions will focus on the following six themes:

1. Male and female role models and stereotypes in ancient narratives:

mythological and legendary couples and gender differences in the Babylonian Gilgamesh epic, Homer’s Odyssey epic (Odysseus and Penelope), Hesiod (Prometheus and Pandora), the Hindu Mahabharata epic (Nala and Damayanti), and Ramayana epic (Rama and Sita)

2. The changing religious status of women during three crucial shifts in the world history of religions: the Neolithic, Axial Age, and Modernization breakthroughs

3. Cross-cultural comparison of mother goddesses and female power: the Hindu goddess Devi-Durga in India, the Greek goddess Demeter in Minor Asia, the Shinto goddess Amaterasu in Japan

4. Religious rules and attitudes regarding homosexuality in Buddhism and in Islam

5. The image of Mary in Christianity and Islam: virgin, Madonna, mother, heroine, virtue, saint, queen

6. Virginity

Aanvullende informatie onderwijsvormen

Lectures (HC) and seminars (WC)

Toetsvorm

Written exam

Literatuur

Articles and book chapters will be available via Canvas or the VU library

Aanvullende informatie doelgroep

The course is intended for regular bachelor and minor students and exchange students.

The course is a module in the FRT minor Religious dimensions of global challenges and in the FGW minor Diversity and Gender.

Overige informatie

The course will be offered on campus-only, not online or hybrid; 75% attendance in class required (exceptions considered on request).

Check in advance whether on the written exam date you will still be in the country, as the exam will take place at the VU Amsterdam campus.

Toelichting Canvas

The course manual describing the programme and readings in detail will be put on Canvas.

Research Questions in Bioinformatics

Vakcode	XB_0073
Studiepunten	3.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. H. Mouhib
Examinator	dr. H. Mouhib
Betrokken Docenten	dr. H. Mouhib, dr. Q. Peng MSc, dr. J.M. Urtasun Elizari
Onderwijsvormen	Study Group, Computer lab, Lecture

Doel vak

In this course you will familiarize yourself with current research topics in the field of Bioinformatics using academic literature.

After completing this course, the student;
- Can read and comprehend scientific writings and answer questions about the text. [Knowledge and understanding]
- Can connect figures and tables to methods and results. [Applying knowledge and understanding]
- Is able to explain and defend the methods used in a research paper. [Communication]
- Is able to think critically about the methods used to answer the research question. [Making judgements]
- Has a better understanding of the research field of bioinformatics and can see the relevance of the three disciplines (molecular biology, math and programming) for this area of research. [Knowledge and understanding]
- Is able to reflect on their own frame of reference, and demonstrate awareness of their own perspective being not necessarily a universal perspective. [Lifelong learning skills]

Inhoud vak

We will read and discuss a cutting edge topic in Bioinformatics using a scientific review, a publication of a Bioinformatics method and a paper that applies this method to answer a biological question.

Aanvullende informatie onderwijsvormen

In this course we use the flipped classroom approach. This means students will spend about two hours reviewing materials before each of the scheduled discussion classes.

Furthermore, there are two computer practical sessions and one seminar with student presentations.

Toetsvorm

The learning goals will be tested by in class quizes, discussions, presentations, and a digital

exam. The discussion classes are mandatory.

Graded assignments: ~40% of final grade

Exam: ~60% of final grade

You are allowed to resit the exam if you passed all other parts of the

course. The resit will be either oral or written after the end of the

course.

Literatuur

The course material will be provided through Canvas.

Aanvullende informatie doelgroep

This course is a compulsory part of the minor 'Bioinformatics and Systems Biology' and is only available for students in this minor.

Overige informatie

This course is taught in English.

Revalidatie

Vakcode	B_REVAL
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	prof. dr. T.W.J. Janssen
Examinator	prof. dr. T.W.J. Janssen
Betrokken Docenten	prof. dr. T.W.J. Janssen, dr. S. de Groot
Onderwijsvormen	Practicum, Hoorcollege

Doel vak

Na het volgen van deze cursus

• Is de student bekend met relevante begrippen, concepten en modellen

uit de revalidatie van mensen met een fysieke beperking

• Toont de student inzicht in de problematiek van speciale groepen in de

context van revalidatie.

• Is de student in staat tot een kritische analyse van een probleem uit

de revalidatie, arbeidsreïntegratie of hulpmiddelproblematiek.

Inhoud vak

Revalidatie is te omschrijven als 'het gecoördineerd en gecombineerd

gebruik van maatregelen op medisch, sociaal, arbeidstechnisch en

onderwijskundig terrein die de persoon met een fysieke beperking op de voor hem/haar optimale

plaats in de samenleving moet helpen'. Bij uitstek een

multidisciplinaire teamprestatie. In deze cursus zullen verschillende

aspecten van deze multidisciplinaire aanpak besproken worden, waarbij de

verschillende disciplines aan bod komen bij het revalidatieproces van

o.a. mensen met een dwarslaesie, niet-aangeboren hersenletsel,

amputatie, neurodegeneratieve aandoening, of hartaandoening.

Daarnaast zal de vraag worden gesteld welke consequenties een

fysieke beperking heeft voor o.a. arbeidsparticipatie en

hulpmiddelgebruik en komen onderwerpen zoals sport en bewegen, rolstoelmobiliteit en gangbeeldanalyse aan bod. De (mogelijke) rol van de bewegingswetenschapper

binnen de revalidatie zal ook bediscussieerd worden.

Aanvullende informatie onderwijsvormen

Deze module bestaat uit twee onderdelen: enerzijds een reeks bijeenkomsten (hoorcolleges, een workshop en een bezoek aan een revalidatiecentrum) anderzijds is er een groepsopdracht. De cursusomvang is 6 ects (168u), waarvan de uren per student als volgt zijn verdeeld over beide onderdelen: collegebijeenkomsten (15x2u), workshops en bezoek revalidatiecentrum (8u), tentamen (2u), de uitwerking van de groepsopdracht (82u), plus tot slot de college- en tentamenvoorbereiding (44u). De groepsopdracht wordt uitgevoerd in viertallen, waarin gewerkt wordt aan een casus in de context van de revalidatie en/of arbeidsreintegratie. De opdracht wordt afgerond met een werkstuk en een referaat tijdens een reeks afsluitende colleges.

Toetsvorm

Toetsing vindt plaats aan de hand van de praktijkopdracht (werkwijze en

verslag) en een afsluitend schriftelijk meerkeuzetentamen. Beide

onderdelen tellen voor 50% in het eindoordeel, waarbij de deelcijfers

niet lager mogen zijn dan een 5.0. De collegestof en

hand-outs en een aantal hoofdstukken uit het boek Revalidatie voor

Volwassenen vormen het tentamenmateriaal.

Literatuur

J.H.B. Geertzen, S. Rietman, G.G. Vanderstraeten, R.J. Berduszek, K. Oostra, B. Fard. Revalidatie voor

volwassenen. Jaar 2022. ISBN 9789023258889.

Handouts en reader.

Overige informatie

Er wordt uitgegaan van latente kennis rond revalidatie op het nivo van het 1ste & 2 de jaar van de opleiding bewegingswetenschappen (Inleiding Bewegen en Gezondheid, Pathologie van het Bewegen en de readers)

Afwijkende intekenprocedure

De indeling van werkgroepen vindt

plaats via Canvas.

Robot Interaction

Vakcode	XB_0040
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. D.F. Preciado Vanegas
Examinator	dr. D.F. Preciado Vanegas
Betrokken Docenten	prof. dr. E.A. Konijn, dr. J.F. Hoorn, dr. D.F. Preciado Vanegas, prof. dr. P.T.J.M. Vossen, drs. ing. M.H.J. Smakman
Onderwijsvormen	Study Group, Lecture

Doel vak

After successful completion of this course, students are expected to be able to:

· Articulate the core themes and challenges in the field of Social Robotics. (Knowledge and insight, Applying knowledge and insight, Judgement,)

· Understand the multi-disciplinary contributions that develop Social Robots. (Knowledge and insight, Applying knowledge and insight, Judgement,)

· Describe and evaluate main theories from various field to understand how humans interact and communicate with social robots. (Knowledge and insight, Applying knowledge and insight, Judgement,)

· Understand, describe and compare screen-based versus robot-based interaction. (Knowledge and insight, Applying knowledge and insight, Judgement,)

· Discuss and compare key findings in fundamental areas of Robot Interaction, such as typology of robots; application areas; relating to robots; psychological aspects; language understanding in robots; robots outperforming humans while being worse communicators; modeling theories for performance in robots; ethical considerations. (Knowledge and insight, Applying knowledge and insight, Judgement,)

· Develop a good understanding of up-to-date and interdisciplinary scientific research in the field and in various application areas. (Knowledge and insight, Applying knowledge and insight, Judgement,)

· Understand and apply an academic multi-disciplinary attitude, academic writing style, search for insights in complex phenomena, theoretical and empirical approaches. (Knowledge and insight, Applying knowledge and insight, Judgement, Communication, Learning skills)

· Apply an analytical approach to model communication in robots. (Knowledge and insight, Applying knowledge and insight, Judgement,)

· Exercise reverse engineering, building scenarios, interaction design, concurrent algorithmic approaches. (Knowledge and insight, Applying knowledge and insight, Judgement,)

Inhoud vak

What does it mean when a robot steps out of your computer in physical space and starts interacting with humans as if it were a social entity itself? This course will address various aspects that come into play when physically present robots interact with humans. This course focuses mainly on social robots, humanoids in particular, suited for communicative purposes. We will discuss both sides of the coin: perspectives from the robot-side as well as perspectives from the user-side. Differences and similarities with common computer-based AI (cf. avatars, chatbots) are discussed in comparison to human-robot interaction, exemplified by main application areas (e.g.,

eldercare, education). Basic models and relevant aspects of the psychology of the user and ethical aspects are also discussed. If robots start talking to humans by themselves, how can they understand each other? Lectures will therefore also address the basics of the understanding of language (e.g., Natural Language Processing (NLP)). How to translate theoretical models into effective computer models is another challenge that will not only be lectured on but also practiced. A team of lecturers with the respective expertise will lecture on the wide variety of topics in this course. Several practical tutorials/workshops accompany the lectures to gain hands-on experience regarding the complexities of human-robot interaction.

Aanvullende informatie onderwijsvormen

Lectures on a wide variety of topics related to Robot Interaction, two times a week.
Several practical tutorials/workshops will accompany the lectures as a 3rd meeting in the week. Details will be presented in the schedule/syllabus on Canvas.

Toetsvorm

Examination: final examination consisting of a multiple choice test of 40 questions and 5 open essay questions; the combined grade counts for 80% of the final grade for this course.

Assignments related to the workshops are also graded; the averaged grade for assignments counts for 20% of the final grade for this course. Some assignments are individually, and some are done in group.

To pass the course, each assignment needs to be passed. Also, to pass the exam, the separate multiple choice and essay question sections must be passed (55%) separately to pass the exam.

There is only a resit for the examination.

Literatuur

For this course, you will read a relatively large number of journal articles, book chapters, and conference proceedings. The reading list will be announced on Canvas (in the syllabus) posted on Canvas before the start of the course. Access to journal articles/ book chapters / conference papers via Canvas, or else via the VU-library, check LibSearch, e-resources (e.g., select search in PsycInfo, EBSCOhost), else Google Scholar, or http://dx.doi.org

Aanvullende informatie doelgroep

Bachelor Artificial Intelligence (Socially Aware Computing track)

Overige informatie

A detailed syllabus will be posted on Canvas in advance (approx. a week before the start of this course).

Toelichting Canvas

Check regular updates on Canvas.

Robot Law and Artificial Intelligence

Vakcode	R_RLAI
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	M. van der Werf
Examinator	M. van der Werf

Betrokken Docenten	dr. mr. M. van der Linden, prof. mr. A.R. Lodder, I. Bratu, Z.B. Cooper
Onderwijsvormen	Study Group, Tutorial, Lecture

Doel vak

Robot Law and Artificial Intelligence focuses on the societal impact of technological constructs such as intelligent software and robots. The student will learn and understand the profound influence that the autonomous and intelligent technological constructs may have on society, as well as the ethical consequences and legal implications thereof. The student will be able to develop an academic, sound judgement on the future of a robotic society from an ethical and legal perspective. The student will be able to analyze and critically evaluate the legal-ethical dimensions of issues relating to the use of intelligent software and robots.

Course objectives:

At the end of the course, students are:

- aware of the contextual demands (not to stifle technological development by enacting restrictive legislation, while at the same time protecting fundamental rights and freedoms of a democratic society) law has to deal with;
- able to assess the legal and societal aspects of a problem in an integrated way and critically reflect on possible approaches and solutions;
- able to formulate their own well-founded opinion on the challenges posed by particular applications of robots and AI.

Eindtermen: 3, 4 en 15.

Inhoud vak

Robots and Artificial Intelligence used to belong to science fiction movies and stories. Also, they were discussed in theoretical academic and popular articles. In recent years both Robots and Artificial Intelligence gradually but strongly are moving away from theory and entering our daily lives. This course focuses on those practical developments, and what role law and ethics play. We do not limit ourselves to present technology, but include prophecies on how society may change in the future and what we can and should do about it.

Aanvullende informatie onderwijsvormen

Lectures and tutorials.

Toetsvorm

Assignment plus exam

Literatuur

Made available via Canvas

Aanvullende informatie doelgroep

Apart from regular students, the course is also available for:
Students from other universities/faculties
Contractor (students who pay for one course).

Robotics

Vakcode	XB_0120
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. K.S. Luck
Examinator	dr. K.S. Luck
Betrokken Docenten	dr. K.S. Luck, dr. ir. K. Baraka
Onderwijsvormen	Lecture, Practical

Doel vak

Knowledge and insight: At the end of the course the student will be familiar with the basics of robotics and robot learning, such as kinematics, motor control and how to apply machine learning algorithms to tasks in robotics.

Applying knowledge and insight & Learning skills: The student will be able to understand how to solve standard control problems with mathematical tools, such as forward and inverse kinematics/dynamics and end-effector-control. The student will be able to solve such problems in simulated and in real-world robots.

Judgment: Furthermore, the student will understand when to use AI-driven robot learning methodologies and when to use analytical and closed-form solutions for robot control.

Communication: The student will be capable of explaining the benefits and drawbacks of the use of AI and analytical solutions to non-experts.

Inhoud vak

The first half of the course will cover the basics of robotics and robot control. Topics will include forward/inverse kinematics, modeling of robots, Denavit-Hartenberg, the basics of dynamics, trajectory planning and optimization and control algorithms.

In the second half, the course will give an introduction into robot learning and embodied AI, i.e., the combination of artificial intelligence & machine learning with robotics. This will include the discussion of model-free and model-learning approaches for robotic tasks, their disadvantages and advantages over control methods.

Aanvullende informatie onderwijsvormen

Two Lectures & one Practical session per week. Attendance of lectures is not mandatory but highly recommended: While lecture slides will be made available, the lecture may also make use of black- and/or white-boards.

The course will also include practical (programming) assignments in Python/ROS.

Toetsvorm

The grade will be determined 70% through the final exam, and 30% through practical assignments.

A resit will only be offered for the exam, but not for the practical assignments.

Literatuur

Recommended Literature to supplement the course:

Mark W. Spong, Seth Hutchinson, M. Vidyasagar, “Robot Modeling and Control”, Wiley

B. Siciliano, L. Sciavicco. Robotics: Modelling, Planning and Control, Springer

C.M. Bishop. Pattern Recognition and Machine Learning, Springer

R. Sutton, A. Barto. Reinforcement Learning - An Introduction, MIT Press

Aanbevolen voorkennis

It is recommended to take the ‘Reinforcement Learning’ course of the Situated AI minor in parallel or prior to this course.

Good knowledge and command of topics taught in the courses ‘Statistical Methods for AI’ and ‘Linear Algebra for AI’ is recommended. Good command of Python is necessary for the practicals

Schrijvers aan het werk I: 1100 - 1720 (minor)

Vakcode	L_AABAALG093
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Nederlands
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	prof. dr. J.M. Koppenol
Examinator	prof. dr. J.M. Koppenol
Betrokken Docenten	prof. dr. J.M. Koppenol
Onderwijsvormen	Practicum, Werkcollege

Doel vak

Wanneer je het vak Schrijvers aan het werk 1 hebt gevolgd, kun je:

- een typering geven van de literatuur uit de Middeleeuwen, rederijkerstijd, ‘Gouden Eeuw’ en vroege Verlichting, door uit elk tijdvak een aantal representatieve auteurs, teksten, genres, stilistische kenmerken en literaire fenomenen te noemen.

-teksten uit de periode tot 1720 interpreteren, op basis van poeticale opvattingen en doelstellingen van de auteurs, met gebruikmaking van literair-historische begrippen en kennis over de grammatica en woordenschat uit deze periode.

- van een aantal representatieve teksten een samenvatting geven en de inhoud in verband brengen met de (cultuur)historische verbanden waarin deze teksten functioneerden.

- toelichten in welke vorm en op welke wijze teksten in de Middeleeuwen en Vroegmoderne tijd werden vastgelegd, overgeleverd en gerecipieerd.

- een basaal eigen onderzoek uitvoeren volgens de wetenschappelijke onderzoekscyclus (een eenvoudige onderzoeksvraag formuleren, literatuur zoeken en het onderzoek uitvoeren en daar op een verantwoorde manier schriftelijk verslag van doen).

- het wetenschappelijke ‘apparaat’ (woordenboeken, digitale middelen, o.a. MNW, WNT, STCN, BNTL, Biografisch Portaal, enz.) adequaat toepassen bij het zelf uitvoeren van een onderzoek.

- uitleggen hoe canonisatie en literatuurgeschiedschrijving bijdragen aan een steeds wisselend beeld van de literatuur van Middeleeuwen, Rederijkerstijd, Renaissance en vroege Verlichting.

- reflecteren op de afstand tussen je eigen schrijverschap en het auteurschap van auteurs uit het verleden door oude literaire vormen zelf toe te passen.

Inhoud vak

In deze module maken de studenten kennis met de Nederlandse literatuur van de Middeleeuwen tot de vroege 18de eeuw. Die literatuur is voor een deel vreemd en anders dan de moderne letterkunde, tegelijkertijd is er veel herkenbaar en zelfs actueel. In dit college, dat de periode 1100-1720 bestrijkt, worden de Middeleeuwen, rederijkerstijd, renaissance en het classicisme behandeld aan de hand van een syllabus en een keuze aan teksten. Uit de Middeleeuwen komen onder meer de ridderroman en religieuze lyriek aan bod. De rederijkerij is een cultureel wapen in de handen van met name de stedelijke burgerij. Vanaf 1550 begint de renaissance, auteurs gaan zich richten naar (meestal aan de oudheid ontleende) voorschriften, voorbeelden en theorieën. Tegelijkertijd vinden er belangrijke ontwikkelingen plaats in de politieke en religieuze situatie in de Nederlanden, die leiden tot de vorming van de Republiek. In de zeventiende eeuw, de Gouden Eeuw, groeit ons land uit tot een wereldmacht. Bij de behandeling van de teksten ligt de nadruk, behalve op de literaire ontwikkelingen, ook steeds op de maatschappelijke rol die teksten in de samenleving hebben vervuld, van statusverhoging tot politiek wapen, en van vermaak tot emancipatoire kracht. Aan de hand van een individuele praktijkopdracht krijgt de student een basisvaardigheid in het omgaan met de wetenschappelijk hulpmiddelen voor het vak. Ook maken de studenten creatieve opdrachten waarin ze hun kennis van de oudere literatuur verwerken.

Aanvullende informatie onderwijsvormen

Werkcolleges, in combinatie met individuele studie van een reeks teksten

en wetenschappelijke artikelen;

Individueel werkstuk (na instructie); creatieve opdrachten.

Toetsvorm

De toetsing bestaat uit drie onderdelen:

1. een schriftelijk tentamen over de collegestof en de gelezen primaire en secundaire teksten (50%);

2.een verslag van een onder begeleiding uitgevoerd onderzoek naar een 17de-eeuws gedicht (30%)

3. drie creatieve opdrachten, op te nemen in het portfolio, waaronder:

- adaptatie / moderniseringsopdracht Middeleeuwen (10%)

- poëticaopdracht (10%)

Alle onderdelen moeten met een voldoende worden afgesloten. Verder is een actieve deelname aan de colleges vereist (colleges moeten zijn voorbereid, opdrachten gemaakt; wie niet actief deelneemt, kan een inhaalopdracht krijgen, bij herhaling volgt uitsluiting van het college).

Vereiste voorkennis

Geen

Literatuur

Syllabus (beschikbaar op Canvas), met overzichten, tekstfragmenten en opgaven. Daarnaast een leeslijst met teksten, die op Canvas bekend zal worden gemaakt in de studiehandleiding. Een aanbevolen leerboek voor de oudere taalfasen van het Nederlands is: M.A. Mooijaart en M.J. van der Wal, Nederlands van Middeleeuwen tot Gouden Eeuw (Nijmegen, 2008 of latere druk).

Aanvullende informatie doelgroep

Eerstejaars studenten Creatief Schrijven: Nederlands; Premasterstudenten Nederlandse letterkunde en het literaire veld.

Afwijkende intekenprocedure

Premasterstudenten hoeven niet zelf in te tekenen voor vakken want dit doet het onderwijsbureau.

Schrijvershuisbezoeken

Vakcode	L_NNBAALG002
Studiepunten	6.00
Periode	P2
Vakniveau	300

Onderwijstaal	Nederlands

Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	prof. dr. J.H.C. Bel
Examinator	prof. dr. J.H.C. Bel
Betrokken Docenten	prof. dr. J.H.C. Bel
Onderwijsvormen	Werkcollege, Excursie

Doel vak

Op basis van interviews met een aantal hedendaagse schrijvers en de analyse van een van hun recente literaire werken:

- Kennismaken met literatuuropvattingen van deze schrijvers;

- Kennismaken met secundaire literatuur over het begrip poëtica en vaardigheid ontwikkelen in het kritisch hanteren van dit literatuurwetenschappelijke concept;

- Leren reflecteren op het scheppingsproces van een literair werk.

Inhoud vak

Onder leiding van de ‘vrije schrijver’ en Jacqueline Bel wordt een bezoek gebracht aan vier schrijvers. Elk van de bezoeken wordt in de daaraan voorafgaande week grondig voorbereid op basis van de lectuur van een of meer werken van deze auteur. Telkens vormt één werk, in combinatie met het zoeklicht 'poëtica', het uitgangspunt voor deze bezoeken. Vragen die aan de orde komen zijn: wat is de literatuuropvatting van deze schrijver? Welke kwesties houden hem/haar bezig? Hoe gaat de schrijver te werk? In hoeverre is het schrijven voor hem of haar een beroep?

Aanvullende informatie onderwijsvormen

Werkcolleges van 2 uur per week en huisbezoeken (daarvoor is een middag gereserveerd) onder leiding van de 'vrije schrijver' en Jacqueline Bel. Er worden vier schrijvers bezocht. De namen worden zo spoedig mogelijk bekend gemaakt.

Toetsvorm

Actieve participatie en deelopdrachten (moet voldoende zijn). Colleges moeten altijd grondig zijn voorbereid conform de instructies uit de studiehandleiding. Afrondend eindwerkstuk (100%) of schriftelijk tentamen (100%). Beide toetsvormen moeten voldoende zijn.

Literatuur

Een werk van de 'vrije schrijver' en van de vier schrijvers aan wie een huisbezoek gebracht wordt; secundaire literatuur over deze schrijvers en secundaire literatuur over poëtica-onderzoek (Van den Akker/Dorleijn, Sötemann).

Aanvullende informatie doelgroep

De minor staat open voor alle studenten die hun eerste studiejaar hebben afgerond.

Overige informatie

Aanwezigheid verplicht (80%). Alleen met een gegronde reden kan iemand afwezig zijn. Is een student door omstandigheden vaker afwezig dan toegestaan dan kan dit eventueel worden gecompenseerd met een extra opdracht. Het college is onderdeel van de minor 'Aan de slag met literatuur'.

Schrijverspalet: retorica

Vakcode	L_AABAALG092
Studiepunten	3.00
Periode	P1
Vakniveau	200

Onderwijstaal	Nederlands
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	prof. dr. J.H.C. Bel
Examinator	prof. dr. J.H.C. Bel
Betrokken Docenten	prof. dr. J.H.C. Bel, dr. J. Grave
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

De student kan

op basis van technieken uit de retorica een essay schrijven

een interview voorbereiden en houden

kan de literatuuropvatting construeren van enkele contemporaine schrijvers en journalisten.

Inhoud vak

Hoe kun je je publiek boeien vanaf de eerste zin? Hoe schrijf je een pakkende tekst? Deze vragen houden de literatuur sinds de klassieke oudheid bezig en de antwoorden zijn ook buiten de literatuur relevant. Immers, een advocaat moet zijn gehoor weten te overtuigen van de onschuld van zijn cliënt of van verzachtende omstandigheden. En een CEO moet zijn bedrijf goed weten te ‘verkopen’. Na enkele introducerende colleges over de klassieke retorica waarbij je je traint in het schrijven van een zakelijk gestructureerde tekst, is er elke week een bekende schrijver of journalist te gast op college: studenten lezen en analyseren vooraf een van zijn of haar teksten, bereiden een interview voor en gaan tijdens het bezoek van de schrijver met hem/haar in gesprek over hun werk. Als gasten worden zowel creatieve schrijvers (o.a. de Vrije Schrijver), als bekende journalisten uitgenodigd (bijvoorbeeld Maxim Februari, Elsbeth Etty, Geert Mak, Abdelkader Benali).

Aanvullende informatie onderwijsvormen

werkcollege

Toetsvorm

• Het schrijven van een korte zakelijke tekst (60%) en mondelinge presentatie (40%) met powerpoint en handout. Voor beide onderdelen moet minimaal een 5,5 worden behaald.

* Voorbereiding interview (moet voldoende zijn om het vak te halen, maakt geen onderdeel uit van eindcijfer).

Literatuur

De verplichte primaire en secundaire literatuur wordt via canvas bekend gemaakt aan het begin van de colleges.

Aanvullende informatie doelgroep

Het vak is bestemd voor studenten van de bachelor track Creatief schrijven Nederlands en van de minoren Aan de slag met literatuur en De schrijfacademie.

Seeing is Believing: Film and Religion and the Construction of Sacredness Beyond the Limits of Representation and Articulation.

Vakcode	G_BATRSAL090
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit Religie en Theologie
Vakcoördinator	dr. J.H. Roeland
Examinator	dr. J.H. Roeland

Betrokken Docenten	dr. J.H. Roeland, dr. S. Sremac
Onderwijsvormen	Seminar

Doel vak

Upon completion of the course, students can:

- denote the visual and imaginative instruments for revealing, producing and transmitting sacred and divine realities and experiences in and through religion and film.
- describe and apply a ‘film as religion’ approach to film, with a specific focus on the dimensions of myth, ritual, morality, transcendence and sacredness.
- can explain how filmmakers apply visual tools to deal with ‘the real’ and ‘the unspeakable’ and the limits of articulation and representation.
- reflect on how film viewing generate deep existential experiences and critically deconstruct the efficacy of film experience in relation to the visual choices made by filmmakers.

Inhoud vak

This course explores the limits of truth-telling and representation. Semantic and text-based approaches have a limited capacity to express the lived experience of people. This is perhaps most obvious when it comes to individual or collective trauma; to the void of reality that is beyond expression (the ‘real’ in Lacanian thinking); and to the experienced presence of sacred and divine realities - three experiences that are often referred to as the ‘unspeakable’, as something beyond articulation. In comparison to purely verbal representations of such experiences, film offers a rich repertoire of audiovisual tools that offers many opportunities of representing realities that are difficult to express in language. In this course, such filmic representations are explored by analyzing the cinematographic and narrative elements of film. In addition, the course discusses the limits of filmic representation as well, for film has its limits too when it comes to traumatic and sacred experiences: they can be not only unspeakable, but also ‘unimaginable’. However, film can at the very least probe the boundaries of what can be expressed; filmic methods and techniques can be mobilized to show what it means not to be able to imagine, not to be able to speak. In the third place, this course not only explores the representation of the ‘unspeakable’; it also applies a film as religion approach that enables students to understand the ‘religious work’ of film: how film may mediate (reveal, produce and transmit) sacred and divine realities and experiences. Students learn to unravel the representation and mediation of the unspeakable by carefully analyzing the cinematographic and narrative elements of film.

Aanvullende informatie onderwijsvormen

Lectures, film analysis classes.

Toetsvorm

Individual assignments (20%). The course is concluded with team assignment (80%), namely a cinematographic and narrative analysis of a film chosen by the team members.

Sensomotorische Coordinatie

Vakcode	B_SENSOCOR
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. K. van der Kooij
Examinator	dr. C.E. Peper
Betrokken Docenten	prof. dr. J.E. Bos MSc, dr. K. van der Kooij, dr. C.E. Peper
Onderwijsvormen	Werkcollege, Practicum, Hoorcollege

Doel vak

Algemene doelstellingen:

• Oriëntatie op het studieterrein van sensomotorische coördinatie: de student is bekend met het soort vragen dat in het onderzoek op dit terrein wordt bestudeerd;

• Aan het eind van de cursus heeft de student basale kennis van twee benaderingen op dit terrein (de neurowetenschappen en de psychologie) en van feiten en begrippen die binnen deze benaderingen centraal staan;

• De student kan de verkregen kennis en inzicht van sensomotorische coördinatie toepassen op concrete problemen, en kan hierbij samenwerken met medestudenten.

Specifieke doelstellingen:

De student heeft kennis van en inzicht in

• fundamentele problemen in de studie van sensomotorische coördinatie;

• neurofysiologische achtergronden van de sensomotorische coördinatie;

• psychologische achtergronden van de sensomotorische coördinatie;

• basisprincipes van de coördinatiedynamica.

Inhoud vak

Bij bewegen staan we zelden stil. We lopen, fietsen, spreken, schrijven, vangen, springen, slaan en schoppen alsof het niets is. Toch gaat het hier, als je er even over nadenkt, om vrij opzienbarende prestaties. Het menselijk lichaam telt meer dan 600 spieren en meer dan 100 gewrichten: daar kunnen heel wat bewegingen mee gemaakt worden, maar hoe maken we juist die ene, gewenste beweging?

Dankzij het zenuwstelsel zijn onze bewegingen in de regel goed gestuurd en gecoördineerd, tenzij we te veel hebben gedronken of lijden aan een ziekte die de motoriek ondermijnt. De vraag die in deze cursus centraal staat is hoe de sturing en coördinatie van bewegingen tot stand komen, en welke rol onze sensorische systemen daarbij spelen.

De cursus biedt een brede en gevarieerde inleiding in dit veelzijdige onderzoeksterrein. Naast een algemene introductie in de centrale thema's, wordt met name aandacht besteed aan de neurofysiologische en psychologische achtergronden van bewegingscoördinatie. Hierbij komt ook de relatie tussen waarnemen en bewegen ruimschoots aan bod. De stof wordt geïllustreerd aan de hand van concrete voorbeelden van zowel alledaagse situaties als bepaalde ziektebeelden.

Aanvullende informatie onderwijsvormen

28 uur/ 14 hoorcolleges 2 uur / 1 vragenuurtje 10 uur/ 5 web-labs (incl. voorbereiding) 4 uur/ 2 werkcolleges 2 uur/ voorbereiding werkcolleges 95 uur/zelfstudie (incl. college- en tentamenvoorbereiding) 3 uur / tentamen 2 uur/ 1 practicum (alleen voor reguliere eerstejaars) 2 uur/ voorbereiding practicum (alleen voor reguliere eerstejaars) 20 uur/ werkstuk schrijven (alleen voor reguliere eerstejaars) 24 uur/ extra zelfstudie-opdracht, incl. tentamenvoorbereiding (alleen voor recidivisten, minor- en bijvakstudenten) De contacturen bestaan uit 14 hoorcolleges, 1 practicum, 2 werkcolleges, 1 optioneel werkcollege en een vragenuurtje. Let op: Reguliere eerstejaarsstudenten voeren andere extra opdrachten uit (practicum & werkstuk) dan recidivisten, minor- en bijvakstudenten (zelfstudie-opdracht).

- De hoorcolleges hebben tot doel de stof in de te bestuderen literatuur en kennisclips nader toe te lichten en met o.a. voorbeelden en korte opdrachten tot leven te brengen. Aanwezigheid bij de hoorcolleges is niet verplicht, maar de inhoud van de colleges maakt wel deel uit van de tentamenstof.

- Tijdens de werkcolleges worden een aantal onderwerpen uit de collegestof nader besproken. De werkcolleges worden uitgevoerd in groepjes van ca. 20 studenten.

- Daarnaast wordt de student regelmatig uitgenodigd tot zelfwerkzaamheid aan de hand web-labs, die worden uitgevoerd via Canvas. Deze opdrachten worden niet behandeld tijdens de colleges. Sommige web-labs fungeren primair als een toets van de beheersing van de gedoceerde stof, terwijl in andere web-labs deze stof verder wordt uitgediept. Iedere web-lab is gedurende ongeveer 1 week beschikbaar.

- Reguliere eerstejaarsstudenten voeren als extra opdracht een practicum uit, waarbij zij een aantal coördinatiefenomenen aan den lijve ondervinden en aan de hand van opdrachten bestuderen. Naar aanleiding van dit practicum schrijft de student een individueel werkstuk. Tijdens een optioneel werkcollege kunnen zij feedback krijgen op een tussenversie van dit werkstuk.

- Recidivisten, minor- en bijvakstudenten bestuderen zelfstandig extra leerstof, die met een aantal extra vragen getoetst wordt tijdens het tentamen. Verplichte onderdelen voor reguliere eerstejaars: practicum, 2 werkcolleges, web-labs, werkstuk, tentamen. Verplichte onderdelen voor recidivisten, minor- en bijvakstudenten: 2 werkcolleges, weblabs, zelfstudie-opdracht (incl. toetsing), tentamen.

Toetsvorm

Schriftelijk tentamen met ja/nee-vragen. Het eindcijfer wordt voor 85% bepaald door het cijfer op dit tentamen.

Reguliere eerstejaarsstudenten: De overige 15% wordt bepaald door het cijfer voor het werkstuk. Het cijfer voor het werkstuk dient minimaal een 4,5 te zijn.

Daarnaast zijn uitvoering van de web-labs en actieve deelname aan het practicum en de werkcolleges een voorwaarde om de cursus te kunnen afronden.

Recidivisten, minor- en bijvakstudenten: De zelfstudie-opdracht wordt getoetst d.m.v. aantal een aantal extra, ‘open’ vragen, waarop een kort antwoord moet worden gegeven. Het cijfer voor dit onderdeel van het tentamen bepaalt voor 15% het eindcijfer. Daarnaast zijn uitvoering van de web-labs en actieve deelname aan de werkcolleges een voorwaarde om de cursus te kunnen afronden.

Literatuur

Verplichte literatuur:

- J. Tresilian (2012). Sensorimotor control & learning. An introductionto the behavioral neuroscience of action. Palgrave Macmillan: H1 t/m 4, §5.3, §7.1-2, §8.1, H9, H11, H12. Nadere

specificatie van verplichte paragrafen wordt aangegeven in de cursushandleiding.

- Collegedictaat

Extra verplichte literatuur voor recidivisten, minor- en bijvakstudenten (zelfstudie-opdracht): Hoofdstuk 8 (Locomotion on Legs) uit bovengenoemd boek.

Geadviseerde literatuur:

- Uit bovengenoemd boek: §5.4.2-3, §6.3, §7.3-5 (i.h.b. §7.5.4).

Afwijkende intekenprocedure

De indeling van de werkcollege- en practicumgroepen vindt plaats via Canvas.

Toelichting Canvas

Via Canvas worden de o.a. volgende zaken beschikbaar gesteld en/of

geregeld:

- cursushandleiding

- collegedictaat

- kennisclips (voorafgaand aan elk college)

- college-slides (na afloop van elk college)

- oefententamen & antwoorden (kort voor tentamendatum)

- (her)tentamenantwoorden (na afloop van (her)tentamen)

- web-labs (op de aangegeven momenten)

- inschrijving werkcolleges en practicum

- handleiding werkcolleges en practicum

- formateisen werkstuk

- inleveren werkstuk

- werkstukcijfer

- diverse mededelingen tijdens de cursus

Sexual Health: Threats and Opportunities for minor GAD

Vakcode	AB_1294
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	S.E. Duijs
Examinator	dr. M.E. Muntinga
Betrokken Docenten	S.E. Duijs, dr. M.E. Muntinga, M. Overman
Onderwijsvormen	Study Group, Lecture

Doel vak

The aim of this interdisciplinary course is to address sexual health in its larger sociopolitical, medical and scientific context, and to discuss ways in which public health research and policy can contribute to better sexual health of populations, communities and individuals. This involves learning objectives related to theory, research methodology, health outcomes and health services.
The course brings together insights from the field of public (global) health, medicine, and the social sciences. Students who take this course should be motivated to collaborate with peers with various discpliniary backgrounds, and to think beyond the boundaries of their own academic fields.
Learning objectives are:
1. Students can describe different theoretical perspectives historically used to
understand and investigate sexuality and sexual health, and identify these perspectives in sexual
health research approaches and practices;
2. Students can describe main sexual health outcomes and their patterns and trends; they can apply these outcomes to (vulnerable) social groups such as sex workers, displaced populations, and gender and sexually diverse people in their political-cultural context;
3. Students can critically reflect on the concept of sexual health and sexual health risk;
4. Students can discuss intervention strategies and priorities to promote sexual health of communities, and can explain challenges researchers, policy makers and practitioners face in planning and implementing sexual health programs;
5. Students can formulate a qualitative or quantitative research question into the field of sexual health, systematically search scientific literature, write a literature review in which they answer their research question, and present their findings to their peers;
6. Students can seek out and synthesize different sources of knowledge (scientific, cultural, experiential) to describe current challenges in sexual health, en formulate recommendations for research, policy and practice.

Inhoud vak

We may or may not always notice, but we are surrounded by sexuality. Sex flashes at us from roadside billboards and our television screens. We discuss our own sex lives and those of others in cafes, at sports courts and at work, and sex jokes are part of the repertoire of many a

comedian. In our modern, wired societies, sexually explicit material can be easily accessed on the internet. Sexuality is a huge part of our shared social world.
Sexuality as an academic subject covers more than just the physical act of having sex. When we talk about sexuality, we refer to a range of issues pertaining to, for instance, the body, sexual identities, orientations and expressions, value systems, and knowledge-producing organisations and institutions. Within the institute of science, the study of human sexual life and relationships is called sexology. American scientist Alfred Kinsey (1895 - 1956) is considered a pioneer in the sexology field, and his work has markedly contributed to our understanding of human sexuality – and how to research it. Now a familiar term to many, sexual health as a discipline of study emerged only after Kinsey completed his most famous work ‘Sexual behavior in the human male’ (1953).
Plenty of scientific studies teach us there is a firm connection between sexuality and (public) health. Some individuals experience that consensual and pleasurable sex adds to their mental health, while others stay sexually healthy by not having sex at all. But sex and health are not always positively correlated. Even sexual activity that is pleasurable might lead to adverse health outcomes such as Sexually Transmitted Infections (STI’s) and unwanted pregnancies. Individuals who experience limitations in expressing themselves sexually might receive the medical diagnosis sexual dysfunction. Unfortunately, sexual violence is still a pervasive public health problem without an easy solution.
Because sex and sexuality are and remain important throughout the life course and across cultures, sexual health is not confined to a particular life stage, for instance the reproductive years, or to certain sociocultural groups or identities. The World Health Organisation (WHO) defines sexual health as a state of physical, mental and social well-being in relation to sexuality. It requires a positive and respectful approach to sexuality and sexual relationships, as well as the possibility of having pleasurable and safe sexual experiences, free of coercion, discrimination and violence . This definition emphasizes a positive approach towards sexuality and sexual behaviour, positioning sexual health as a fundamental aspect of human rights and highlighting the potential benefits of sexual activity, such as emotional and physical pleasure, intimacy and individual expression. However, definitions of sexual health that emerged during and after periods marked by the rapid spread of sexually transmitted infections (STI’s) such as HIV/AIDS have highlighted the threats of sexuality, conceptualizing sexual health as, for instance, the absence of disease and illness.
In this interdisciplinary course, sexual health and its themes will be mainly approached from a public health perspective. Different topics related to sexual health will be discussed in more detail in order to come to a better understanding of sexual health in its individual, societal and scientific context. In addition, we will address theories and methodologies of research into sexual health, and examine implications of outcomes of such research for public health policy.

Aanvullende informatie onderwijsvormen

The course consists of lectures (on Wednesday morning), mentor/instructor sessions (on Monday morning), and seminars (on Wednesday afternoon). A proactive attitude toward group participation, reading and writing will enhance your course experience!
Each week, a new theme is introduced that corresponds to the textbook chapters (i.e. Sexual Health. A Public Health Perspective (2012)). Students are encouraged to actively participate in group discussions and exchange scientific, cultural and individual perspectives. When discussing sexuality, mutual respect and deep listening skills are key. We will work in a non-hierarchical setting, in which each individual’s contribution is valued, and learning from each other is considered an essential classroom asset.
Lectures center on sexual health topics in the context of a textbook chapter. For instance, a lecture on sexual violence in the week about sexual health outcomes, or a lecture on research bias in the week about

sexual health methodology.
The seminars center on elaboration and in-depth analysis of the theme within its scientific, societal and individual context. In the Monday sessions, students work on their course assignments. In the Wednesday sessions, students discuss current sexual health topics from a research and policy perspective, elaborate on the information presented in the textbook, and apply their knowledge to public health challenges. We will end the course with a debate on a current sexual health challenge.
Attendance: Lecture attendance is recommended. Mentor session and seminar attendance is obligatory. If a student is absolutely unable to attend, and alternative assignment should be made and handed in before the end of the course.

Toetsvorm

The assessment of the course consists of four parts:
1. Writing assignment – group grade (40%)
2. Written exam in week 8 (open questions) – individual grade (60%)
To complete the course, a minimum score of >5.5 for both the literature review and the exam is required. The grades cannot be compensated.
Paper writing assignment: Students collaborate in groups of 3-5 to write one paper (appr. 5000 words) about a current sexual health theme. Writing tasks should be shared equally among group members.

Literatuur

The literature for this course consists of:
1. The book Sexual Health. A Public Health Perspective (edited by Kaye Wellings, Kirstin Mitchell & Martine Collumbien). ISBN: 9780335244812.
2. Scientific articles

Aanvullende informatie doelgroep

Target group for this interdisciplinary minor course are students from FALW, Biomedical Sciences and other students eligible to participate in the minor Five Big issues in Health, as well as students participating in the minor Gender & Diversity.

Overige informatie

Please note: This course will touch upon topics such as sexual assault, harassment and violence.

As course instructors, we value educational environments in which students' learning can thrive. We are therefore committed to considering different comfort levels with topics that might cause recollection of harmful events or trigger feelings of unsafety. At the same time, we also believe those topics are worth discussing and investigating, because they are essential for our collective understanding of individual and population sexual health and wellbeing – and for developing effective interventions, so that we can move toward equitable health care systems responsive to the needs of people at risk of victimization and survivors.

If you read the above and wonder whether this course is for you, feel free to reach out to one of the instructors. Together, we can discuss your options.

Course instructors:

- Saskia Duijs, instructor and course coordinator

- Marik de Koning, instructor

- Instructor TBA

Examinator: Maaike Muntinga

Afwijkende intekenprocedure

You have to register for the Minor Five Big Issues of Health or the minor Gender and Diversity.

Showed trials: Truth telling in documentary film and court

Vakcode	R_STTT
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Rechtsgeleerdheid
Vakcoördinator	dr. S. Stolk
Examinator	dr. S. Stolk
Betrokken Docenten	dr. S. Stolk, prof. dr. W.G. Werner
Onderwijsvormen	Study Group, Lecture

Doel vak

This course focuses on truth and (visual) evidence in documentary film and court proceedings. More concretely, we explore the truth regimes of documentary film and court proceedings and study what happens when the two meet.

Upon completion of the course, students can:

- Recognize, understand and compare the functions of truth-telling in documentary film and court proceedings.
- Analyse how visual evidence is used in both contexts.
- Critically reflect on the representation of legal proceedings in documentary film.
- Use film to study law and court cases.

The course specifically focuses on international courts and tribunals but the core questions apply to national contexts too.

Inhoud vak

The course revolves around four main questions:

(a) What are the functions of truth telling in documentary film and court proceedings? How is (visual) evidence used in both contexts? We will discuss these questions on the basis of a concrete cases before international courts and documentary films about related topics.

(b) What happens when film footage is used as evidence in a court case? How do truth claims shift? How can primarily artistic products such as documentary films be transformed into legal evidence, and how does this differ from the use of other types of visual evidence? Again, we will discuss these questions in relation to a concrete case.

The function of truth in court is primarily geared towards a decision, and thus sets boundaries to what counts as relevant and what is out of order. In addition, court cases are pre-structured through rules of evidence and procedure, determining what counts as reliable (visual) evidence. Finally, truth in court is most of all articulated verbally. Documentary film, on the other hand, can be used to tell a multitude of truths and knows numerous ways to present and combine different types of evidence. It can leave questions undecided and allows for ambiguity. What is more, documentaries not only tell a truth, they most of all show a truth.

In the course, students are trained to recognize and understand the differences and overlap between these truth and evidence regimes of (documentary) film and law.

Aanvullende informatie onderwijsvormen

Interactive lectures; film screenings; and group discussion.

Toetsvorm

Three short assignments that compare the use of visual materials in court cases and documentary film. The assignments include written and visual components (such as essays, storyboards, film clips etc.)

Vereiste voorkennis

Not required

Literatuur

See course manual published on Canvas.

Aanvullende informatie doelgroep

Apart from regular students, the course is also available for:

- Students from other universities/faculties
- Contractor (students who pay for one course)

Overige informatie

The course is part of the minor ‘Visual Evidence’. The course is open for everyone.

Toelichting Canvas

Canvas is the main communication platform of the course.

Aanbevolen voorkennis

Not applicable. No specific knowledge in the field of law and/or film is required.

Sportpsychologie

Vakcode	B_SPORTPSY
Studiepunten	6.00
Periode	P1
Vakniveau	200
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. Y. Hill
Examinator	dr. Y. Hill
Betrokken Docenten	dr. R.R.D. Oudejans, dr. Y. Hill
Onderwijsvormen	Hoorcollege

Doel vak

Studenten beschikken over kennis van en inzicht in de belangrijkste onderwerpen,stromingen en theorieën van de sportpsychologie.

Inhoud vak

De cursus beoogt de studenten te introduceren in het domein van de

sportpsychologie en hen kennis te laten maken met het gebied van de

exercise psychology. Aan de hand van het boek ‘Sportpsychologie’ vindt

kennismaking plaats met de belangrijkste onderwerpen van de

sportpsychologie. Aan de orde komen onder andere:

- sportpsychologie en de relatie van sportpsychologie met 'de'

psychologie; de ontwikkeling van de sportpsychologie;

- motivatie, attributie en emotie en sport;

- persoonlijkheid en sport;

- mentale vaardigheden en mentale training;

- coaching;

- sportteams;

- agressie, blessures, burn-out, verstoord eetgedrag en 10.000 uur

oefenen;

Per hoofdstuk zal ook een wetenschappelijk artikel in het Engels bestudeerd moeten worden.

Daarnaast wordt kort stilgestaan bij mentale voorstellingen.

Kennismaking met de exercise psychology vindt plaats aan de hand van

het hoofdstuk over "Exercise behavior and adherence" uit het boek

‘Foundations of sport and exercise

psychology’ van Weinberg & Gould, waarbij onder andere aandacht wordt

gegeven aan verschillende modellen van gedragsverandering.

Aanvullende informatie onderwijsvormen

De cursus omvat 12 hoorcolleges van elk twee uur en wordt afgesloten met
een tentamen. De resterende circa 144 uren zijn voor zelfstudie. Twee
van de 12 colleges zijn gastcolleges verzorgd door sportpsychologen
die in de praktijk van de sport werkzaam zijn.

Toetsvorm

Tentamen (waar-onwaarvragen).

Literatuur

- Bakker, F.C., & Oudejans, R.R.D. (2019). Sportpsychologie.
Nieuwegein: Arko Sports Media, 5de herziene druk; of 6de druk;
- Weinberg, R.S. & Gould, D. (2015 of 2019). Foundations of sport
and exercise psychology. Hieruit Hoofdstuk 18 (6de druk pp. 425-456) of
Hoofdstuk 19 (7de druk pp.435-466),
Exercise behavior and adherence. Champaign, IL: Human
Kinetics.
- Aanvullende literatuur wordt aan het begin van de cursus
opgegeven en is opgenomen in de cursushandleiding.

State, Power and Conflict

Vakcode	S_SPC
Studiepunten	6.00
Periode	P1
Vakniveau	100
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. H.J.E.M. Jongen
Examinator	dr. H.J.E.M. Jongen
Betrokken Docenten	dr. H.J.E.M. Jongen, P. Lutz
Onderwijsvormen	Lecture

Doel vak

This course aims to familiarize students with key concepts and approaches in political science and how to apply these in order to gain a better understanding of the recent history of, and contemporary issues in, world politics.

Learning outcomes:

Knowledge and understanding – The student has acquired knowledge and understanding of:

(1) different approaches to studying politics, democracy, power and influence;

(2) what ‘states’ and ‘nations’ are and how the modern nation state (system) came into being;

(3) key approaches in political science and an overview of the discipline and major sub-disciplines;

(4) the different ways in which power can be institutionalized and legitimized;

(5) the main developments in the history of world politics from the Peace of Westphalia to the Iraq War, and the current era of globalization and the power shift to Asia;

(6) the main patterns of cooperation and conflict among states as well as non-state actors.

Skills – The student is able to:

(7) apply the key political science concepts and approaches within the sub-discipline of Comparative Politics and the sub-discipline of International Relations in an analysis of (contemporary) political issues.

Inhoud vak

This course provides a general introduction to major concepts and key approaches in political science. It consists of two parts. In the first part, a critical overview is given of several key concepts and their position within political science such as ‘ideology’, ‘power’, ‘legitimacy’, ‘the state’, ‘the nation’ and ‘democracy’, as well as contending perspectives on conflict and cooperation within modern political systems.

We will also discuss contemporary political phenomena (manifest in many different countries) such as the rise of populism. Thus, the first part of the course primarily focuses on concepts and topics related to the sub-discipline of Comparative Politics. The second part of the course introduces students to contemporary world politics and the main concepts and theories in the sub-discipline of International Relations. It first provides an overview of international political history from the 17th century to the present. We then seek to understand history by identifying recurrent patterns of cooperation and conflict between and within states, looking at the role of both state- and non-state actors. We will also apply some of the concepts that we discussed in the first part of the course.

The second part of the course will be concluded with a discussion of several key features of and developments in contemporary global politics, such as the role of the UN, current international conflicts, the information revolution and the growing web of transnational relations.

Aanvullende informatie onderwijsvormen

Lectures

Toetsvorm

Final exam.

Literatuur

To be announced in the course manual (see CANVAS).

Aanvullende informatie doelgroep

First-year Bachelor students in Political Science, Pre-Master students in Political Science, Minor students in Political Science

Strategic Management of Technology and Innovation

Vakcode	E_MFEN_SMTI
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. S.A. Rijsdijk
Examinator	dr. S.A. Rijsdijk
Betrokken Docenten	dr. S.A. Rijsdijk, prof. dr. ir. J.J. Berends, H.K. Fults, E.M. Haynes, C.C. van der Geest
Onderwijsvormen	Seminar, Lecture

Doel vak

Academic & research skills:
In this course students learn to critically evaluate innovation management concepts from academic literature and popular management press.

Bridging theory and practice - knowledge:
In this course, students gain theoretical understanding concerning:
- innovation types and the external innovation environment including innovation trajectories, standards, platforms, and ecosystems
- the development of innovation strategies and their operationalization in project selection, collaboration, and protection
- the product development process and organizational conditions for innovation

In this course, students gain insight in:
- the strategic importance of technological innovation for firms and society,
- recent developments in technology and innovation

Bridging theory and practice - application
In this course, students develop skills to analyze real life cases.

Inhoud vak

This course focuses on the strategic management of technology and innovation. Innovation refers to the development and implementation of new products, services, processes and business models and many of those innovations are enabled by technological developments. Innovation is crucial for business organizations to stay competitive in ever changing markets. In this course, students learn to understand and apply basic theories behind the processes of technology-based innovation within organizations and their environments, the development of innovation strategies, and the organizational implementation of innovation strategies. Theoretical understanding is applied in a simulation game and real life cases focusing on managerial dilemmas in the management of innovation.

Aanvullende informatie onderwijsvormen

Lectures
Tutorials

Toetsvorm

Individual assignment
Group assignments
Written exam

Literatuur

- Schilling, M. (2019). Strategic management of technological innovation (6th ed). Boston: McGraw-Hill.
- Selection of academic articles (listed in course manual)
- Lectures, tutorials, and lecture slides/knowledge clips

Sub-Saharan Africa and the World, 1800-present

Vakcode	L_GABAGES226
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. W. Mwatwara
Examinator	dr. W. Mwatwara
Betrokken Docenten	dr. W. Mwatwara
Onderwijsvormen	Lecture

Doel vak

• To give an intensive introduction to an unfamiliar period and/or approach to the study of history;

• To offer experience in the history and debates about sub-Saharan Africa and the world since 1800;

• To develop skills in analysing historiography; and

• To develop core skills such as: bibliographical search techniques; source analysis; essay writing; giving presentations; and, undertaking independent research.

Students who complete this module successfully will:

• Acquire insight into the history, historiography, and debates about the historical relationship between sub-Saharan Africa and the world since 1800;

• Comprehend how incorporation into/ participation in global networks has shaped sub-Saharan African societies over time.

• Work with primary and secondary sources, applying appropriate concepts and (digital) methodological tools;

• Effectively communicate their ideas in oral form, participating actively in class, preparing group presentations, and coordinating class discussions;

• Develop their academic writing skills, by preparing a book review, a primary source analysis, and a research paper based on primary and secondary sources.

Inhoud vak

Africans and people for whom Africa has been a critical point of reference are actors on the world scene and have been for a long time. For this reason, sub-Saharan African countries, institutions, people, and economies are intertwined with the world in many ways. Its recent history has mostly been characterized by, among other things, violent conflict, abject poverty and governmental mismanagement. Despite its long painful and tragic engagement with the world, sub-Saharan Africa is probably one of the most misunderstood and misrepresented regions in the modern world. This module utilizes sub-Saharan Africa as a starting point from which to explore the spatial, technological, political and socio-economic location of Africa in global networks since 1800, and demonstrate how this has had an impact on the African condition. It seeks to inculcate a productive understanding of Africa's place (past and present) in the world. Special attention is given to the dynamics that produced generations of African facilitators, as Joseph Miller has argued, "from the suppliers of slaves through post-colonial "gatekeeping" elites brokering flows of funds and commodities between national economies and international capital, often to their and their clients' personal advantage." An understanding of the relationship between sub-Saharan Africa and the world enriches understanding of both global connections and specific regional, sub-regional and local contexts. Crucially, it encourages students to assess how Africans shaped as well as experienced such engagements.

Aanvullende informatie onderwijsvormen

2 Seminars (4 hours per week) plus work in archives, independent and assisted by teachers (scheduled moments)

Toetsvorm

• Participation in class (10%)

• Research portfolio (40%), including

➢ a book review (20%)

➢ a primary source analysis (10%)

➢ a research pitch: an oral presentation of the research paper (10%)

• Research paper (50%)

Format: The teacher will provide a list of texts that students can select. Students are also allowed to suggest their own preferred texts. Within the broader theme, the reading list will be adapted to fit the interests and needs of the students who participate.

Meetings: weekly, by appointment.

Reading list as discussed with students (ca. 600 pages)

Sustainability and Environmental Change

Vakcode	AB_1230
Studiepunten	6.00
Periode	P1
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. F. Riva
Examinator	dr. F. Riva
Betrokken Docenten	A.C. Cassanti, dr. F. Riva, dr. P. Scussolini
Onderwijsvormen	Seminar, Lecture

Doel vak

In this course students learn about the environment’s pivotal role in

achieving sustainable solutions for human development, mainly focused on global environmental problems. After this course, students:

1. can explain key concepts from the natural sciences relevant for

sustainability studies;

2. can characterize key components of the environment, namely water, land and atmosphere, and can explain key processes affecting their characteristics;

3. can explain the role of the environment in socio-environmental

systems;

4. can identify methods to quantify the state of the environment, and

analyze environmental change;

5. are familiar with data and trends for relevant the environmental

conditions for specific sustainability challenges.

Inhoud vak

The environment plays a crucial role in supporting societies, for example by providing materials, energy, food, clean air, and clean water. Environmental conditions change over space and time, influenced by both natural and

human factors. In this course students learn about the environment's pivotal role in achieving sustainable solutions for human development. Starting from the key environmental components water, land and atmosphere, we characterize environmental change and how that leads to other environmental and societal changes. Methods to assess environmental change are addressed and students identify for their specific case studies what strengths, opportunities, weaknesses, and threats are associated to the 'planet dimension'. The course comprises interactive lectures and exercises and is evaluated through an assignment and a written exam.

Aanvullende informatie onderwijsvormen

The course is organized in thematic weeks, which provide students with an understanding of the specifics of the dimensions water, land and atmosphere, how these can be studied and how they interact. Each week has 1 to 2 lectures, in parallel to which students develop their assignment. Lectures and assignment are supported by in-class discussions, reading material, and exercises.

Toetsvorm

The course will be evaluated through

1) A 3-minute speed talk (presentation) describing a sustainability challenge in quantitative terms (30% of final grade)

2) A closed-book written exam (70% of final grade).

Note:

- An overall minimum grade of 5.5 is required to pass the course.

- A minimum grade of 5.0 for the exam is required to pass the course.

- There is one resit opportunity for the exam (will be discussed,

typically a few months after).

- Assignments with a grade lower than 5.5 can be improved once, after

which the maximum grade that can be obtained for the assignment is 6.0.

Literatuur

The course builds on a series of selected articles, relative to the lecture content. Required and recommended readings are listed at the

learning environment.

Moreover we make use of:

- Open data sources, educational software packages, websites, videos etc

Aanvullende informatie doelgroep

The course is primarly aimed at students following the minor Sustainability: Global Challenges, Interdisciplinary Solutions, but is open to all 3rd year BSc students.

Overige informatie

The course is coordinated by Dr. Federico Riva

Core lecturers:

- Dr. Federico Riva

- Dr. Paolo Scussolini

Sustainable Business Processes

Vakcode	E_MFS_SBP
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. ir. D.A.M. Inghels
Examinator	dr. ir. D.A.M. Inghels
Betrokken Docenten	dr. ir. D.A.M. Inghels, S. Hülagü
Onderwijsvormen	Study Group, Lecture

Doel vak

After successfully completing the course Sustainable Business Processes you are able to
• Analyse business and societal problems taking into account the interests of different stakeholders (economic, ecological, societal, and others)
• Quantify the economic, ecological, and societal objectives for business process cases and find optima using multi-objective linear optimization
• Apply systems thinking using System Dynamics to examine the dynamic behaviour of business and societal processes
• Understand the transition from a linear to a circular economy and its implications for managing business processes
• We put emphasis on applying the methodologies to real-life cases

Inhoud vak

This course aims to introduce students to operationalizing sustainability in business and societal problems. We define sustainability as the combined economic, environmental, and social optimum of business process alternatives that take into account constraints, such as technological limits or legislation, also known as the triple bottom line (TBL) approach of People-Planet-Profit optimization. Life Cycle Assessment (LCA) is presented as a methodology to quantify the environmental impact of products and processes and Analytic Hierarchy Process (AHP) to quantity social impact.
Multi-Criteria Decision Analysis is introduced as a concept to operationalize the TBL approach for practical business process problems. Next, we discuss systems thinking using Systems Dynamics for understanding and evaluating the complex and interactive behavior of systems, such as sustainable supply chains. Finally, the transition from a linear to a circular economy and its effects on business processes is discussed.

Aanvullende informatie onderwijsvormen

12 Lectures and 6 computer tutorials of each 2 hours

Toetsvorm

Written exam – Individual assessment;
Two written assignments – Group assessment that may be corrected depending on the individual contribution

Vereiste voorkennis

Students should have a basic knowledge of business processes or operations management and business mathematics. Students are expected to have a basic understanding of MS Excel before the start of the course.

Literatuur

Mandatory readings:

(1) Inghels, D. (2020). Introduction to Modeling Sustainable Development in Business Processes - Theory and Case Studies. Springer Nature Switzerland AG. Online: https://www.springer.com/gp/book/9783030584214#aboutBook
(2) Fiksel, J. (2015). Resilient By Design: Creating Businesses That Adapt and Flourish 35 in a Changing World, ISBN: 978-1-61091-684-4 (Print) 978-1-61091-588-5 (Online), Online: http://link.springer.com.vu-nl.idm.oclc.org/book/10.5822/978-1-61091-588

Aanvullende informatie doelgroep

This course is part of the Minor Sustainability and Innovation.

Overige informatie

During the course, models will be built using software tools. Students should be interested in applying mathematics using software tools. Moreover, working out two written assignments in groups appeals to your commitment to contributing to the success of the group result.

Talent and Talent Identification

Vakcode	B_TALIDENT
Studiepunten	6.00
Periode	P3
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	dr. D.L. Mann
Examinator	dr. D.L. Mann
Betrokken Docenten	dr. D.L. Mann
Onderwijsvormen	Hoorcollege

Doel vak

By the end of this course, the aim is that you will be able to design a basic evidence-based system of talent identification and development for an applied organisation (e.g., in sport, music, or another area of expertise).

We will achieve this aim by learning to:
1. Evaluate the relative contributions of ‘nature’ and ‘nurture’ in the development of talent;
2. Understand how talent develops and apply that knowledge to identify the stage of development at which talented athletes should be identified;
3. Identify the impact that environmental factors such as an individual’s date or place of birth can have in influencing the development of talent;
4. Critically evaluate the limitations of existing measures of talent and suggest newer measures that overcome those limitations;
5. Evaluate whether talent identification systems used by applied sporting organisations (both in the Netherlands and overseas) adhere to best scientific practice.

Inhoud vak

The ability to identify and develop talent in potentially skilled athletes is a central role for many coaches, scientists, and sporting administrators. National and professional sporting organisations invest substantial amounts of time and money in establishing systems designed to identify and nurture future talent, yet there is still considerable doubt about which approach is most likely to be successful, and how effective these systems may be.

This course on Talent and Talent Identification will assess what it takes to become talented in sport and other areas of expertise (e.g., music, dance), and will uncover what we know about the ideal conditions for developing skill. The course will address the new and developing area of research that seeks to evaluate existing talent identification systems and to develop newer, more evidence-based procedures for identifying and developing talent. Further, guest lecturers from professional sporting organisations will reveal the systems that they have in place to identify and nurture talent.

Aanvullende informatie onderwijsvormen

The course consists of 12 lectures (mandatory attendance, 18 hours in total), in addition to the expectations of self-study (approximately 114 hours), at least one assignment (approximately 10 hours total), and a final assessment (assignment or exam).

The course is taught primarily in English, though some readings and guest lectures may be in Dutch.

Toetsvorm

Assessment comprises at least one assignment and may include a final exam.

Literatuur

Textbook: Baker J., Cobley S., Schorer, J. (2020) Talent identification and development in sport. International perspectives. Routledge: Abingdon, Oxon

Aanvullende informatie doelgroep

The course is taught in the Department of Human Movement Sciences but is open to students of all backgrounds with an interest in talent identification in sport and other areas of expertise.

Text Mining for AI

Vakcode	XB_0085
Studiepunten	6.00
Periode	P5
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. I. Markov
Examinator	dr. I. Markov
Betrokken Docenten	prof. dr. P.T.J.M. Vossen, dr. I. Markov
Onderwijsvormen	Lecture, Seminar

Doel vak

Knowledge and understanding: at the end of the course, students will be

familiar with basic knowledge of some of the core aspects of Natural

Language Processing, Linguistics and Text Mining:

rule-based systems, machine learning, deep learning, text classification, sentiment

extraction, entity recognition and topic modeling of texts.

Applying knowledge and understanding: students will be able to implement

NLP processing systems and modules and evaluate these.

Making judgements: students will have a basic understanding of the ethical and societal implications of the developments in NLP.

Communication skills: students will be able to write a scientific reports about a specific research question in a group of students.

Learning skills: students will be trained in acquiring a set of complex NLP and text mining topics in a restricted period of time, come up with a research question and perform the necessary (empirical) research.

Basic concepts from Linguistics and foundational concepts from Natural Language Processing. Skills to use, apply and critically assess text mining techniques. Adapt and build text mining techniques to specific target domains and applications.

Inhoud vak

Basic concepts from Linguistics and foundational concepts from Natural Language Processing. Skills to use, apply and critically assess text mining techniques. Adapt and build text mining techniques to specific target domains and applications.

Aanvullende informatie onderwijsvormen

Theoretical lectures and working group sessions

Toetsvorm

Multiple choice exam on theory 60% and the group project report 40%. Both the multiple choice exam and the group project are eligible for resit.

Aanbevolen voorkennis

Programming in Python, using Github

The Developing Brain

Vakcode	AB_1059
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. M.C. van den Oever
Examinator	dr. M.C. van den Oever
Betrokken Docenten	dr. L.N. Cornelisse, prof. dr. B.P.C. Kreukels, dr. M.C. van den Oever, prof. dr. R.E. van Kesteren, B.M.W. Borgman, dr. V.M. Heine, dr. L.R. van der Kallen
Onderwijsvormen	Study Group, Computer lab, Lecture, Practical

Doel vak

Students will learn about mechanisms in the brain that underlie

normal and aberrant brain development, shaping the life of

individuals over time.

Inhoud vak

The focus of this course is on phases of brain development that shape the life of individuals over time. The brain performs differently at various ages. Whereas the young brain is very plastic, the aging brain gradually loses its adaptive capacity. Importantly, early and late brain development are affected by specific genetic factors and vulnerable to changes induced by environmental stimuli. These alterations can result in neurodevelopmental and neurodegenerative disorders.

The course consists of three modules (one week per module), each covering a critical phase of brain development. In module 1, we will focus on early (prenatal) brain development and its relation to brain disorders such as autism. In the second module, we will focus on postnatal brain development and how alterations contribute to issues during childhood and adolescence, such as gender identity and schizophrenia. In the last module, we will discuss cognitive decline associated with normal brain aging as well as specific diseases of aging, such as Alzheimer’s and Parkinson's disease. Notably, each module contains a keynote lecture related to the topic of the module. Keynote lectures are given by renowned experts in their fields and are mandatory for all students.

In addition, students will perform a Science in the Media assignment. Scientific findings are not always accurately covered in main stream media due to various reasons. The goal of this assignment is that students learn to critically read and evaluate a media coverage of a scientific study, and to present their evaluation of the flaws/strengths of a media article in small groups to their peers.

Aanvullende informatie onderwijsvormen

Lectures: 30 h Workgroups: 4 h Q&A sessions: 3 h Self study: ~60 - 120 h (in between lectures, students are expected to spend time on reading of literature to prepare for next lectures/workgroups and to make assignments on canvas).

Attendance at the Science in the Media sessions and keynote lectures is mandatory.

Toetsvorm

Exam (E; multiple choice questions and open questions): 80%
Academic skills assignment (A): 20%
The grade of both tests has to be >5.5 to pass the course.

Students have the option to resit the exam (E).

Vereiste voorkennis

This course is part of the minor Brain and Mind. University students need at least 90 ECTS to be eligible for (courses in) this minor. HBO students can follow (courses in) this minor if they have at least 120 ETCS.

Literatuur

Exam material: Slides of all lectures (will become available on Canvas).
Recommended reading material: will become available on Canvas.
No mandatory literature/books.

Aanvullende informatie doelgroep

Students of the minor Brain & Mind.
Third year bachelor students who are interested in the neurobiology of the brain during prenatal and postnatal development, as well as aging.
Due to curriculum overlap, students from Biomedical Sciences and Health and Life Sciences from the VU cannot follow this course.

Overige informatie

This minor course requires a minimum of 20 participants.

Afwijkende intekenprocedure

Students are assignment to groups on Canvas

Toelichting Canvas

Slides of the lectures and recommended reading material will become available on Canvas in three modules.
Each module ends with quiz questions about the lectures of that module. Quizes have to completed by the student to be able to access the next module.
Keynote lectures will end with an open question that has to be answered on Canvas. Therefore, the student has to make sure that he/she has access to Canvas during the keynote lectures.
Students have to upload their Science in the Media assignment to Canvas.

Aanbevolen voorkennis

This course is part of the minor Brain and Mind. A basic understanding of neurons, neurophysiology and molecular biology (DNA and proteins) is required. For this, we recommend to follow the courses 'Cognitive Neuroscience' and 'Nature versus Nurture' of this minor.

The Economics of Crises

Vakcode	E_MUE_TEC
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	School of Business and Economics
Vakcoördinator	dr. M. Mastrogiacomo
Examinator	dr. M. Mastrogiacomo
Betrokken Docenten	dr. M. Mastrogiacomo
Onderwijsvormen	Study Group, Lecture

Doel vak

The objective of the course is to introduce students to the difference between different types of crises, such as the present one (caused by an exogenous event, a pandemic plus a war and subsequent inflation) and that of 2008-2012 (that was caused by a financial bubble). To do so, we review theory and practice of macroeconomic, fiscal and monetary policy, including regulation of the financial system.

Specific learning outcomes upon completion of this course are:

• Ability to set the present economic situation caused by a pandemic, a war, and high inflation, in a broader contest, evaluating policies to restore public health and economic activities.

• Ability to apply macroeconomic concepts and theories to analyze problems of employment and inflation and measurement of Business Cycles.

• Capability to analyze the role of macroeconomic policymakers in managing money, banking and the business cycles.

• An understanding of the policy problems facing central banks, with a focus on monetary policy and financial stability.

• An understanding of the policy problems facing governments, with a focus on fiscal policy.

• Understand the origin of economic crises, mostly if exogenous (corona crisis, aggression to Ukraine) or debt-driven (financial crisis), and the effects of fiscal and monetary policy.

Inhoud vak

We investigate the origin and approach to the current economic swings and the previous (financial) crisis looking at these issues from different perspectives. So, for the present crisis, we debate mostly the policy response of national governments, the ECB and the EU, and the effect of the preset geopolitical crisis on inflation and growth. For the previous crisis we discuss the common narrative (the banks were guilty) but also the micro view on the role of indebtedness (we did it!). Which explanations will convince you the most? The course discusses Macroeconomic policies intended to affect and dampen the business cycles linked to a crisis. We discuss the roles of different authorities in conducting macroeconomic policies aimed at preventing a crisis or at restoring the economic activity after a crisis hits. Relevant topics are for instance: economic activity after a pandemic, money creation, control of the interest rates, central banking and the stabilizing role of fiscal policy with a specific focus on health care expenditures.

Aanvullende informatie onderwijsvormen

In this course the tutorials have COMPULSORY attendance. This course consists of a series of lectures and tutorials in small groups. Assuming that by the time this course starts didactics will be allowed on campus, the lectures (each week 2 lectures of 2 hours each) aim at informing students about the content of the course. The traditional set-up is interrupted by short discussions using different multimedia devices. The tutorials (small size, each week 1 lecture of 2 hours each) aim at stimulating active participation of the students, in group but also individually. These are typically arranged in the form of debates, round table discussions or the simulation of an academic conference. Participation in the tutorial is compulsory. Students prepare individual assignments; these are (partly) handed in before the tutorial (deadlines change depending on the tutorial, check on Canvas). Group discussions on the reading material, as well as presentations in which insights are applied and reflected on, are partly evaluated during tutorials. Solving problem sets, and discussing economic data are part of the tutorials too.

Toetsvorm

Exams will take place around mid-December and a re-sit opportunity is offered in the second half of March. For details, please check information disseminated on VUNet or posted on the website rooster.vu.nl. Exams last approximately 2 hours. Course grade is determined as follows: exam grade needs to be 5.0 or higher – in that case: 30% from hand-in assignments, 70% from written exam; else: exam grade. To those who participate into less than 4 (compulsory) tutorials and/or do not deliver their tutorial work, one point will be subtracted from the tutorial grade. Not attending the tutorials makes it nearly impossible to pass this course.

Vereiste voorkennis

Textbook:

• Acemoglu, Daron, David Laibson and John A. List, 2016, Economics. Harlow, Essex: Pearson Education Ltd. ISBN13: 978-1-292-07920-2 This book should be familiar from earlier Minor Economics courses on Fundamentals of Microeconomics and Development of Macroeconomic Thought.

The Law and Politics of Fencing the Use of Force

Vakcode	S_LPFUF
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	prof. dr. W.M. Wagner
Examinator	prof. dr. W.M. Wagner
Betrokken Docenten	prof. dr. W.M. Wagner, prof. dr. W.G. Werner, prof. dr. S. Protschky
Onderwijsvormen	Seminar

Doel vak

When finalizing the course, students will have knowledge and understanding of:

- The prospects and problems of international law as an instrument of mitigating and overcoming inter-state war;

- The historical context in which certain approaches concerning fencing and the use of force have developed;

- The prospects and problems of systems of collective security;

- The prospects and problems of combining international law and political science in studying international security.

Inhoud vak

The aim of this course is to introduce students to the changing international rules and regulations on the use of armed force from the perspectives of international law, history and political science/international relations. Over the course of human history, the nature of war and armed conflict has been changing frequently and dramatically. In addition to technology, these developments have been driven by changing ideas about just causes and legitimate ways of using armed force. Limiting the human costs of war has become an ever more powerful motive in designing and modifying the rules governing the use of force. The prime instrument of constraining the use of armed force has been international law. The course discusses the most important developments in the laws of armed conflict since the late Middle Ages, including just war theory, collective security and humanitarian interventions from an interdisciplinary perspective that builds on Public International Law and Political Science/International Relations. This interdisciplinary perspective allows a comprehensive understanding of the achievements and shortcomings in the laws and politics of constraining the use of force. Milestones under discussion include early modern concepts of just war, the balance of power system of the 19th century, colonial warfare, the League of Nations, the United Nations system and wars of decolonization.

Aanvullende informatie onderwijsvormen

This course will be taught in interactive seminars on campus.

Toetsvorm

Final exam

Literatuur

Articles and book chapters from a variety of authors.

Aanvullende informatie doelgroep

3rd year bachelor students in the minor Peace and Conflict Studies.

Toelichting Canvas

further information will be provided on canvas

The Law of Artificial Intelligence

Vakcode	XB_0084
Studiepunten	6.00
Periode	P6
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Bètawetenschappen
Vakcoördinator	dr. T.A.M. Schrepel LLM
Examinator	dr. T.A.M. Schrepel LLM
Betrokken Docenten	dr. T.A.M. Schrepel LLM
Onderwijsvormen	Lecture, Seminar

Doel vak

At the end of the course, the students will:

- understand the legal, ethical, and societal implications of AI developments. (Knowledge and understanding)

- be able to write informed opinions about AI legislation. (Applying knowledge and understanding) (Communication) (Judgement)

- be trained in acquiring a set of complex legal AI-related topics in a restricted period of time. (Knowledge and understanding)

- be able to work together as a team to identify not only the technical but also the legal issues with the design of AI systems. (Applying knowledge and understanding)

- be able to share their findings through presentations, videos, and writings. (Applying knowledge and understanding) (Communication)

- be familiar with basic knowledge about the law that applies to AI systems, including the law specifically focusing on AI systems such as the EU AI Act, and other more general laws such as data protection law, intellectual property law, competition law, liability, and contract law. (Knowledge and understanding)

- be able to identify legal risks when designing AI systems. (Applying knowledge and understanding)

- be in a position to design AI systems compliant with the law. (Applying knowledge and understanding)

- be able to use legal rules to protect their AI systems. (Applying knowledge and understanding)

- be able to reflect critically on the impact of legal rules on technology. (Applying knowledge and understanding) (Judgement)

- be able to express their own motivated opinions and contribute to the writing of better legal rules. (Applying knowledge and understanding) (Communication)

Inhoud vak

For long, lawyers were primarily seen as a nuisance by software developers, maybe best illustrated by the first lawyer who appeared around 2010 at the board of directors meeting of Github. During a discussion, he noticed some legal issues and cautiously interrupted: “I know you do not want to hear this from a lawyer, but…”. He could not continue, for another board member aggressively said: “… then shut the f**** up.” A similar attitude used to be common for AI people. Developers and designers were concentrating on the possible, the opportunities; lawyers were focusing on the permissible, or what is not possible—often without trying to understand the technology potential.

Today, the situation is no longer black and white but rather gray. Lawyers understand better technology and development processes, and AI specialists understand the need for compliance with the law. Technology and the law are not systematically seen as antagonists anymore. Against this background, many governments worldwide are launching initiatives to regulate AI. Their objective often is to contain high-risk AI while supporting other AI applications. One such example is the 2021 European Union legislative initiative Proposal for a Regulation laying down harmonized rules on artificial intelligence.

This course explores various ways AI technology is regulated by law and what role ethics play. The law that is being addressed is both the law specifically focusing on AI and other laws relevant to such systems—such as data protection law, intellectual property law, liability, and contract law. And, because the law naturally lags behind technological evolutions, ethics also plays a role.

After this course, the student will understand how the law regulates AI. They will also understand how to design AI systems compliant with the

law and be protected by it. In short, students will learn about the law and how to benefit from it. This is a world-first.

Aanvullende informatie onderwijsvormen

Students will be given four lectures, including two on “law basics” and two on the European AI Act. Teachers will use a Socratic method to create discussions in the classroom.

Students will also take part in two seminars to explore the European AI Act from a practical perspective. They will comment on the AI Act and discuss their inputs with other classmates and professors.

Students will work in small teams their professors will assign.

Toetsvorm

During one of the seminars, students will do a written exam in the classroom (50%).

Students (pairs of two) will record a 2-minute video in which they will

pick one article of the AI Act and explain either why it is not

technically sound or why it creates legal issues (50% of the final

grade)

There will be a resit for the written exam but not for the video.

Literatuur

The literature will be made available through Canvas

Aanvullende informatie doelgroep

Bachelor of Artificial Intelligence (year 2)

The Personal is Political: life-world and biographies of the Othered

Vakcode	L_AABAALG068
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. Y. Saramifar
Examinator	dr. Y. Saramifar
Betrokken Docenten	dr. Y. Saramifar
Onderwijsvormen	Seminar

Doel vak

This course is influenced by, designed and taught according to The

Anti-Racist Writing Workshop by Felicia Rose Chavez. Therefore, the

learning outcomes are proposed in the form of a communal act and growth:

Upon the completion of this course, students:

1- have engaged in a supportive peer community with an interest in social justice and intersectional feminism.

2- have engaged with political undertones of one another's biography and enhance their appreciation of auto/biographical genre

3- have selected, thought through and written from a living archive of multicultural texts and objects that best inform their individual projects.

4- have curated their definitions of political, personal, gender and social justice according to their lived experiences

5- I have experimented with storytelling through intersectional feminism.

6- have learned why the category called human must be abandoned.

Inhoud vak

"The Personal is political", from the late 1970s women's movement to engage with auto/biographies, storytelling and personhood. The course brings them forth to discuss how sexualities, pleasures, bodies, racialization, illnesses and disabilities intersect historically and become multidimensions of gendered differences and natureculture divide? This course explores how these multidimensions shape everyday lives and biographies of politically engaged personalities, famous or unknown, by studying the genre of political biography, autobiography and life writing.

This course is designed for interactive dialogues between peers (including the teacher) as well as watching films, documentaries, and YouTube channels, followed by writing and research assignments. Students are offered three modes of assessments

• Group Writing (inducing media content such Vlog),

• Individual Writing (Including media content such as Vlog),

The writing assessments shall be based on a biographical research project of students' choosing and the teacher will be more than happy to think along with students on possible topics. Students shall present their findings at the end of the course.

Aanvullende informatie onderwijsvormen

Classes (twice weekly)

Toetsvorm

Modes of assessment:

1- Class Participation (Social Media discussions, In-class dialogues, Attendance) 20%
2- Individual/Group presentations 10%
3- Conceptual/Theoretical Essay 20%
4- Final Paper 50%

Literatuur

Literature: Please refer to the course manual on Canvas

Aanvullende informatie doelgroep

Who may take the course: BA 3 Students History (Geschiedenis), History and International Studies, Humanities, Social Sciences, Philosophy, and Medical Studies; Exchange students with a background or anyone with interest in Humanities

Overige informatie

This course is part of the Minor Gender and Diversity.

Aanbevolen voorkennis

Academic skills course (ACVA) passed.

Theorizing Gender and Intersectionality

Vakcode	W_TGI
Studiepunten	6.00
Periode	P1+2+3
Vakniveau	200
Onderwijstaal	Engels
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. J.M. Halsema
Examinator	dr. J.M. Halsema
Betrokken Docenten	dr. J.M. Halsema
Onderwijsvormen	Lecture

Doel vak

Students of this course will acquire the abilities:

1. to recognize and interpret the main theories, concepts and ideas in

gender studies and intersectionality theory;

2. to apply these theories, concepts and ideas to their major;

3. to critically question texts from the perspective of gender, race,

and sexuality;

4. to perceive a problem from different disciplinary perspectives;

5. to define and defend an argument and to communicate the research

conclusions verbally and in writing to an educated audience of different

disciplines.

Inhoud vak

This course introduces into the critical perspectives developed in gender studies and intersectionality theory. The key themes and debates that form the backbone of the minor Gender and Diversity will be introduced in the lectures: the sex/gender debate, social constructivism, sexual difference, performativity, the intersection of race and gender, and of gender and sexuality, the notion ‘intersectionality’.

The aim of the course is to gain an intersectional and multidisciplinary perspective: in the course we will focus upon the interrelations and divergences between gender, race and sexuality and discuss these from the perspective of the various disciplines of the students.

The course does not only aim at introducing these debates theoretically, but at stimulating reflection upon their disciplines by the students.

Aanvullende informatie onderwijsvormen

The main part of the course takes place in September and October (period 1), in the form of lectures and seminars. In period 2 we will have a few meetings to discuss the setup of the final symposium. At the end of period 3 students organize a final Gender and Diversity symposium, that closes of the course and the minor. Part of the program of the symposium is the presentation of a poster in which they demonstrate to be able to analyze a subject/text/film/book in their own field from the angle of gender and intersectionality.

Toetsvorm

· Multidisciplinary group paper (group assignment, 40 % of your final grade) (assessment of learning objectives 2, 3, 4, 5) of 5000 words. Each student writes an individual chapter which is clearly marked. You will receive a grade for your chapter and for the paper as a whole. Your grade is the average of the grade for the paper as a whole and of the individual part.

· Written exam about the texts (Individual assignment: 50%) (assessment of learning objective 1). You need at least 5,5 to pass the course.

· Poster presentation at the final symposium (individual assignment: 10 %) (assessment of learning objective 2 and 5).

Literatuur

- Book "Gender Theory in Troubled Times," Kathleen Lennon & Rachel

Alsop, Cambridge & Medford: Polity Press, 2020 (Purchase yourself)

- Texts of: Sara Ahmed, Anne Fausto-Sterling, Simone de Beauvoir, Kimberle Crenshaw,

Judith Butler, bell hooks, Gloria Wekker, Raewyn Connell, and others.

Aanvullende informatie doelgroep

The course is part of the university minor Gender and Diversity.

Aanbevolen voorkennis

Required is a bachelor 2 level.

Toegepaste Inspanningsfysiologie

Vakcode	B_TIF
Studiepunten	6.00
Periode	P2
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit der Gedrags- en Bewegingswetenschappen
Vakcoördinator	D. Houtkamp MSc
Examinator	D. Houtkamp MSc
Betrokken Docenten	prof. dr. H.A.M. Daanen, dr. R.C.I. Wust, D. Houtkamp MSc, dr. R.P. Goulding
Onderwijsvormen	Practicum, Hoorcollege

Doel vak

Het uitbreiden van inspanningsfysiologische kennis en het toepassen daarvan op vraagstukken binnen de sport en

gezondheid.

Inhoud vak

De verhoogde energiebehoefte van het musculaire systeem als gevolg van fysieke activiteit vraagt van verschillende fysiologische mechanismen een zodanige actie dat homeostase van het interne milieu behouden blijft. Het cardiovasculaire en respiratoire systeem spelen hierin een cruciale rol. De mogelijkheid van deze systemen om in te spelen op de belasting bepaalt in hoge mate de inspanningstolerantie en/of gezondheid van het individu. Er zijn vele factoren die het functioneren van het cardiovasculaire en respiratoire systeem beïnvloeden. Te denken valt aan trainingstoestand, voeding, klimaat, hypo- en hyperbare omstandigheden en sportspecifieke omstandigheden. Daarnaast hebben chronische aandoeningen aan de verschillende systemen grote invloed op de inspanningstolerantie. Ten grondslag aan het functioneren van het musculaire-, cardiovasculaire- en respiratoire systeem liggen de moleculair biologische processen die aanmaak en afbraak van eiwitten reguleren. Inzicht in deze processen maakt duidelijk hoe training en adaptatie aan veranderende omstandigheden werkt. Ook klimatologische omstandigheden hebben invloed op het functioneren van de mens.

In deze cursus wordt aandacht besteed aan factoren die de inspanningstolerantie bepalen, de moleculair biologische processen die trainingseffecten reguleren en de thermofysiologie. De aandacht zal liggen op hoe deze kennis toegepast kan worden binnen sport en

gezondheid. De cursus bevat practica waarin de student vertrouwd wordt gemaakt met de interpretatie van integratieve cardiopulmonaire

inspanningstesten en de thermofysiologie.

Aanvullende informatie onderwijsvormen

De stof wordt aangeboden in de vorm van hoorcolleges in combinatie met practica. Totaal 168 uur, waarvan 42 uur hoorcollege, 12 uur practicum, 111 uur zelfstudie en 3 uur tentamen.

Toetsvorm

Het tentamen is een schriftelijke tentamen met open en gesloten vragen.
De practica zijn verplicht.

Vereiste voorkennis

De kennis van de volgende vakken wordt als bekend verondersteld:

- Inleiding inspanningsfysiologie
- Training en prestatie

Literatuur

McArdle, Katch and Katch. Exercise Physiology: Nutrition, energy and human performance. Williams & Wilkins, ISBN 1-6083-1859-1, 8th or

9th edition, 2014/2022.

Literatuur aangeboden via Canvas.

Afwijkende intekenprocedure

De indeling van practica vindt plaats via Canvas.

Aanbevolen voorkennis

De student moet beschikken over basiskennis van de inspanningsfysiologie (energiesystemen, cardio-pulmonair systeem, training).

Urban Studies

Vakcode	S_UBS
Studiepunten	6.00
Periode	P2+3
Vakniveau	300

Onderwijstaal	Engels
Faculteit	Faculteit der Sociale Wetenschappen
Vakcoördinator	dr. R.A. Gigengack
Examinator	dr. R.A. Gigengack
Betrokken Docenten	dr. A. Smienk, dr. R.A. Gigengack
Onderwijsvormen	Study Group, Lecture

Doel vak

Knowledge and understanding:

1. Students gain knowledge and understanding of the most important research themes, concepts, and debates within the field of urban anthropology.

2. Students can recognize how urban development, inequality and insecurity are spatialized, materialized, reproduced, challenged, and represented in cities of the Global North and the Global South.

Application:

3. Students can apply key concepts from urban anthropology to an urban case study in Amsterdam or another city in the Netherlands.

4. Students hone their qualitative research skills and become attuned to the ethnographic sensitivities that urban settings require.

Making judgements:

5. Students develop a critical attitude towards social, spatial, and material urban dynamics and can formulate their own questions on urban

processes.

6. Students are better able to reflect on their own role as researchers operating in an urban setting.

Communication:

7. Students can report on their respective research projects in verbal and written form.

Learning skills:

8. Students have learned to work in small research teams to carry out a short-term ethnographic research project.

Inhoud vak

Cities are not just static locations; they are also thought-provoking processes. Across the Global North and South, cities shape – and are shaped by – the way people, nature, resources, infrastructures, and ideas flow and settle. Drawing on interdisciplinary and anthropological debates on ‘the urban’, this course delves into the social, spatial, material, and cultural processes that drive and result from patterns of urban development, inequality, and insecurity. These themes are explored through case studies that raise questions on how and why resources and risk are distributed unequally across and within different urban environments. Additionally, we also engage the everyday modes of resistance and negotiation that emerge in response to social and spatial forms of inequality. During the course, students will have the opportunity to conduct their own urban research projects in relation to themes such as housing, gentrification, sustainability, and labour.

Aanvullende informatie onderwijsvormen

Lectures and tutorials

Toetsvorm

Written exam and research project in teams.

Literatuur

To be announced on Canvas.

Aanvullende informatie doelgroep

2nd year students in Cultural Anthropology and Development Sociology;

Students in the Minor Anthropology or the Minor Development and Global Challenges; also open as an elective course for Exchange Students.

Overige informatie

This course fits into several programmes. It is part of the Bachelor Cultural Anthropology and Development Sociology; it is the closing of the theme block “Development”, but in time follows directly on two courses from the theme block “World Making” (in particular Identity, Diversity and Inclusion, and Nation and Migration). The themes of these courses –politics, inequality, development, globalization, diversity, identity, migration– all return in Urban Studies. In the same vein, Urban Studies is the closing of the minor Development and Global Challenges and the minor Anthropology. While Urban Studies is integrated in all these programmes, the course can also be taken as an elective course of its own.

Note that students are expected to attend three preparatory meetings in November-December (the so-called studielint).

Aanbevolen voorkennis

There are no requirements, but ideally students have completed the courses Political and Economic Anthropology, Development and Globalization, and/or Identity, Diversity and Inclusion.

Verhalen uit het Rijksmuseum

Vakcode	L_AABAALG085
Studiepunten	6.00
Periode	P2
Vakniveau	200
Onderwijstaal	Nederlands
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	prof. dr. J.M. Koppenol
Examinator	prof. dr. J.M. Koppenol
Betrokken Docenten	prof. dr. J.M. Koppenol
Onderwijsvormen	Hoorcollege, Werkcollege, Excursie

Doel vak

Na het volgen van het vak Verhalen uit het Rijksmuseum:

• kan de student de geschiedenis van het museum als instituut resumeren, de belangrijkste ontwikkelingen beschrijven in het kader van de bredere cultuurgeschiedenis en die ontwikkelingen kritisch becommentariëren

• kan de student een tentoonstelling analyseren, aan de hand van de storyline, de uitvoering en de publicitaire middelen

• is de student bekend met en heeft geoefend met meerdere tekstgenres die kunnen reflecteren op de tentoonstellingswereld, zoals de recensie, de wervingstekst en het essay.

• kan de student zelf een storyline ontwerpen en schrijven aan de hand van een object of een tentoonstelling.

Inhoud vak

Musea zijn bijzondere instellingen waarin de bezoekers worden uitgedaagd na te denken over zichzelf en de wereld om hen heen. De betekenis en de boodschap die daarbij wordt uitgedragen is niet statisch, maar een dynamisch geheel. Zo is de museale wereld de afgelopen decennia uitgegroeid tot het toneel van maatschappelijke debatten over identiteit, inclusiviteit, representatie en nationaliteit. In dit college verdiepen de studenten zich in de

geschiedenis, de rol en de betekenis van het museum. De hoofdrol is daarbij weggelegd voor het Rijksmuseum in Amsterdam, dat als nationaal platform een bijzondere

positie inneemt en een uitgesproken taak heeft.

Aan de hand van o.a. theorieën van de filosoof Foucault denken we na over de aard van het museum als uitzonderlijke ruimte, en over positie en de dynamische rol van het museum. Daarnaast komen de geschiedenis, de architectuur, de signatuur en de collectie aan bod, en wordt er ingegaan op de veranderende opvattingen over verzamelingen en tentoonstellingen van het Rijksmuseum. Daarbij is er aandacht voor fenomenen als canonvorming, culturele identiteit en de vraag in hoeverre er aandacht

wordt besteed aan de multiculturele samenleving. Welke rol spelen de vaderlandse geschiedenis en de zeventiende-eeuwse schilderkunst in de collectievorming? Hoe zit het met roofkunst? Welke didactische en/of kritische rol speelt het museum?

Studenten leren nadenken over de manier waarop musea hun verhaal vertellen. Hierbij spelen narrativiteit en storytelling een rol: elke tentoonstelling vertelt een verhaal – de keuze van schilderijen en voorwerpen, de wijze waarop ze zijn geordend en de volgorde waarin de bezoekers erlangs worden geleid, aan de hand van de begeleidende teksten, zijn van invloed op het oordeel van de bezoeker en op de boodschap die hij of zij uit de tentoonstelling destilleert. Studenten bezoeken meerdere musea, analyseren de gebruikte narratieven en oefenen zich in het zelf ontwikkelen van een ‘story line’.

Aanvullende informatie onderwijsvormen

hoor/werkcollege (2 uur per week), museumbezoeken, excursies.

Toetsvorm

De toetsing bestaat uit de volgende onderdelen:

• het maken van een tentoonstellingsplan (10%);

• schrijfopdracht: een wervingstekst of creatieve tekst, plus een kritische beschouwing over de waarde van AI (30%);

• afrondend werkstuk over een zelf te kiezen onderwerp een aspect van de collegethematiek wordt uitgewerkt, met gebruikmaking van tenminste drie academische bronnen (60%).

Een deel van de opdrachten wordt in het college gemaakt. Voor alle onderdelen moet minimaal een 5,5 worden behaald.

Literatuur

De te lezen primaire en secundaire teksten worden voor aanvang van de cursus via Canvas bekendgemaakt.

Aanvullende informatie doelgroep

Dit vak is onderdeel van de minor 'De Schrijfacademie'. Deze minor staat open voor alle studenten die hun eerste studiejaar hebben behaald, van binnen en van buiten de VU. Dit vak is ook als los vak te volgen.

Overige informatie

Deze cursus is een verplicht onderdeel van de minor ‘De Schrijfacademie’. Daarnaast volgen de studenten ‘Meesterwerken uit de

literatuur 1 en 2 (3 EC en 6 EC), ‘Multatuli multicultureel’ (6 EC), 'De schrijversparade’ (3 EC) en ‘Creatief schrijven’ (6 EC).

Wereldliteratuur 1

Vakcode	L_AABAALG091
Studiepunten	3.00

Periode	P1
Vakniveau	200
Onderwijstaal	Nederlands
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. J. Grave
Examinator	dr. J. Grave
Betrokken Docenten	dr. E.M. van Opstall, dr. J. Grave
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

Kennismaking met enkele belangrijke periodes en stromingen binnen de Europese literatuur vanaf de Oudheid tot en met de negentiende eeuw aan de hand van klassiek geworden meesterwerken. Aan het eind van de cursus kun je:

- canoniseringsprocessen beschrijven;

- aan de hand van klassiek geworden literaire werken enkele belangrijke periodes en stromingen binnen de (Europese) literatuur vanaf de Oudheid tot heden noemen;

- de inhoud van klassiek geworden literaire werken samenvatten;

- de receptie van klassiek geworden literaire werken beschrijven;

Inhoud vak

Elke week, dat wil zeggen vóór elk college, leest de student een ‘literair meesterwerk’ met de bijbehorende secundaire literatuur. Bij de bespreking van elke tekst gaan we in op de redenen waarom deze als een meesterwerk wordt beschouwd: status in het literaire veld, literaire prijzen, aandacht in wetenschappelijke studies en receptie. Soms zal de docent naast het te lezen meesterwerk ook enkele fragmenten uit andere canonieke teksten uit de betreffende periode/stroming bespreken.

Aanvullende informatie onderwijsvormen

Twee colleges per week: een hoor- en een werkcollege of twee werkcolleges.

Toetsvorm

Tentamen na periode 1 (minimaal 5,5) en tussentijdse schriftelijke opdrachten (moeten voldoende zijn). De opdrachten dienen te worden ingeleverd voor aanvang van het betreffende college. De opdrachten tellen voor 10% mee van het eindcijfer. Verder: actieve participatie, blijkend uit inbreng tijdens de discussie over de te lezen teksten.

Literatuur

Na een algemene inleiding over theoretische kwesties als periodisering, classificering en canonvorming worden vijf werken besproken. Bij het publiceren van de studiegids waren nog niet alle keuzes van de docenten voor een meesterwerk bekend. De definitieve lijst zal zo snel mogelijk op Canvas bekend worden gemaakt, maar in elk geval worden de volgende werken gelezen: Thomas More (Utopia), Ivan Toergenjev (Eerste liefde) en Gustave Flaubert (Madame Bovary – in de vertaling van Hans van Pinxteren). De teksten moeten in Nederlandse vertaling gelezen worden. De te lezen secundaire literatuur wordt via Canvas bekendgemaakt.

Aanvullende informatie doelgroep

De minor staat open voor alle studenten die hun eerste studiejaar hebben afgerond. Dit vak is ook los te volgen.

Deze module is een verplicht onderdeel van de minoren Nederlands ‘Aan de slag met literatuur’ en ‘De schrijfacademie’. Een vervolg op dit college is ‘Wereldliteratuur 2’ (periode 2+3, 6 EC). Daarnaast zijn de onderdelen ‘De schrijversparade’ (periode 1, 3 EC) en ‘Creatief schrijven’ (periode 2, 6 EC) verplichte onderdelen binnen deze minoren. De resterende vakken van deze minoren verschillen van elkaar. Zie verder de studiegids.

Overige informatie

Er is verplichte aanwezigheid (minimaal 80% aanwezigheid).

Dit onderdeel vormt inhoudelijk één geheel met Wereldliteratuur 2, maar kan zelfstandig gevolgd worden. In dit

eerste deel komt de literatuur tot en met de negentiende eeuw aan bod; in het tweede deel staan de twintigste en de eenentwintigste eeuw centraal.

Wereldliteratuur 2

Vakcode	L_AABAALG094
Studiepunten	6.00
Periode	P2+3
Vakniveau	300
Onderwijstaal	Nederlands
Faculteit	Faculteit der Geesteswetenschappen
Vakcoördinator	dr. J. Grave
Examinator	dr. J. Grave
Betrokken Docenten	dr. J. Grave
Onderwijsvormen	Hoorcollege, Werkcollege

Doel vak

Kennismaking met de belangrijkste periodes en stromingen binnen de (grotendeels) Europese literatuur van de twintigste en de eenentwintigste eeuw aan de hand van klassiek geworden meesterwerken. Aan het eind van de cursus kun je:

- canoniseringsprocessen beschrijven;

- aan de hand van klassiek geworden literaire werken enkele belangrijke periodes en stromingen binnen de (Europese) literatuur vanaf de twintigste eeuw tot heden noemen;

- de inhoud van klassiek geworden literaire werken samenvatten;

- de receptie van klassiek geworden literaire werken beschrijven.

Inhoud vak

Elke week, dat wil zeggen vóór elk college, lees je een literair werk uit de wereldliteratuur met bijbehorende secundaire literatuur. Bij de bespreking van elke tekst gaan we in op de redenen waarom deze als een meesterwerk wordt beschouwd: status in het literaire veld, literaire prijzen, aandacht in wetenschappelijke studies, receptie. Soms zal de docent naast het te lezen meesterwerk ook enkele fragmenten uit andere (canonieke) teksten uit de betreffende periode/stroming bespreken.

Aanvullende informatie onderwijsvormen

Twee colleges per week (hoor- en werkcollege of twee werkcolleges), met opdrachten en discussie, eigen presentatie van een werk uit de wereldliteratuur in periode 3. Het rooster voor deze presentaties (twee colleges per week) wordt op Canvas bekendgemaakt.

Toetsvorm

Schriftelijk tentamen aan het eind van periode 2 (70% van het eindcijfer). Voor elk college levert de student de leesopdrachten in (ze tellen voor 10% mee). Mondelinge presentatie in periode 3 (20% van het eindcijfer). Voor het schriftelijk tentamen en de mondelinge presentatie moet minimaal een 5,5 worden behaald.

Vereiste voorkennis

Het vak Werelditeratuur 2 is verplicht voor studenten die de hele minor ‘Aan de slag met literatuur’ of ‘De schrijfacademie’ volgen.

Literatuur

In dit college worden zes teksten besproken (en mogelijk nog fragmenten van andere teksten). Bij het publiceren van de studiegids waren nog niet alle keuzes van de docenten voor de tekst bekend. De definitieve lijst zal zo snel mogelijk op Canvas bekend worden gemaakt, maar in elk geval komen de volgende werken aan bod: Carry van Bruggen (*Eva*), Frans Kellendonk (*Mystiek lichaam*), Maria Dermoût (De tienduizend dingen) en W.G. Sebald (*Austerlitz*).

De teksten moeten in het Nederlands gelezen worden. De te lezen secundaire literatuur wordt via Canvas

bekendgemaakt.

Aanvullende informatie doelgroep

De minor staat open voor alle studenten die hun eerste studiejaar hebben afgerond. Dit vak is ook los te volgen.

Overige informatie

Deze module is een verplicht onderdeel van de minoren ‘De schrijfacademie’ en ‘Aan de slag met literatuur’. Het college is een vervolg op Wereldliteratuur 1.

Er is verplichte aanwezigheid (minimaal 80%). Dit onderdeel vormt inhoudelijk één geheel met Wereldliteratuur 1, maar kan zelfstandig gevolgd worden. In het eerste deel komt de literatuur van de Klassieke Oudheid tot en met de negentiende eeuw aan bod; in dit tweede deel staan de twintigste en de eenentwintigste eeuw centraal.

Aanbevolen voorkennis

Het vak Werelditeratuur 2 is verplicht voor studenten die de hele minor ‘Aan de slag met literatuur’ of ‘De schrijfacademie’ volgen. Het vak Wereldliteratuur 1 strekt tot aanbeveling voor studenten die deze module los willen volgen, maar is niet verplicht voor deze studenten.